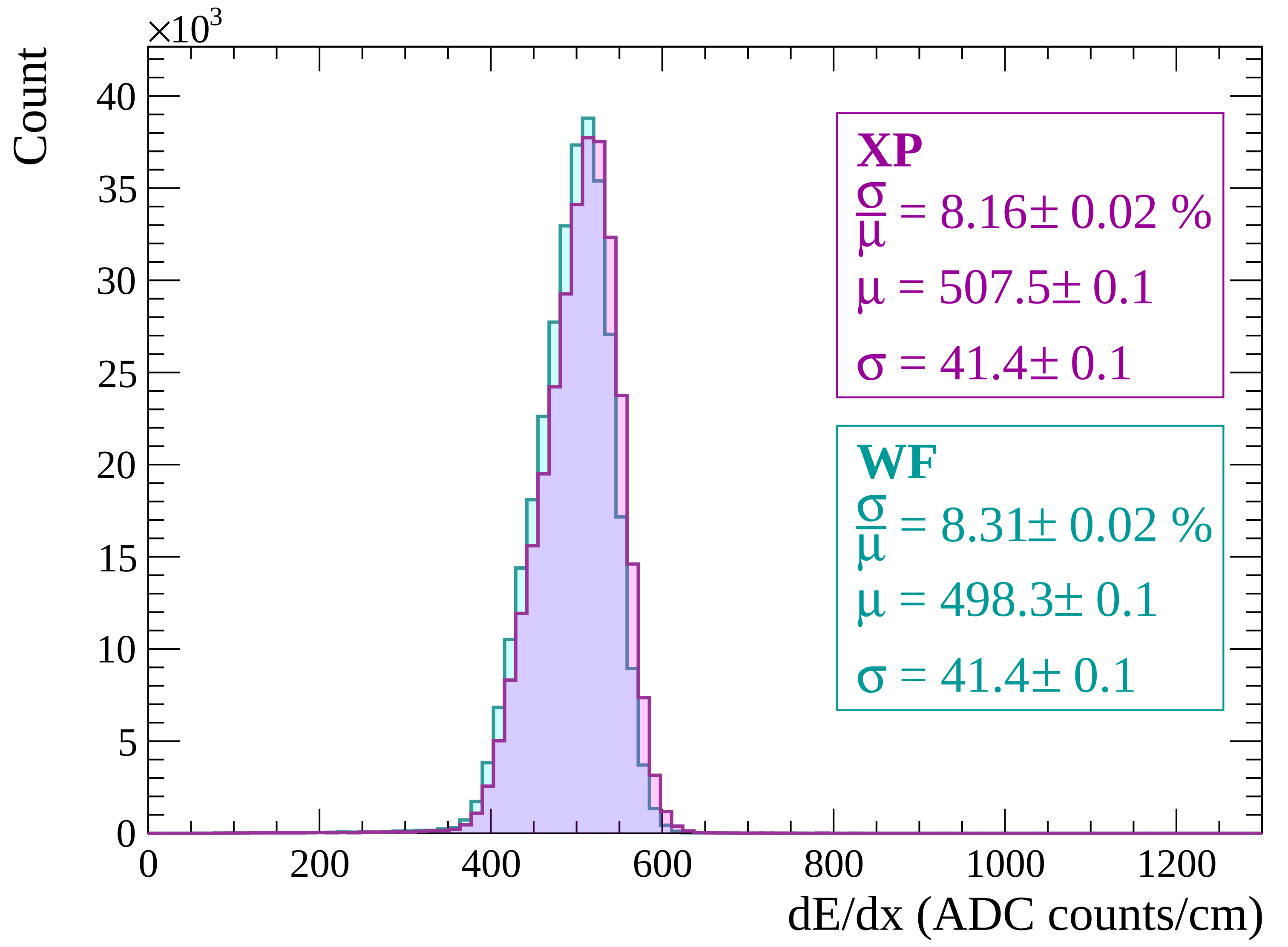
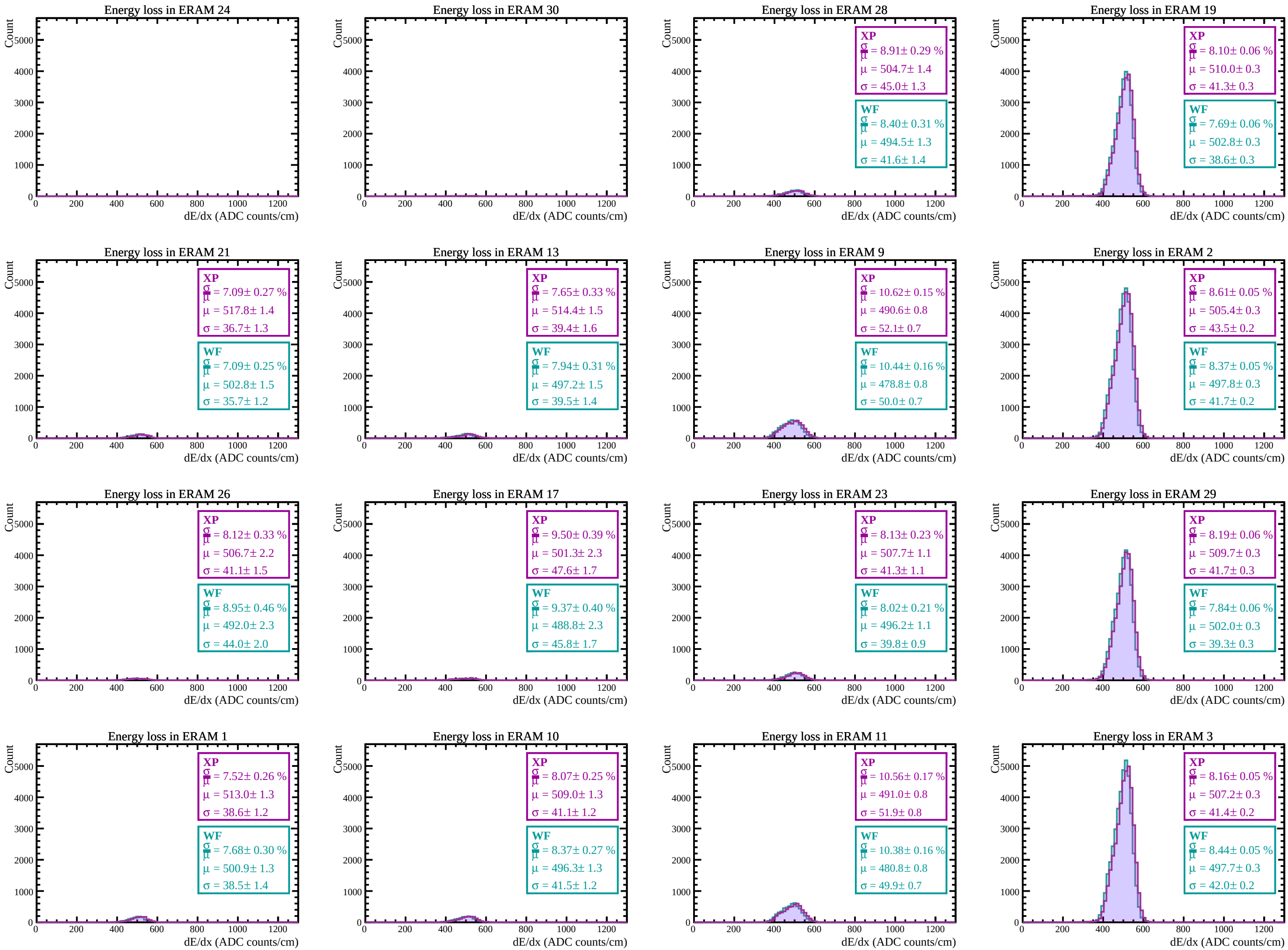
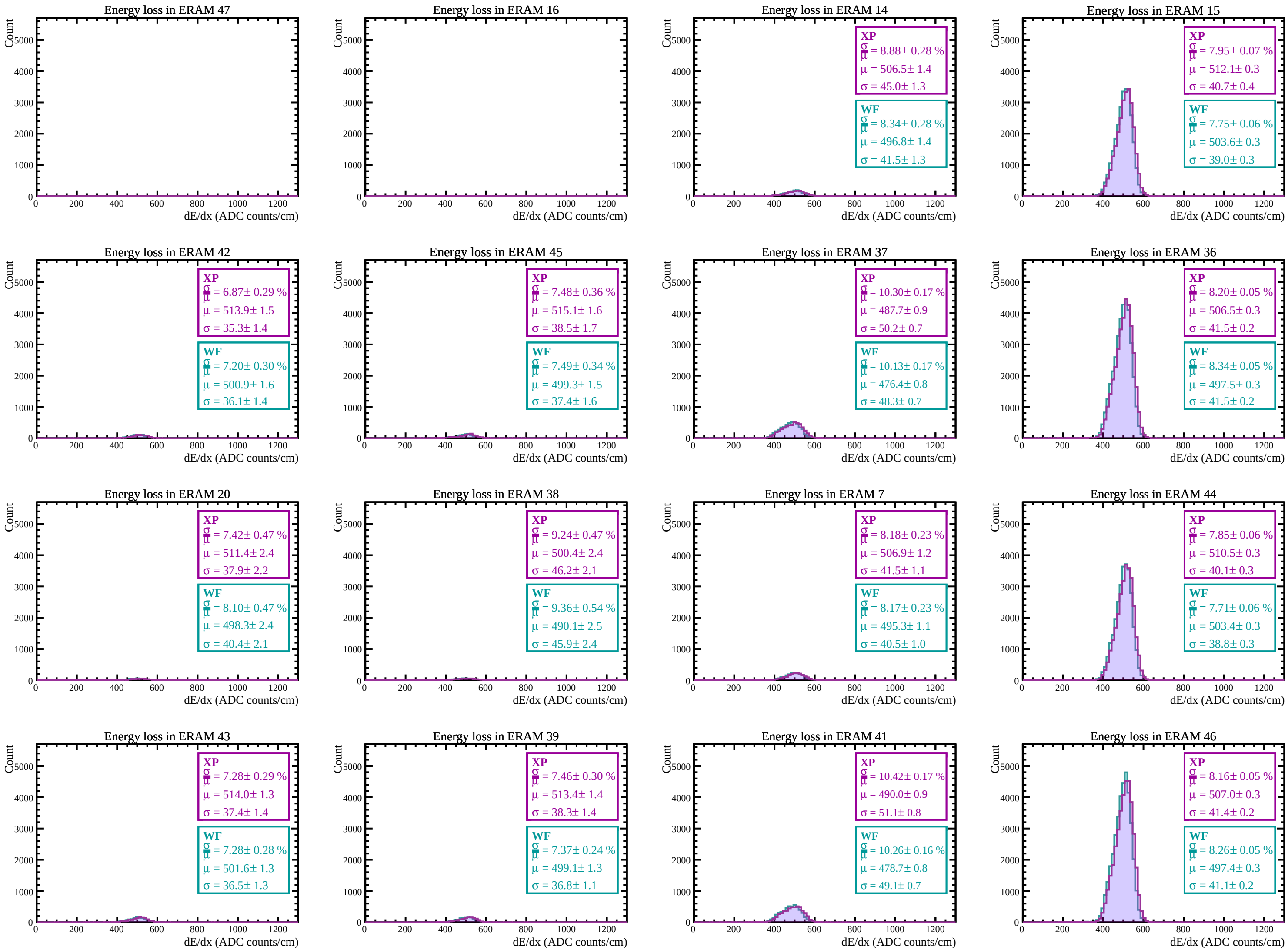
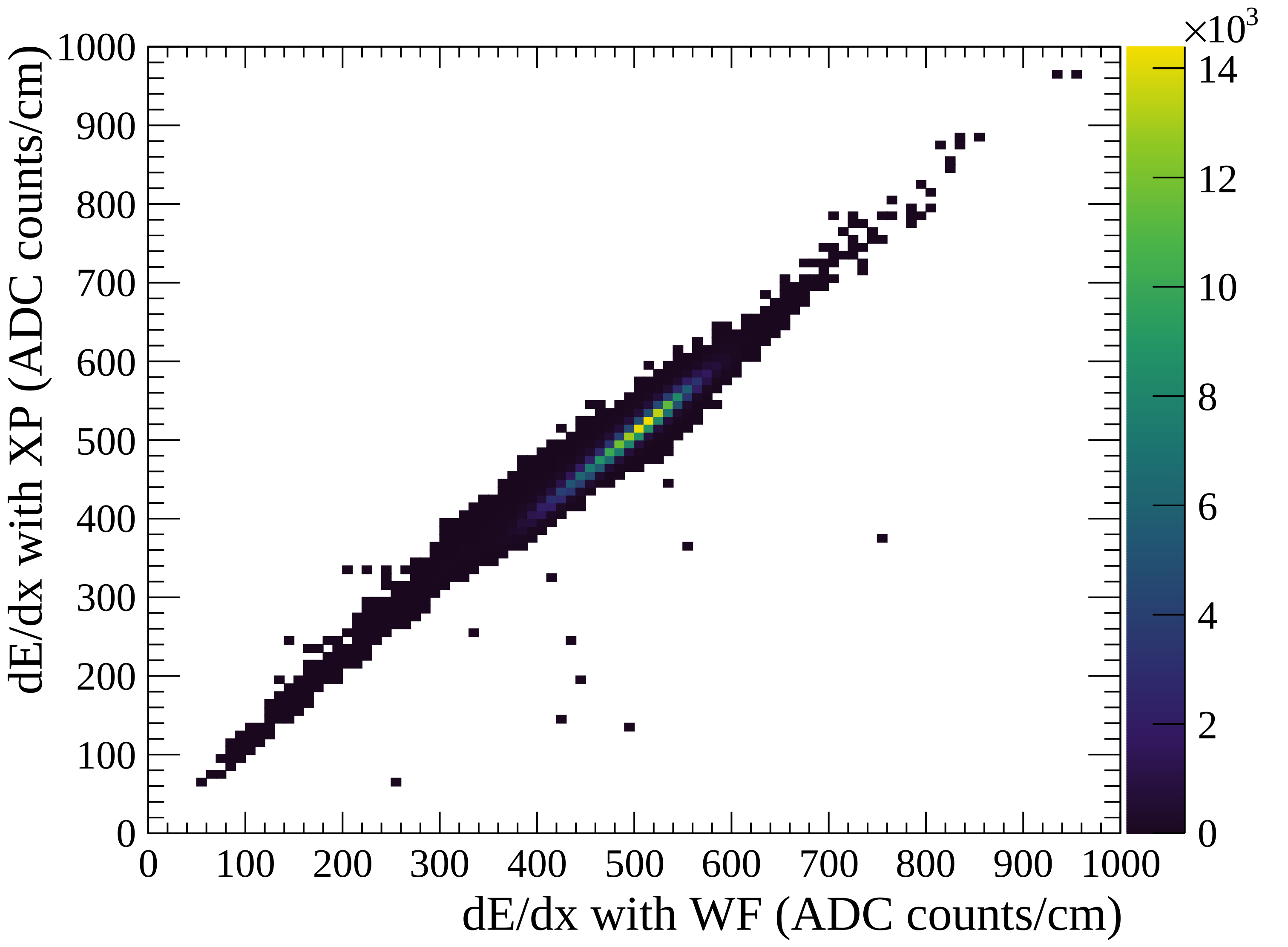


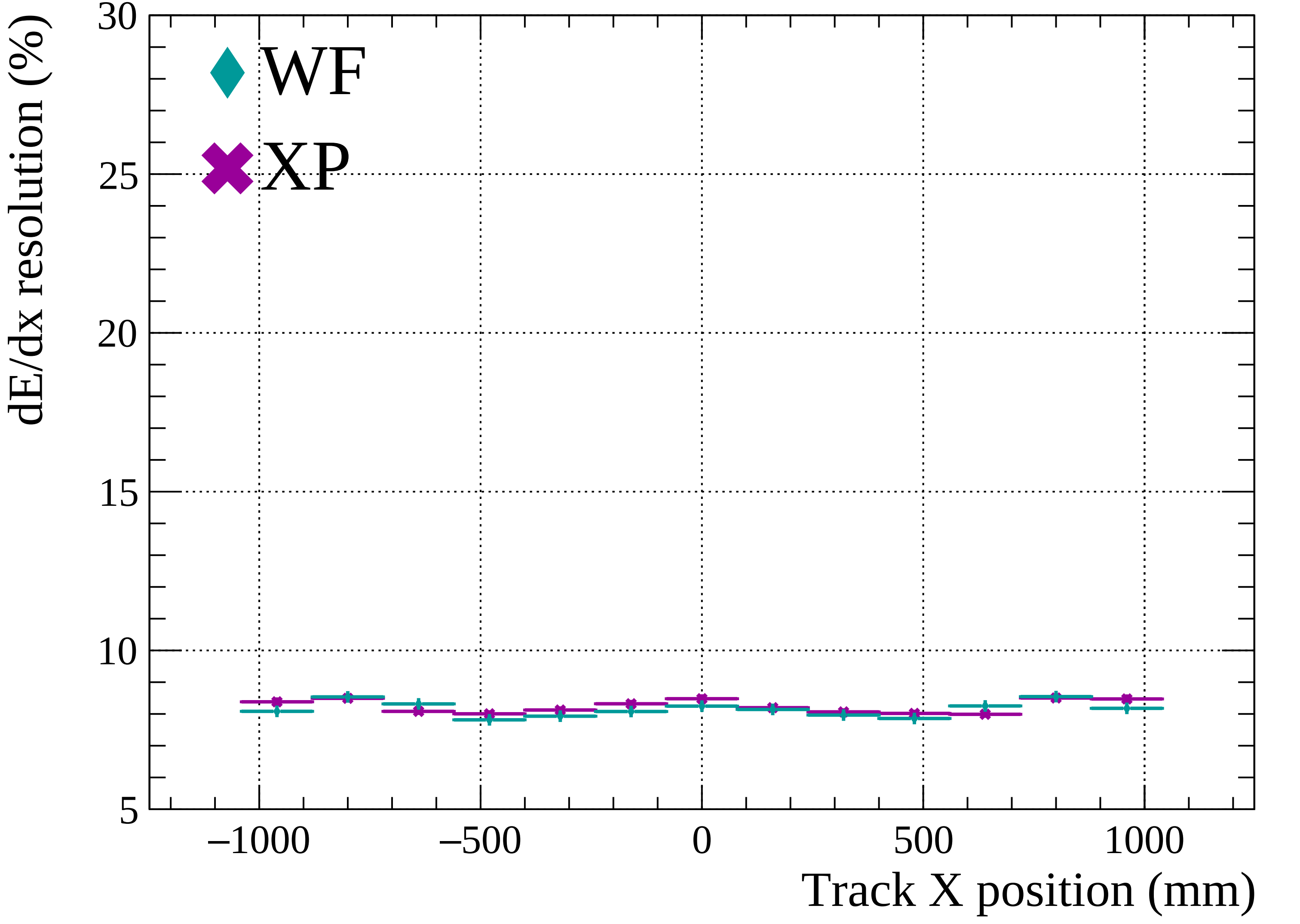
Count
×10^3
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.16± 0.02 %
μ = 507.5± 0.1
σ = 41.4± 0.1
WF
σ/μ = 8.31± 0.02 %
μ = 498.3± 0.1
σ = 41.4± 0.1


Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
XP
σ/μ = 8.91± 0.29 %
μ = 504.7± 1.4
σ = 45.0± 1.3
WF
σ/μ = 8.40± 0.31 %
μ = 494.5± 1.3
σ = 41.6± 1.4
Energy loss in ERAM 19
XP
σ/μ = 8.10± 0.06 %
μ = 510.0± 0.3
σ = 41.3± 0.3
WF
σ/μ = 7.69± 0.06 %
μ = 502.8± 0.3
σ = 38.6± 0.3
Energy loss in ERAM 21
XP
σ/μ = 7.09± 0.27 %
μ = 517.8± 1.4
σ = 36.7± 1.3
WF
σ/μ = 7.09± 0.25 %
μ = 502.8± 1.5
σ = 35.7± 1.2
Energy loss in ERAM 13
XP
σ/μ = 7.65± 0.33 %
μ = 514.4± 1.5
σ = 39.4± 1.6
WF
σ/μ = 7.94± 0.31 %
μ = 497.2± 1.5
σ = 39.5± 1.4
Energy loss in ERAM 9
XP
σ/μ = 10.62± 0.15 %
μ = 490.6± 0.8
σ = 52.1± 0.7
WF
σ/μ = 10.44± 0.16 %
μ = 478.8± 0.8
σ = 50.0± 0.7
Energy loss in ERAM 2
XP
σ/μ = 8.61± 0.05 %
μ = 505.4± 0.3
σ = 43.5± 0.2
WF
σ/μ = 8.37± 0.05 %
μ = 497.8± 0.3
σ = 41.7± 0.2
Energy loss in ERAM 26
XP
σ/μ = 8.12± 0.33 %
μ = 506.7± 2.2
σ = 41.1± 1.5
WF
σ/μ = 8.95± 0.46 %
μ = 492.0± 2.3
σ = 44.0± 2.0
Energy loss in ERAM 17
XP
σ/μ = 9.50± 0.39 %
μ = 501.3± 2.3
σ = 47.6± 1.7
WF
σ/μ = 9.37± 0.40 %
μ = 488.8± 2.3
σ = 45.8± 1.7
Energy loss in ERAM 23
XP
σ/μ = 8.13± 0.23 %
μ = 507.7± 1.1
σ = 41.3± 1.1
WF
σ/μ = 8.02± 0.21 %
μ = 496.2± 1.1
σ = 39.8± 0.9
Energy loss in ERAM 29
XP
σ/μ = 8.19± 0.06 %
μ = 509.7± 0.3
σ = 41.7± 0.3
WF
σ/μ = 7.84± 0.06 %
μ = 502.0± 0.3
σ = 39.3± 0.3
Energy loss in ERAM 1
XP
σ/μ = 7.52± 0.26 %
μ = 513.0± 1.3
σ = 38.6± 1.2
WF
σ/μ = 7.68± 0.30 %
μ = 500.9± 1.3
σ = 38.5± 1.4
Energy loss in ERAM 10
XP
σ/μ = 8.07± 0.25 %
μ = 509.0± 1.3
σ = 41.1± 1.2
WF
σ/μ = 8.37± 0.27 %
μ = 496.3± 1.3
σ = 41.5± 1.2
Energy loss in ERAM 11
XP
σ/μ = 10.56± 0.17 %
μ = 491.0± 0.8
σ = 51.9± 0.8
WF
σ/μ = 10.38± 0.16 %
μ = 480.8± 0.8
σ = 49.9± 0.7
Energy loss in ERAM 3
XP
σ/μ = 8.16± 0.05 %
μ = 507.2± 0.3
σ = 41.4± 0.2
WF
σ/μ = 8.44± 0.05 %
μ = 497.7± 0.3
σ = 42.0± 0.2
Count
dE/dx (ADC counts/cm)

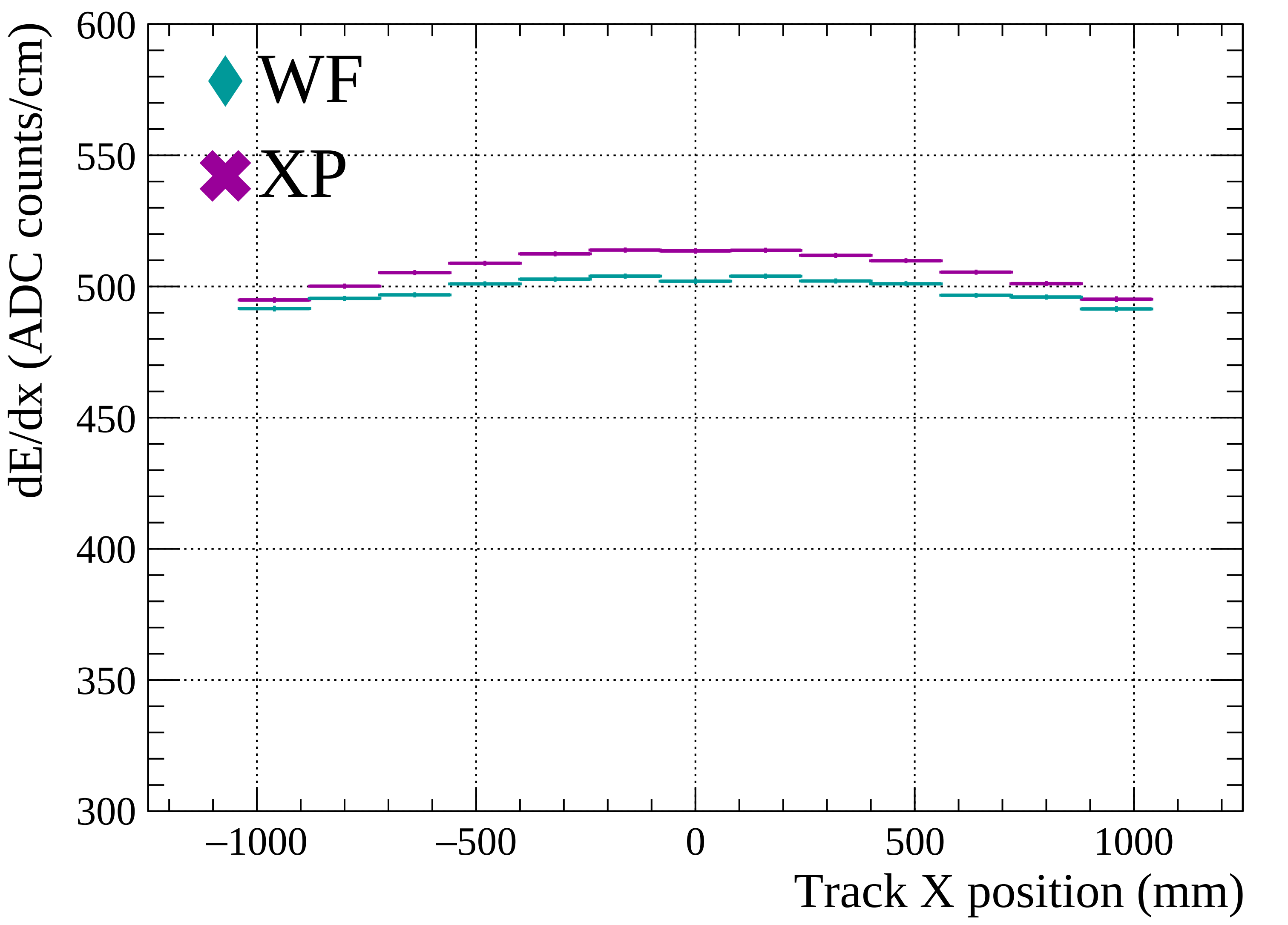

Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
XP
σ/μ = 8.88± 0.28 %
μ = 506.5± 1.4
σ = 45.0± 1.3
WF
σ/μ = 8.34± 0.28 %
μ = 496.8± 1.4
σ = 41.5± 1.3
Energy loss in ERAM 15
XP
σ/μ = 7.95± 0.07 %
μ = 512.1± 0.3
σ = 40.7± 0.4
WF
σ/μ = 7.75± 0.06 %
μ = 503.6± 0.3
σ = 39.0± 0.3
Energy loss in ERAM 42
XP
σ/μ = 6.87± 0.29 %
μ = 513.9± 1.5
σ = 35.3± 1.4
WF
σ/μ = 7.20± 0.30 %
μ = 500.9± 1.6
σ = 36.1± 1.4
Energy loss in ERAM 45
XP
σ/μ = 7.48± 0.36 %
μ = 515.1± 1.6
σ = 38.5± 1.7
WF
σ/μ = 7.49± 0.34 %
μ = 499.3± 1.5
σ = 37.4± 1.6
Energy loss in ERAM 37
XP
σ/μ = 10.30± 0.17 %
μ = 487.7± 0.9
σ = 50.2± 0.7
WF
σ/μ = 10.13± 0.17 %
μ = 476.4± 0.8
σ = 48.3± 0.7
Energy loss in ERAM 36
XP
σ/μ = 8.20± 0.05 %
μ = 506.5± 0.3
σ = 41.5± 0.2
WF
σ/μ = 8.34± 0.05 %
μ = 497.5± 0.3
σ = 41.5± 0.2
Energy loss in ERAM 20
XP
σ/μ = 7.42± 0.47 %
μ = 511.4± 2.4
σ = 37.9± 2.2
WF
σ/μ = 8.10± 0.47 %
μ = 498.3± 2.4
σ = 40.4± 2.1
Energy loss in ERAM 38
XP
σ/μ = 9.24± 0.47 %
μ = 500.4± 2.4
σ = 46.2± 2.1
WF
σ/μ = 9.36± 0.54 %
μ = 490.1± 2.5
σ = 45.9± 2.4
Energy loss in ERAM 7
XP
σ/μ = 8.18± 0.23 %
μ = 506.9± 1.2
σ = 41.5± 1.1
WF
σ/μ = 8.17± 0.23 %
μ = 495.3± 1.1
σ = 40.5± 1.0
Energy loss in ERAM 44
XP
σ/μ = 7.85± 0.06 %
μ = 510.5± 0.3
σ = 40.1± 0.3
WF
σ/μ = 7.71± 0.06 %
μ = 503.4± 0.3
σ = 38.8± 0.3
Energy loss in ERAM 43
XP
σ/μ = 7.28± 0.29 %
μ = 514.0± 1.3
σ = 37.4± 1.4
WF
σ/μ = 7.28± 0.28 %
μ = 501.6± 1.3
σ = 36.5± 1.3
Energy loss in ERAM 39
XP
σ/μ = 7.46± 0.30 %
μ = 513.4± 1.4
σ = 38.3± 1.4
WF
σ/μ = 7.37± 0.24 %
μ = 499.1± 1.3
σ = 36.8± 1.1
Energy loss in ERAM 41
XP
σ/μ = 10.42± 0.17 %
μ = 490.0± 0.9
σ = 51.1± 0.8
WF
σ/μ = 10.26± 0.16 %
μ = 478.7± 0.8
σ = 49.1± 0.7
Energy loss in ERAM 46
XP
σ/μ = 8.16± 0.05 %
μ = 507.0± 0.3
σ = 41.4± 0.2
WF
σ/μ = 8.26± 0.05 %
μ = 497.4± 0.3
σ = 41.1± 0.2
Count
dE/dx (ADC counts/cm)



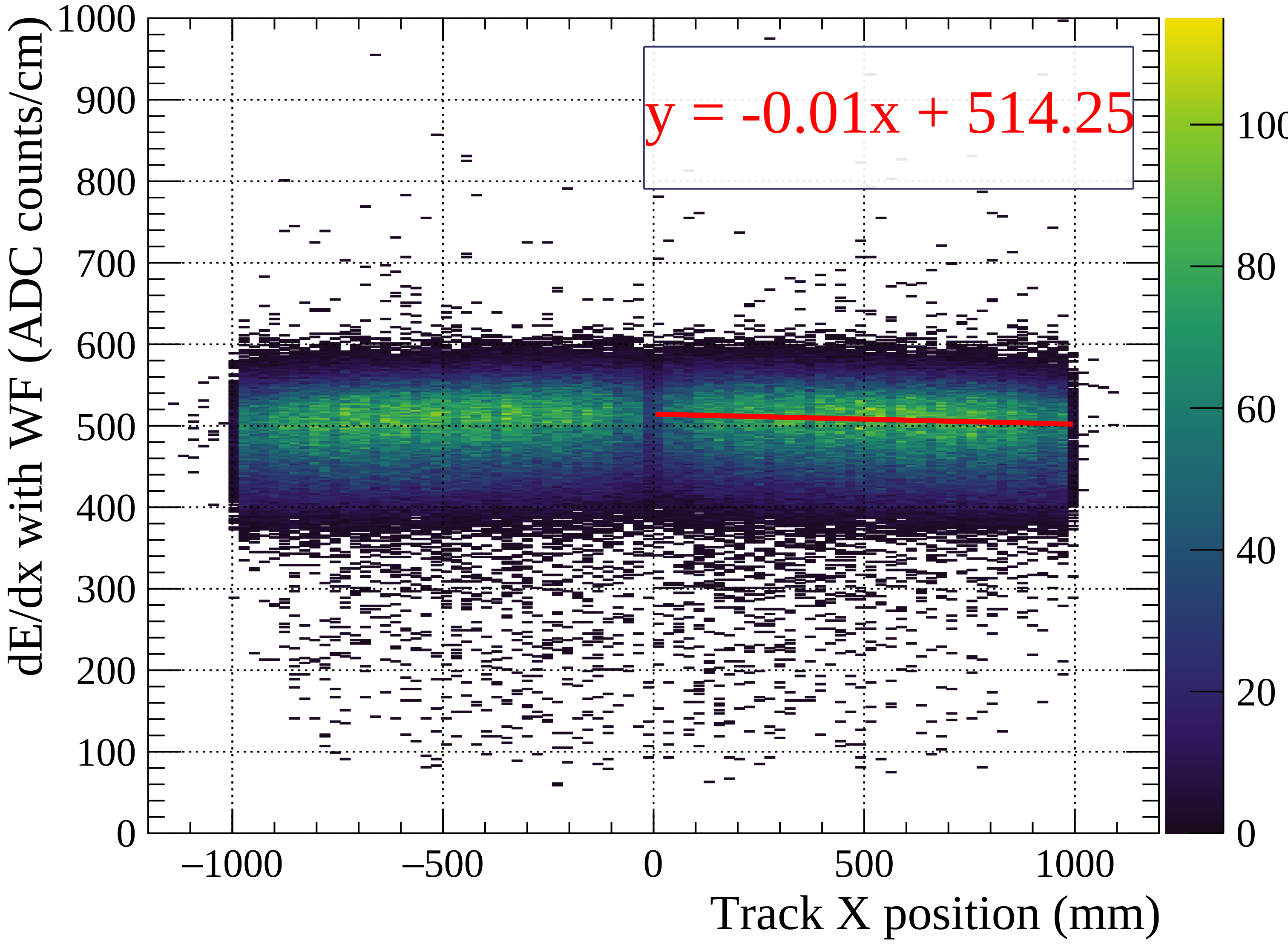
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3
0
100
200
300
400
500
600
700
800
900
1000
0
2
4
6
8
10
12
14



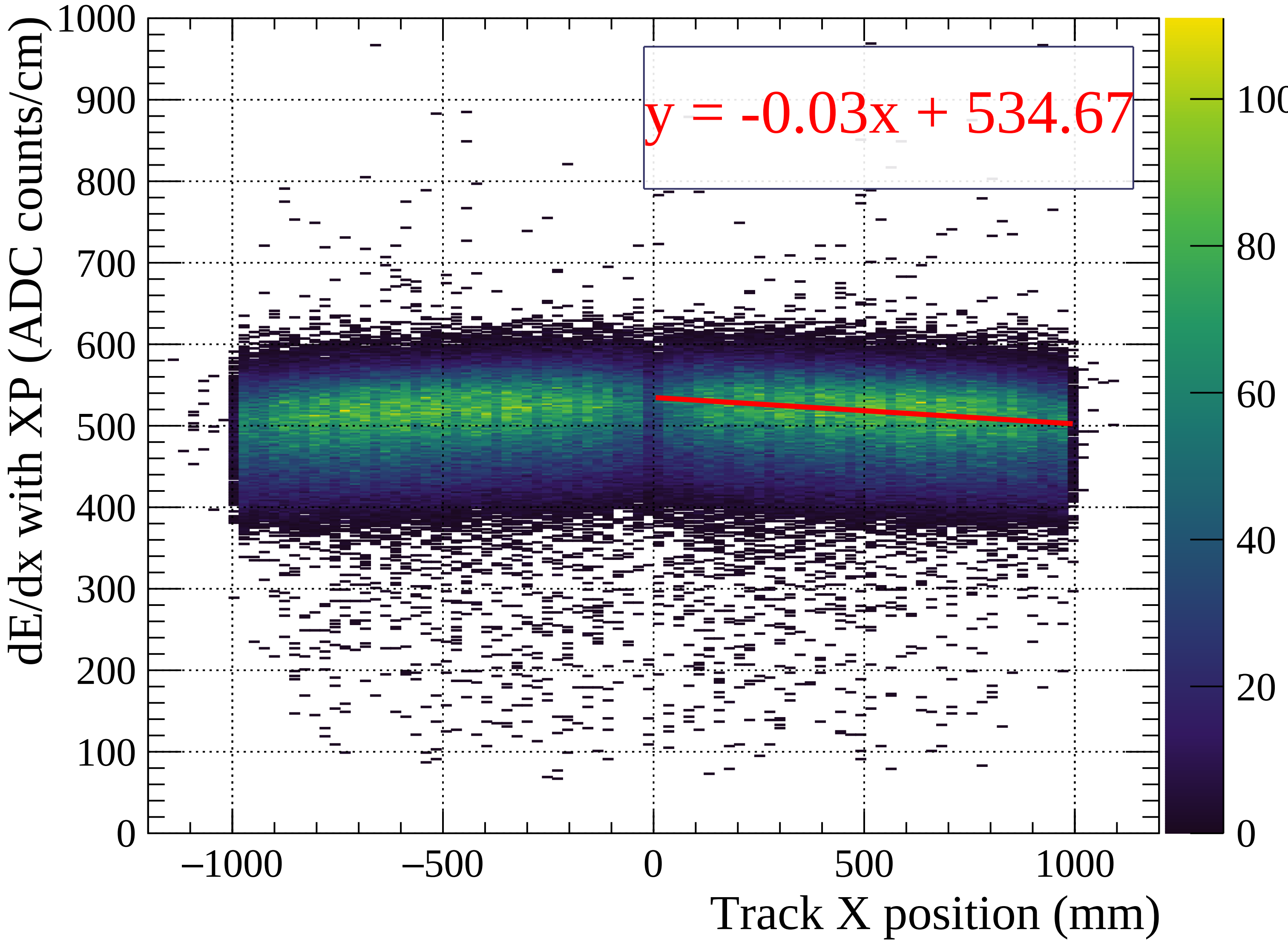
WF
XP
dE/dx resolution (%)
Track X position (mm)
30
25
20
15
10
5
−1000
−500
0
500
1000

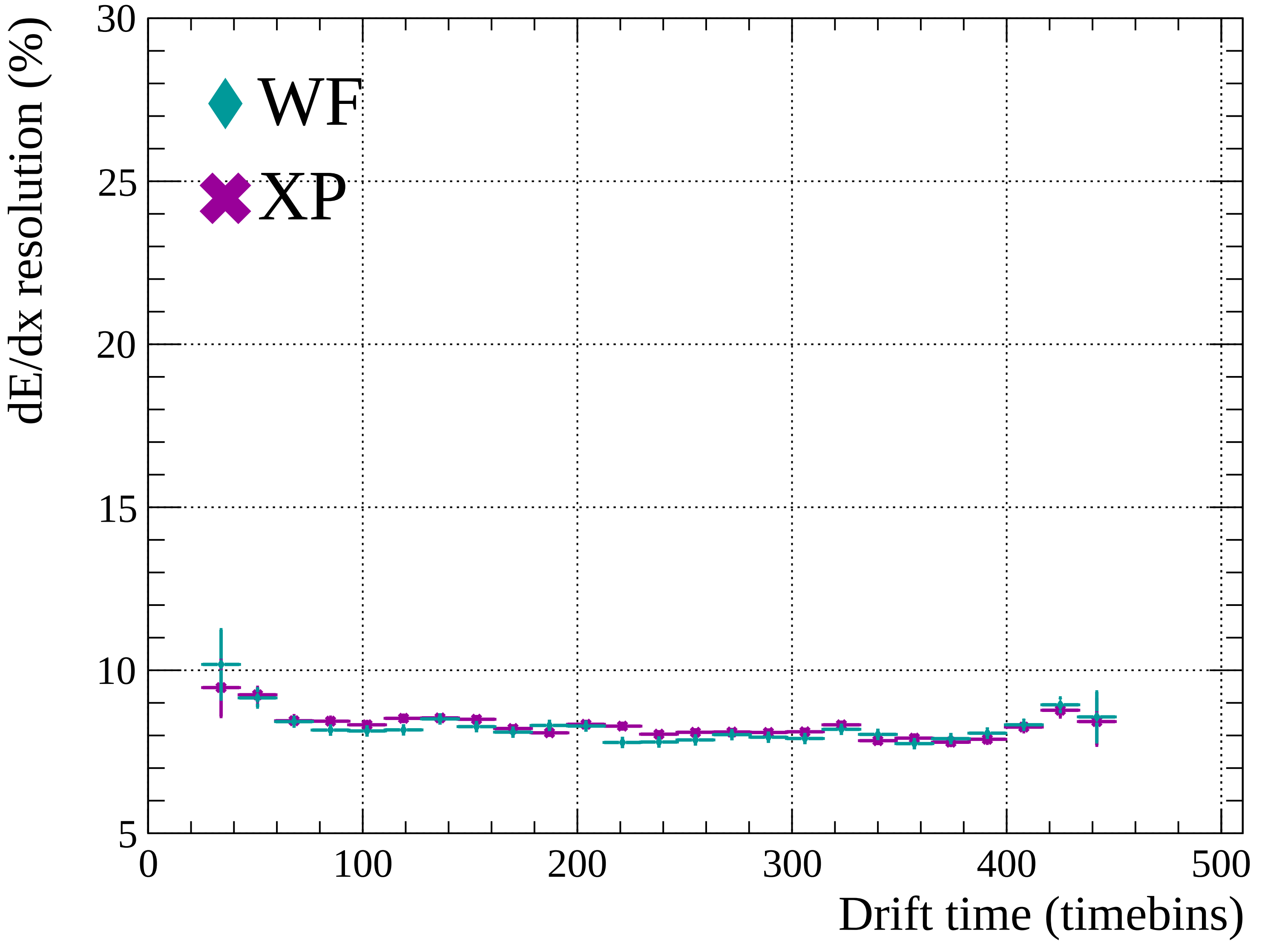


dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−1000
−500
0
500
1000
Track X position (mm)

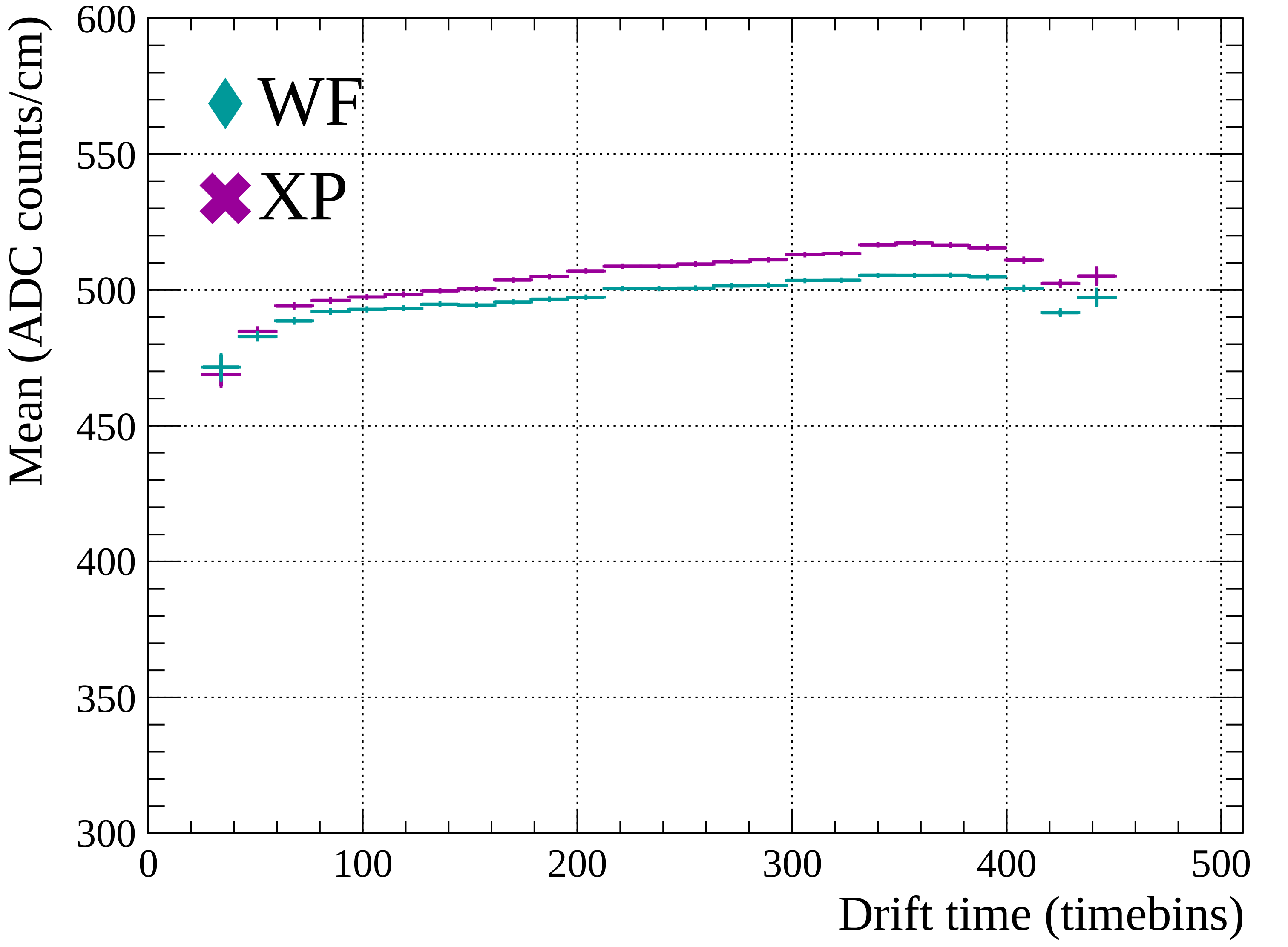


y = -0.01x + 514.25
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
20
40
60
80
100

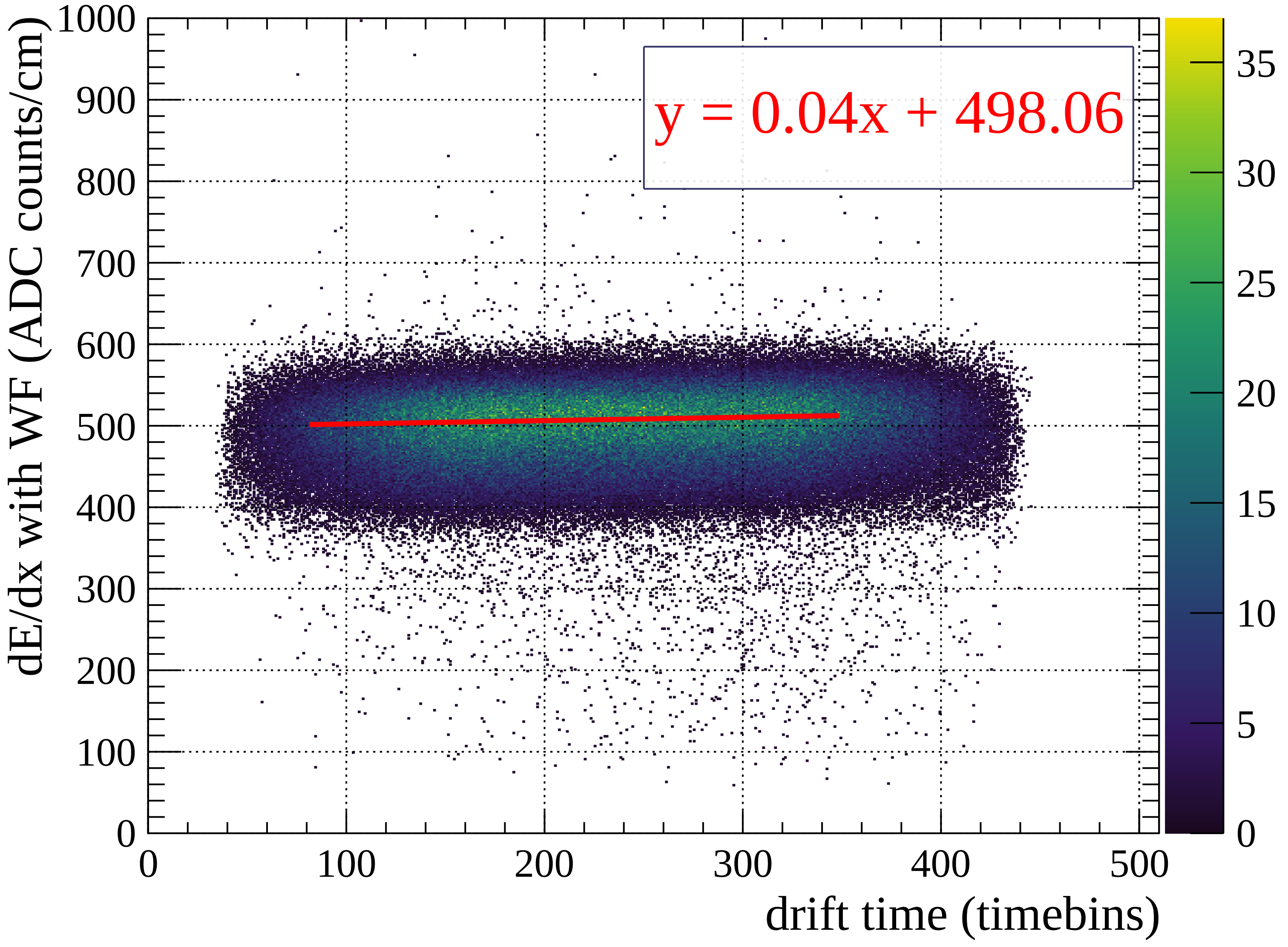

y = -0.03x + 534.67
dE/dx with XP (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
20
40
60
80
100

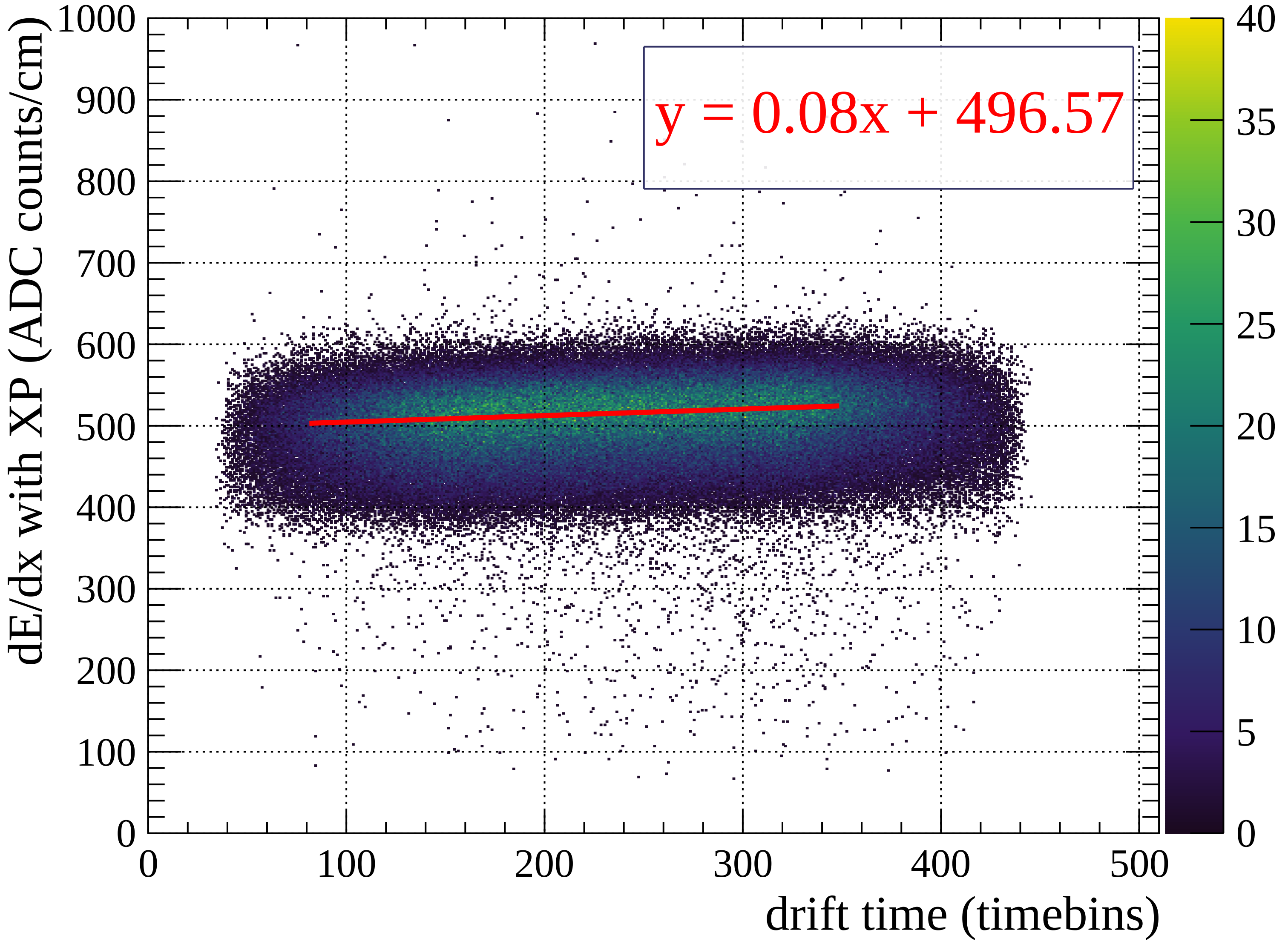


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
100
200
300
400
500
Drift time (timebins)

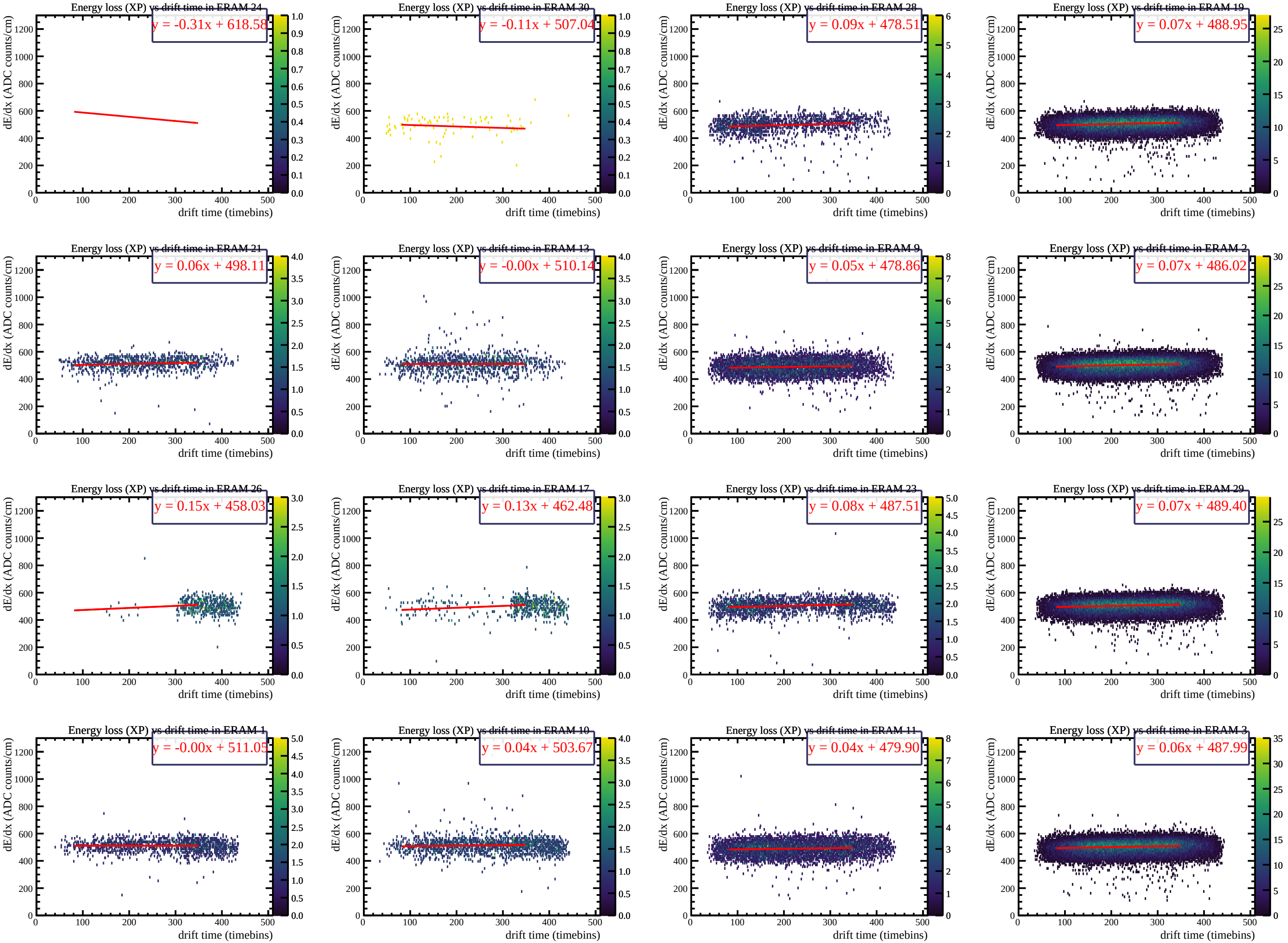


Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
100
200
300
400
500
Drift time (timebins)

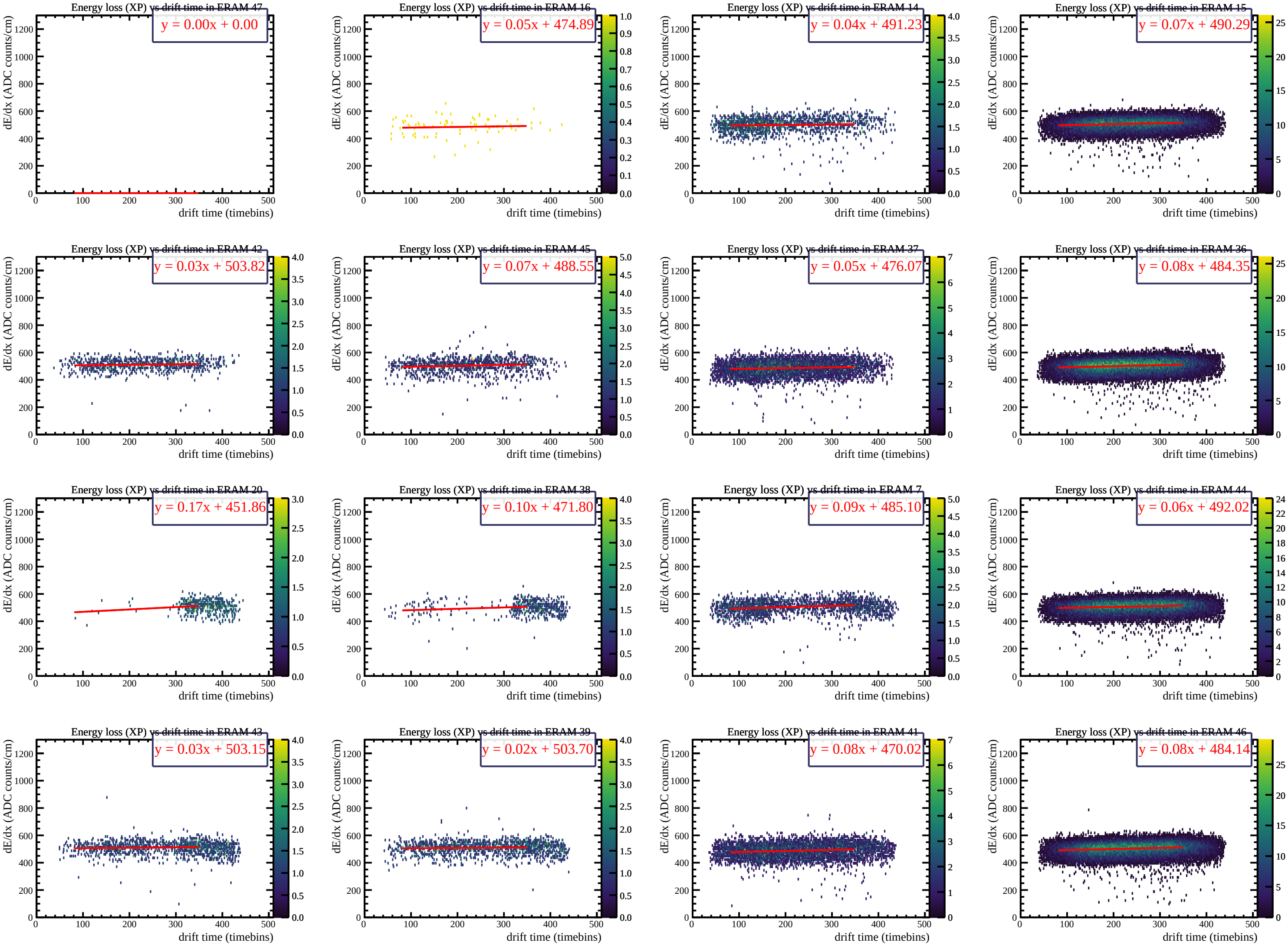

y = 0.04x + 498.06
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30
35

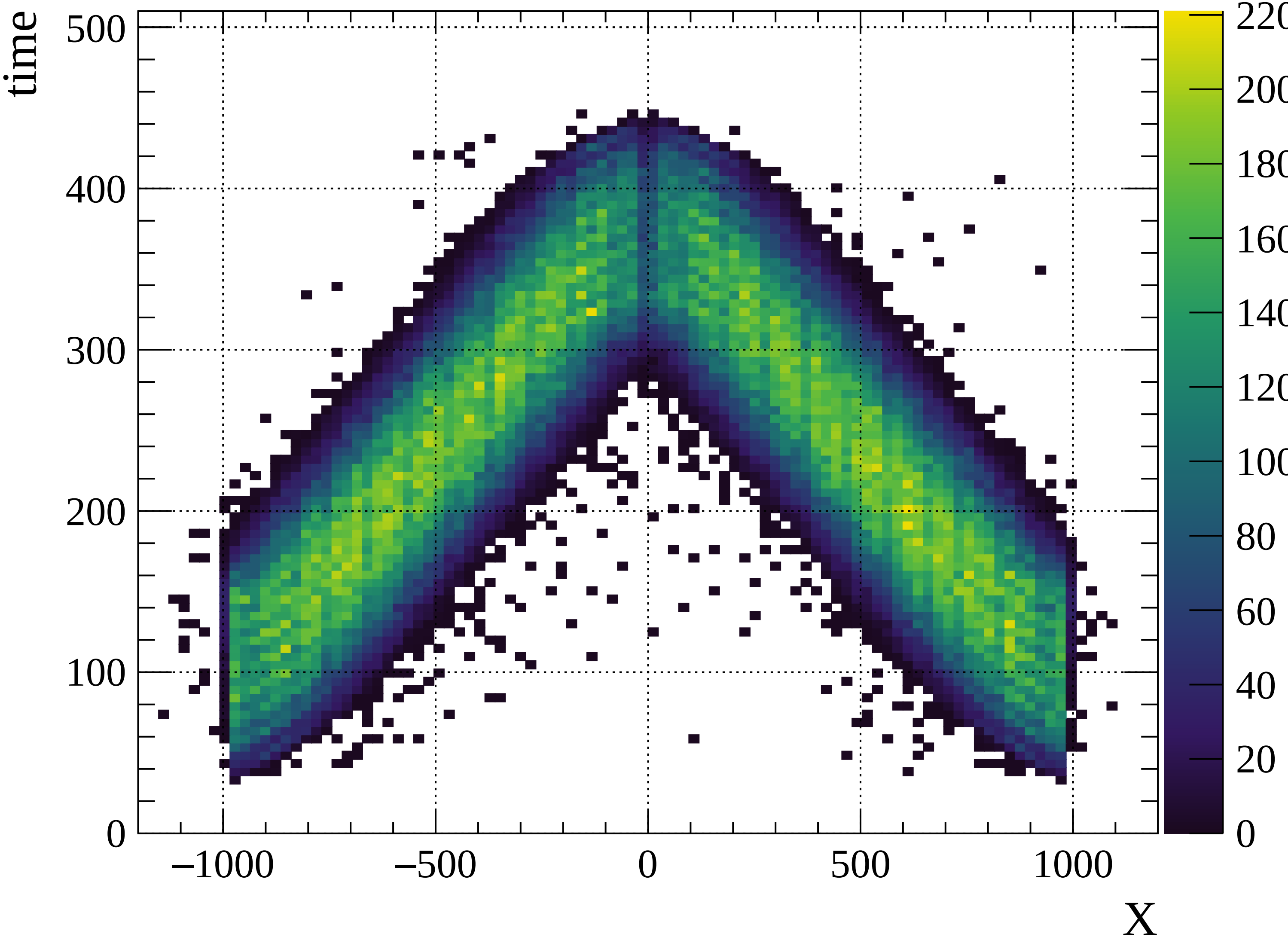


y = 0.08x + 496.57
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30
35
40

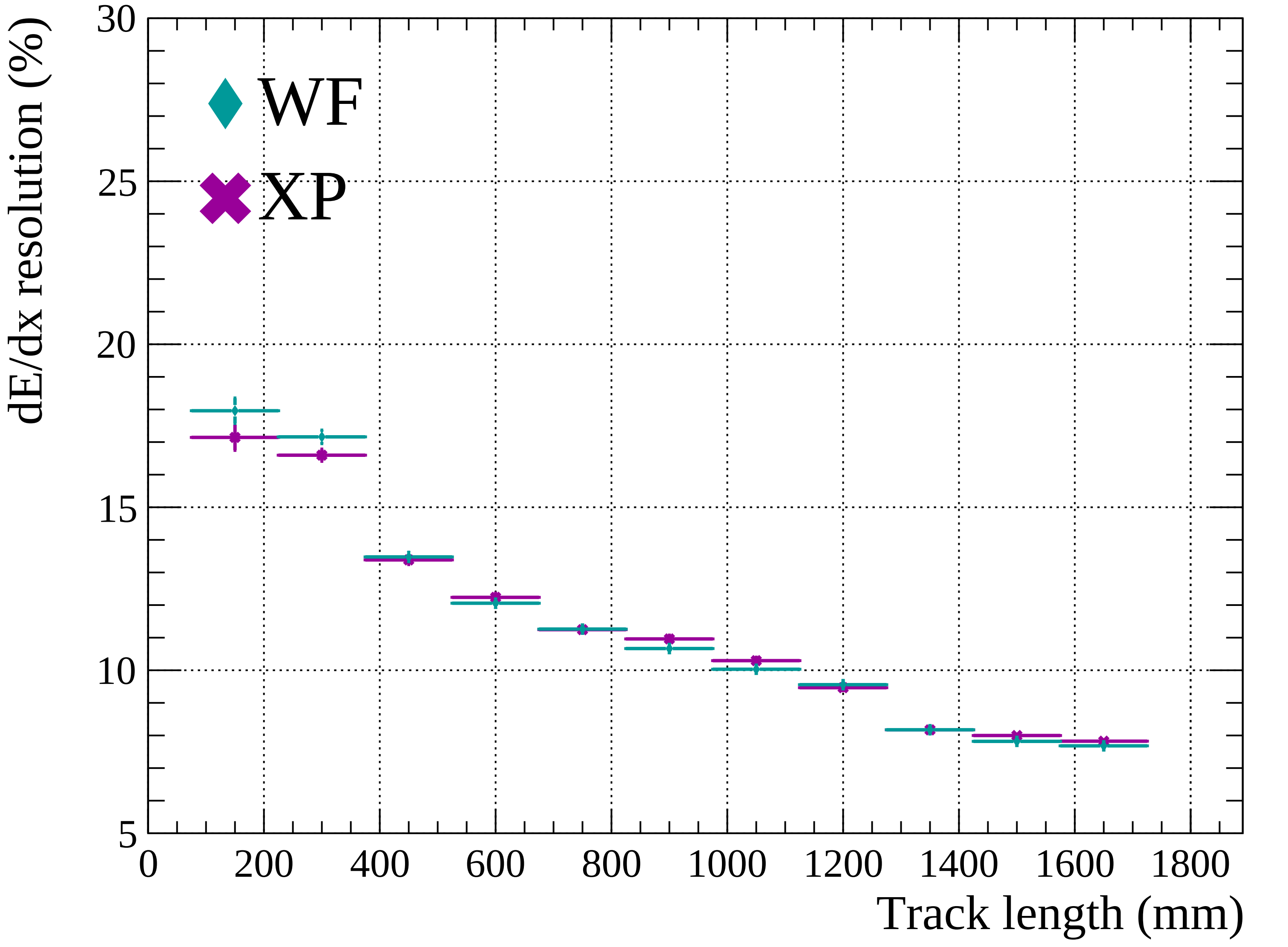


Energy loss (XP) vs drift time in ERAM 24
y = -0.31x + 618.58
Energy loss (XP) vs drift time in ERAM 30
y = -0.11x + 507.04
Energy loss (XP) vs drift time in ERAM 28
y = 0.09x + 478.51
Energy loss (XP) vs drift time in ERAM 19
y = 0.07x + 488.95
Energy loss (XP) vs drift time in ERAM 21
y = 0.06x + 498.11
Energy loss (XP) vs drift time in ERAM 13
y = -0.00x + 510.14
Energy loss (XP) vs drift time in ERAM 9
y = 0.05x + 478.86
Energy loss (XP) vs drift time in ERAM 2
y = 0.07x + 486.02
Energy loss (XP) vs drift time in ERAM 26
y = 0.15x + 458.03
Energy loss (XP) vs drift time in ERAM 17
y = 0.13x + 462.48
Energy loss (XP) vs drift time in ERAM 23
y = 0.08x + 487.51
Energy loss (XP) vs drift time in ERAM 29
y = 0.07x + 489.40
Energy loss (XP) vs drift time in ERAM 1
y = -0.00x + 511.05
Energy loss (XP) vs drift time in ERAM 10
y = 0.04x + 503.67
Energy loss (XP) vs drift time in ERAM 11
y = 0.04x + 479.90
Energy loss (XP) vs drift time in ERAM 3
y = 0.06x + 487.99
dE/dx (ADC counts/cm)
drift time (timebins)

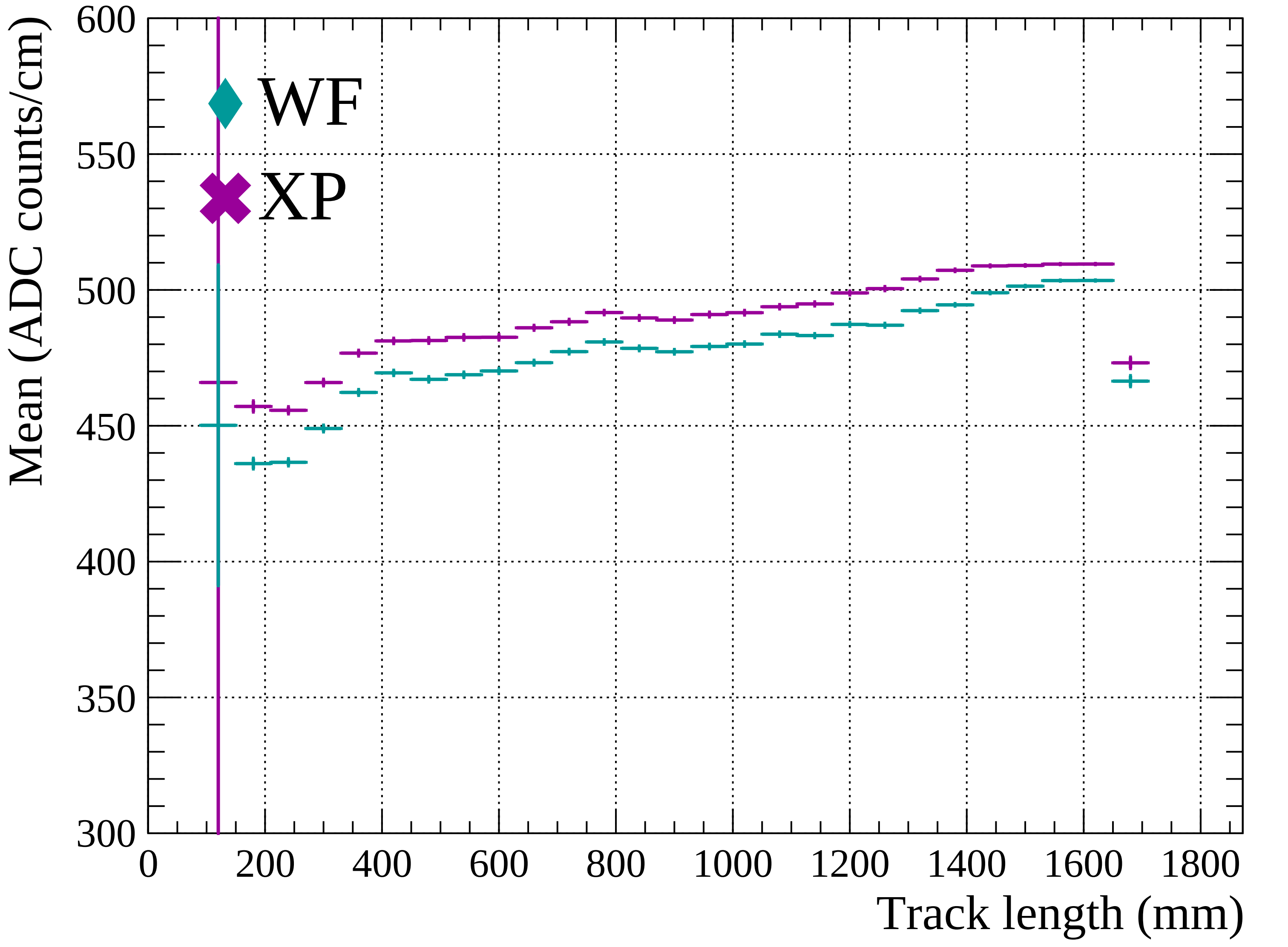

Energy loss (XP) vs drift time in ERAM 47
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 16
y = 0.05x + 474.89
Energy loss (XP) vs drift time in ERAM 14
y = 0.04x + 491.23
Energy loss (XP) vs drift time in ERAM 15
y = 0.07x + 490.29
Energy loss (XP) vs drift time in ERAM 42
y = 0.03x + 503.82
Energy loss (XP) vs drift time in ERAM 45
y = 0.07x + 488.55
Energy loss (XP) vs drift time in ERAM 37
y = 0.05x + 476.07
Energy loss (XP) vs drift time in ERAM 36
y = 0.08x + 484.35
Energy loss (XP) vs drift time in ERAM 20
y = 0.17x + 451.86
Energy loss (XP) vs drift time in ERAM 38
y = 0.10x + 471.80
Energy loss (XP) vs drift time in ERAM 7
y = 0.09x + 485.10
Energy loss (XP) vs drift time in ERAM 44
y = 0.06x + 492.02
Energy loss (XP) vs drift time in ERAM 43
y = 0.03x + 503.15
Energy loss (XP) vs drift time in ERAM 39
y = 0.02x + 503.70
Energy loss (XP) vs drift time in ERAM 41
y = 0.08x + 470.02
Energy loss (XP) vs drift time in ERAM 46
y = 0.08x + 484.14
dE/dx (ADC counts/cm)
drift time (timebins)



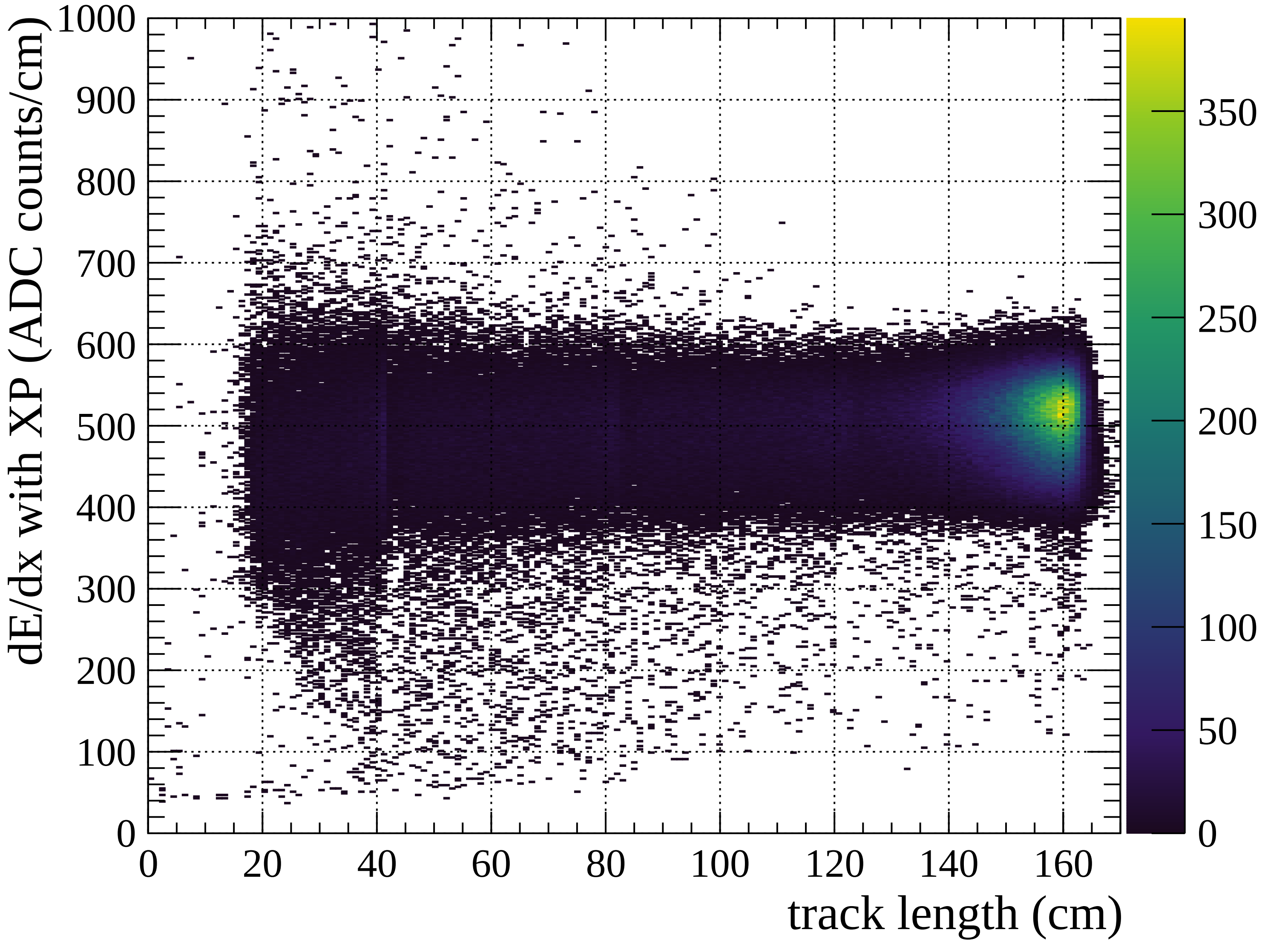
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
0
20
40
60
80
100
120
140
160
180
200
220



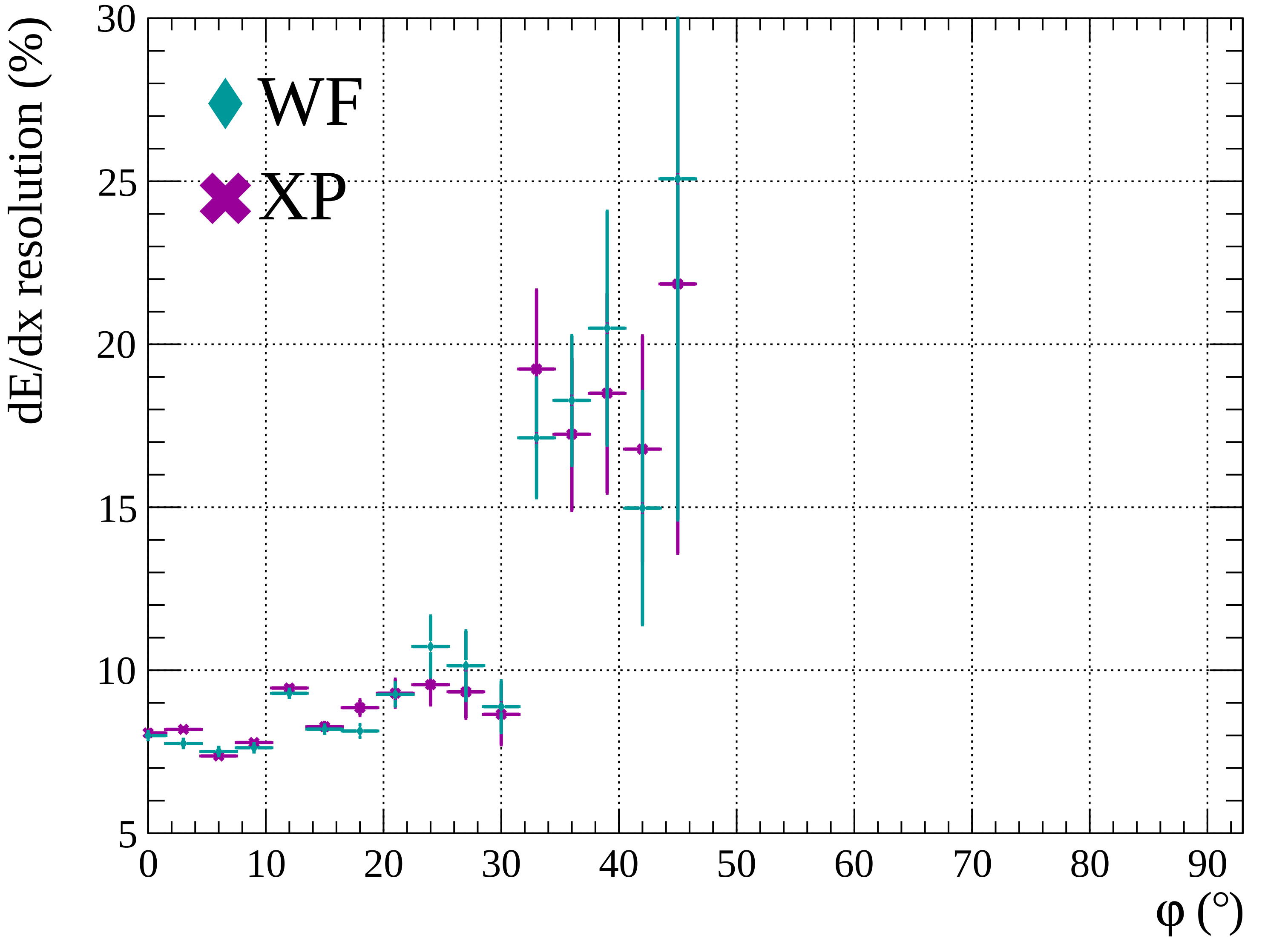
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
Track length (mm)



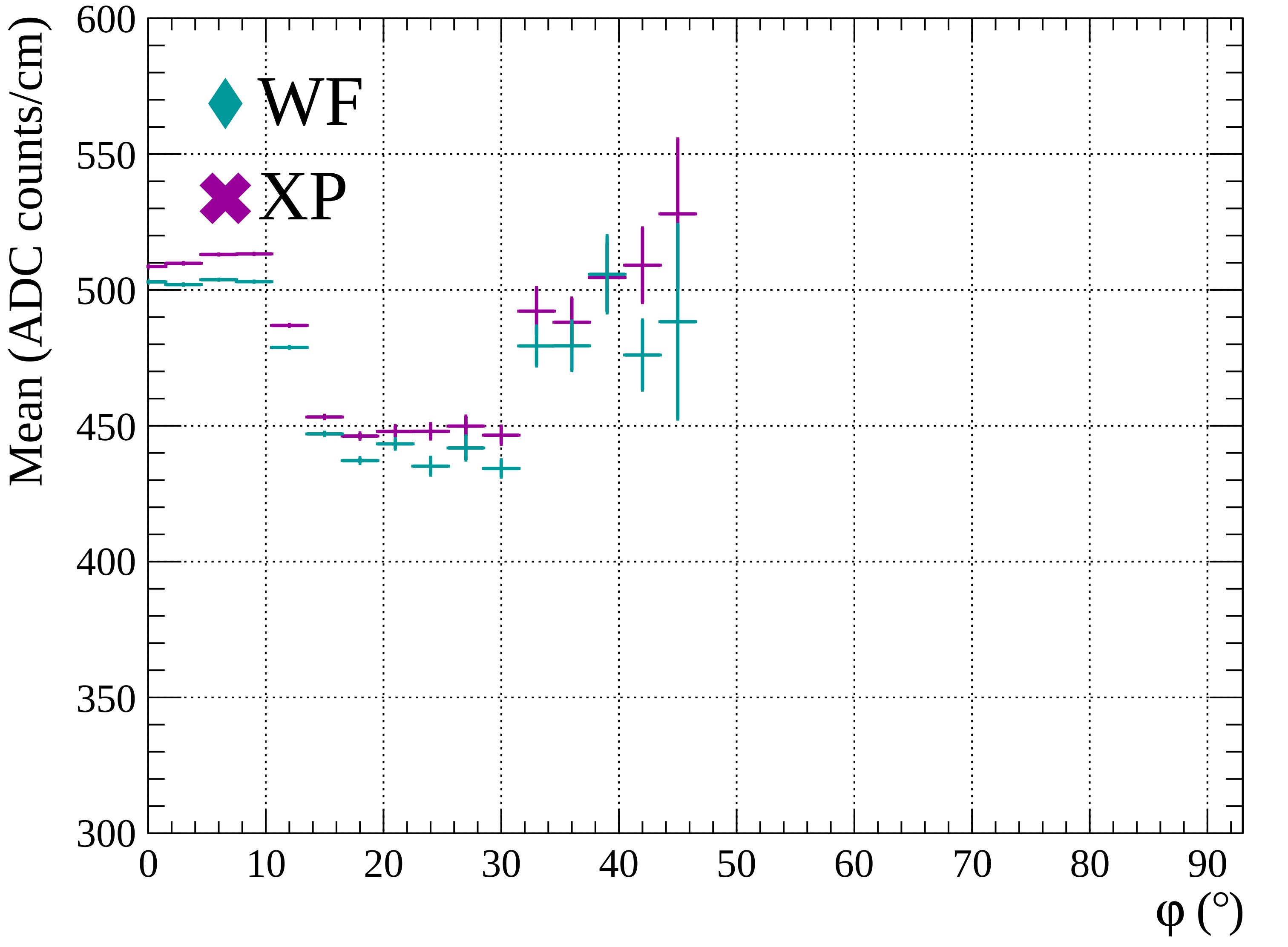
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
Track length (mm)



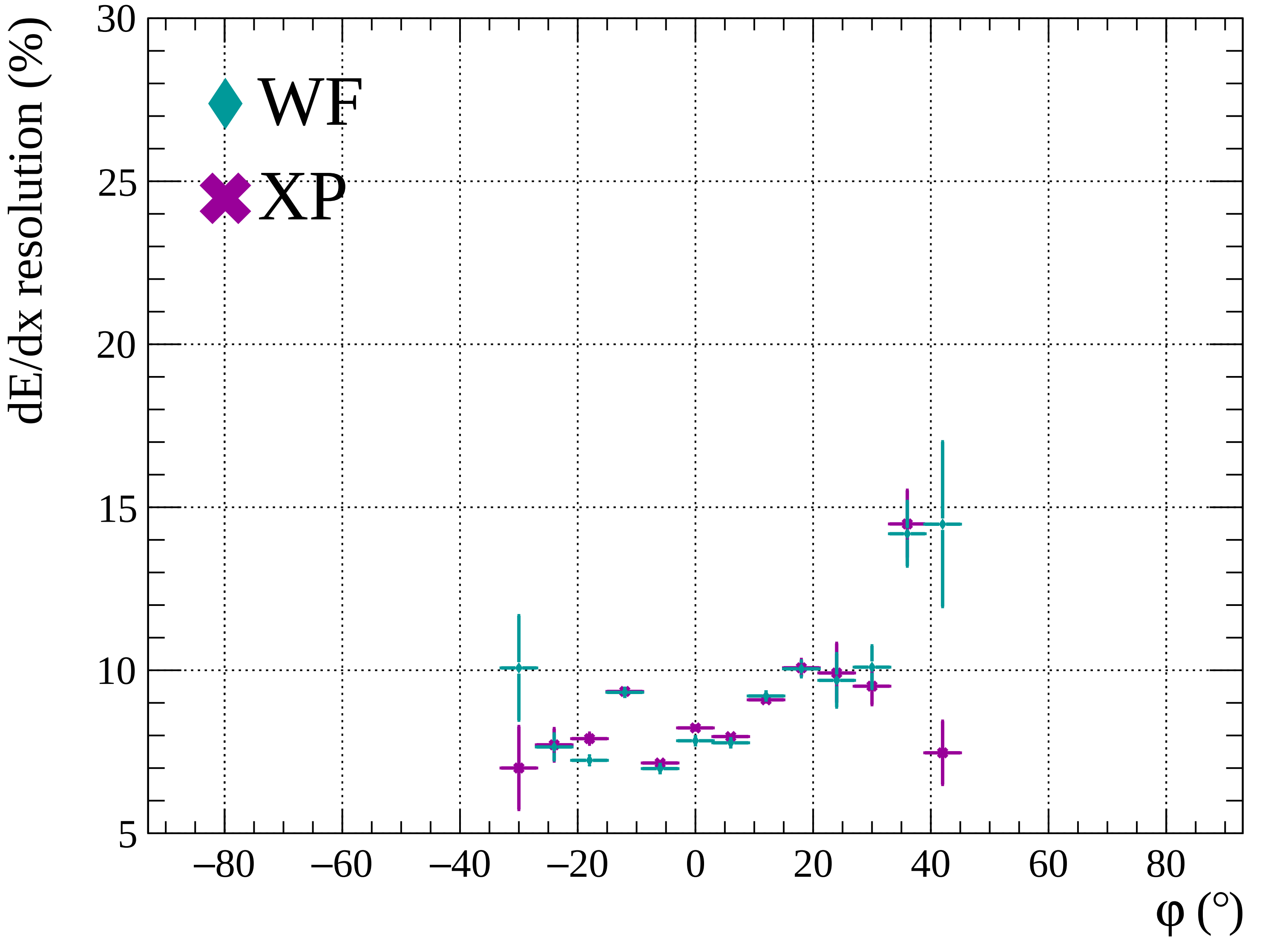
dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
50
100
150
200
250
300
350

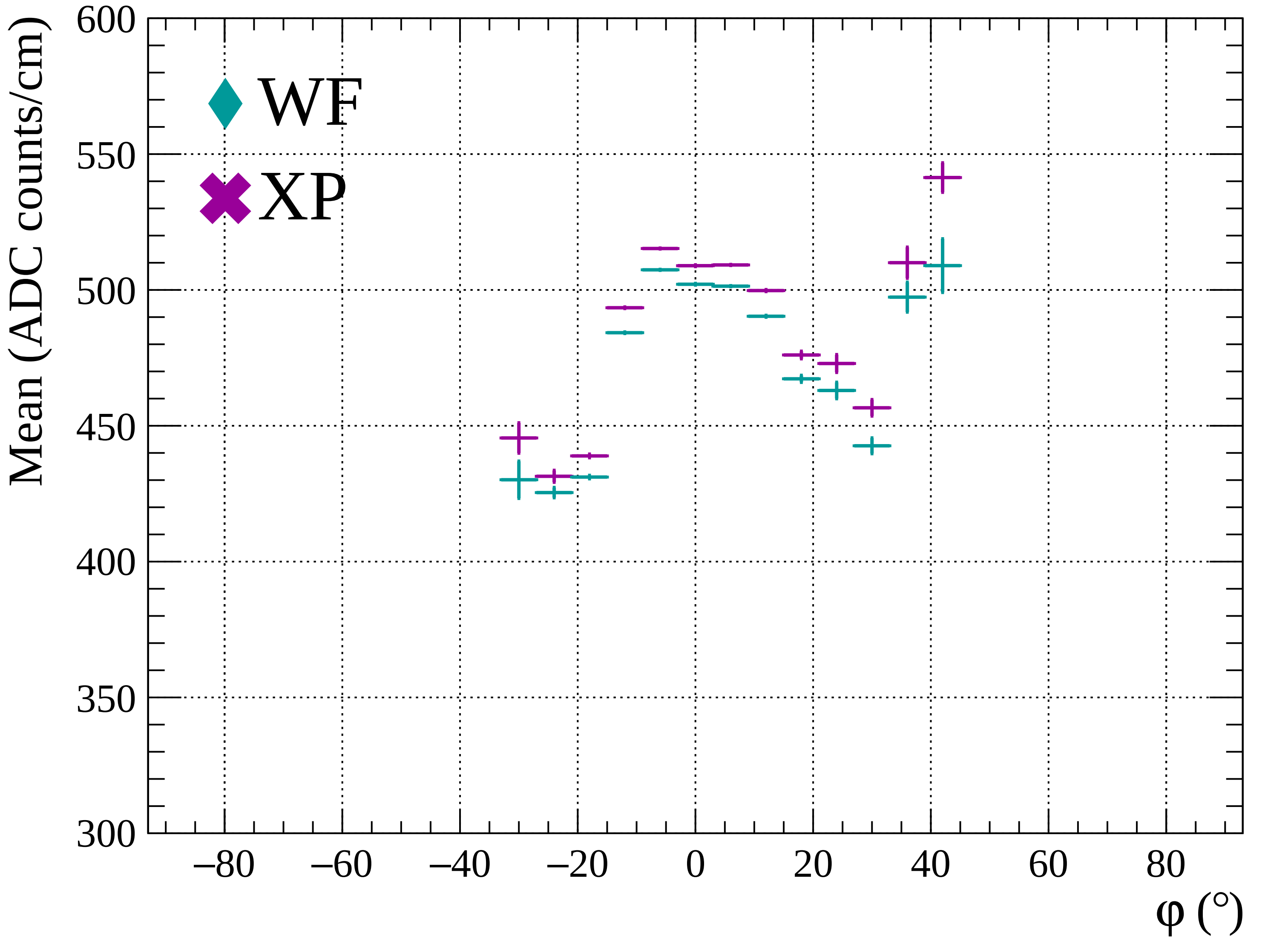


dE/dx resolution (%)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90
φ (°)
WF
XP

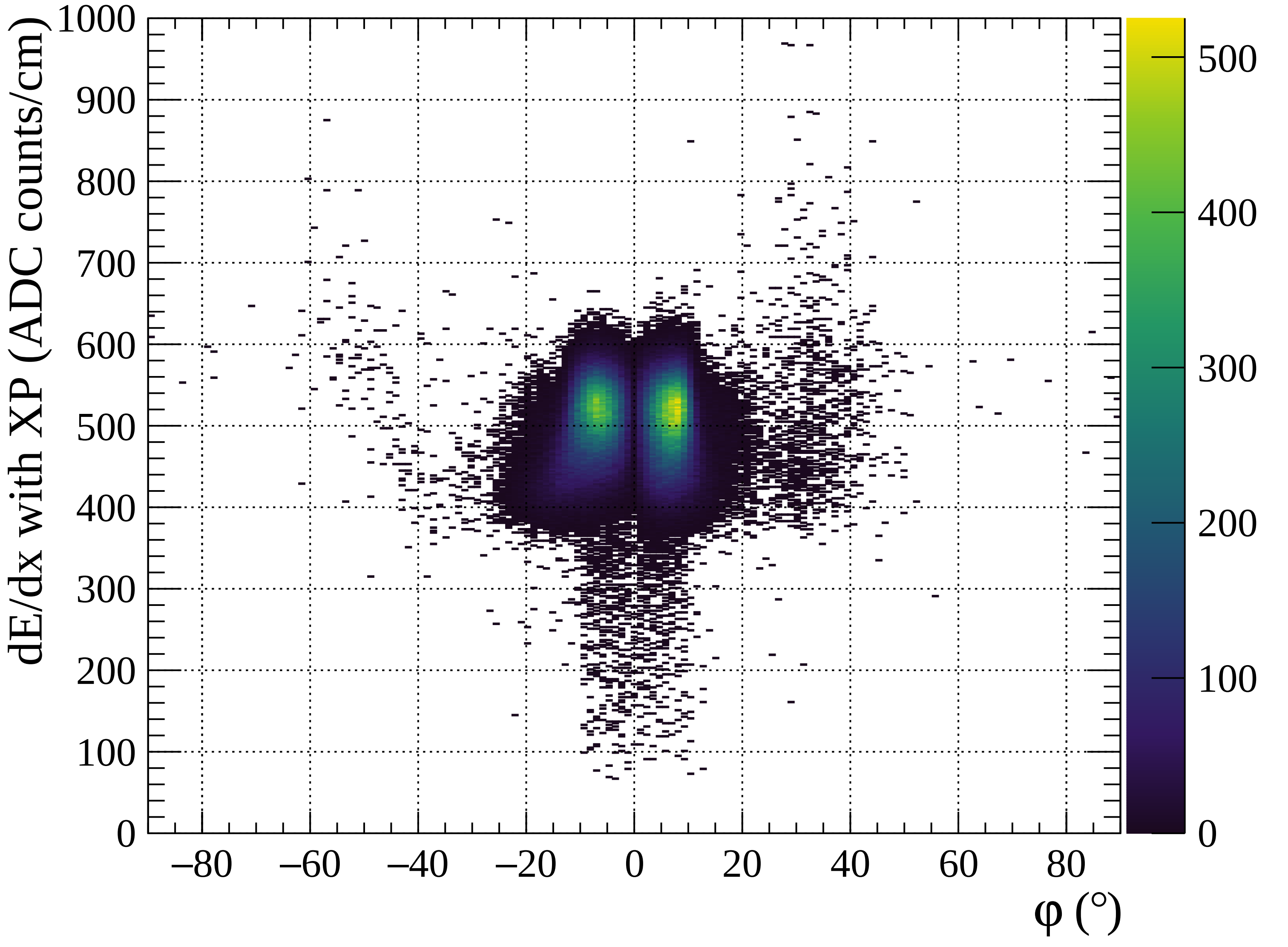


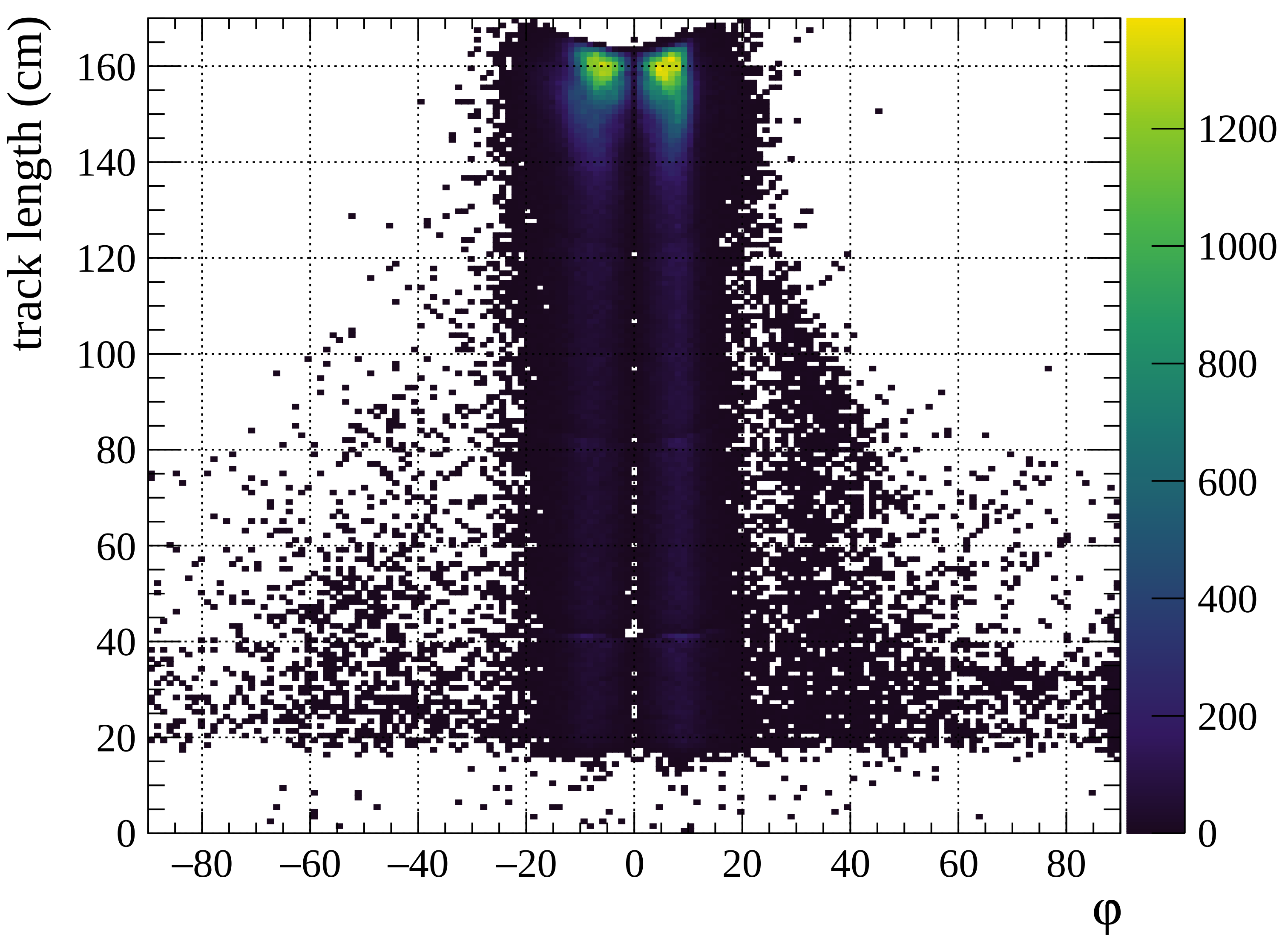
Mean (ADC counts/cm)
WF
XP
φ (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90

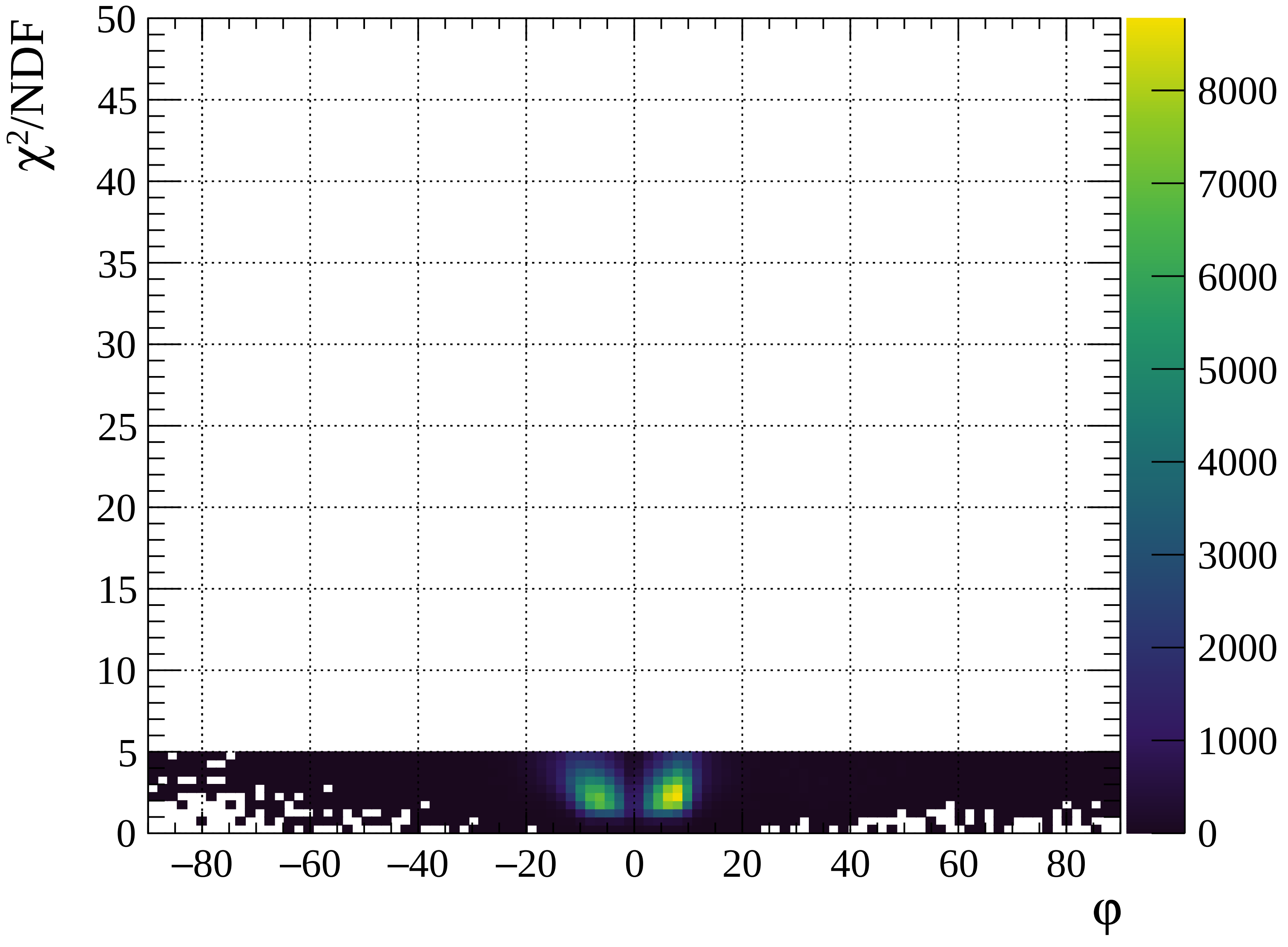

dE/dx resolution (%)
φ (°)
WF
XP
5
10
15
20
25
30
−80
−60
−40
−20
0
20
40
60
80



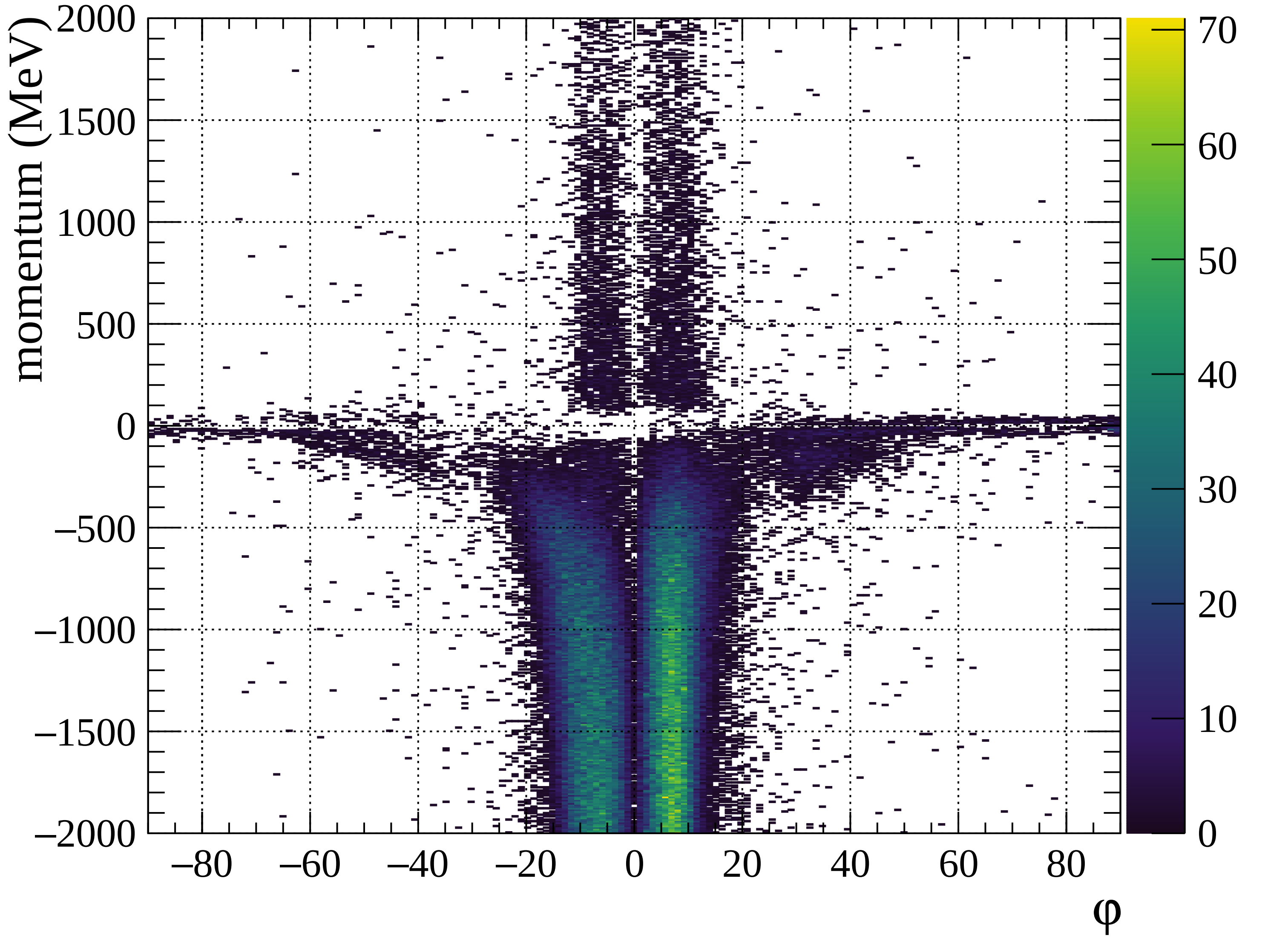
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)


dE/dx with XP (ADC counts/cm)
φ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
500
400
300
200
100
0

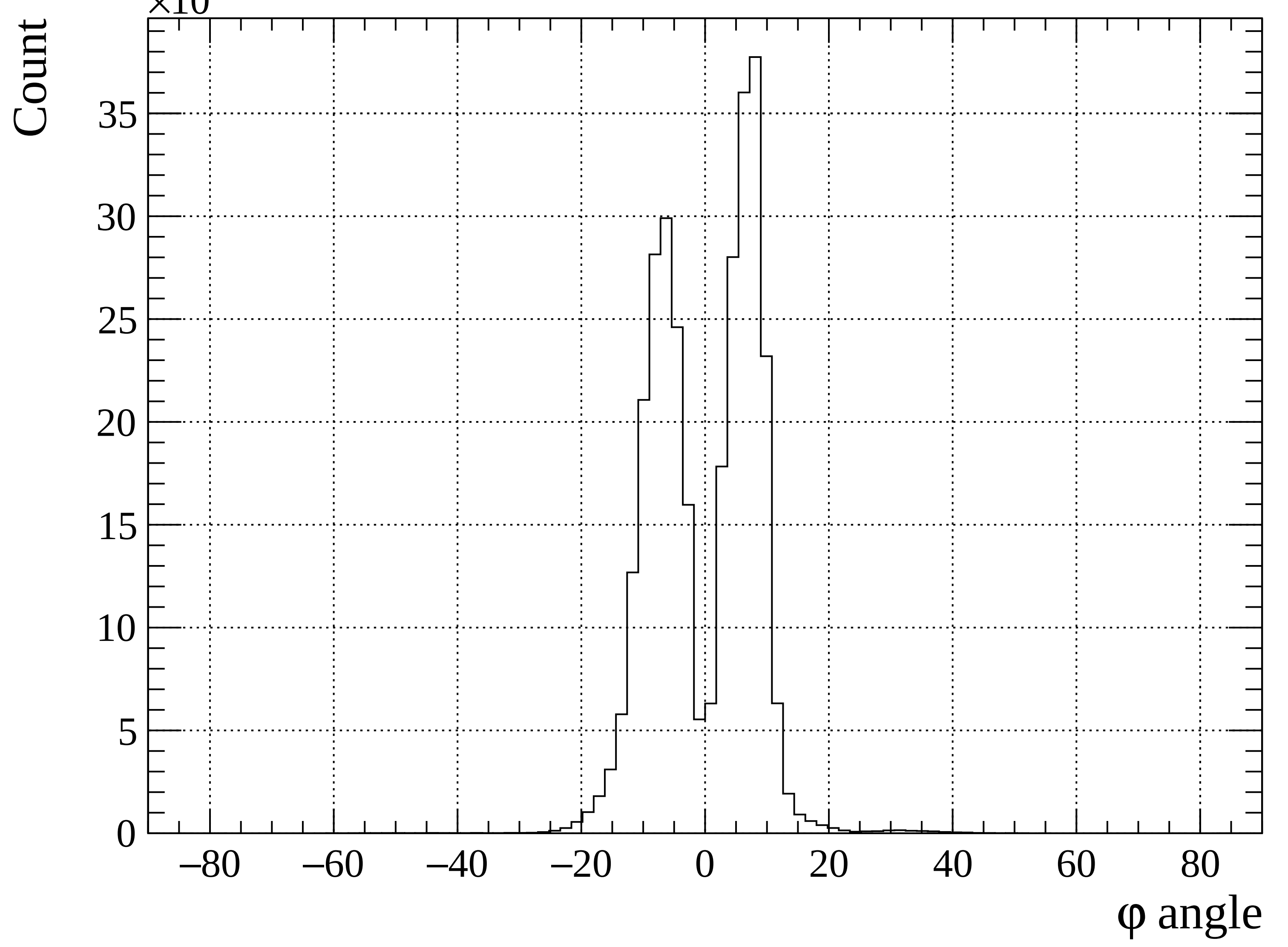

track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
200
400
600
800
1000
1200



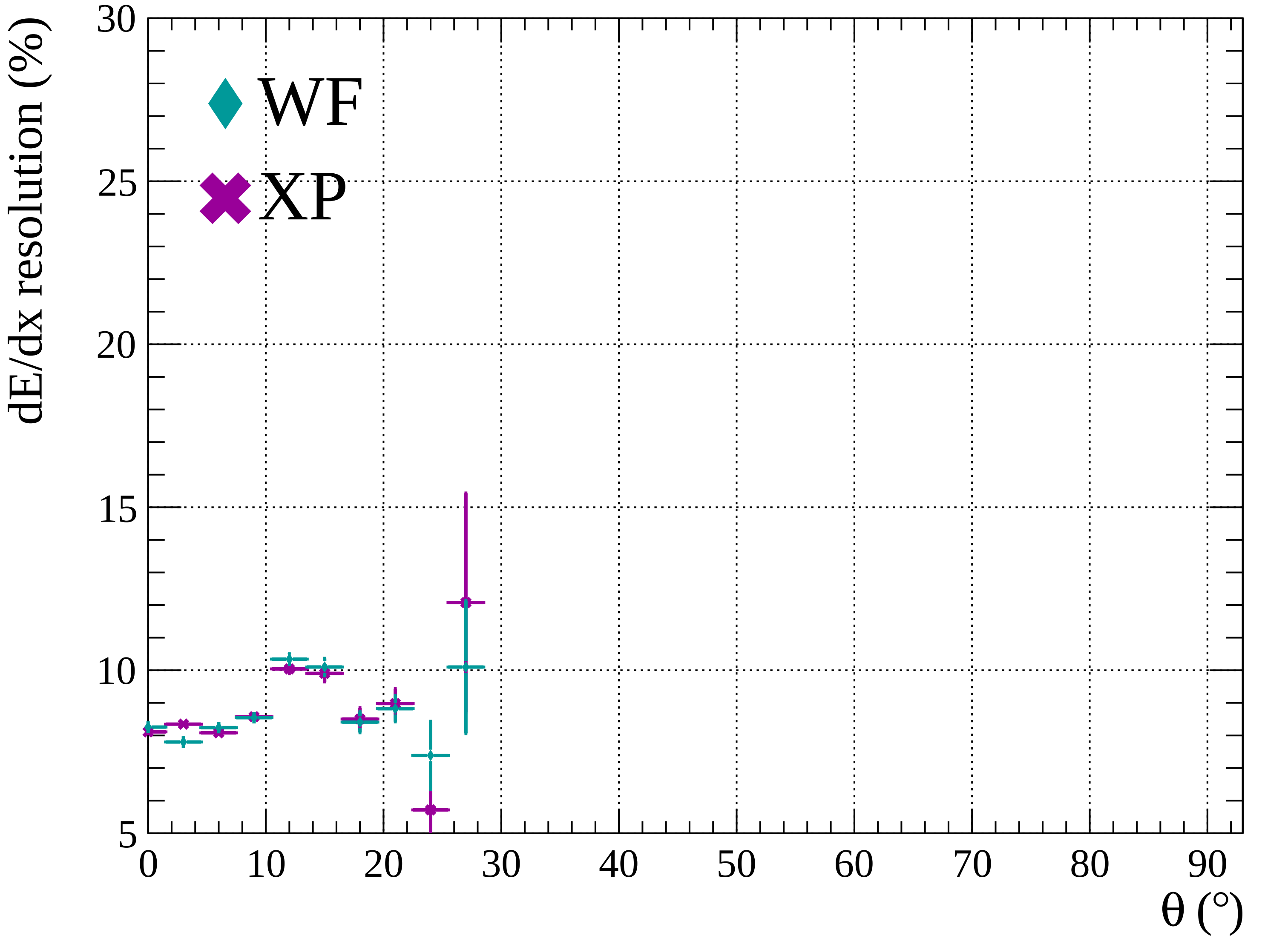
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
8000
7000
6000
5000
4000
3000
2000
1000
0



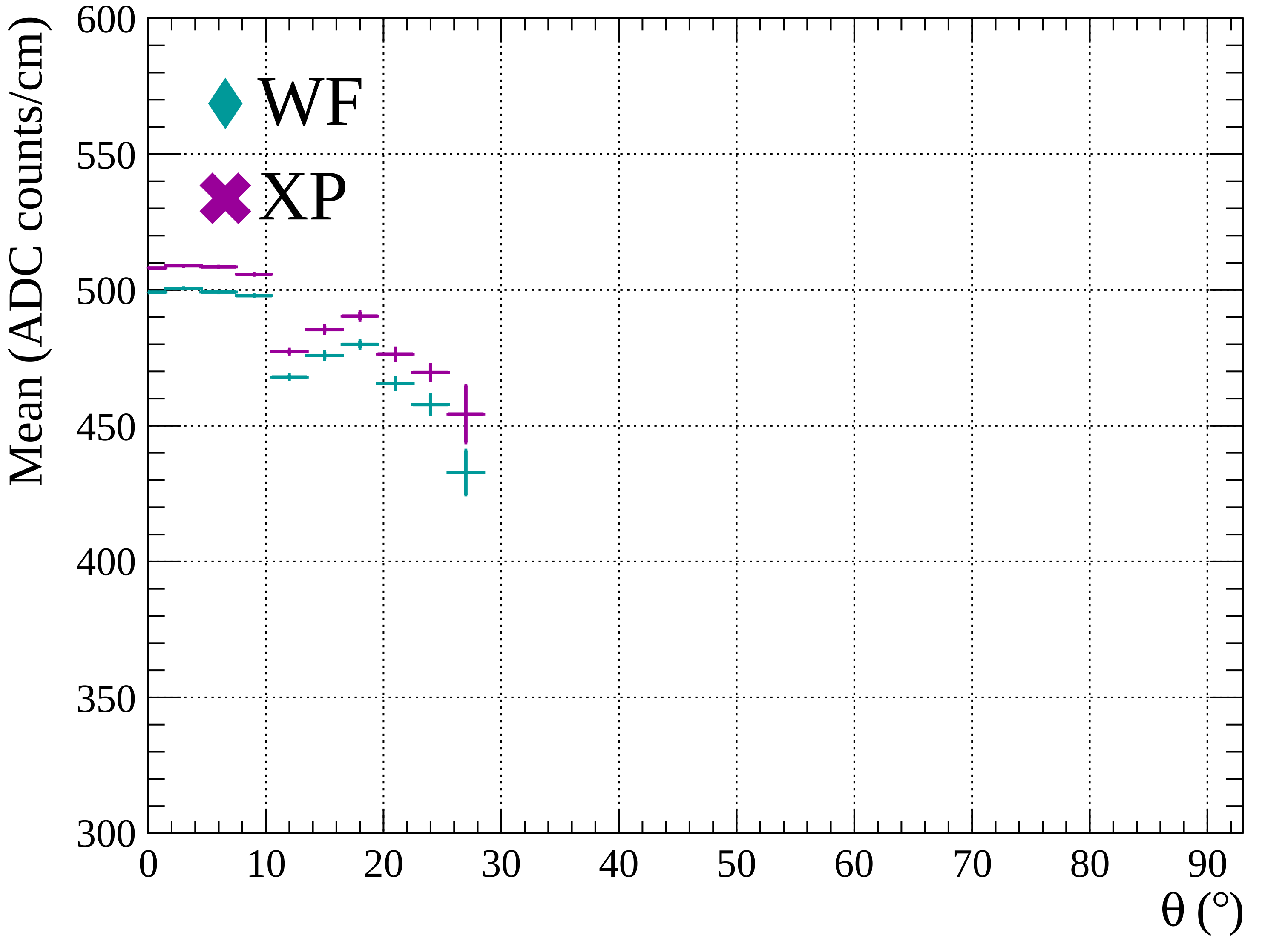
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
70
60
50
40
30
20
10
0



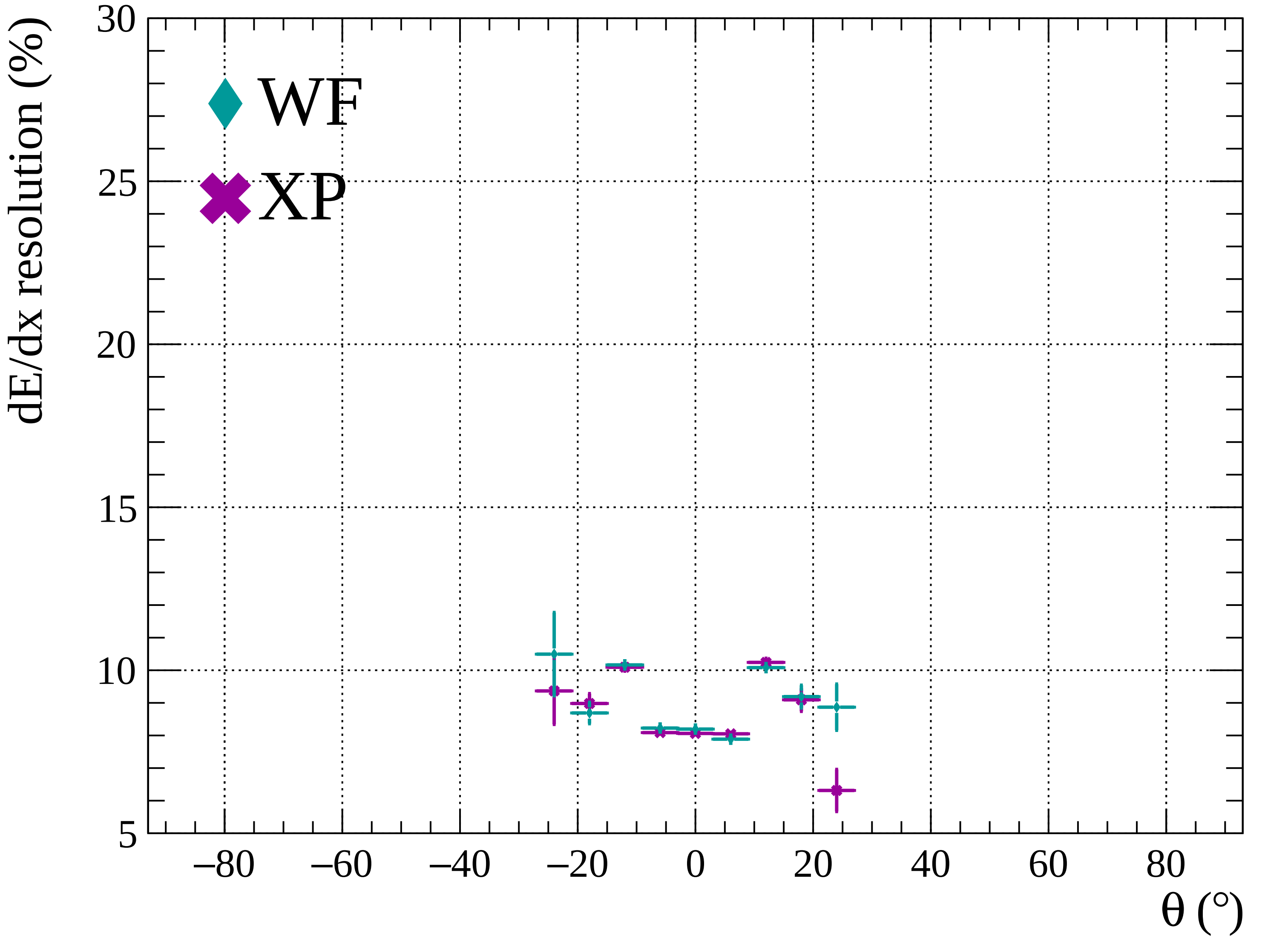
×10
Count
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



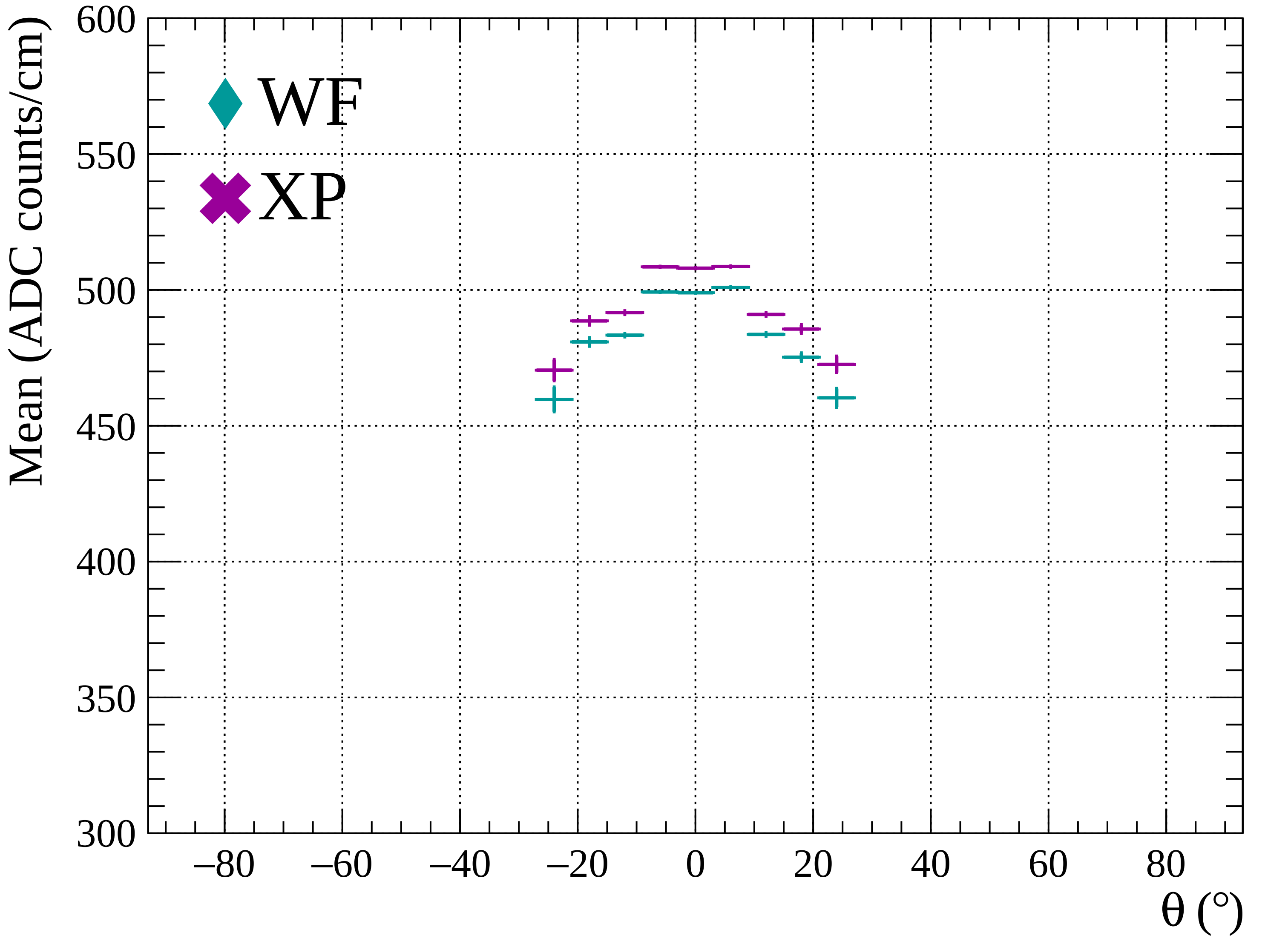
dE/dx resolution (%)
θ (°)
WF
XP
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



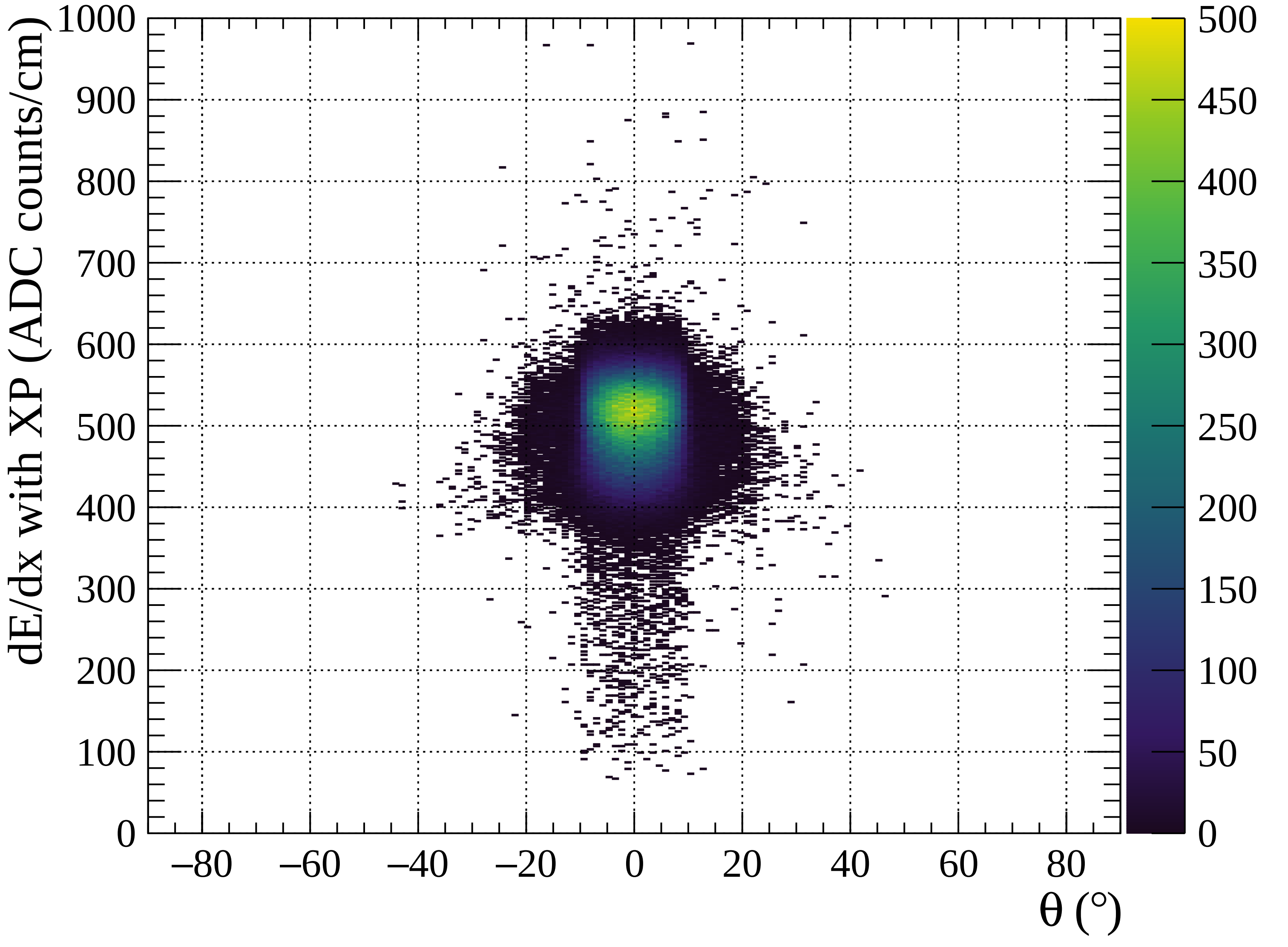
WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90

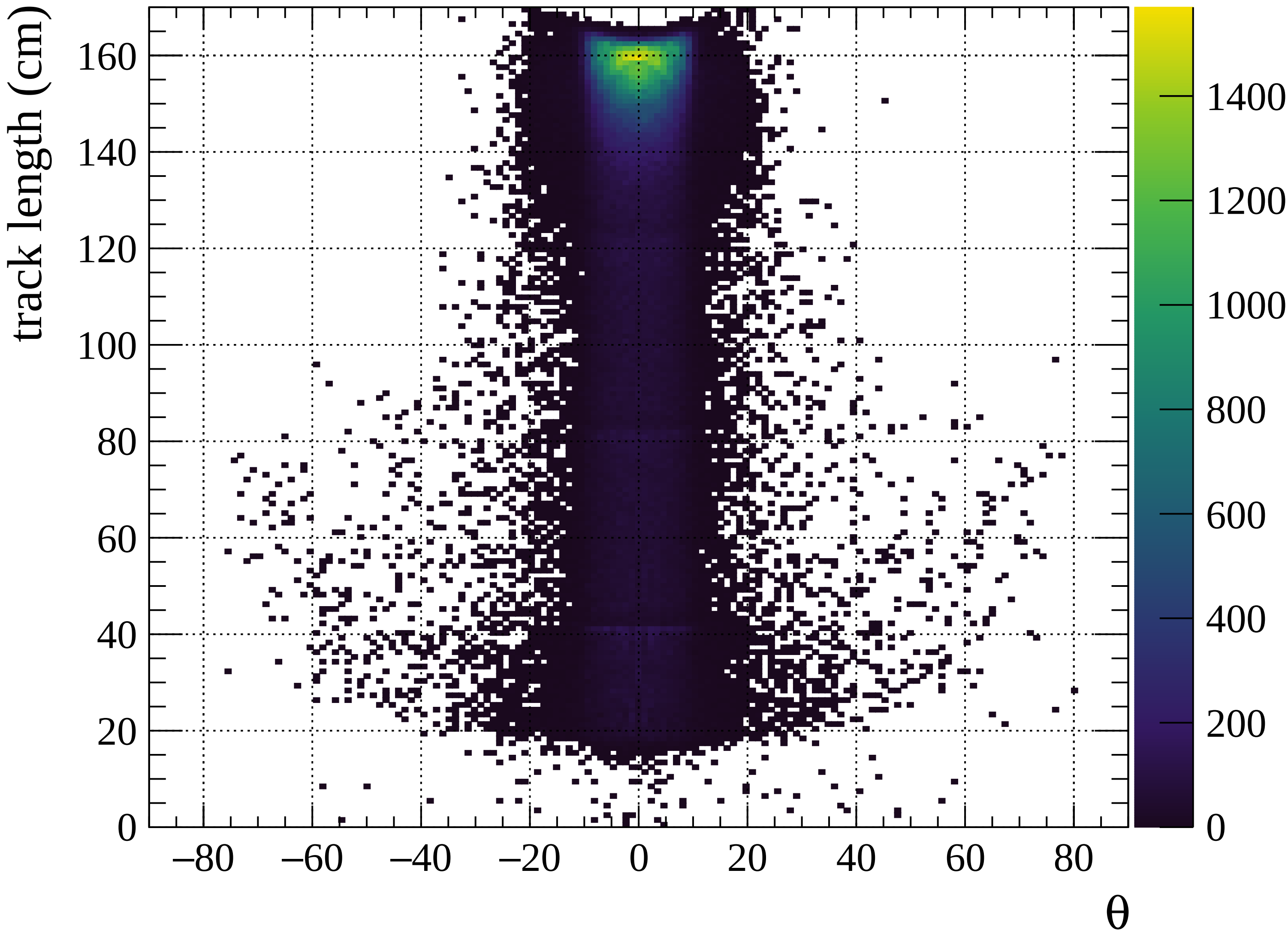


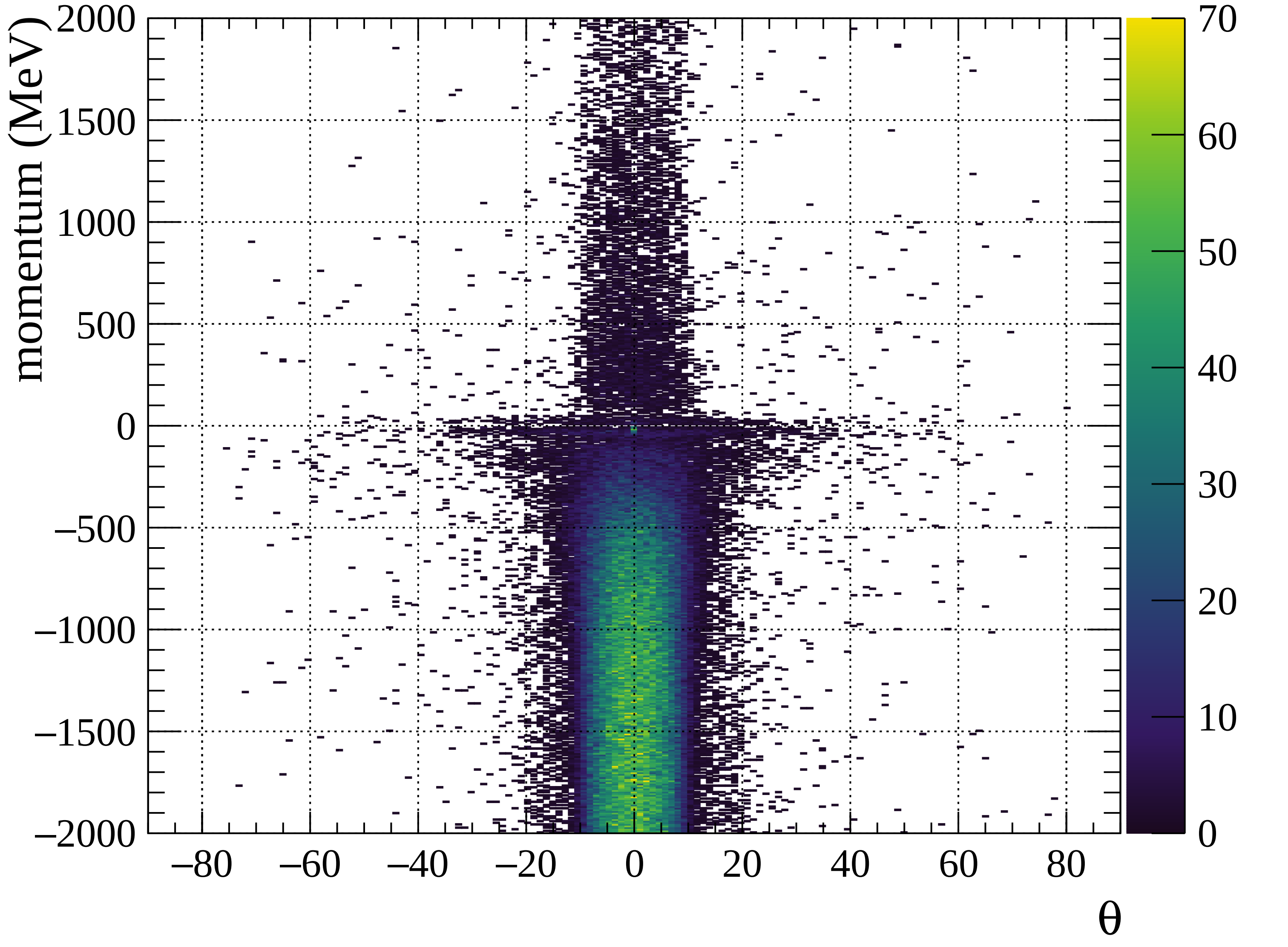
dE/dx resolution (%)
WF
XP
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80



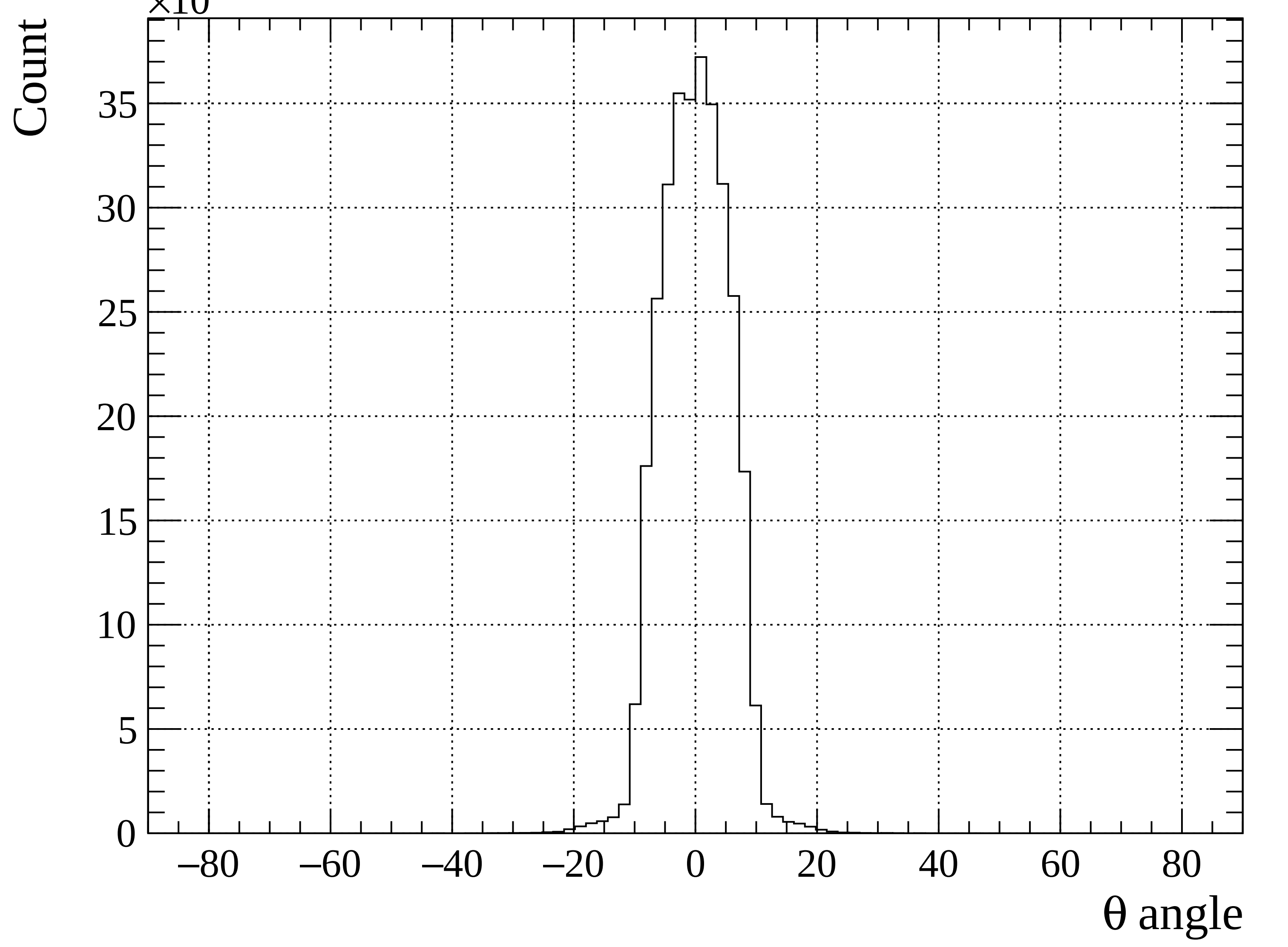
WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80


dE/dx with XP (ADC counts/cm)
θ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
500
450
400
350
300
250
200
150
100
50
0

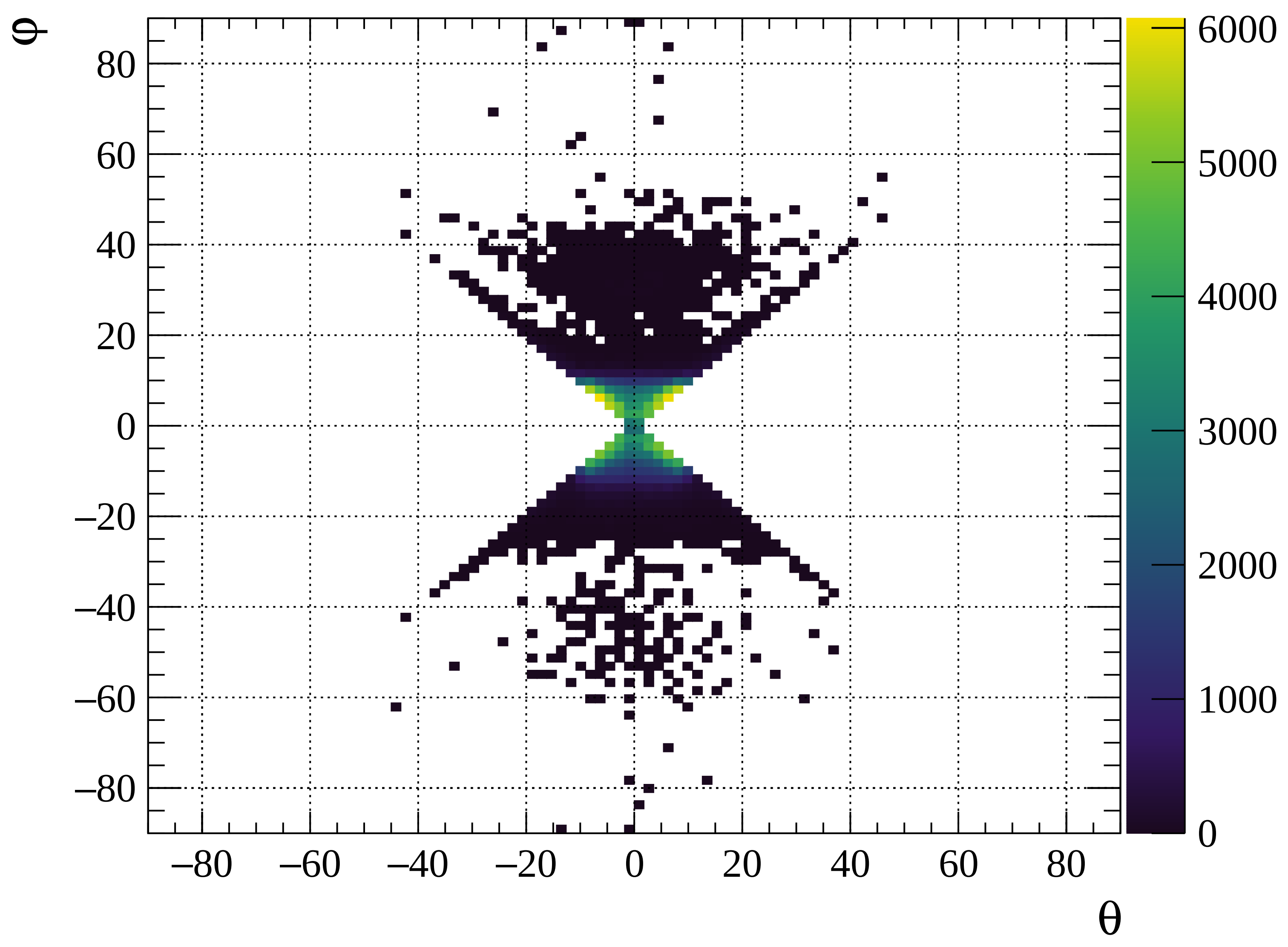

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
1400
1200
1000
800
600
400
200
0



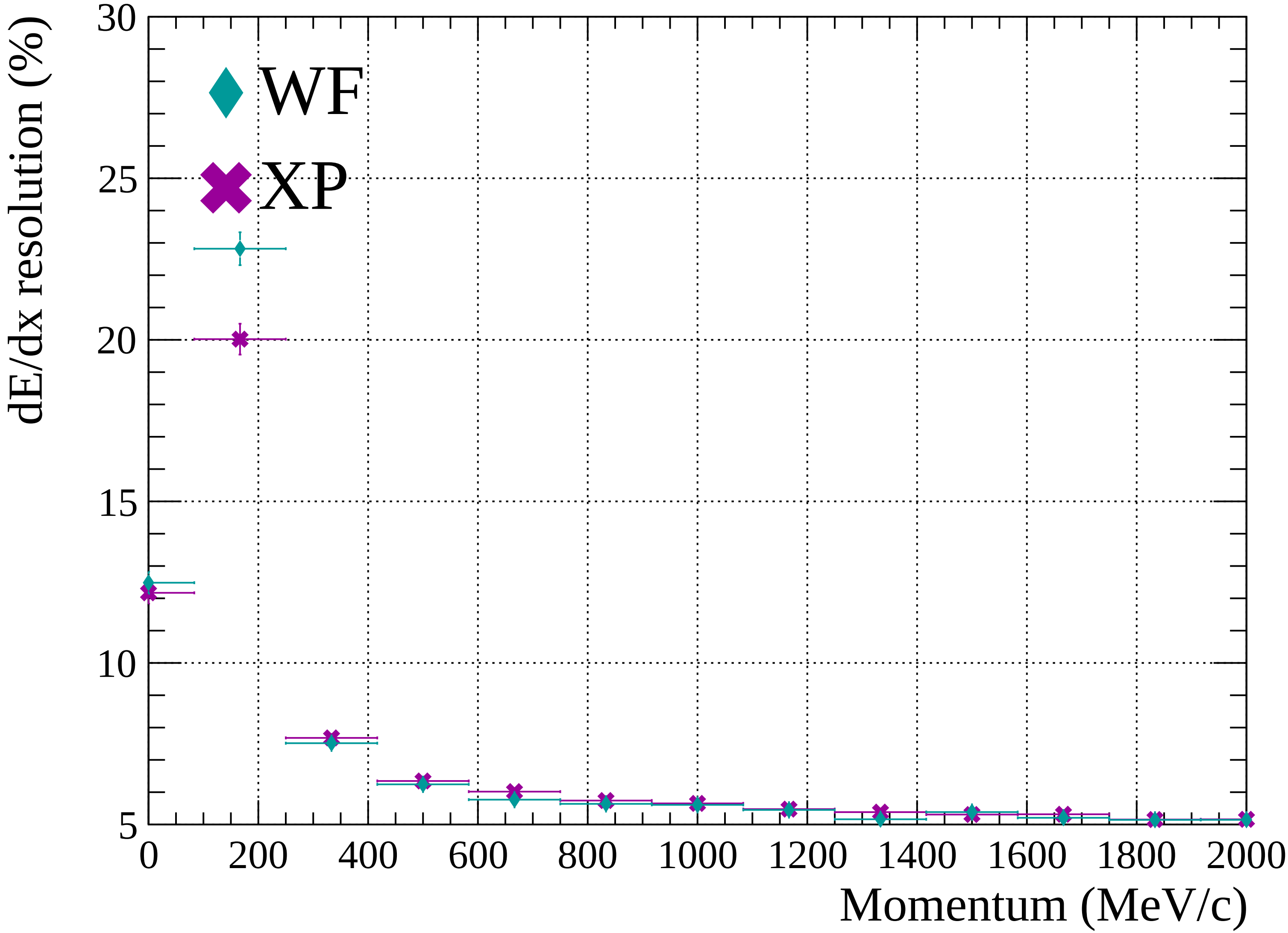
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
70
60
50
40
30
20
10
0



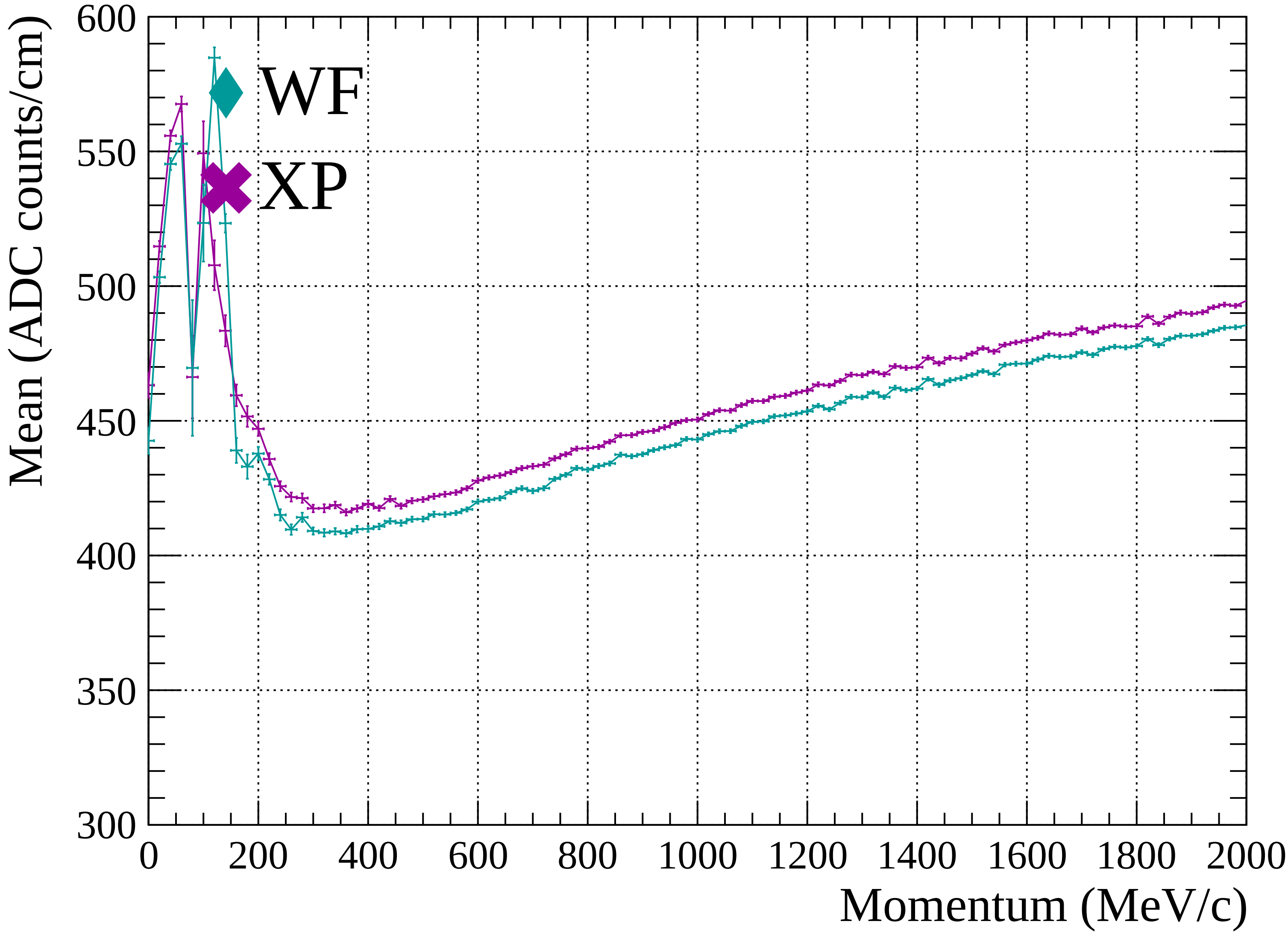
×10
Count
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



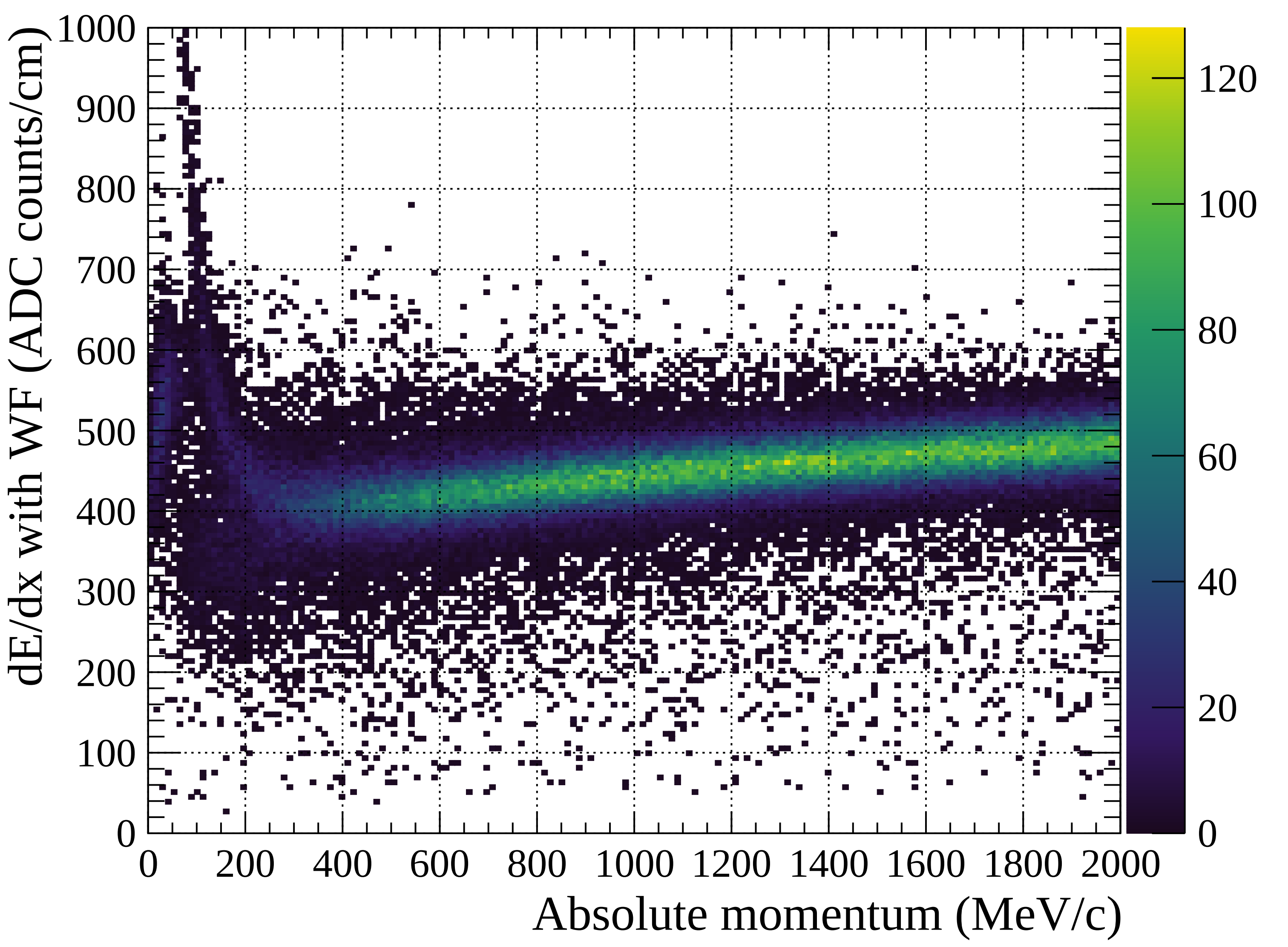
φ
80
60
40
20
0
−20
−40
−60
−80
θ
6000
5000
4000
3000
2000
1000
0



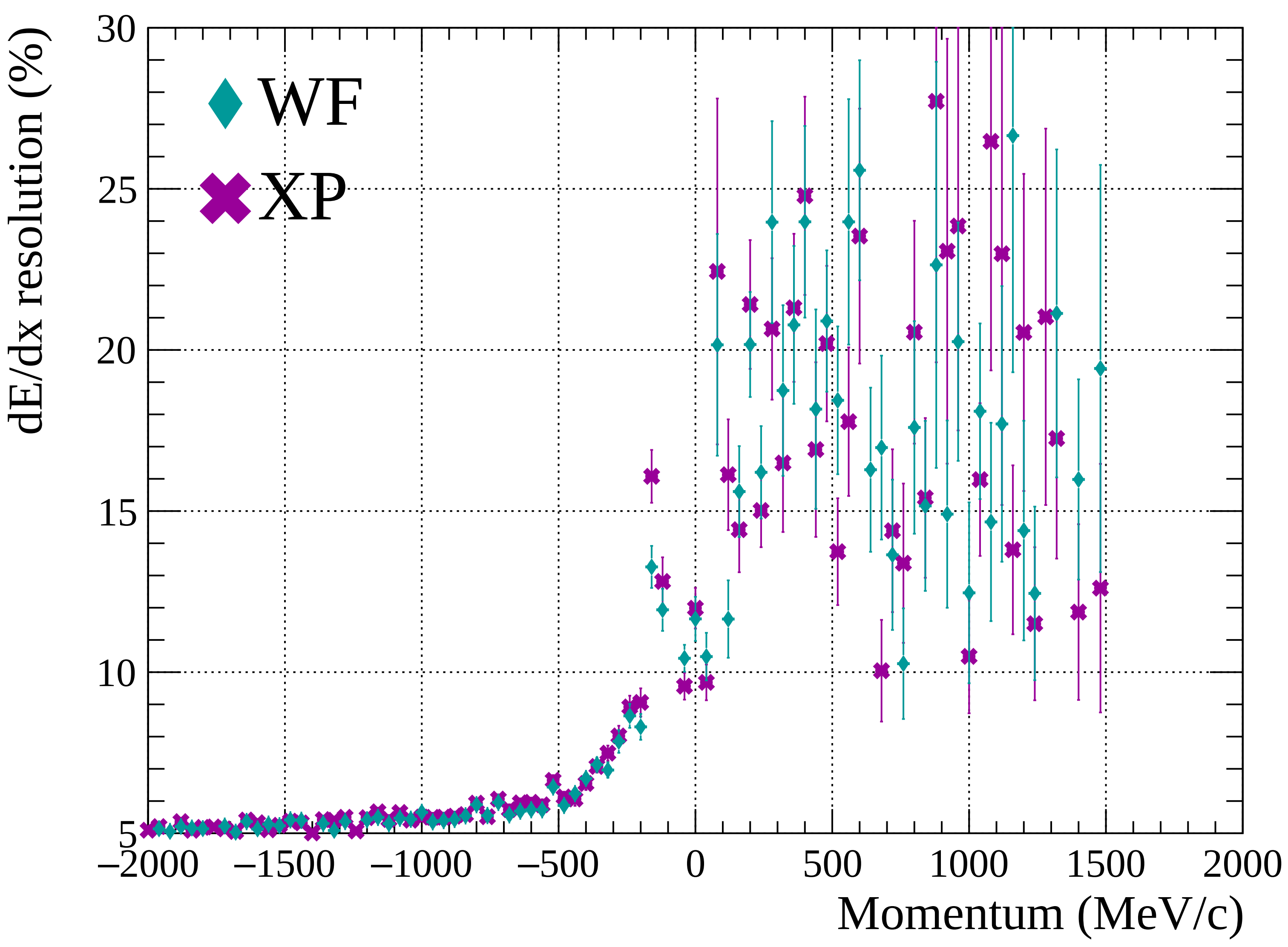
WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000

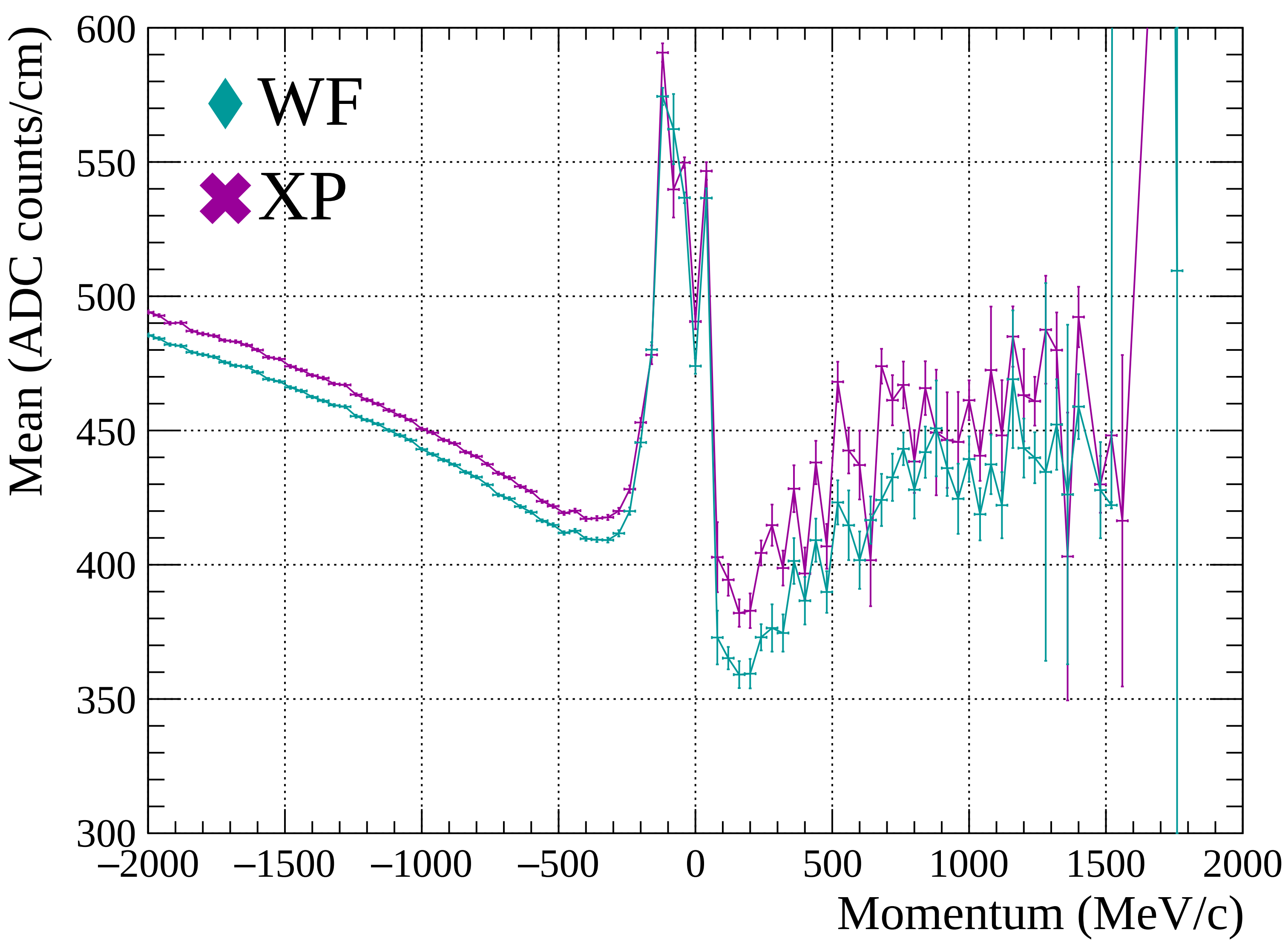


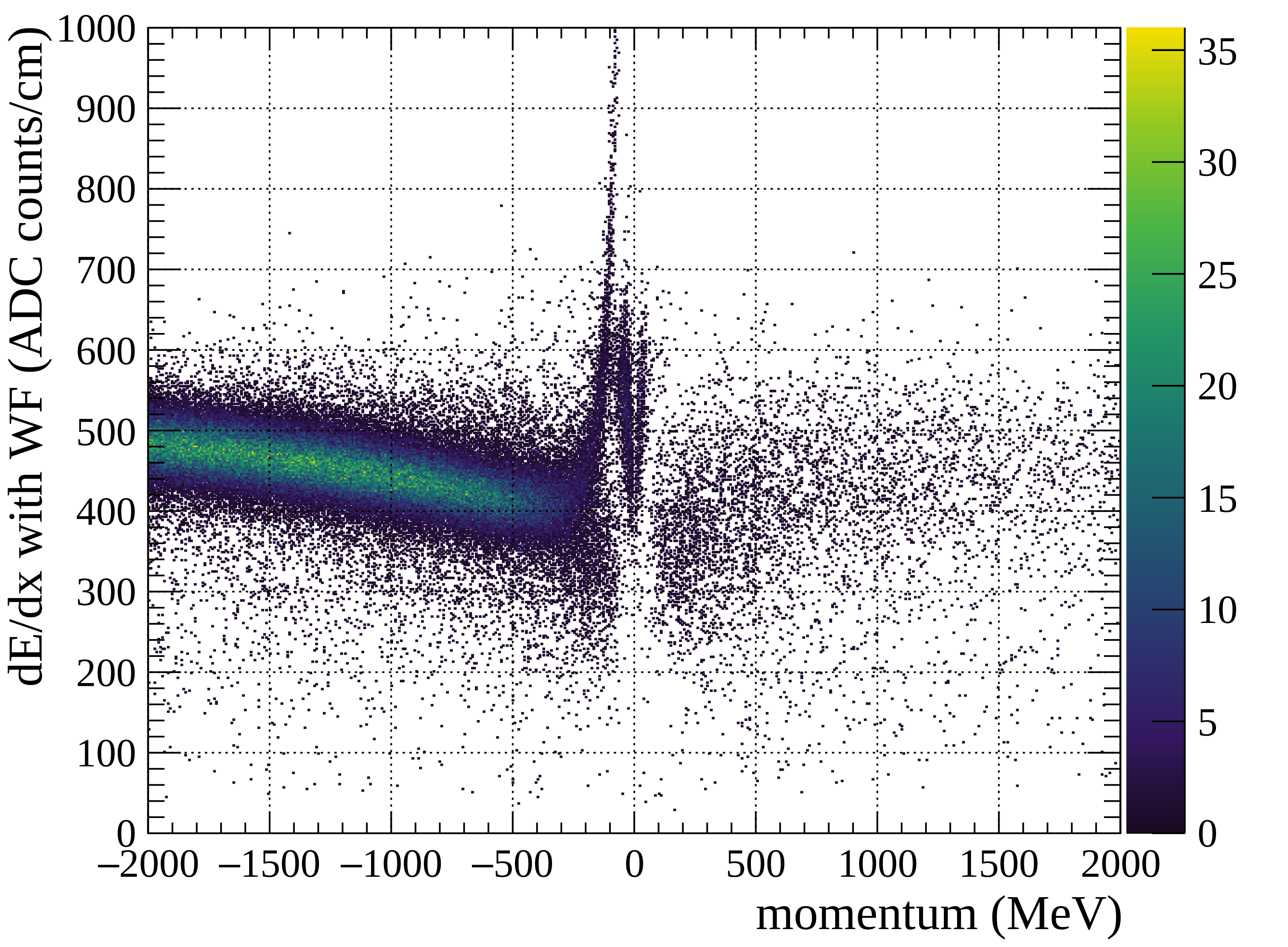
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000



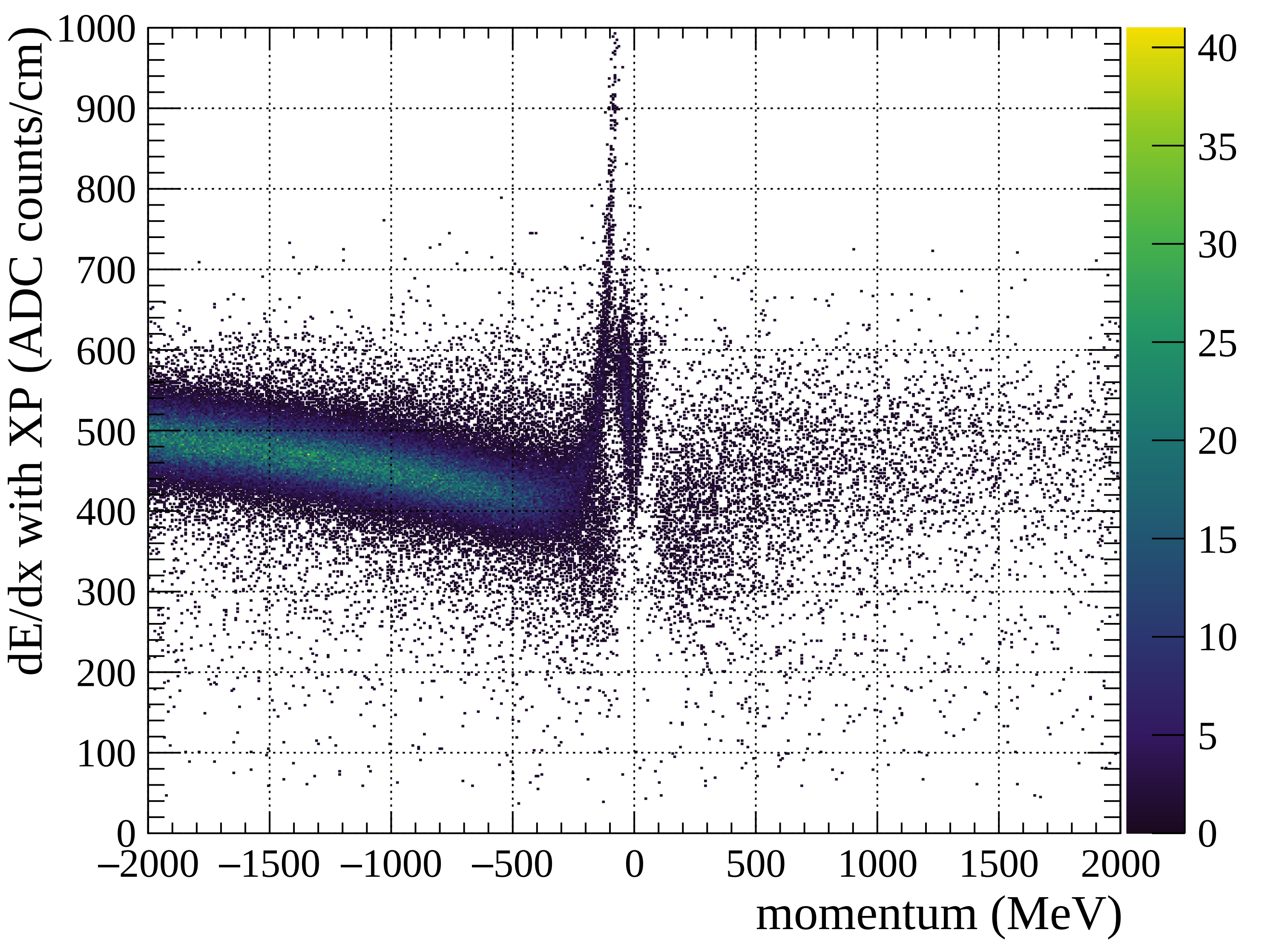
dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
20
40
60
80
100
120


dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

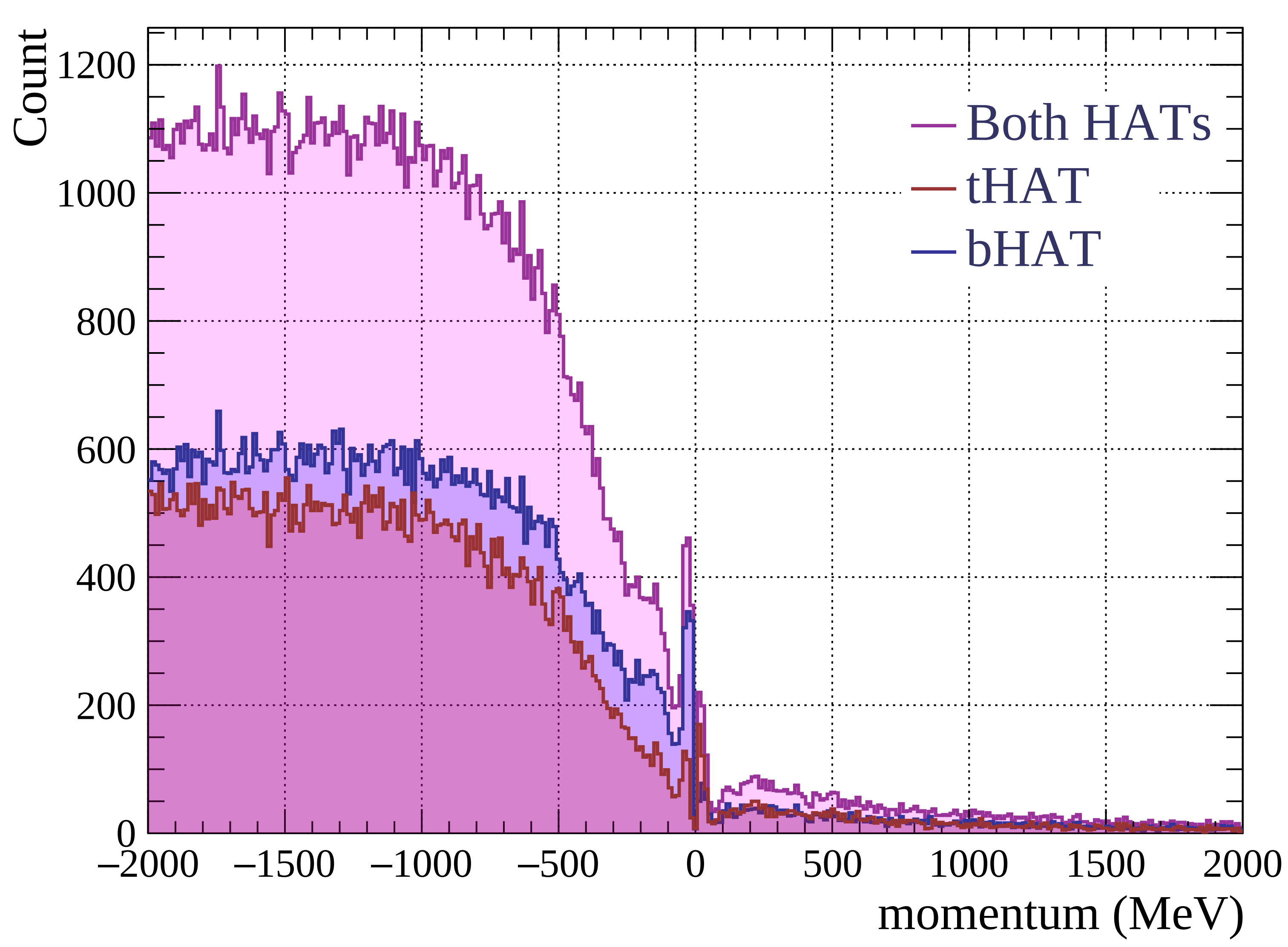

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)

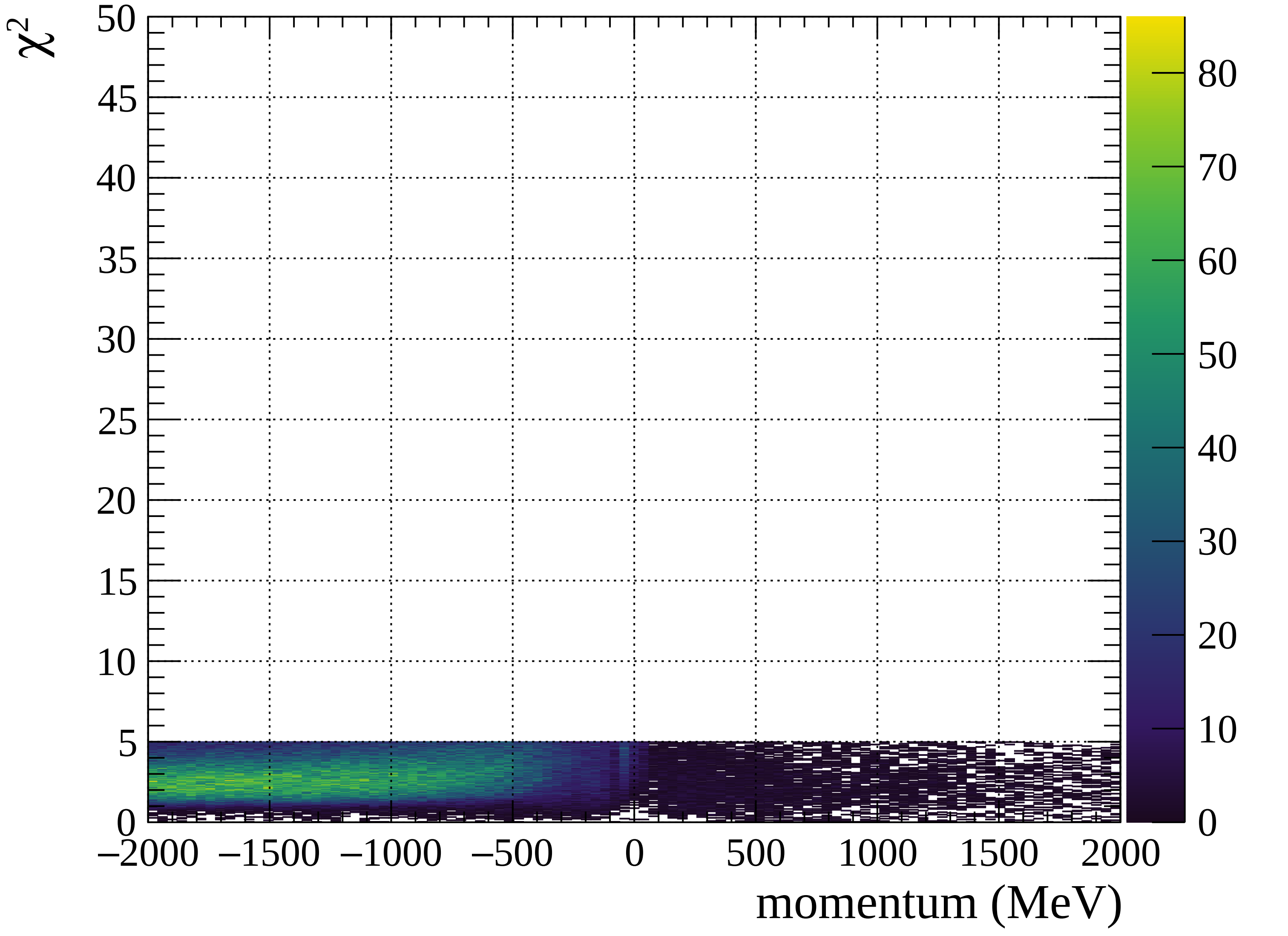


dE/dx with WF (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
35
30
25
20
15
10
5
0

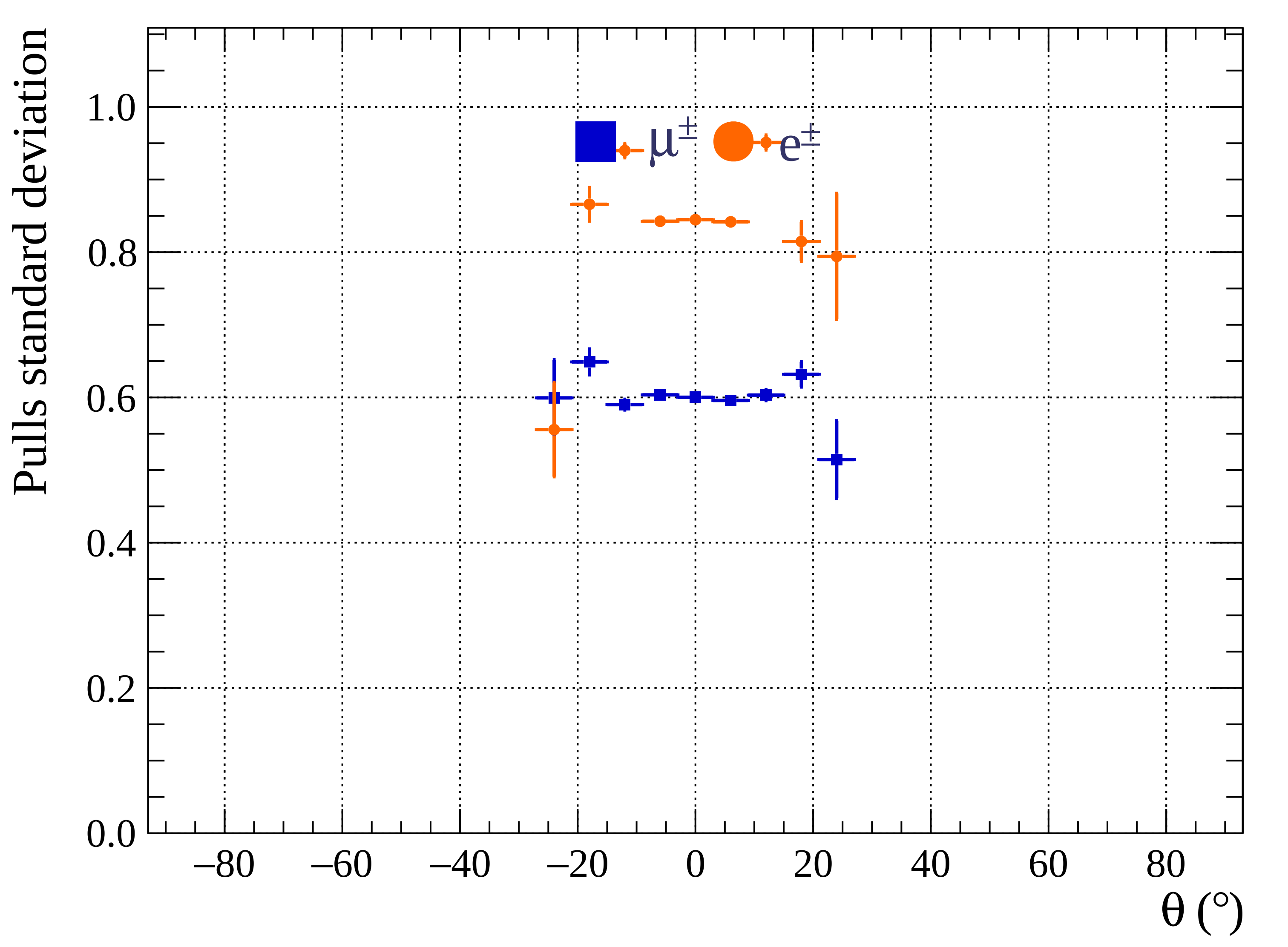


dE/dx with XP (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
40
35
30
25
20
15
10
5
0

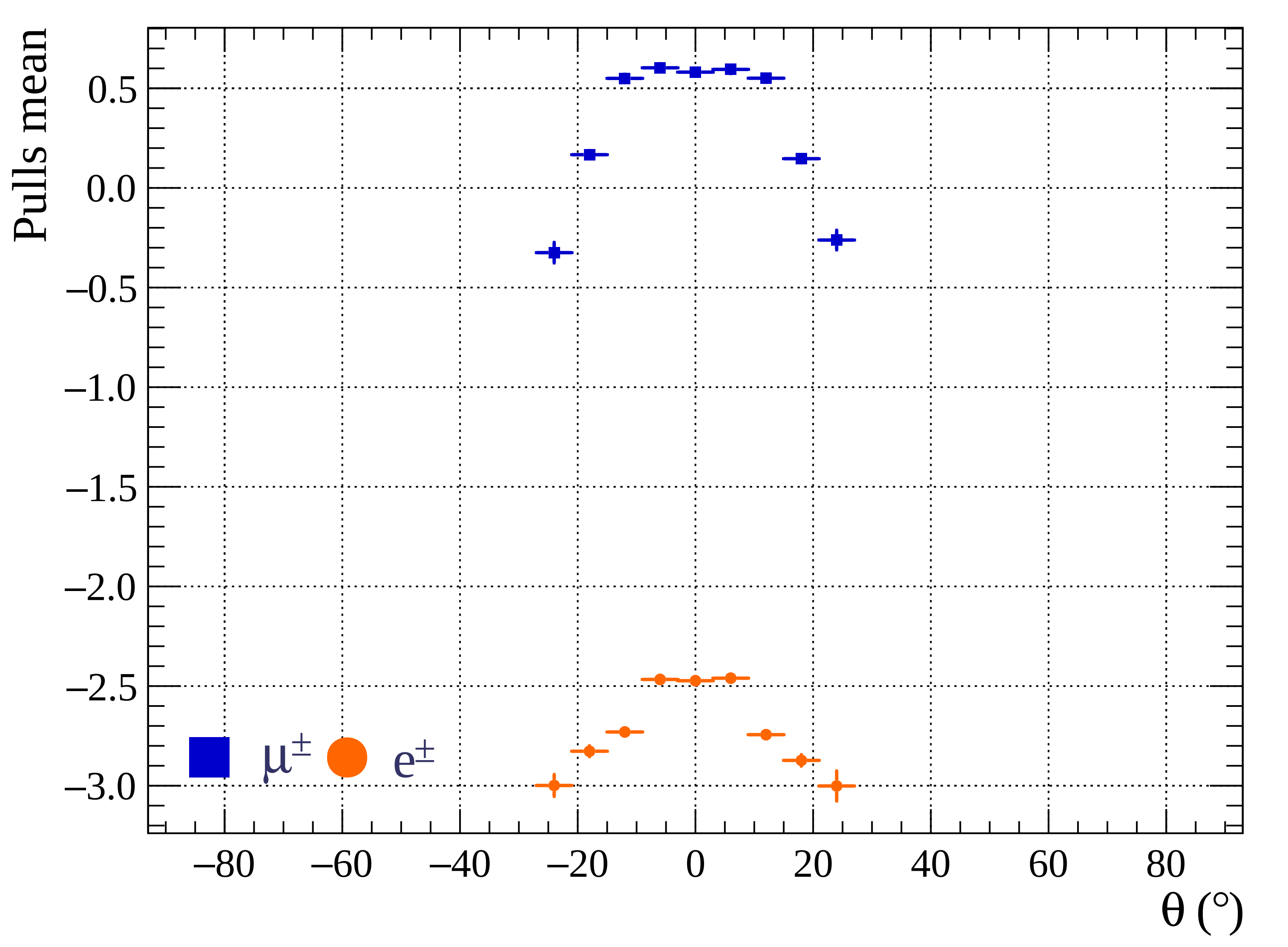

Count
1200
1000
800
600
400
200
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT

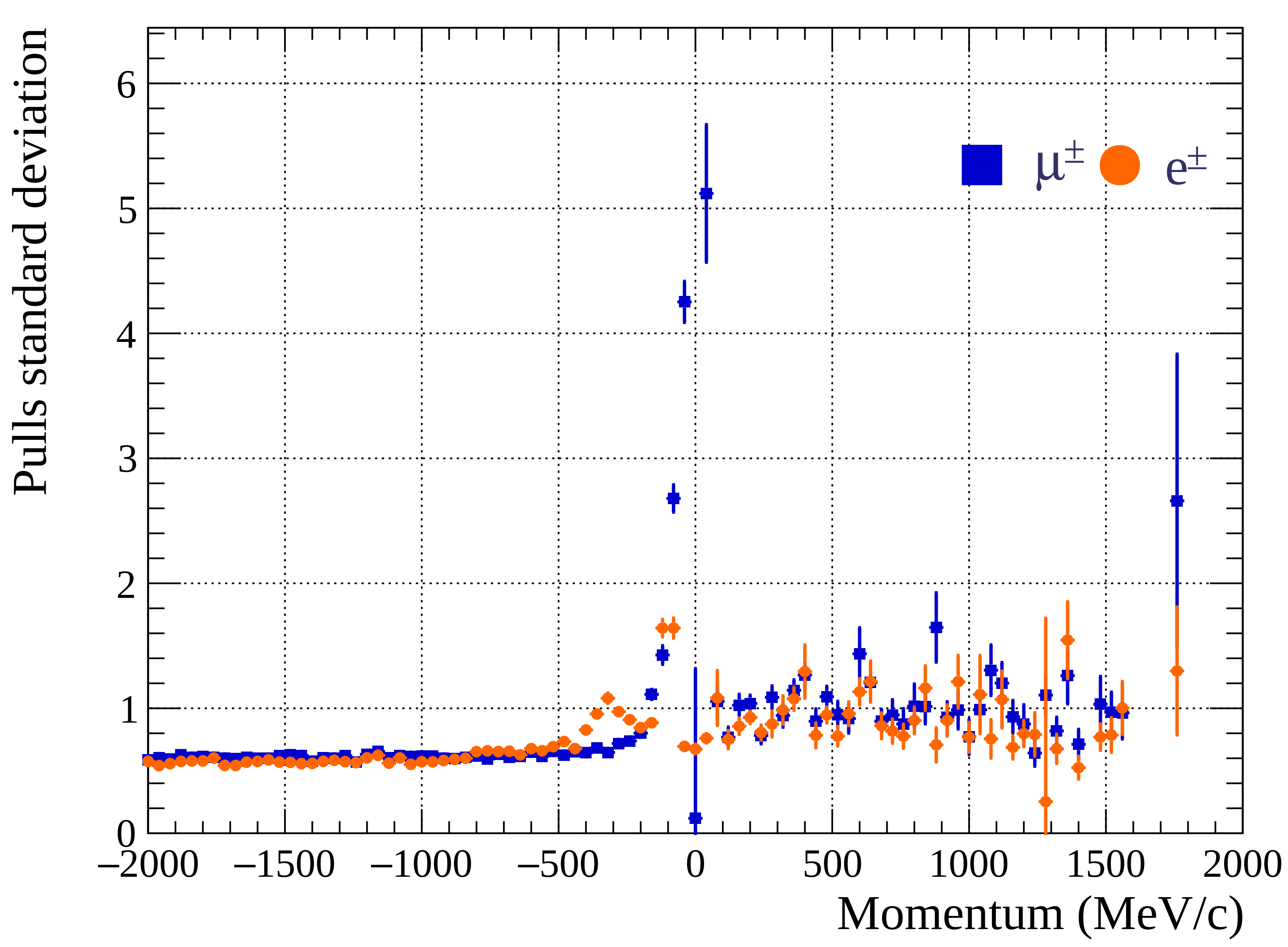


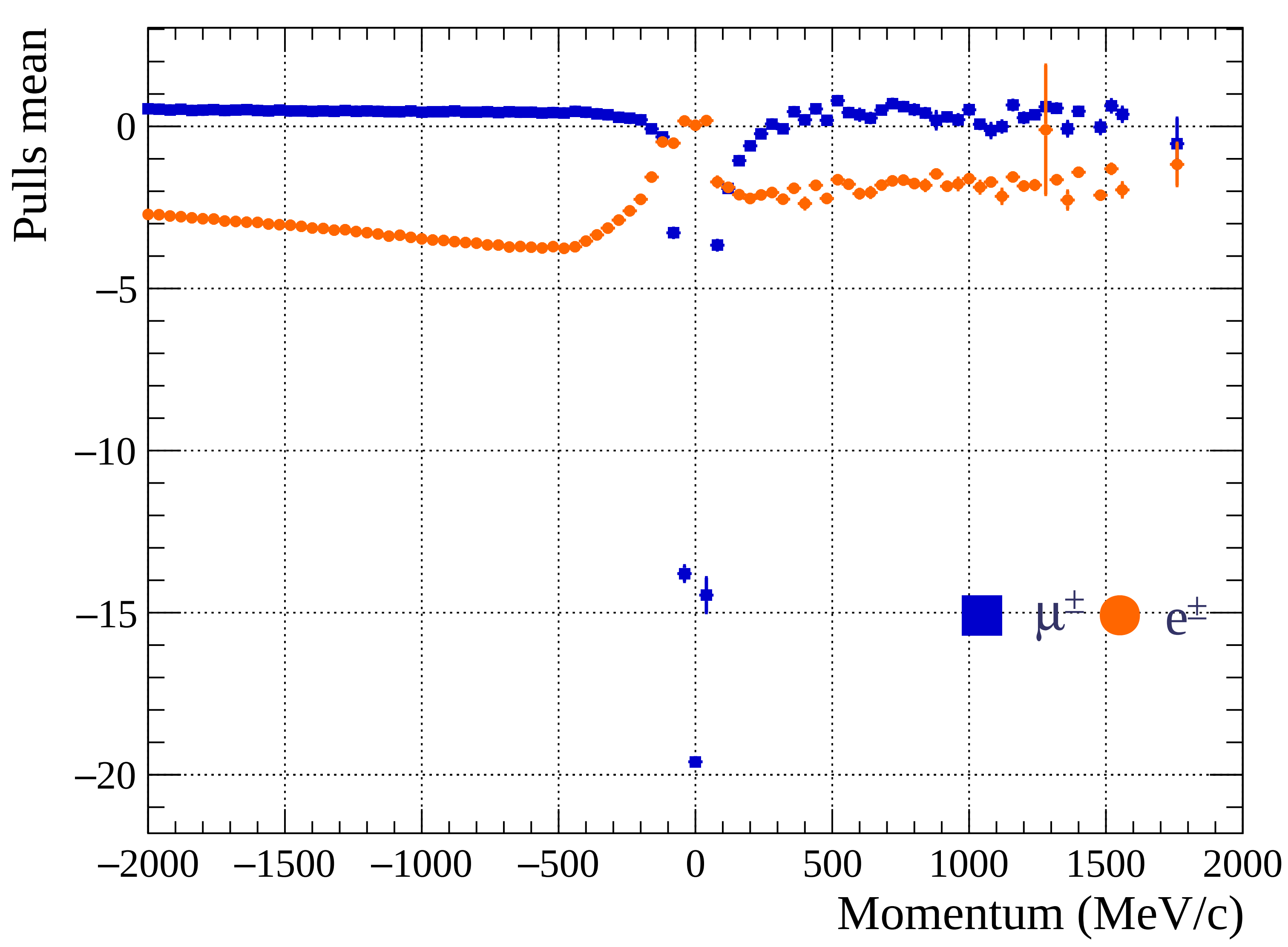
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
80
70
60
50
40
30
20
10
0



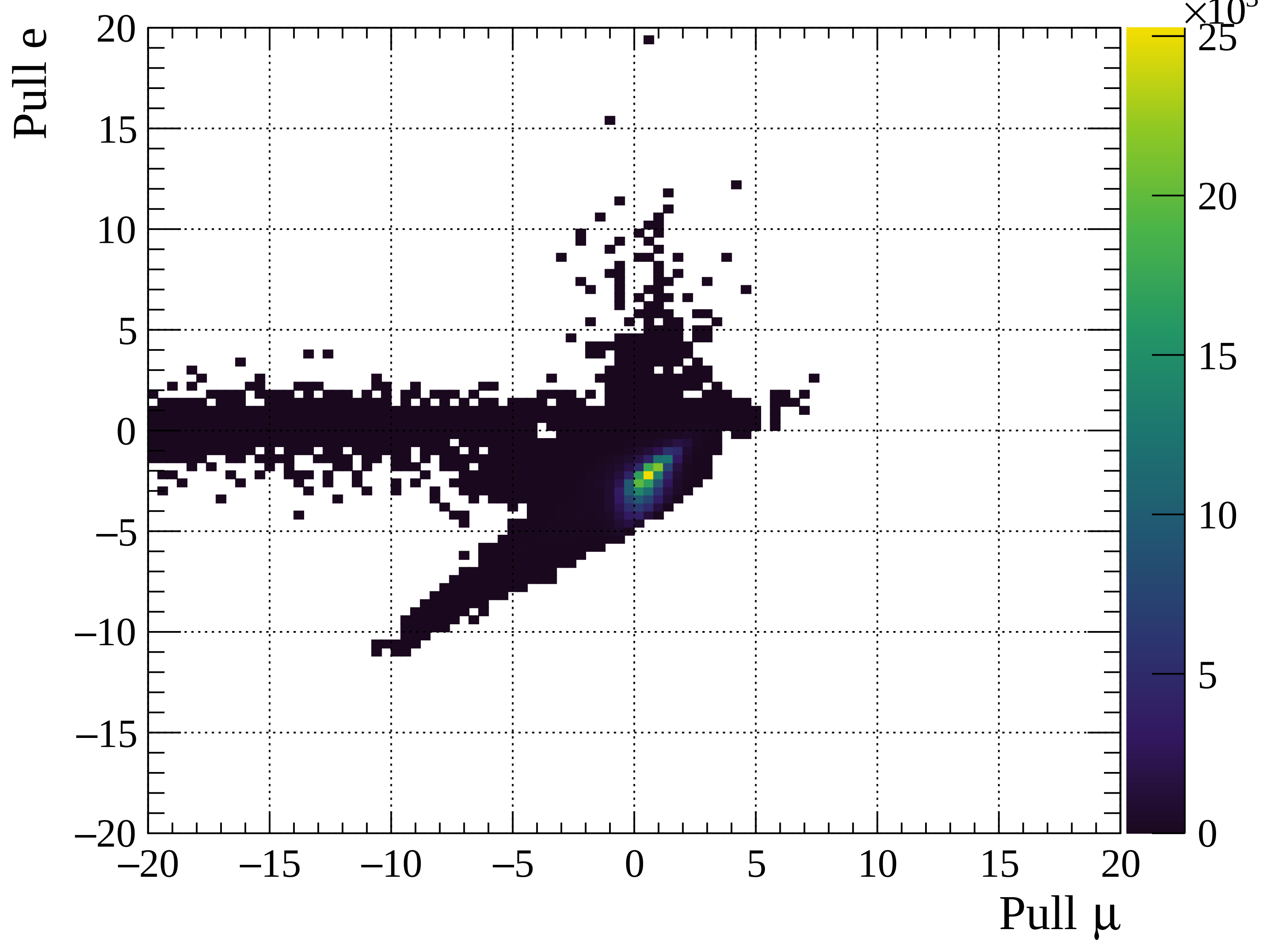
Pulls standard deviation
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

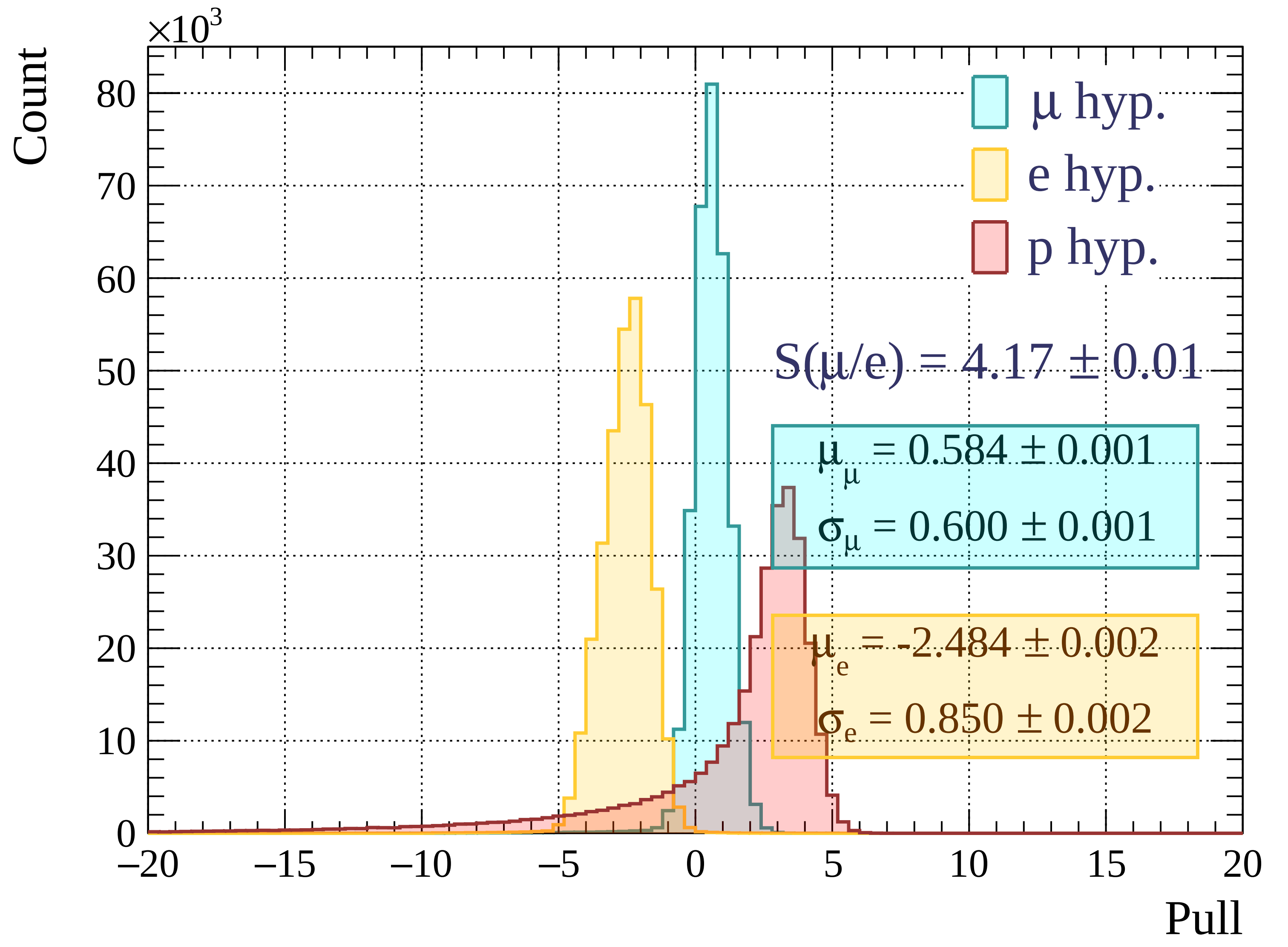

Pulls mean
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±


Pulls standard deviation
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$
0
1
2
3
4
5
6
−2000
−1500
−1000
−500
0
500
1000
1500
2000

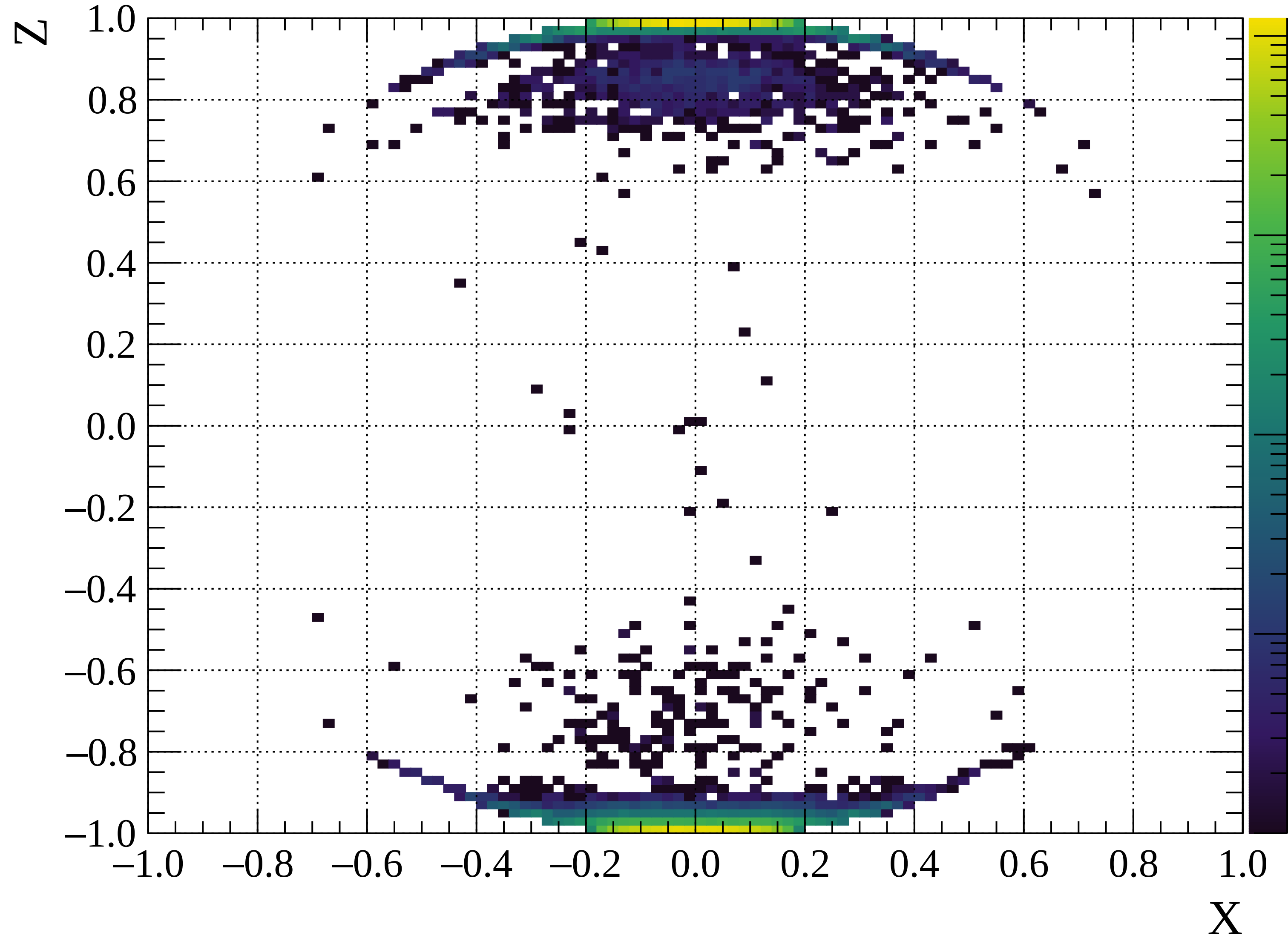


Pulls mean
0
−5
−10
−15
−20
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±

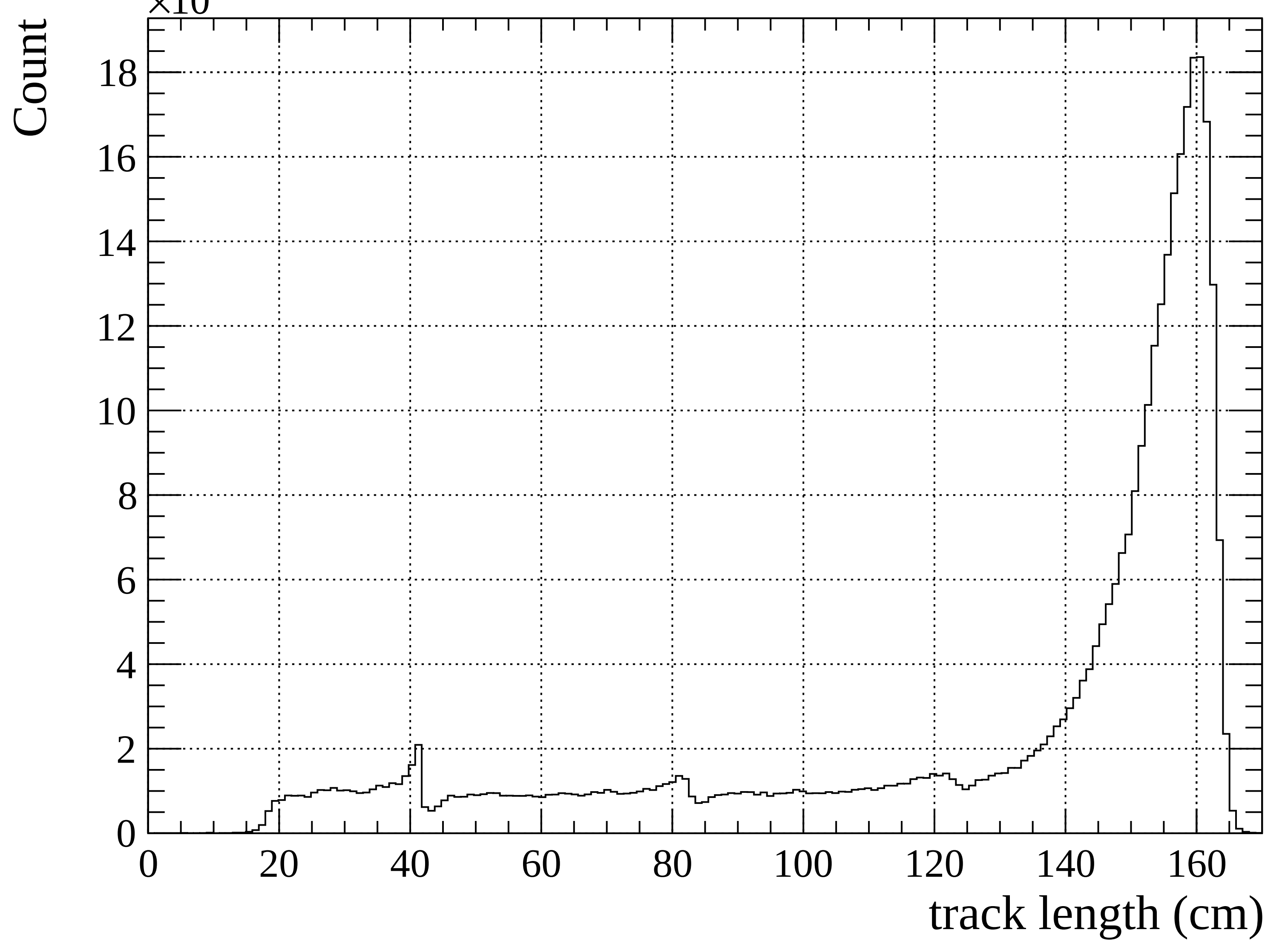

Pull e
Pull μ
$\times 10^3$
−20
−15
−10
−5
0
5
10
15
20
25

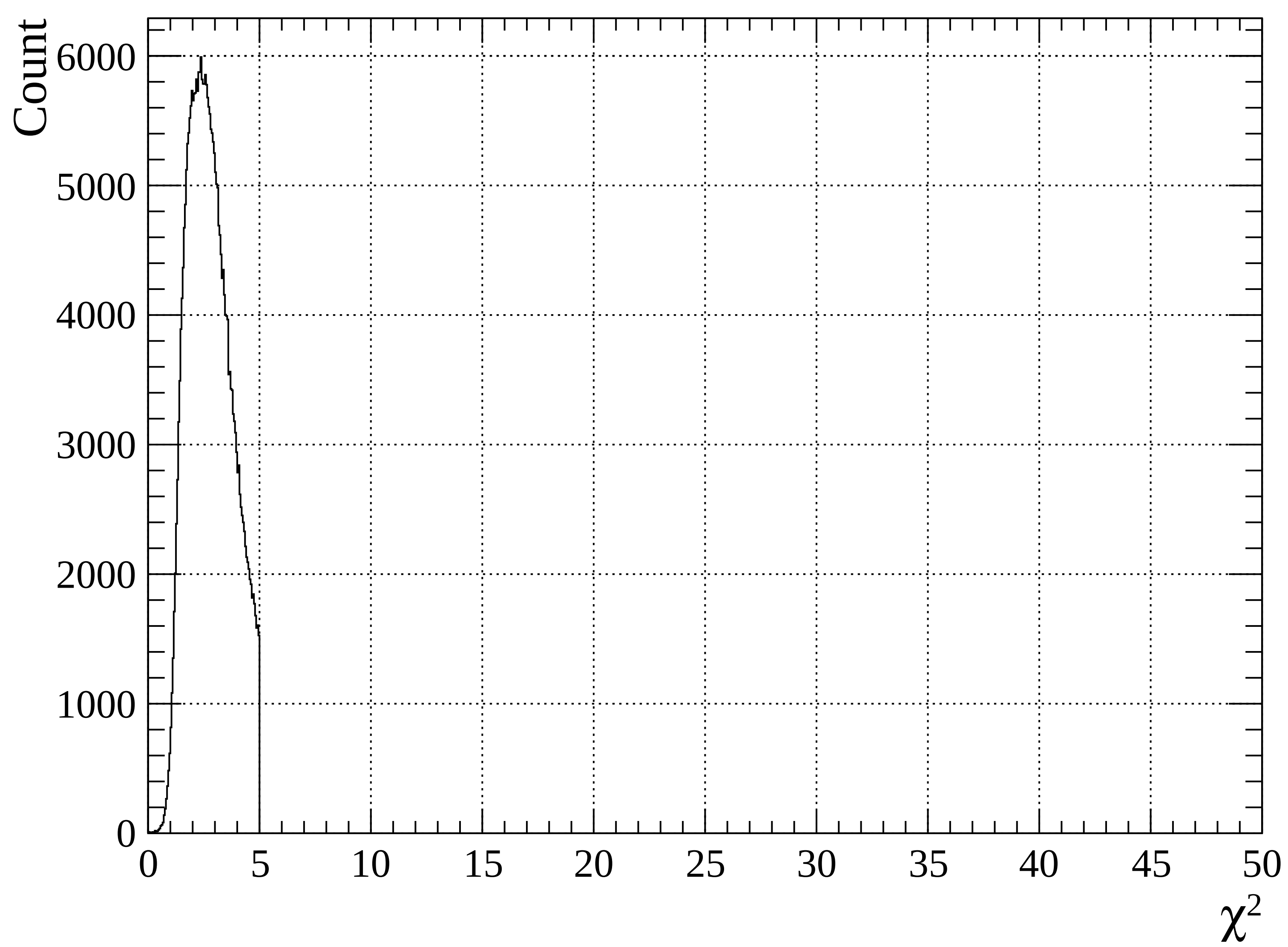


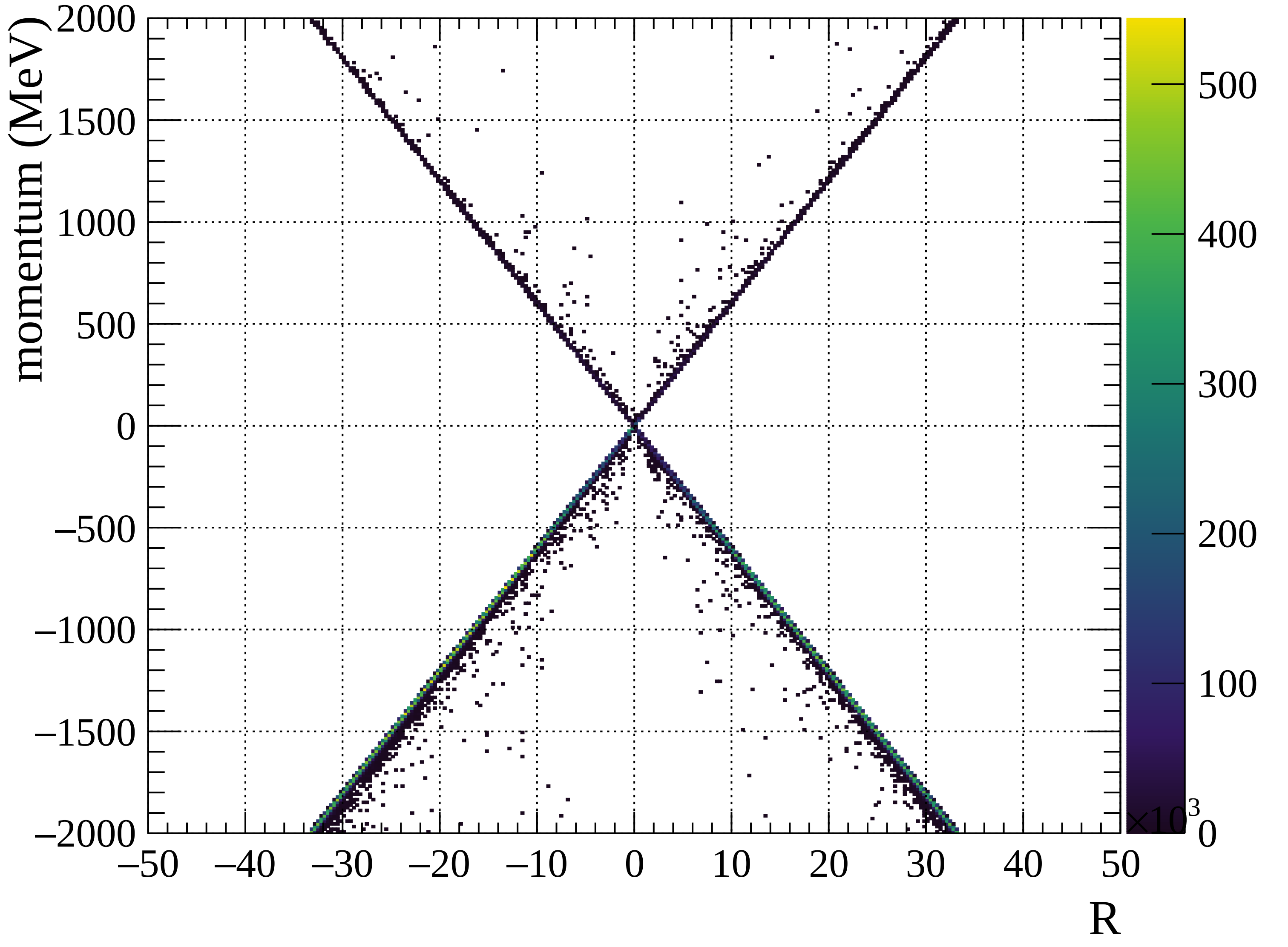
×10³
Count
80
70
60
50
40
30
20
10
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 4.17 ± 0.01
μ_μ = 0.584 ± 0.001
σ_μ = 0.600 ± 0.001
μ_e = -2.484 ± 0.002
σ_e = 0.850 ± 0.002



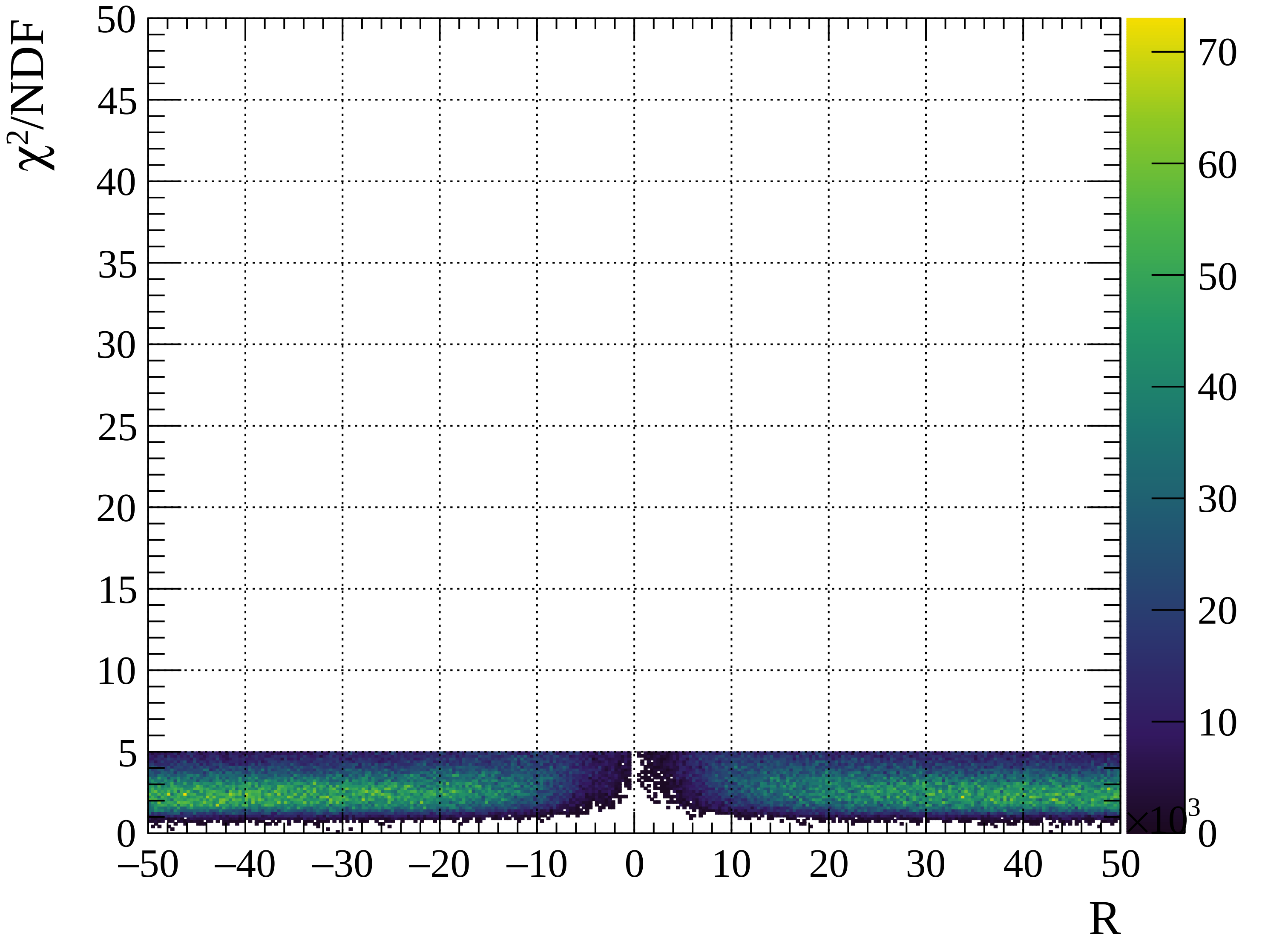
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X


×10
Count
0
2
4
6
8
10
12
14
16
18
0
20
40
60
80
100
120
140
160
track length (cm)

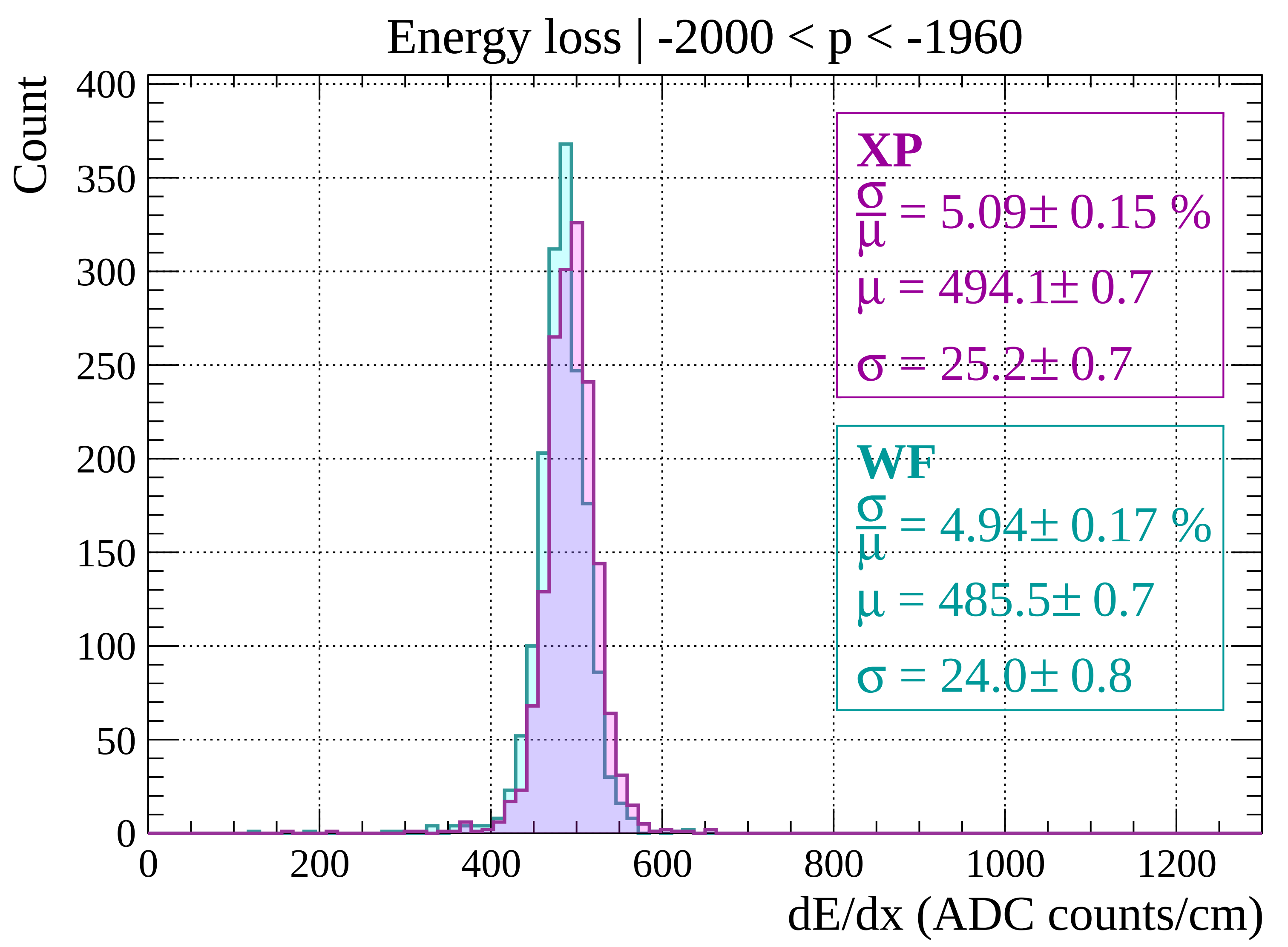

Count
6000
5000
4000
3000
2000
1000
0
0
5
10
15
20
25
30
35
40
45
50
χ^2



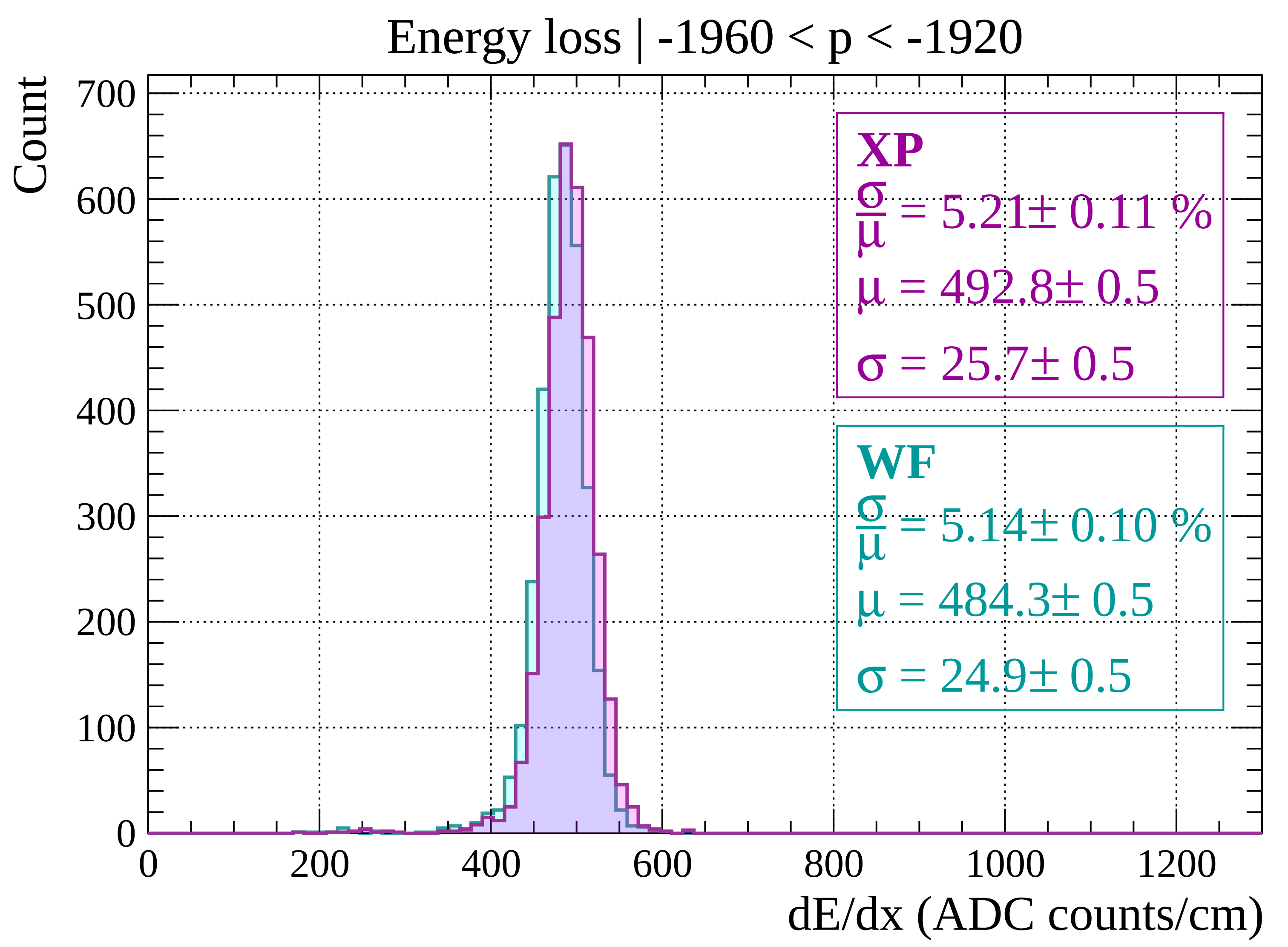
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
500
400
300
200
100
0
$\times 10^{3}$



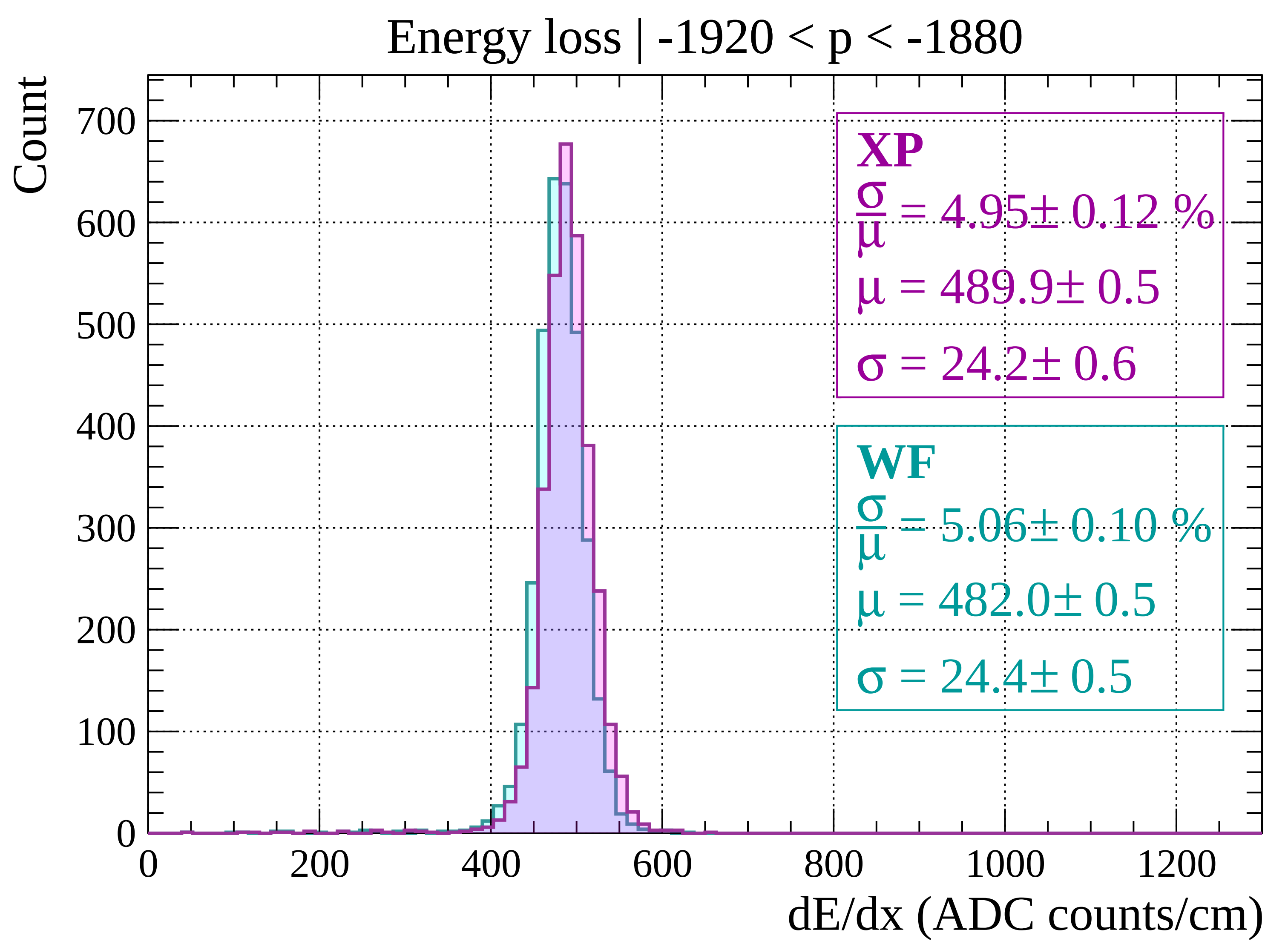
χ^2/NDF
R
$\times 10^3$
0
10
20
30
40
50
60
70
−50
−40
−30
−20
−10
0
10
20
30
40
50
0
5
10
15
20
25
30
35
40
45
50



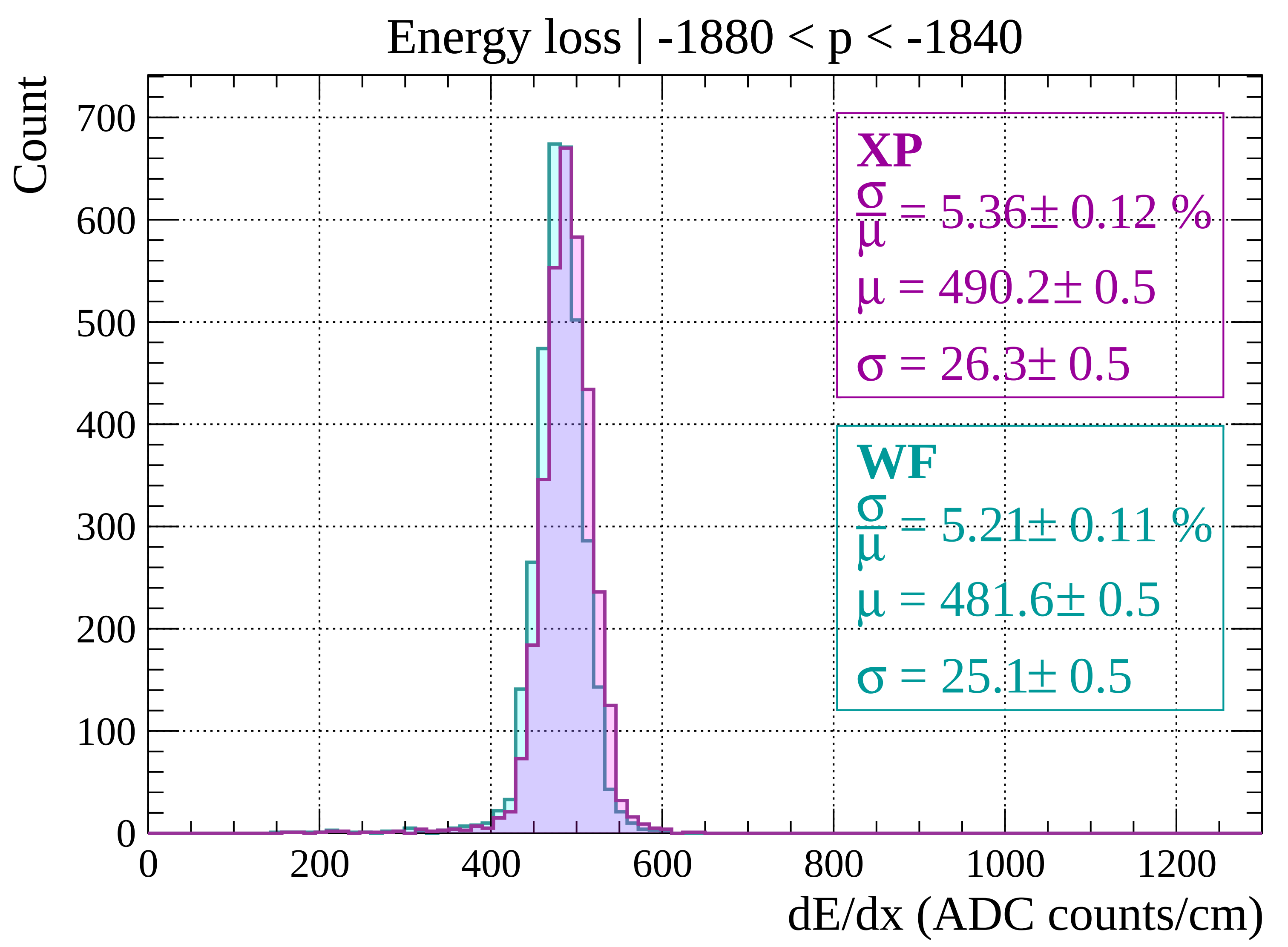
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.09± 0.15 %
μ = 494.1± 0.7
σ = 25.2± 0.7
WF
$\frac{\sigma}{\mu}$ = 4.94± 0.17 %
μ = 485.5± 0.7
σ = 24.0± 0.8
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

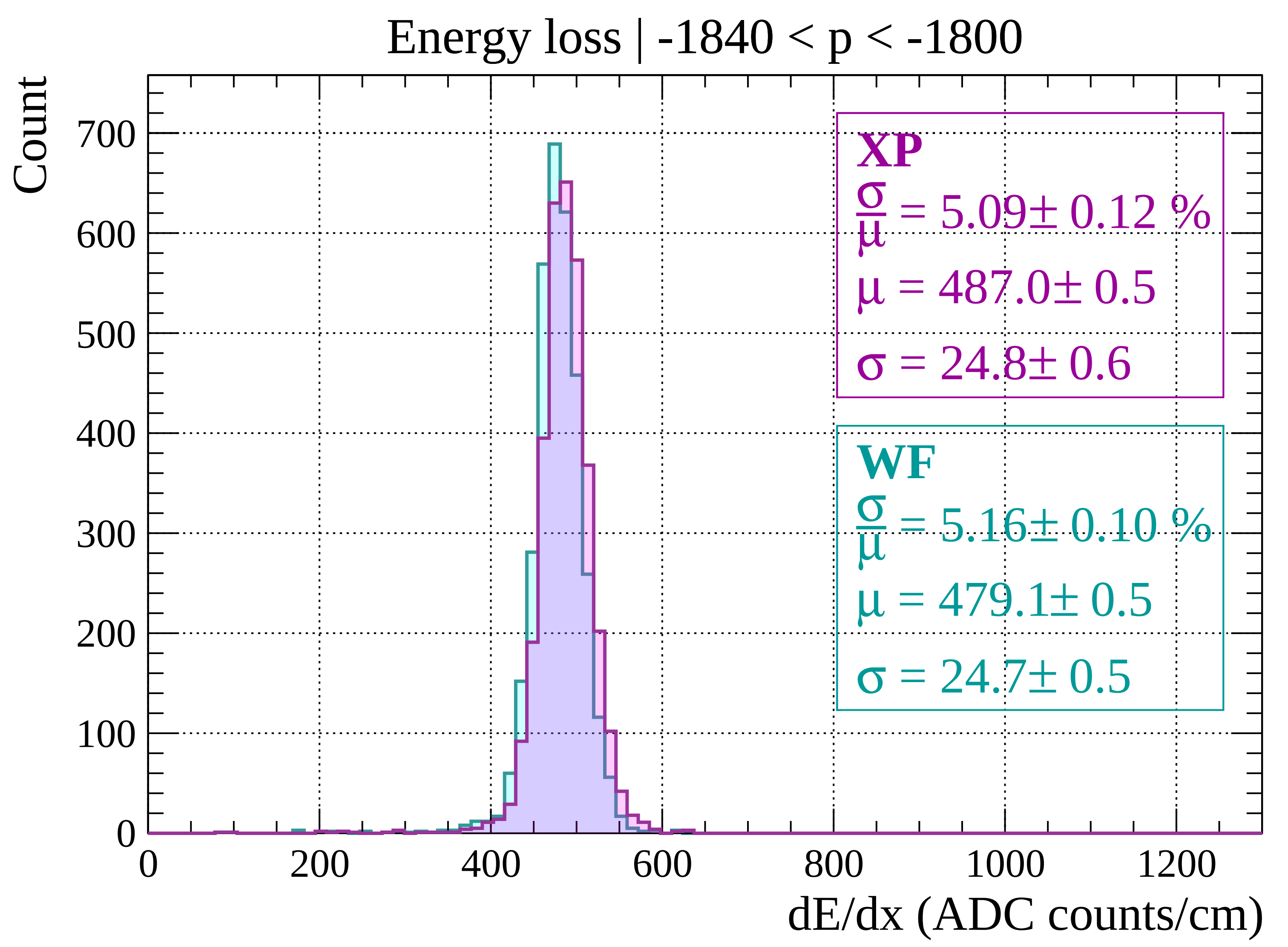


Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.21± 0.11 %
μ = 492.8± 0.5
σ = 25.7± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.14± 0.10 %
μ = 484.3± 0.5
σ = 24.9± 0.5

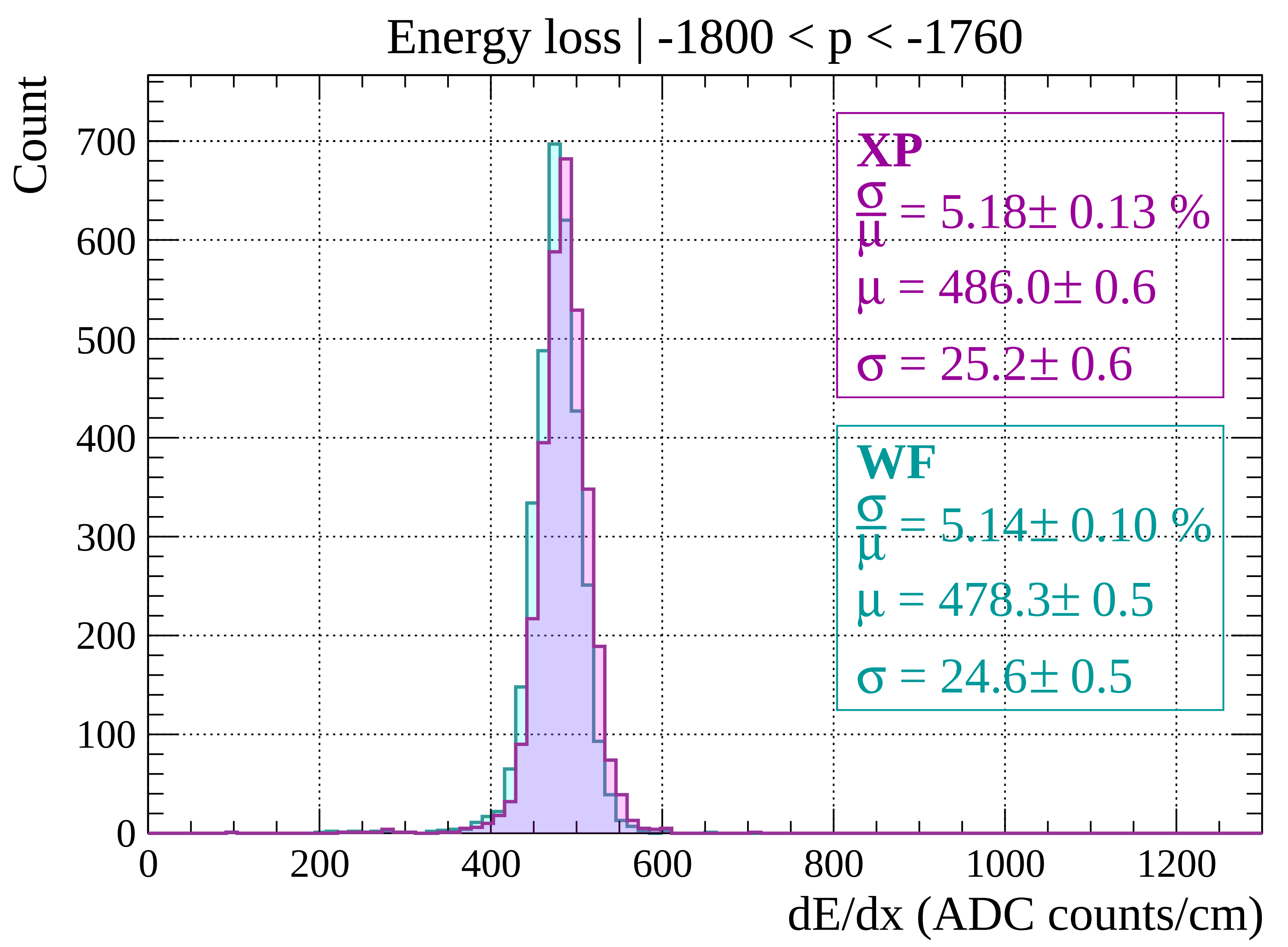


Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 4.95± 0.12 %
μ = 489.9± 0.5
σ = 24.2± 0.6
WF
σ/μ = 5.06± 0.10 %
μ = 482.0± 0.5
σ = 24.4± 0.5
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

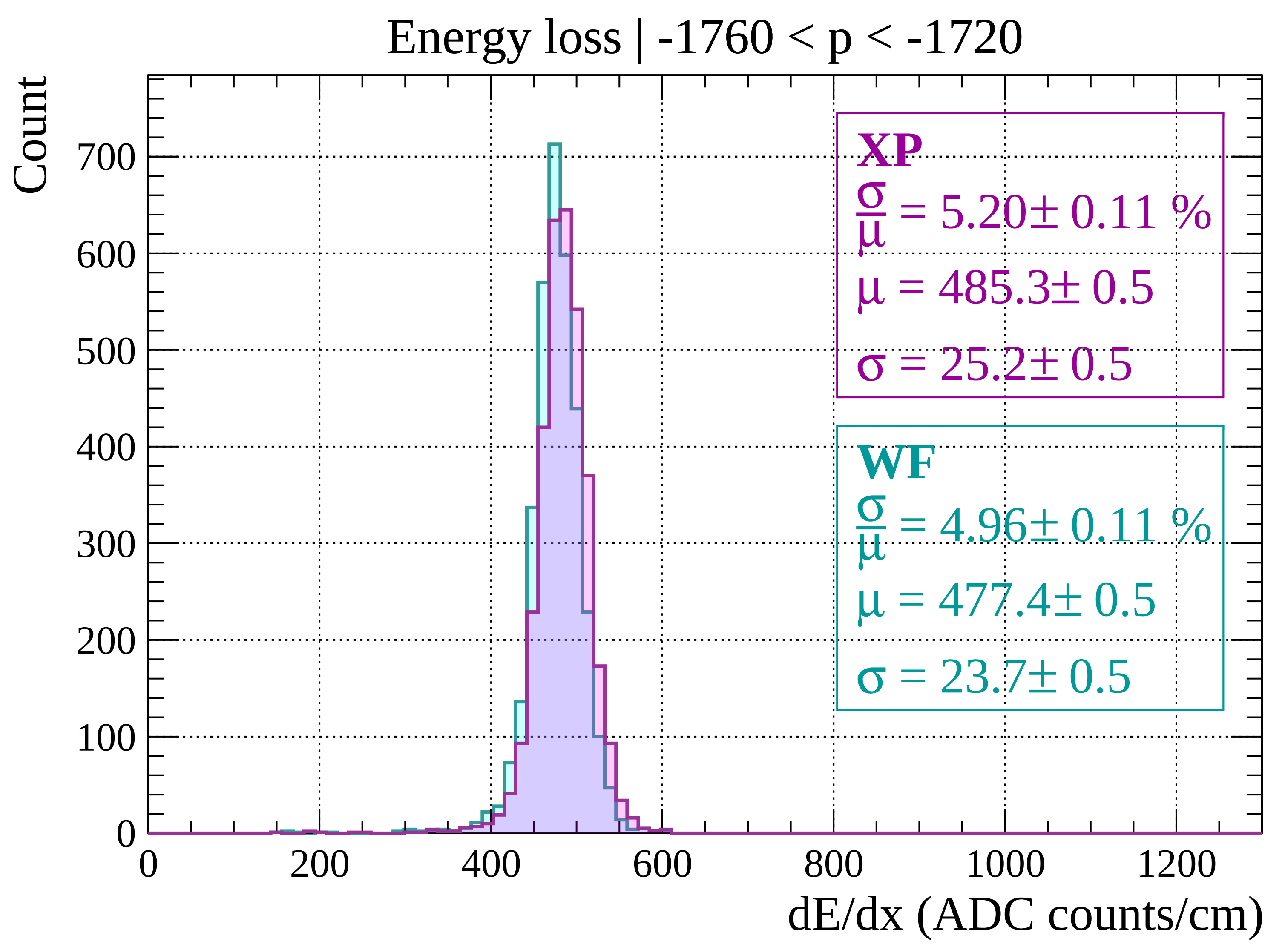

Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.36± 0.12 %
μ = 490.2± 0.5
σ = 26.3± 0.5
WF
σ/μ = 5.21± 0.11 %
μ = 481.6± 0.5
σ = 25.1± 0.5



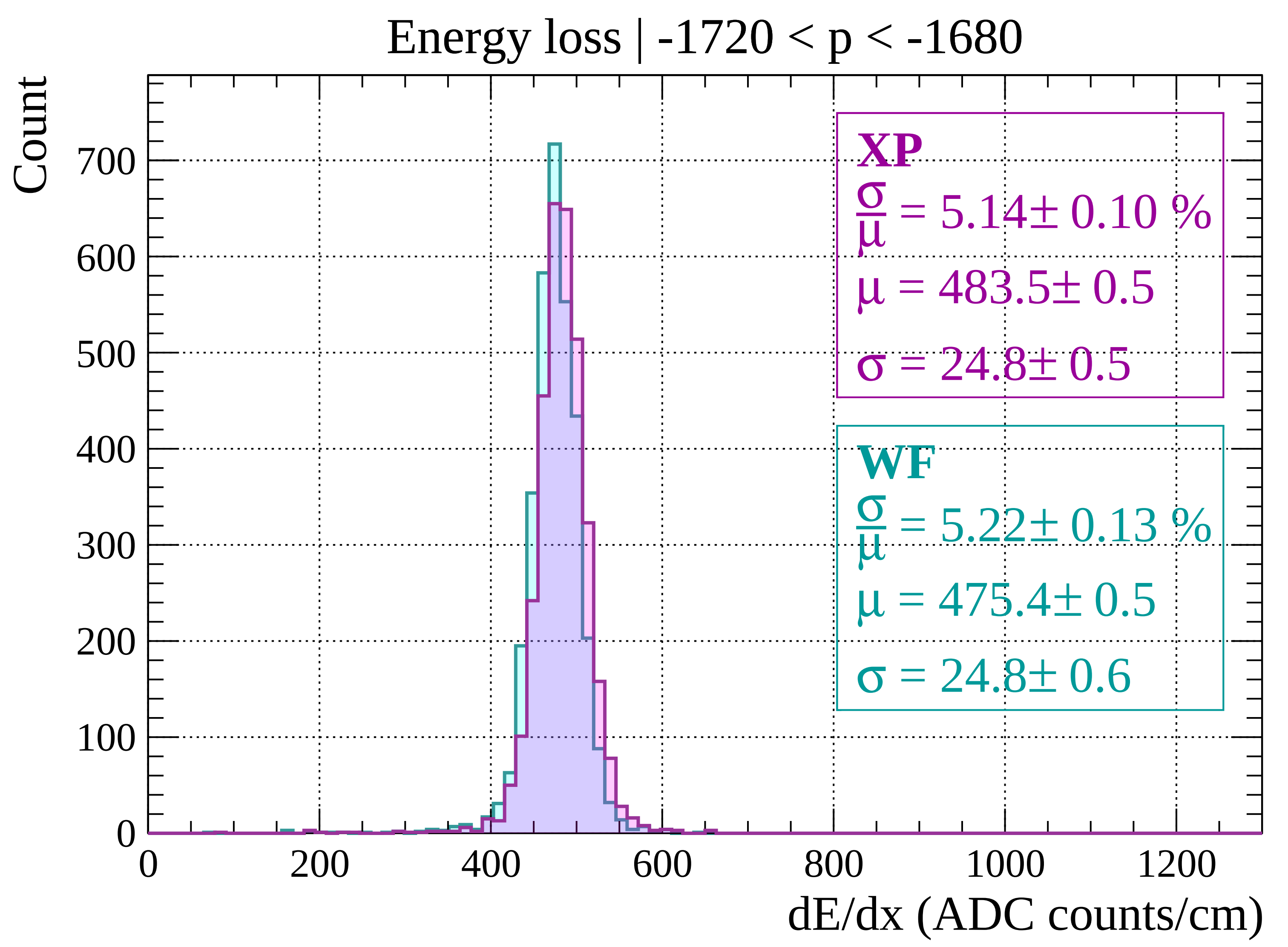
Energy loss | -1840 < p < -1800
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.09± 0.12 %
μ = 487.0± 0.5
σ = 24.8± 0.6
WF
σ/μ = 5.16± 0.10 %
μ = 479.1± 0.5
σ = 24.7± 0.5



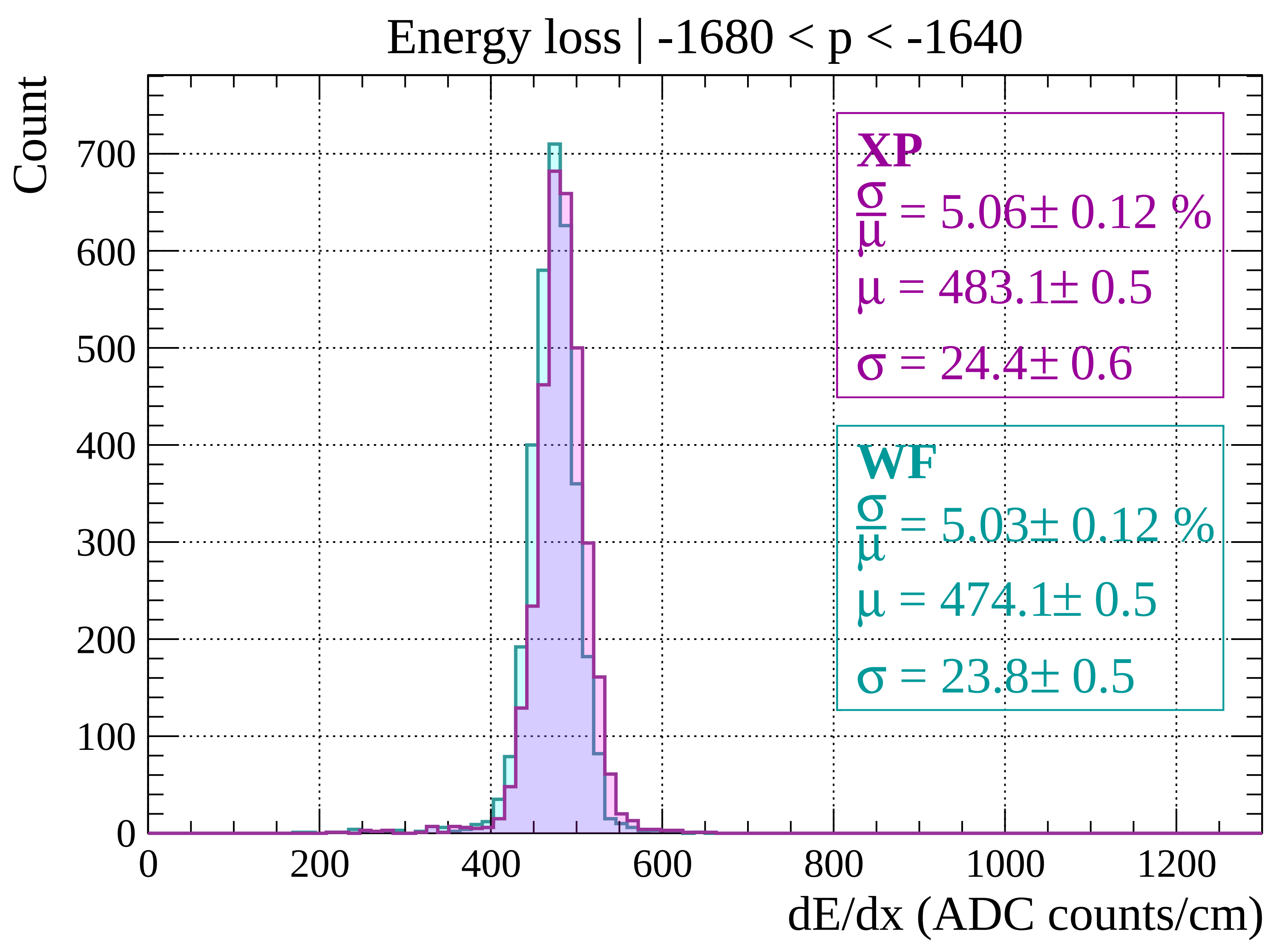
Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP
$\frac{\sigma}{\mu} = 5.18 \pm 0.13$ %
$\mu = 486.0 \pm 0.6$
$\sigma = 25.2 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 5.14 \pm 0.10$ %
$\mu = 478.3 \pm 0.5$
$\sigma = 24.6 \pm 0.5$

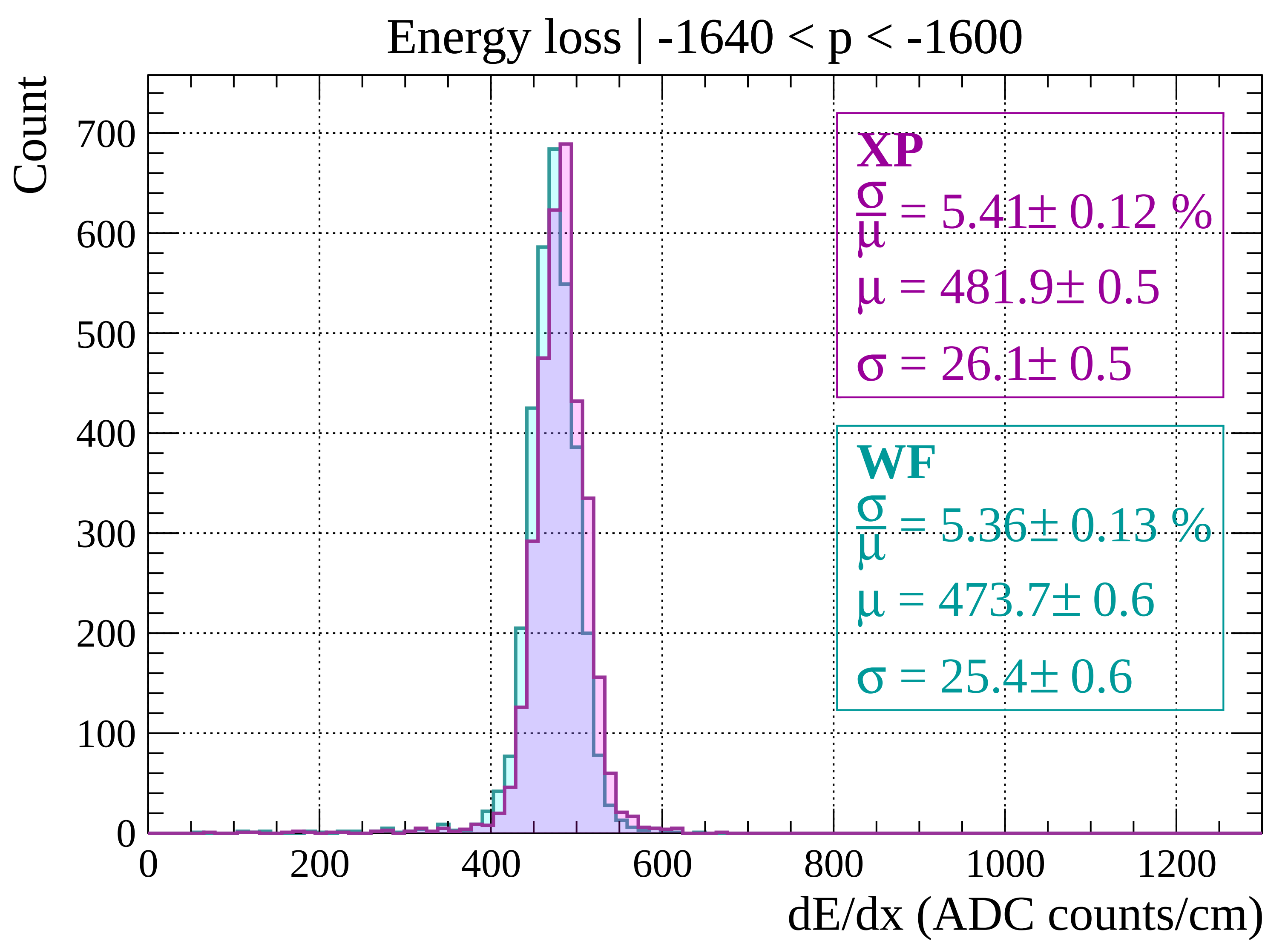


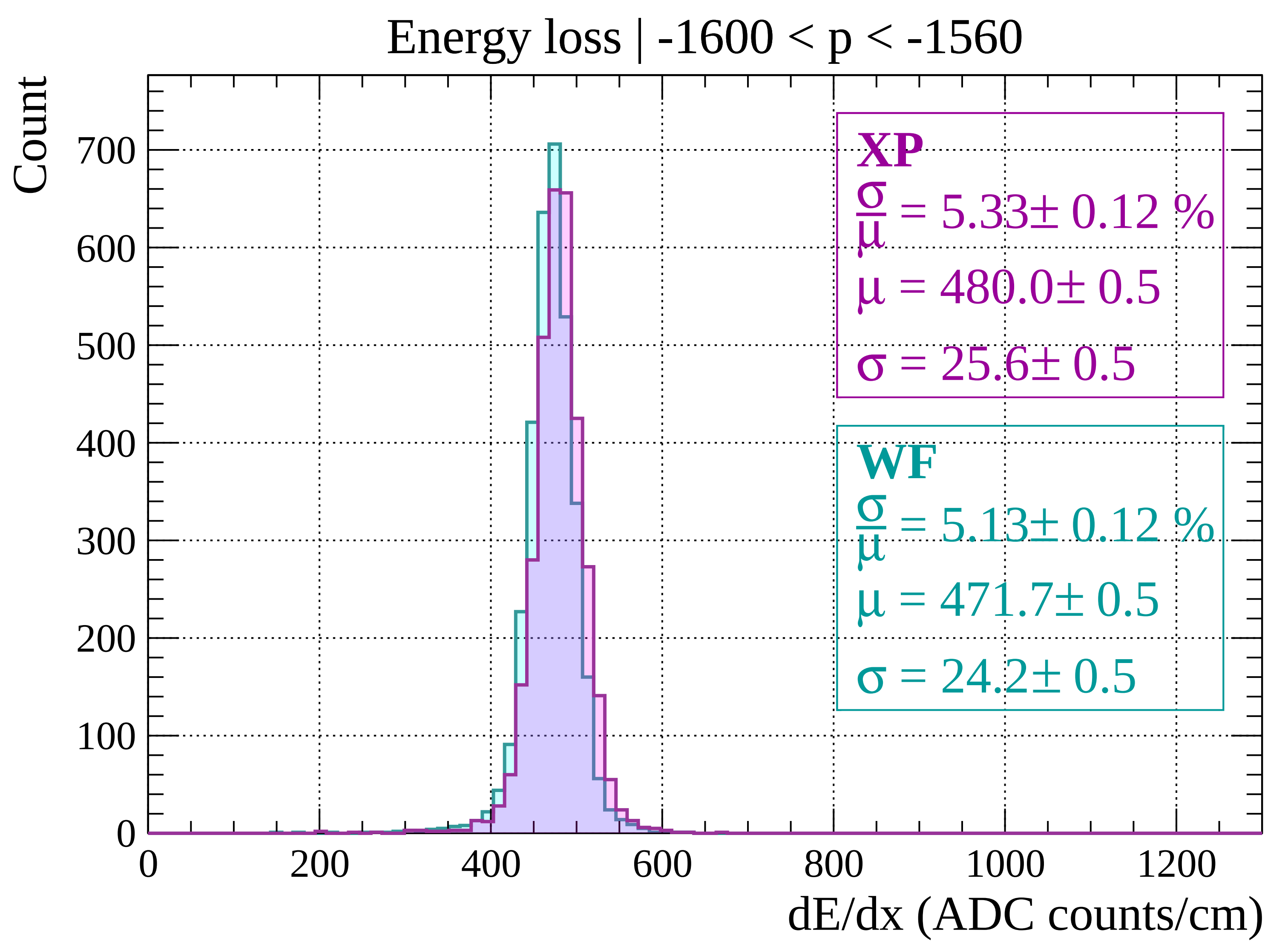
Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 5.20± 0.11 %
μ = 485.3± 0.5
σ = 25.2± 0.5
WF
σ/μ = 4.96± 0.11 %
μ = 477.4± 0.5
σ = 23.7± 0.5



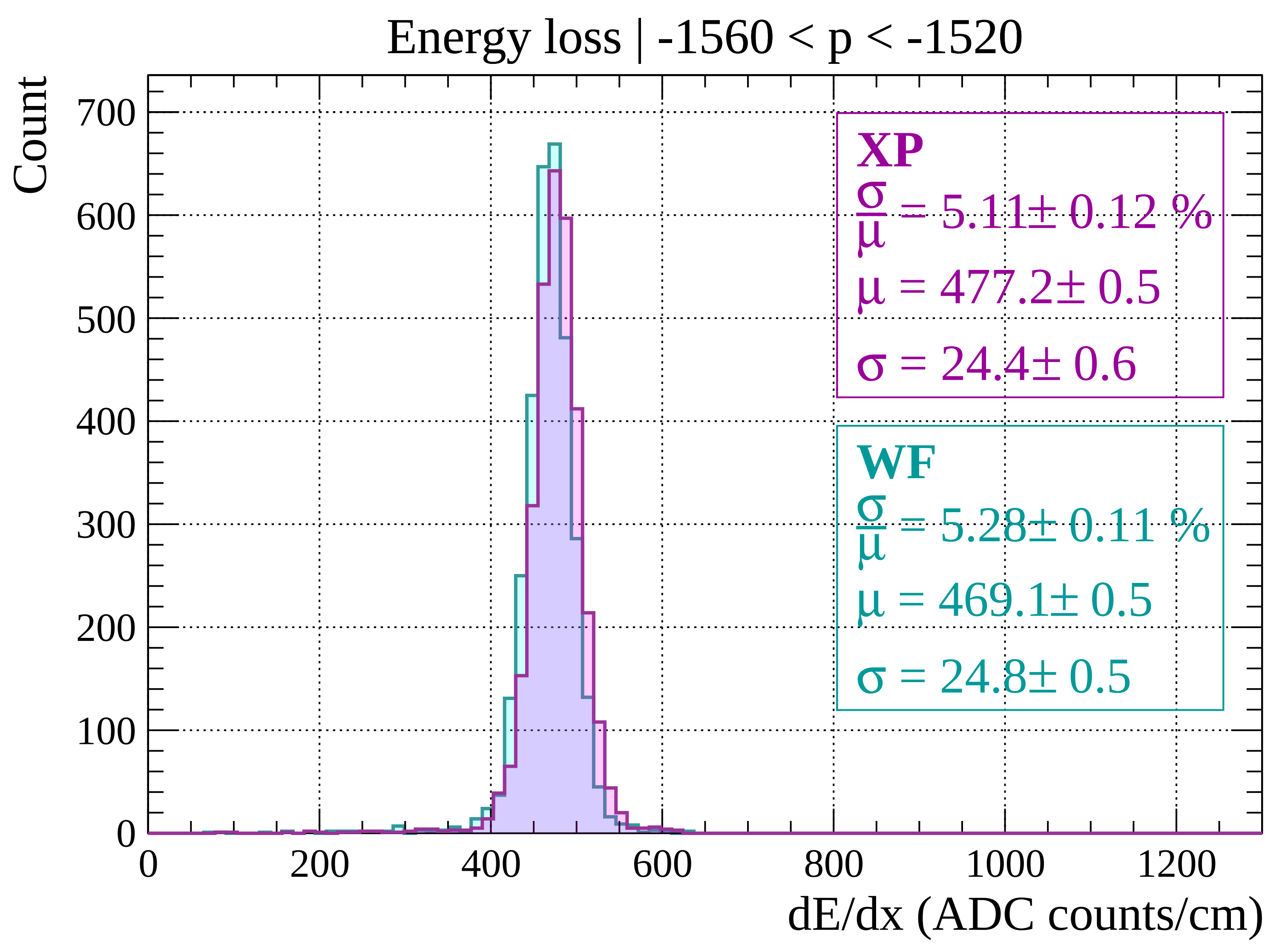
Energy loss | -1720 < p < -1680
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.14± 0.10 %
μ = 483.5± 0.5
σ = 24.8± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.22± 0.13 %
μ = 475.4± 0.5
σ = 24.8± 0.6


Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.06± 0.12 %
μ = 483.1± 0.5
σ = 24.4± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.03± 0.12 %
μ = 474.1± 0.5
σ = 23.8± 0.5

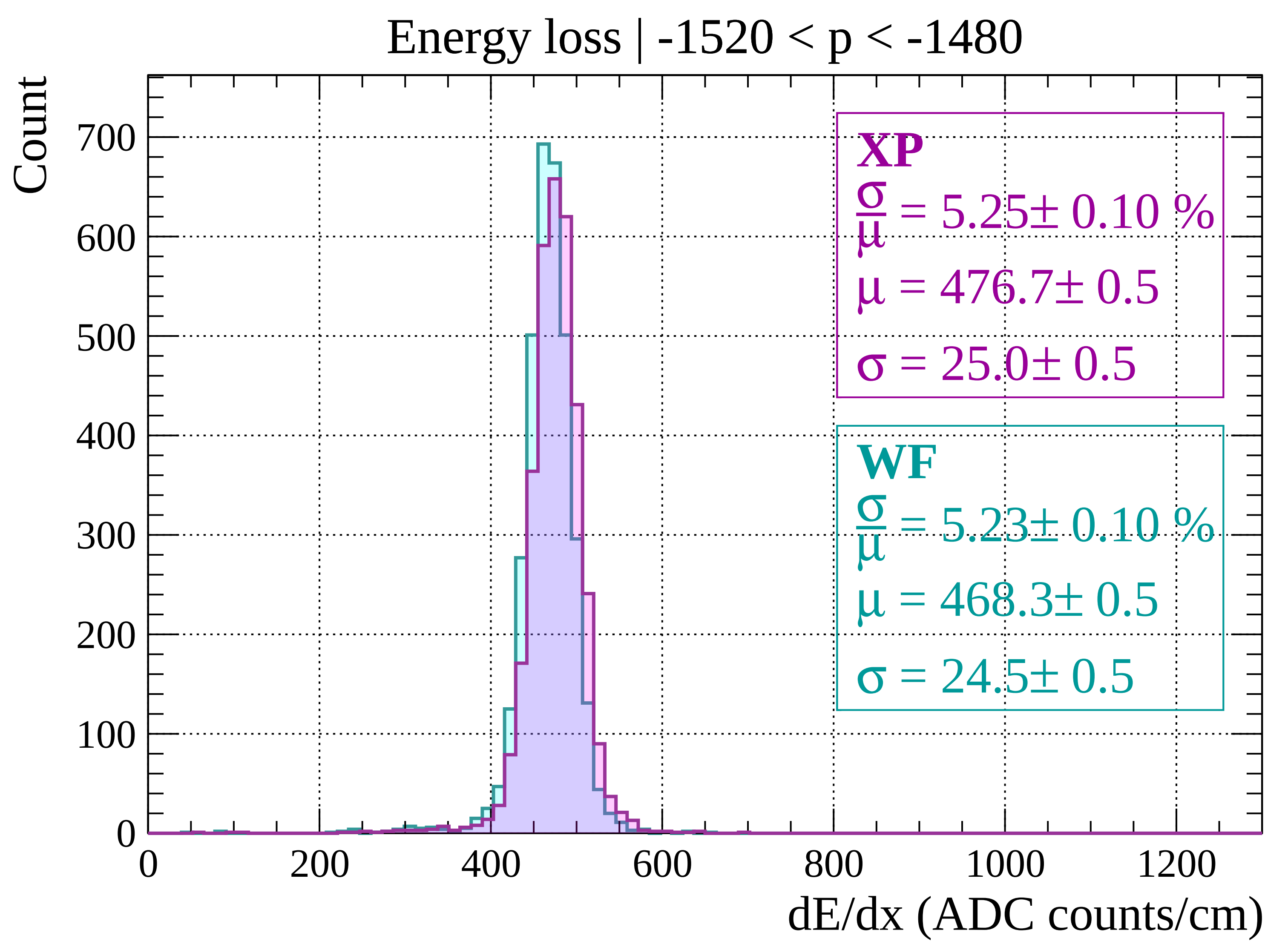

Energy loss | -1640 < p < -1600
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.41± 0.12 %
μ = 481.9± 0.5
σ = 26.1± 0.5
WF
σ/μ = 5.36± 0.13 %
μ = 473.7± 0.6
σ = 25.4± 0.6

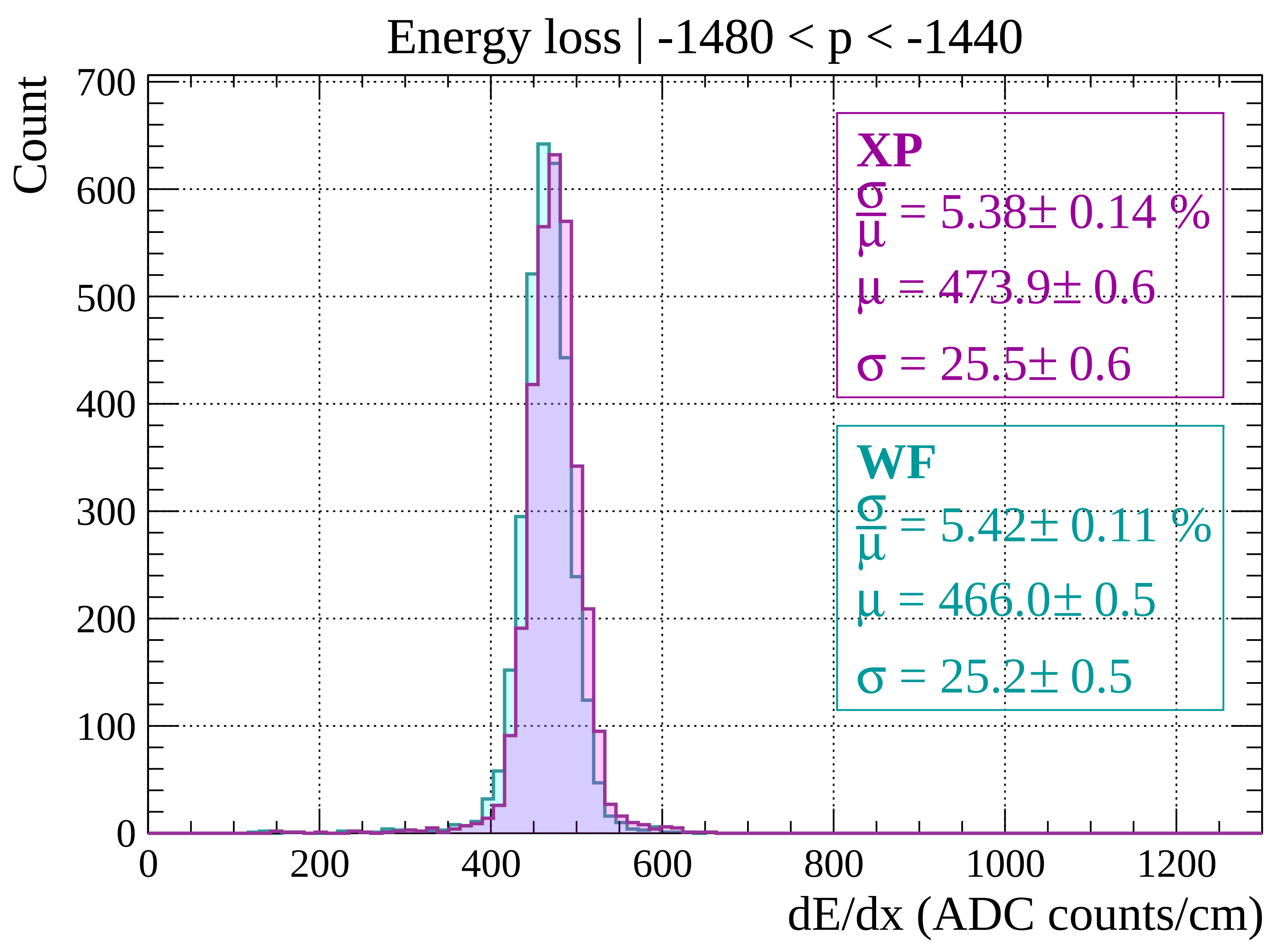


Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.33± 0.12 %
μ = 480.0± 0.5
σ = 25.6± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.13± 0.12 %
μ = 471.7± 0.5
σ = 24.2± 0.5

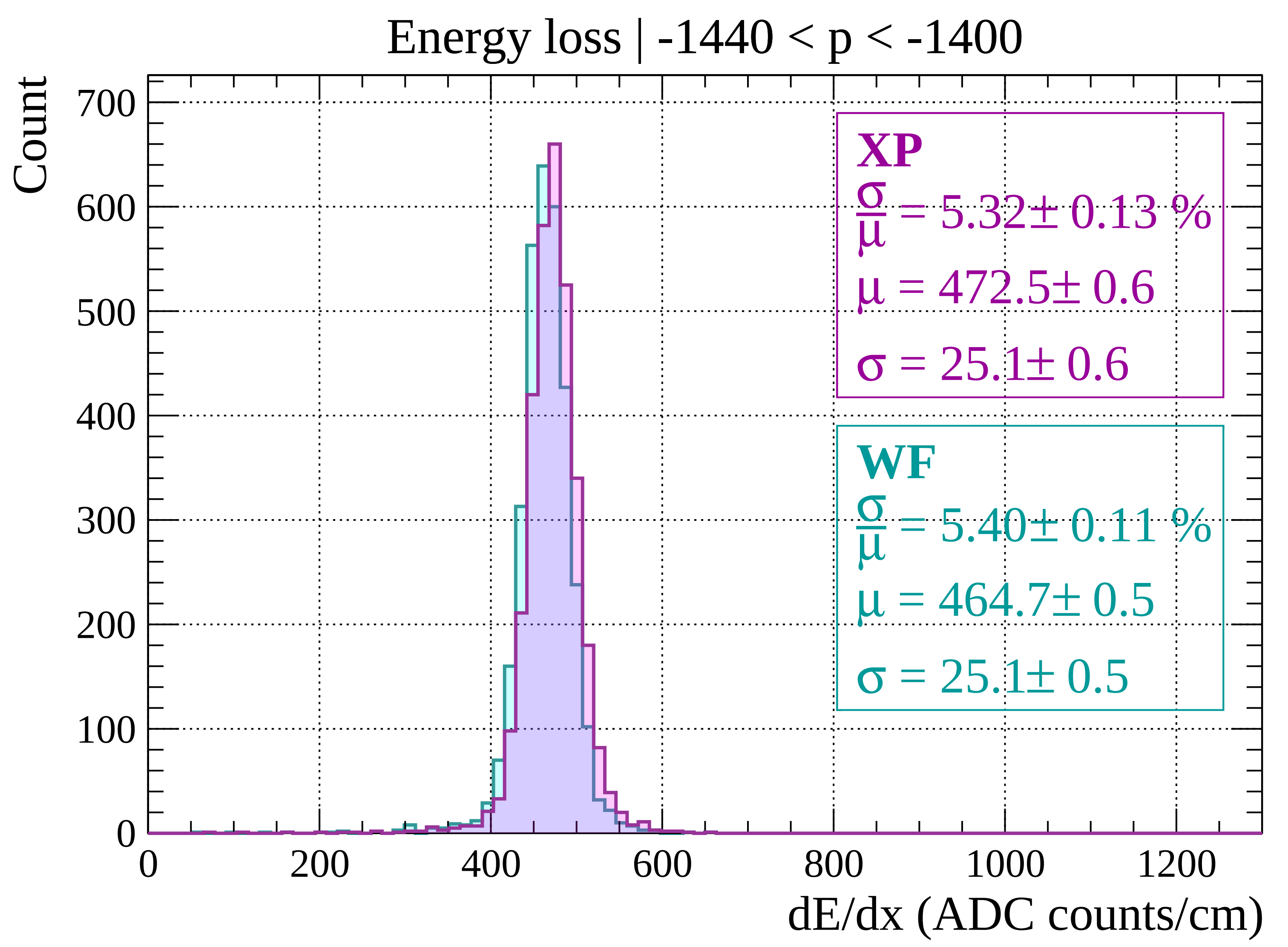


Energy loss | -1560 < p < -1520
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.11± 0.12 %
μ = 477.2± 0.5
σ = 24.4± 0.6
WF
σ/μ = 5.28± 0.11 %
μ = 469.1± 0.5
σ = 24.8± 0.5

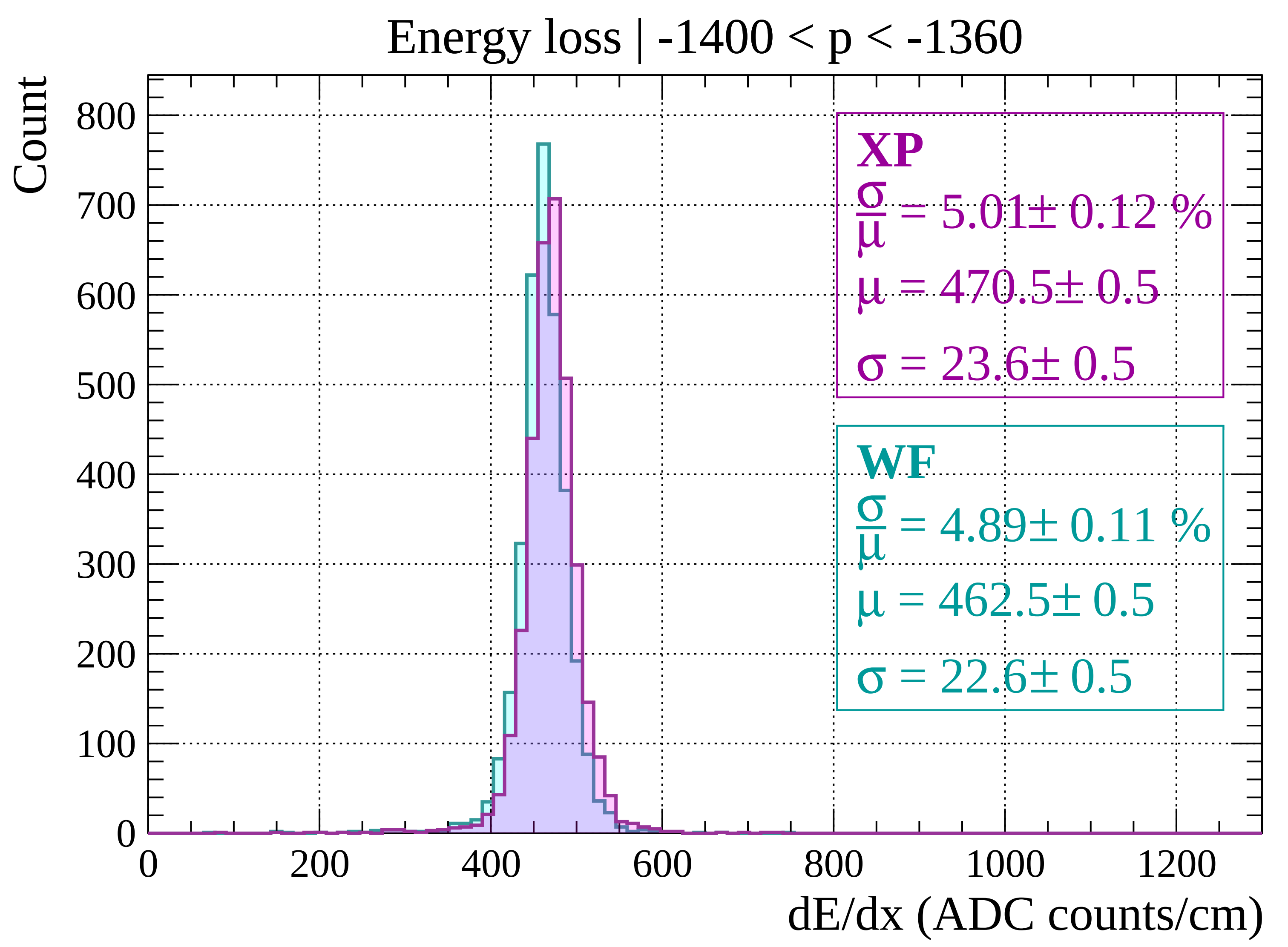

Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 5.25± 0.10 %
μ = 476.7± 0.5
σ = 25.0± 0.5
WF
σ/μ = 5.23± 0.10 %
μ = 468.3± 0.5
σ = 24.5± 0.5

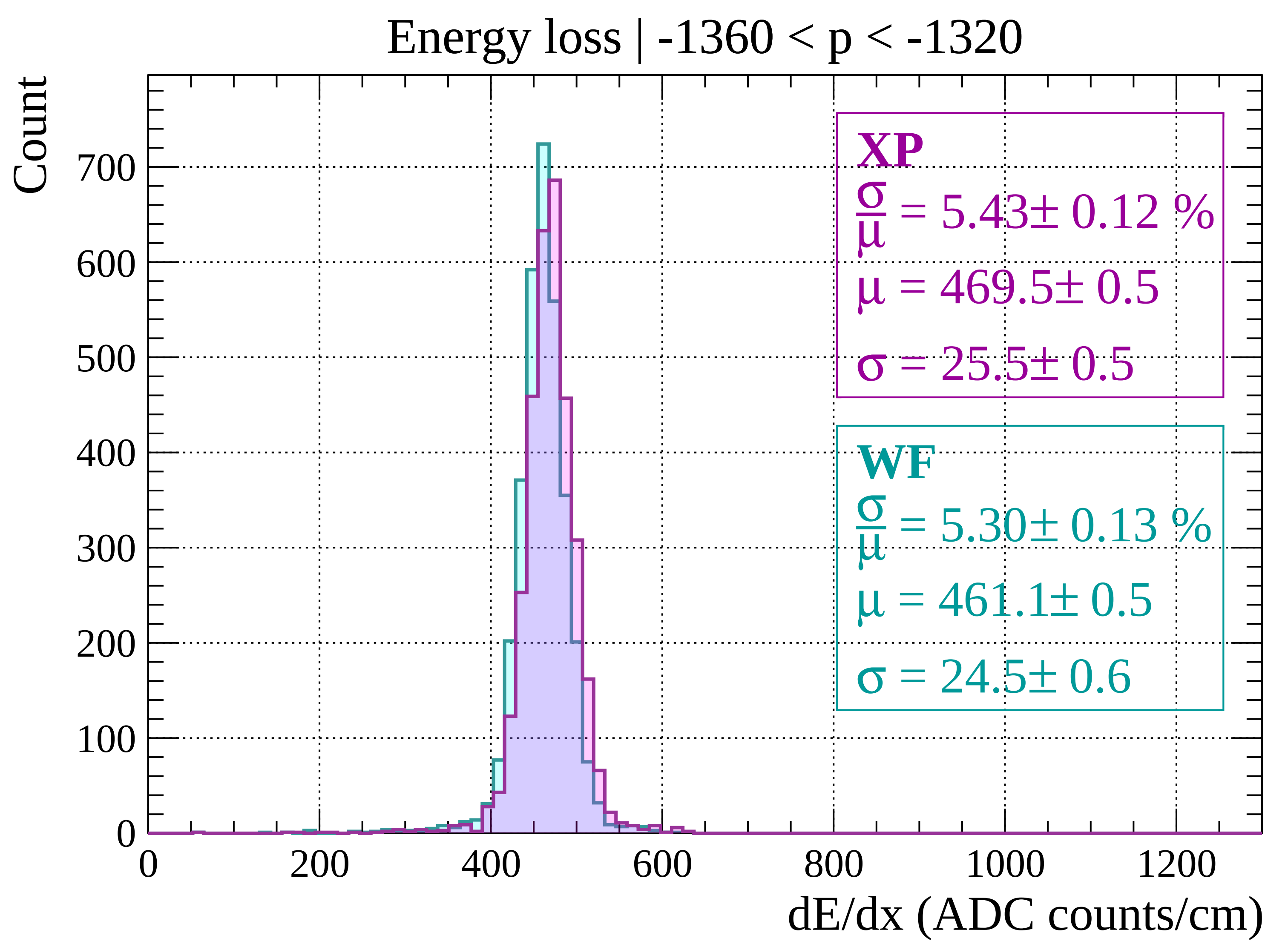


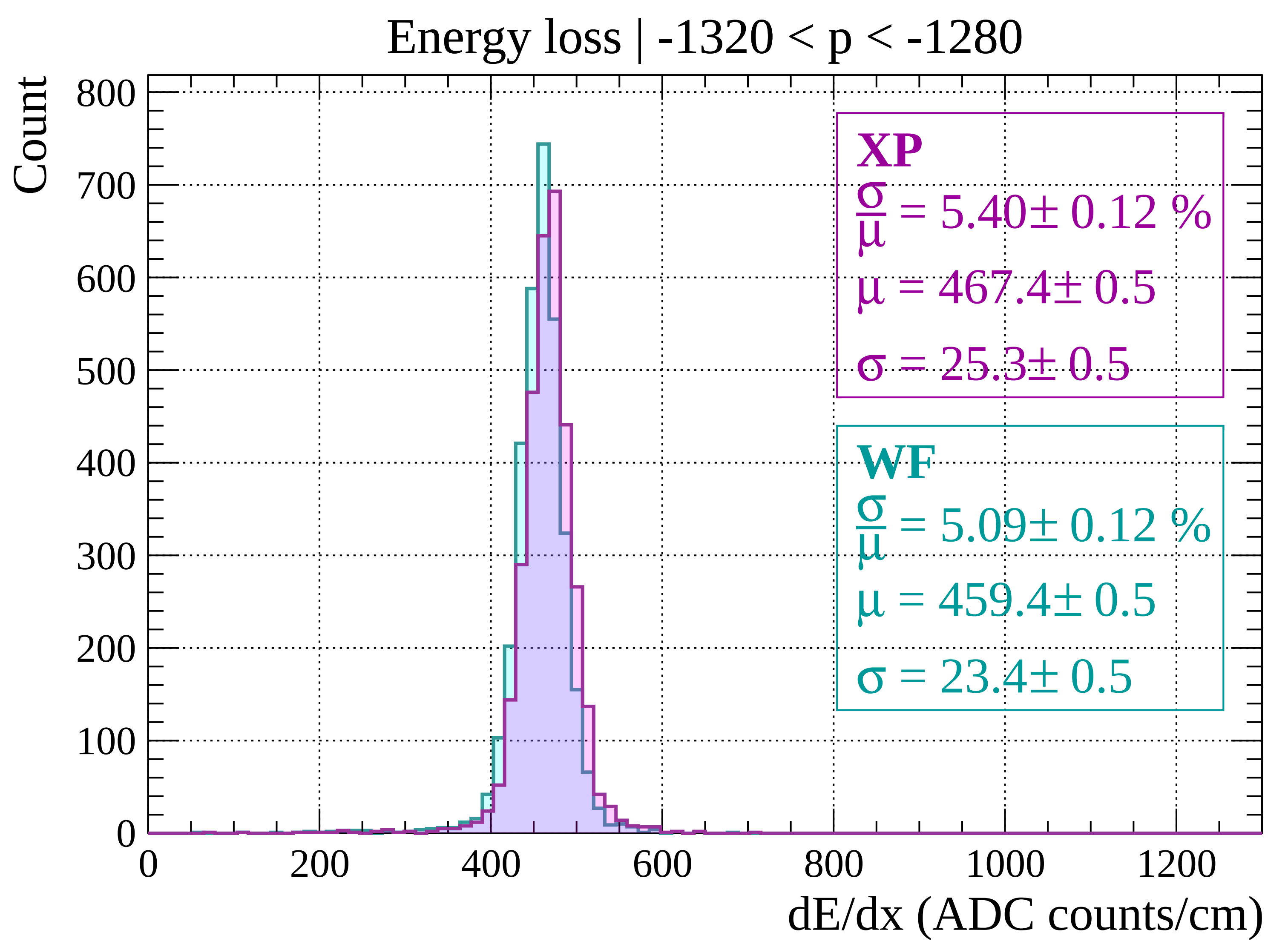
Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 5.38 \pm 0.14$ %
$\mu = 473.9 \pm 0.6$
$\sigma = 25.5 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 5.42 \pm 0.11$ %
$\mu = 466.0 \pm 0.5$
$\sigma = 25.2 \pm 0.5$

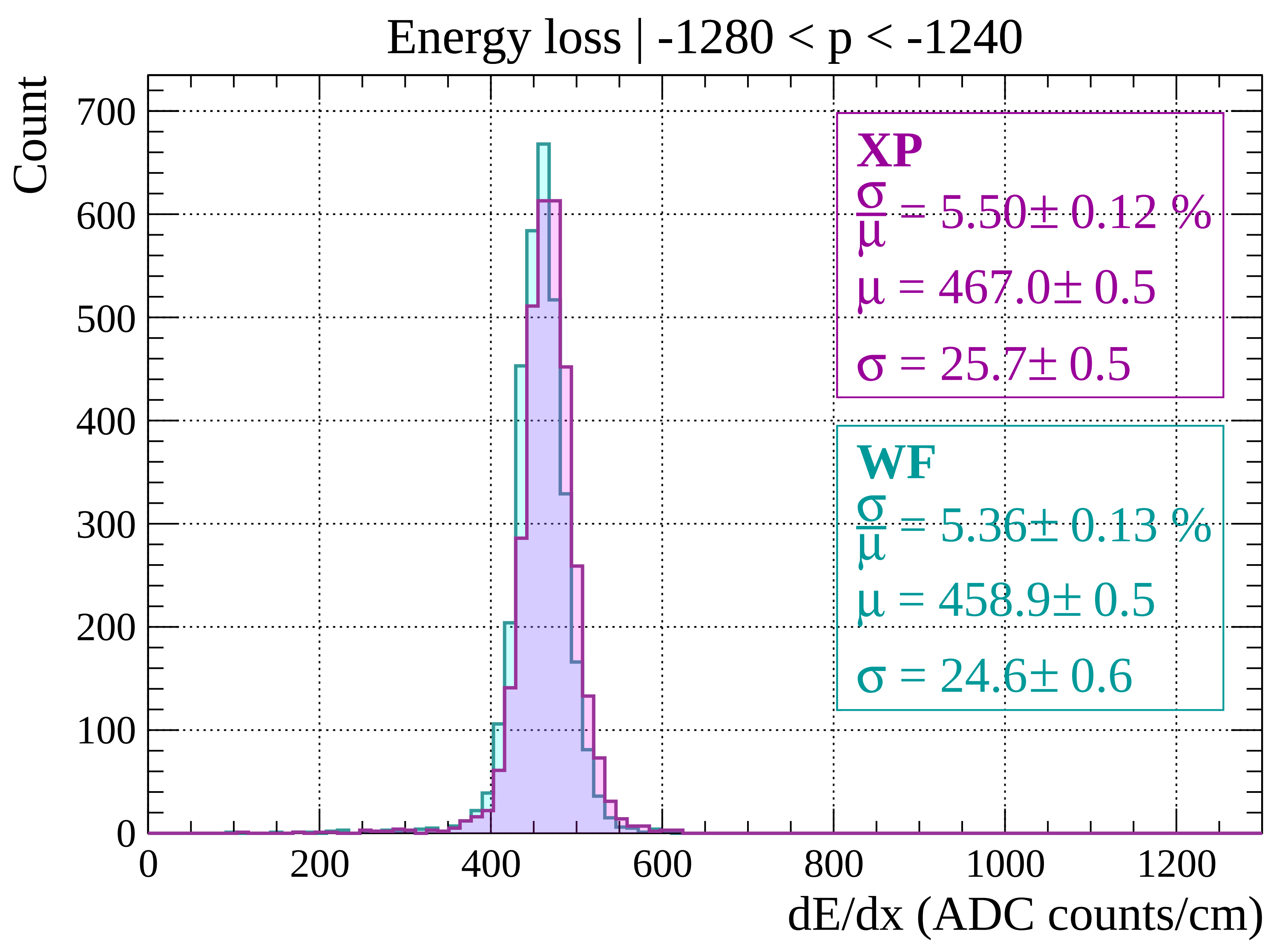


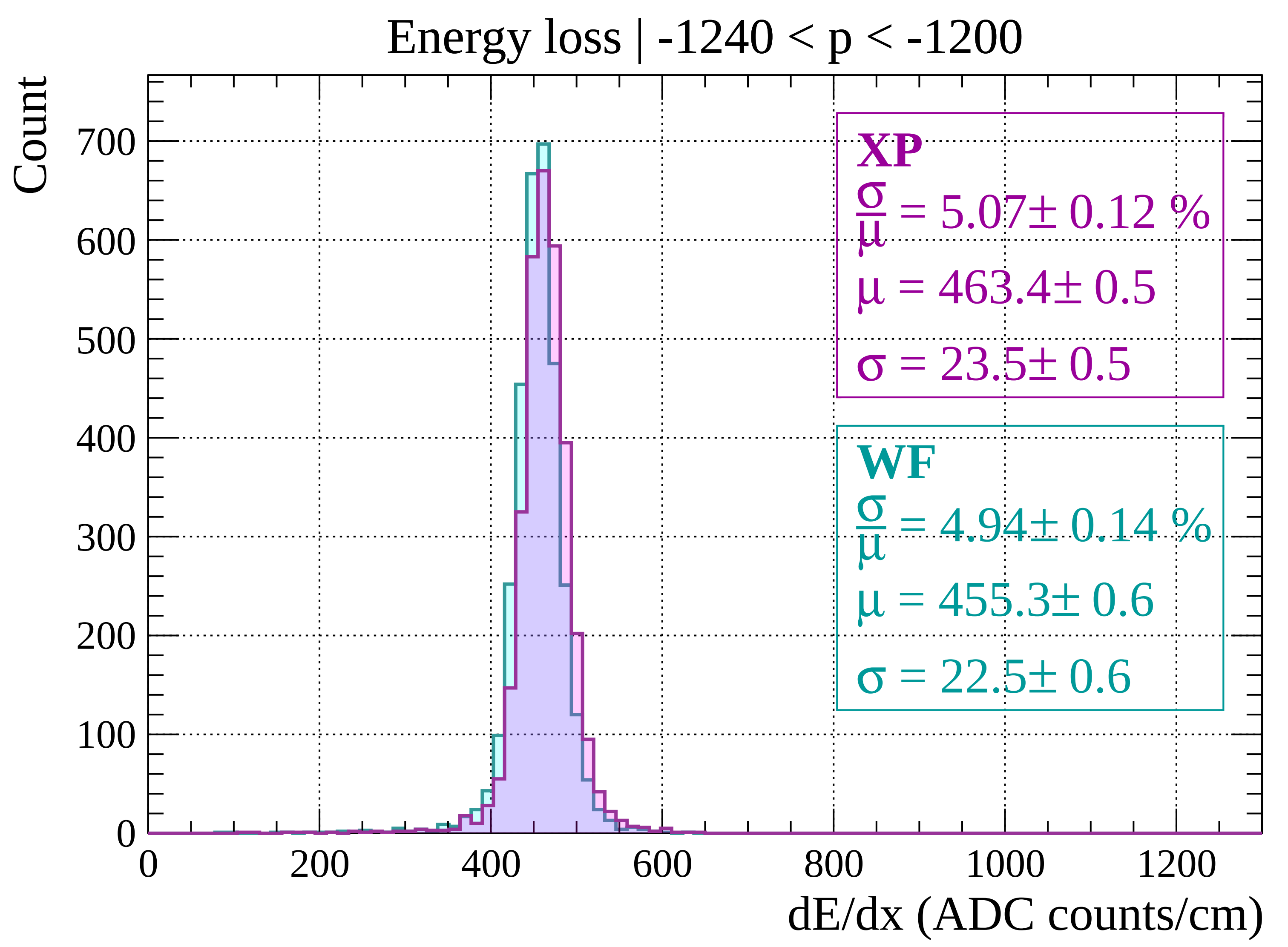
Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.32± 0.13 %
μ = 472.5± 0.6
σ = 25.1± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.40± 0.11 %
μ = 464.7± 0.5
σ = 25.1± 0.5

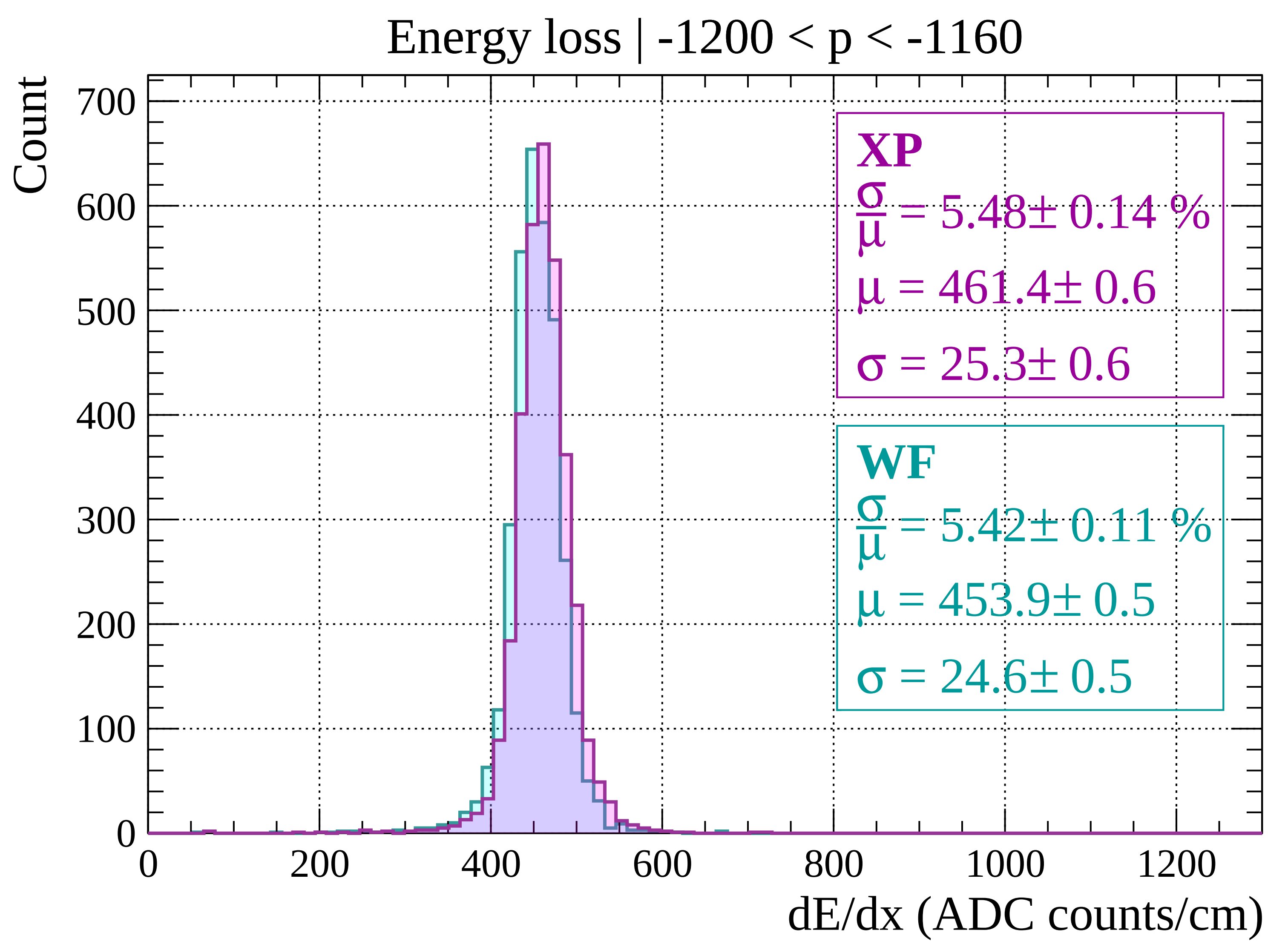

Energy loss | -1400 < p < -1360
Count
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.01± 0.12 %
μ = 470.5± 0.5
σ = 23.6± 0.5
WF
σ/μ = 4.89± 0.11 %
μ = 462.5± 0.5
σ = 22.6± 0.5

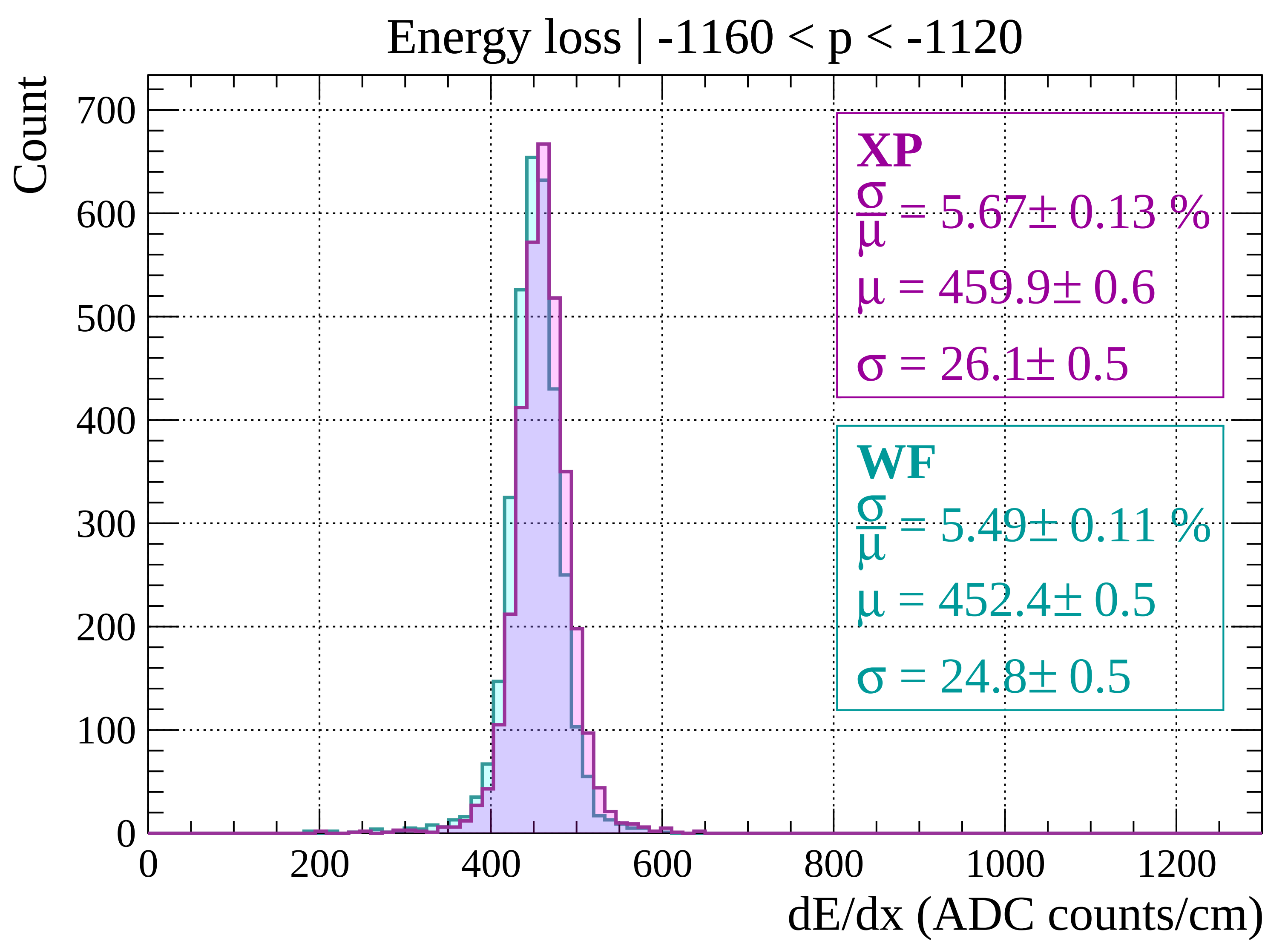

Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.43± 0.12 %
μ = 469.5± 0.5
σ = 25.5± 0.5
WF
σ/μ = 5.30± 0.13 %
μ = 461.1± 0.5
σ = 24.5± 0.6
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

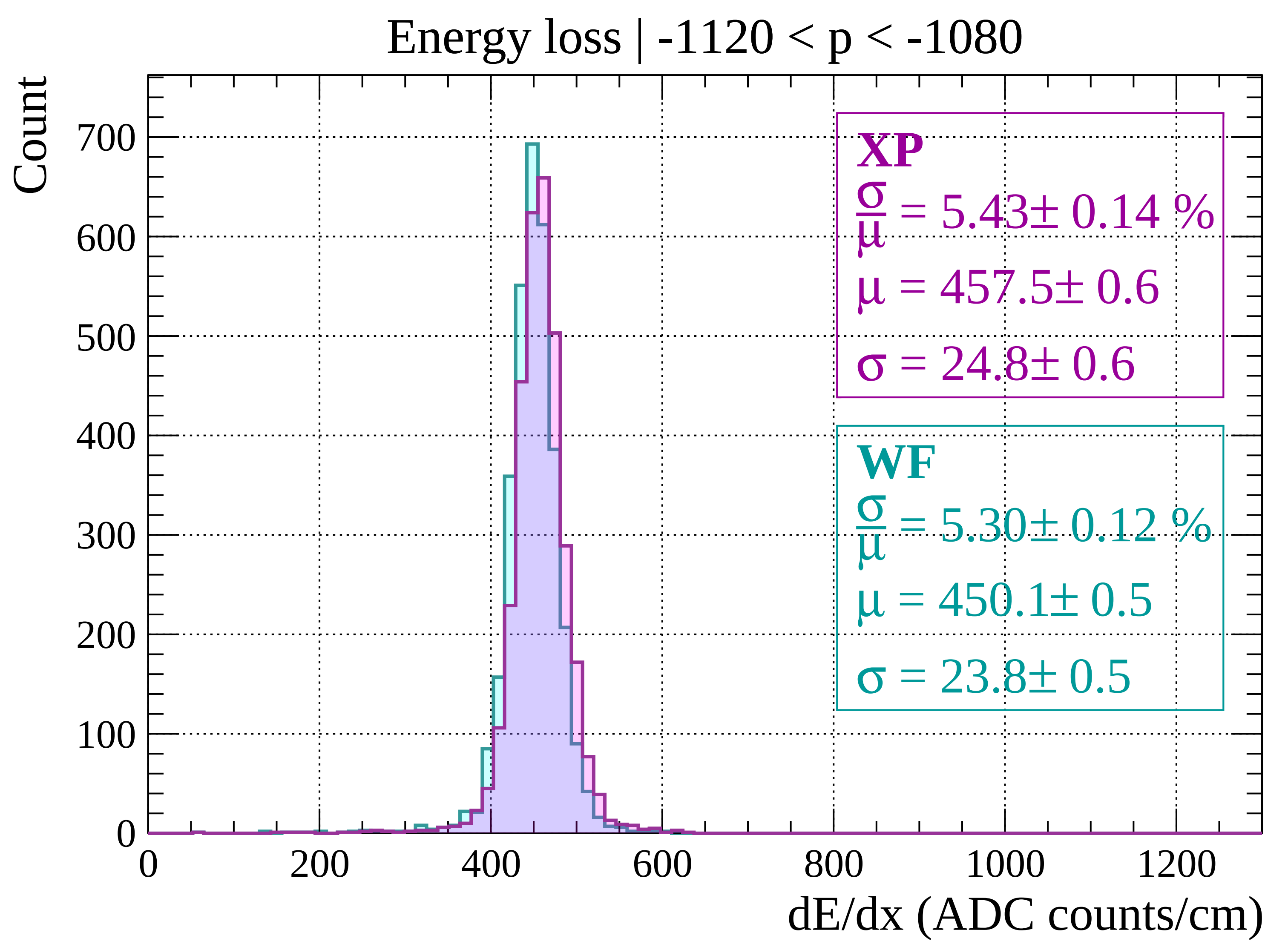

Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 5.40± 0.12 %
μ = 467.4± 0.5
σ = 25.3± 0.5
WF
σ/μ = 5.09± 0.12 %
μ = 459.4± 0.5
σ = 23.4± 0.5


Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.50± 0.12 %
μ = 467.0± 0.5
σ = 25.7± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.36± 0.13 %
μ = 458.9± 0.5
σ = 24.6± 0.6

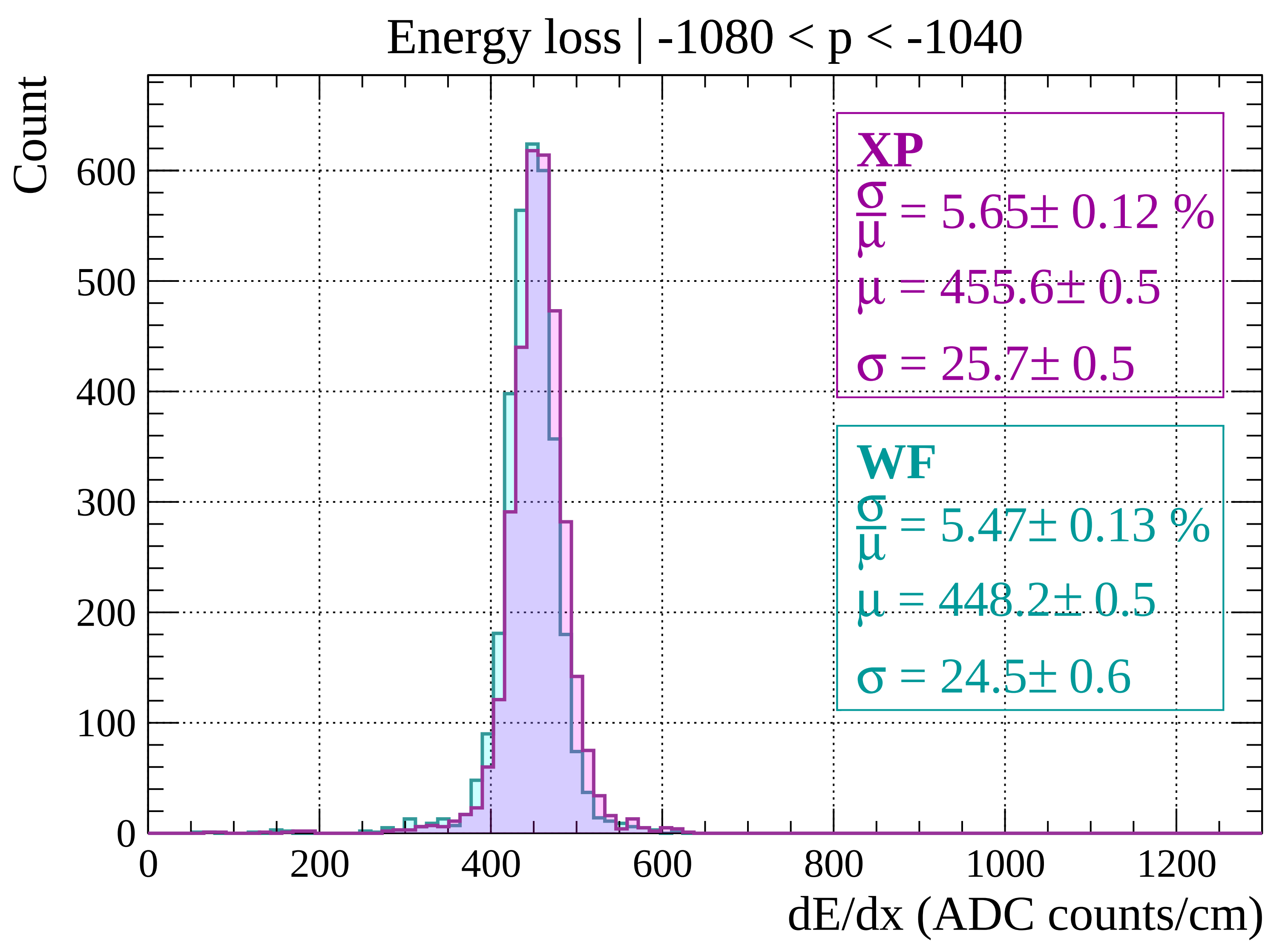

Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 5.07± 0.12 %
μ = 463.4± 0.5
σ = 23.5± 0.5
WF
σ/μ = 4.94± 0.14 %
μ = 455.3± 0.6
σ = 22.5± 0.6


Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.48± 0.14 %
μ = 461.4± 0.6
σ = 25.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.42± 0.11 %
μ = 453.9± 0.5
σ = 24.6± 0.5

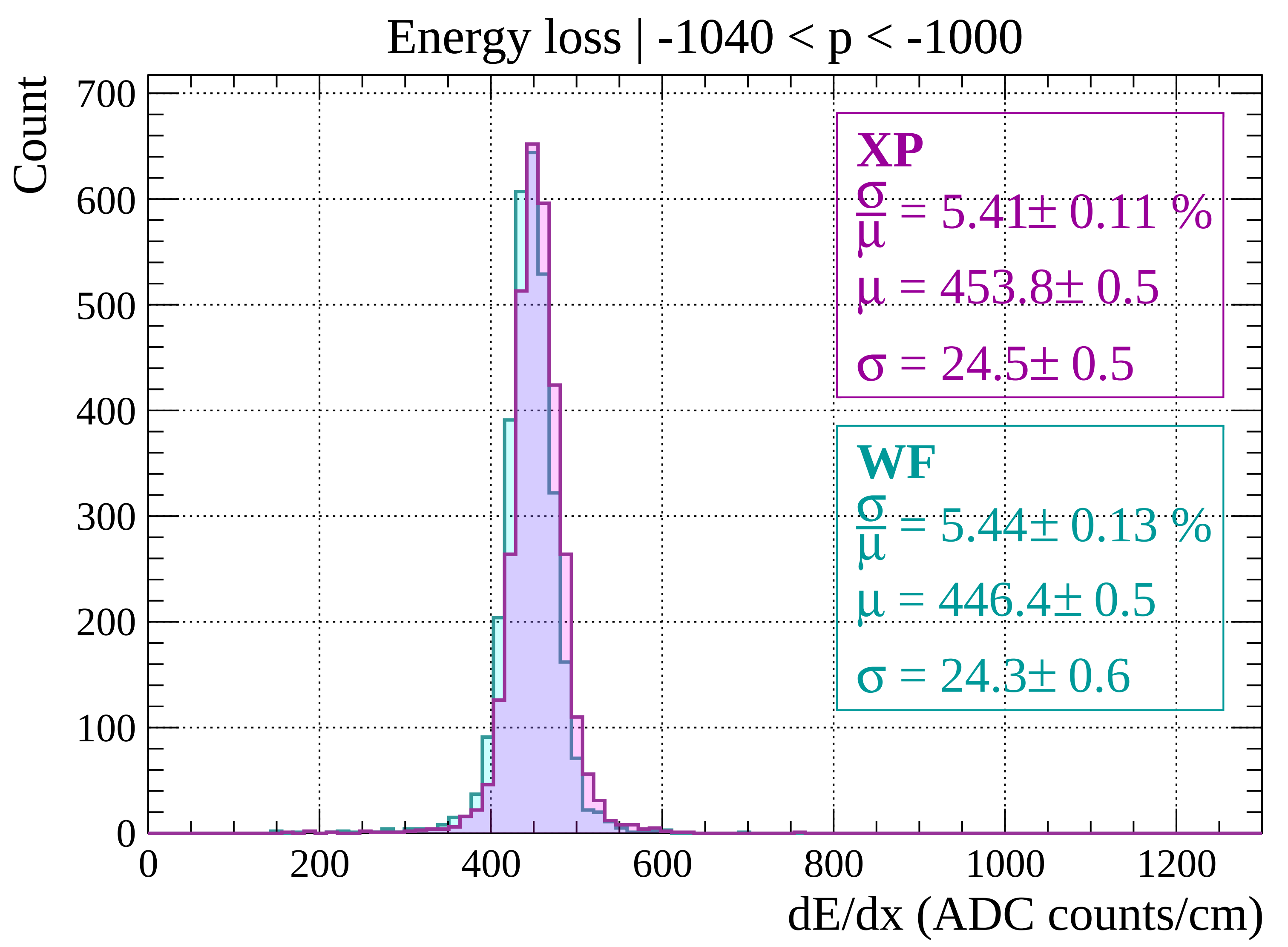

Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.67± 0.13 %
μ = 459.9± 0.6
σ = 26.1± 0.5
WF
σ/μ = 5.49± 0.11 %
μ = 452.4± 0.5
σ = 24.8± 0.5
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200



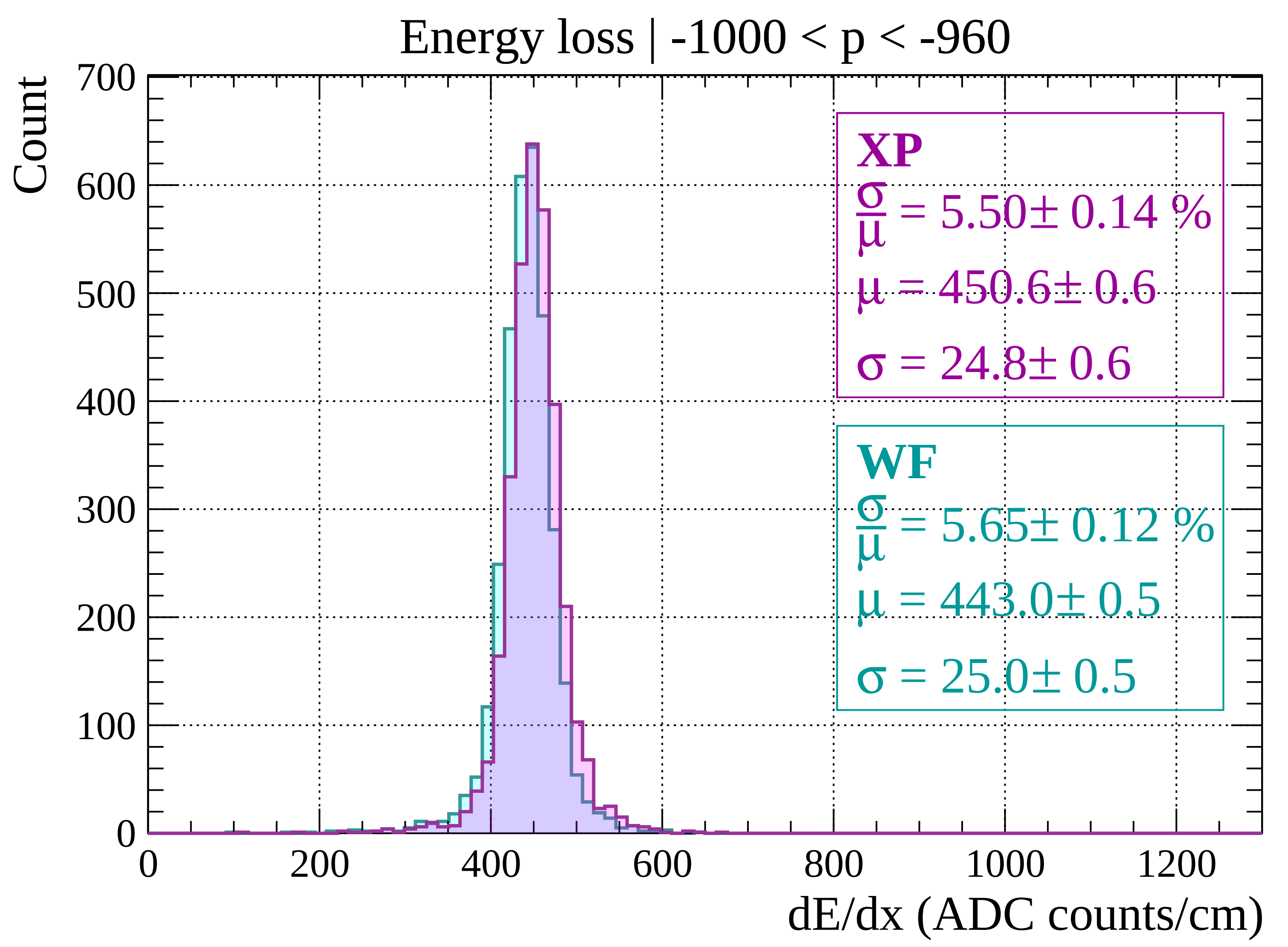
Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 5.43± 0.14 %
μ = 457.5± 0.6
σ = 24.8± 0.6
WF
σ/μ = 5.30± 0.12 %
μ = 450.1± 0.5
σ = 23.8± 0.5



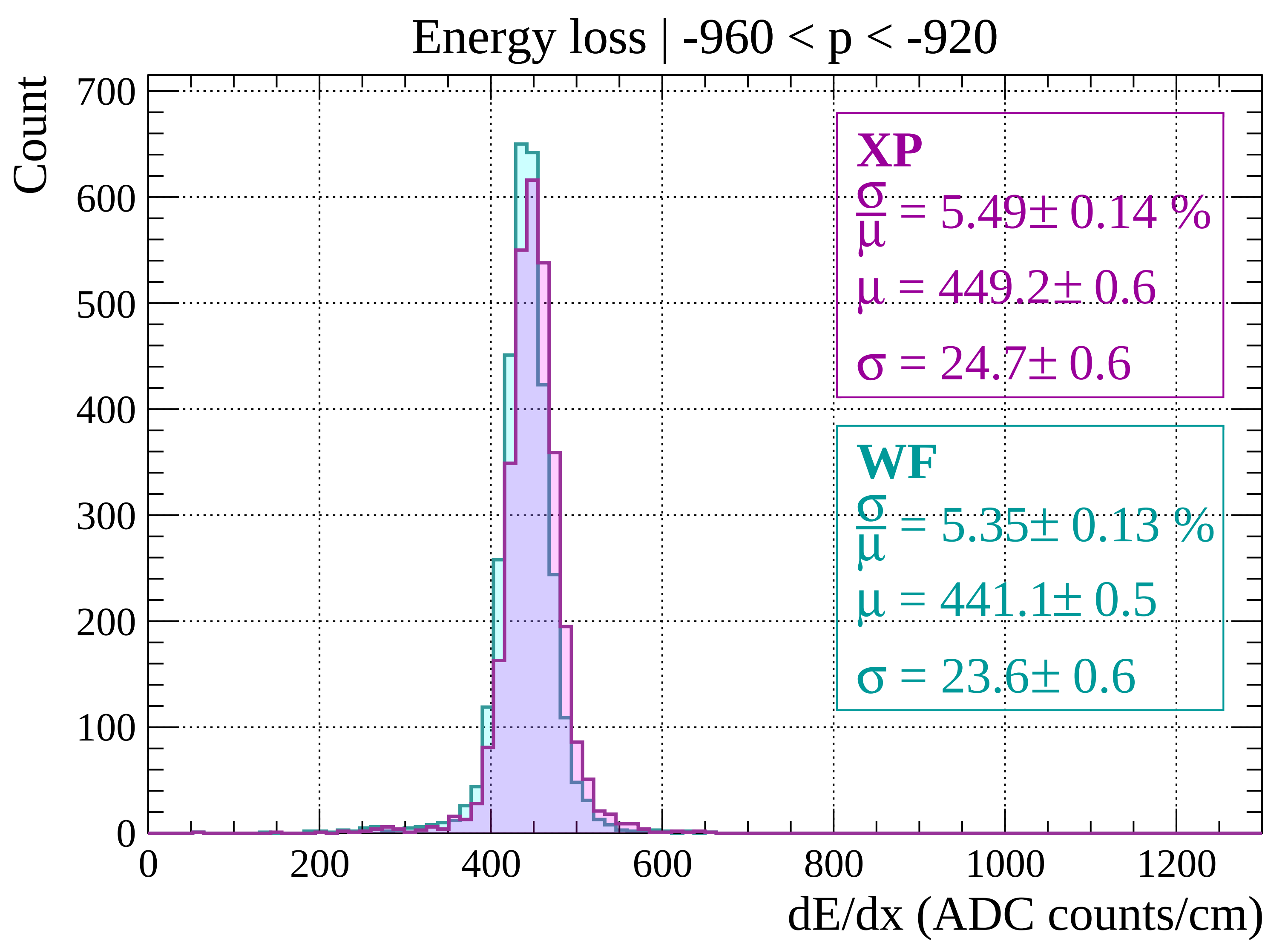
Energy loss | -1080 < p < -1040
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.65± 0.12 %
μ = 455.6± 0.5
σ = 25.7± 0.5
WF
σ/μ = 5.47± 0.13 %
μ = 448.2± 0.5
σ = 24.5± 0.6



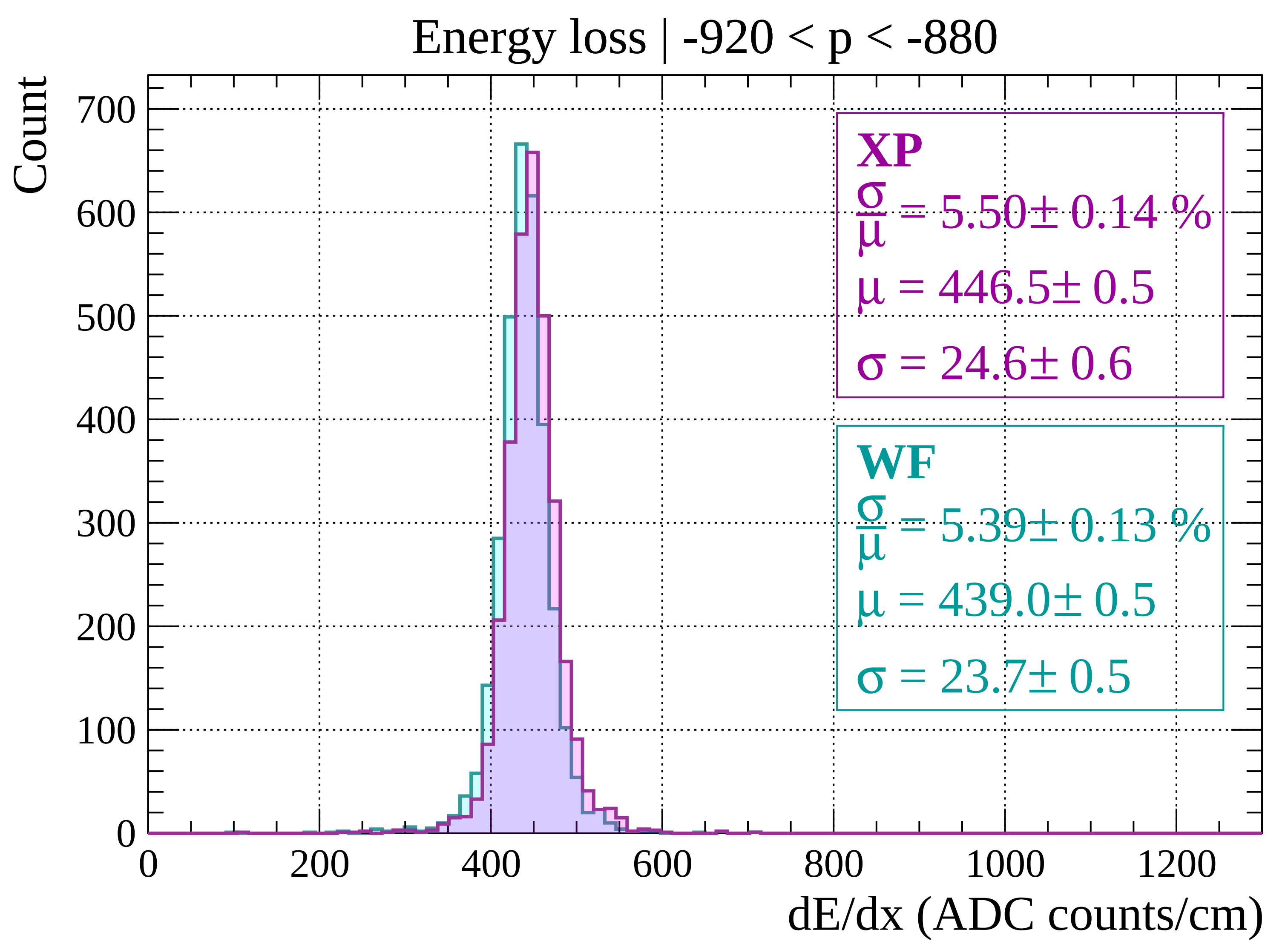
Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.41± 0.11 %
μ = 453.8± 0.5
σ = 24.5± 0.5
WF
σ/μ = 5.44± 0.13 %
μ = 446.4± 0.5
σ = 24.3± 0.6



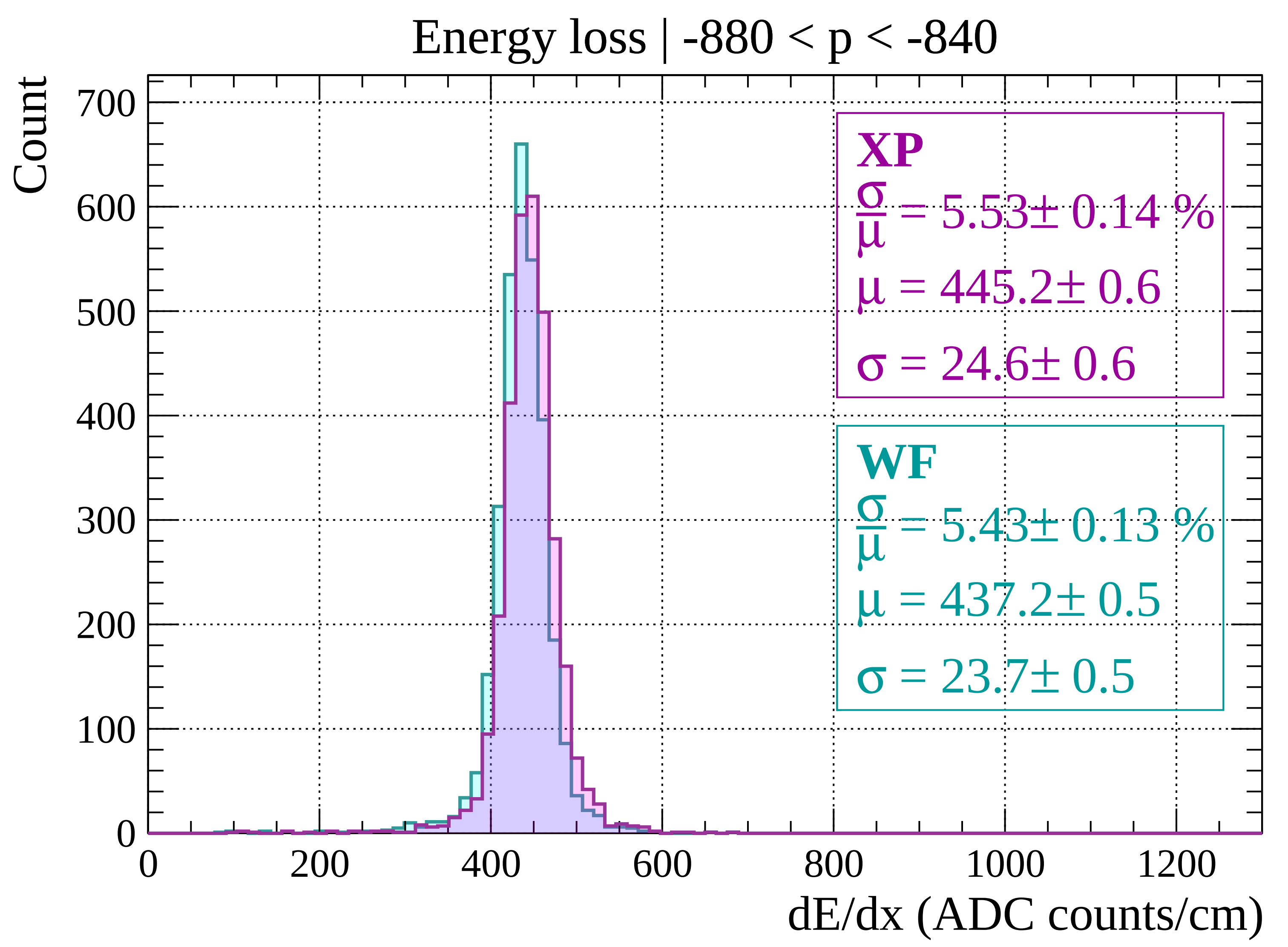
Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.50± 0.14 %
μ = 450.6± 0.6
σ = 24.8± 0.6
WF
σ/μ = 5.65± 0.12 %
μ = 443.0± 0.5
σ = 25.0± 0.5



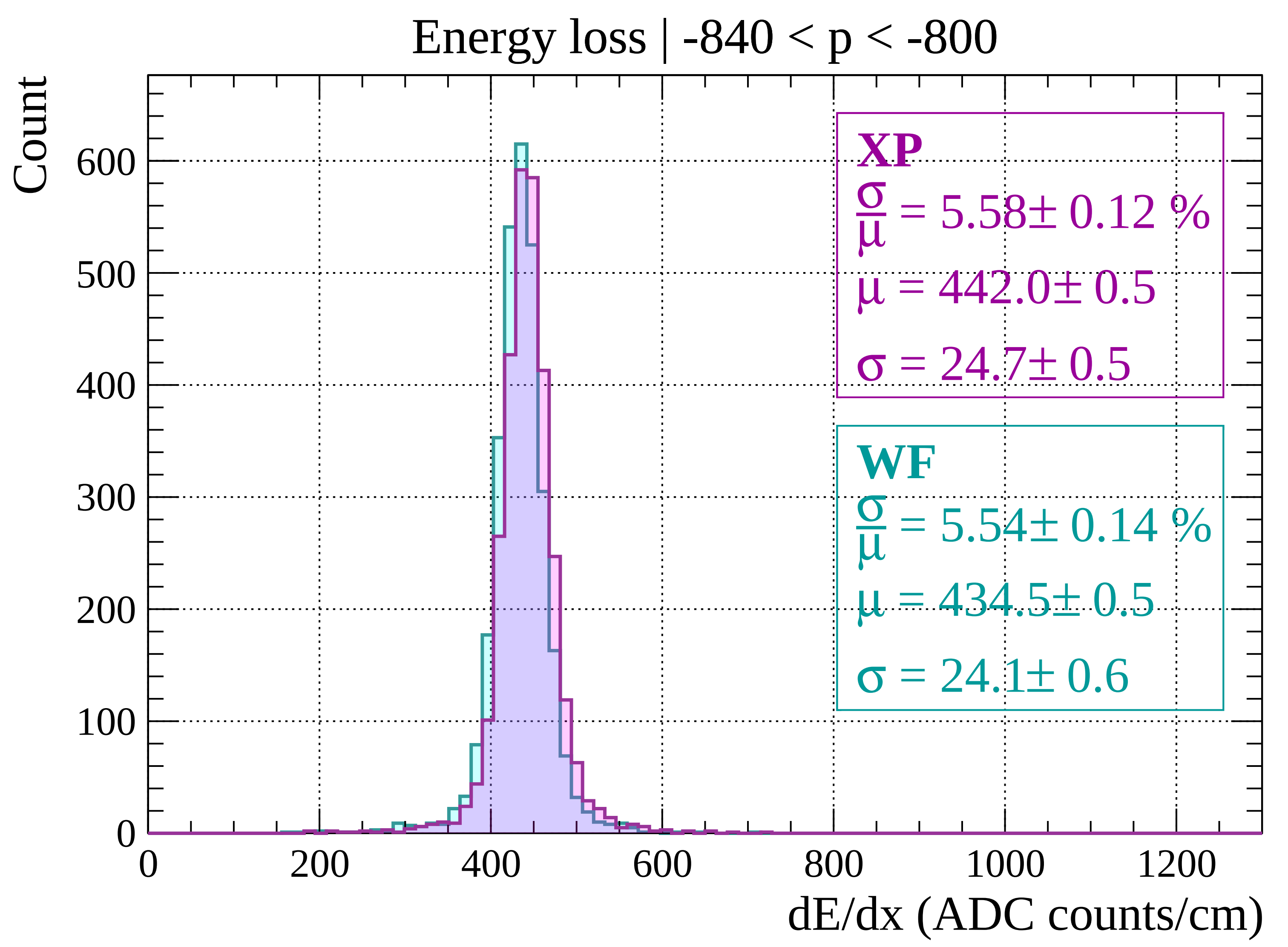
Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.49± 0.14 %
μ = 449.2± 0.6
σ = 24.7± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.35± 0.13 %
μ = 441.1± 0.5
σ = 23.6± 0.6



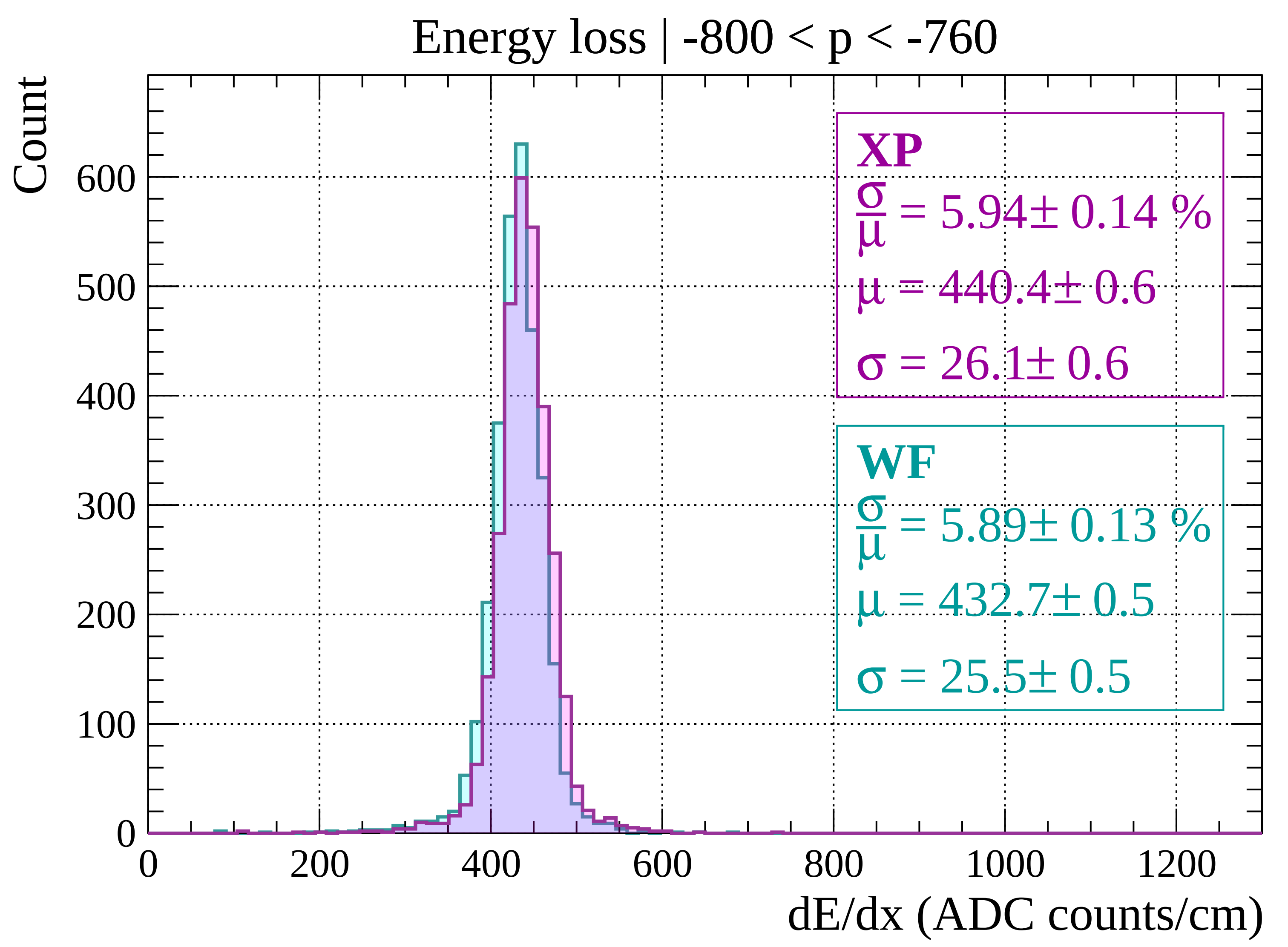
Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 5.50 \pm 0.14$ %
$\mu = 446.5 \pm 0.5$
$\sigma = 24.6 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 5.39 \pm 0.13$ %
$\mu = 439.0 \pm 0.5$
$\sigma = 23.7 \pm 0.5$

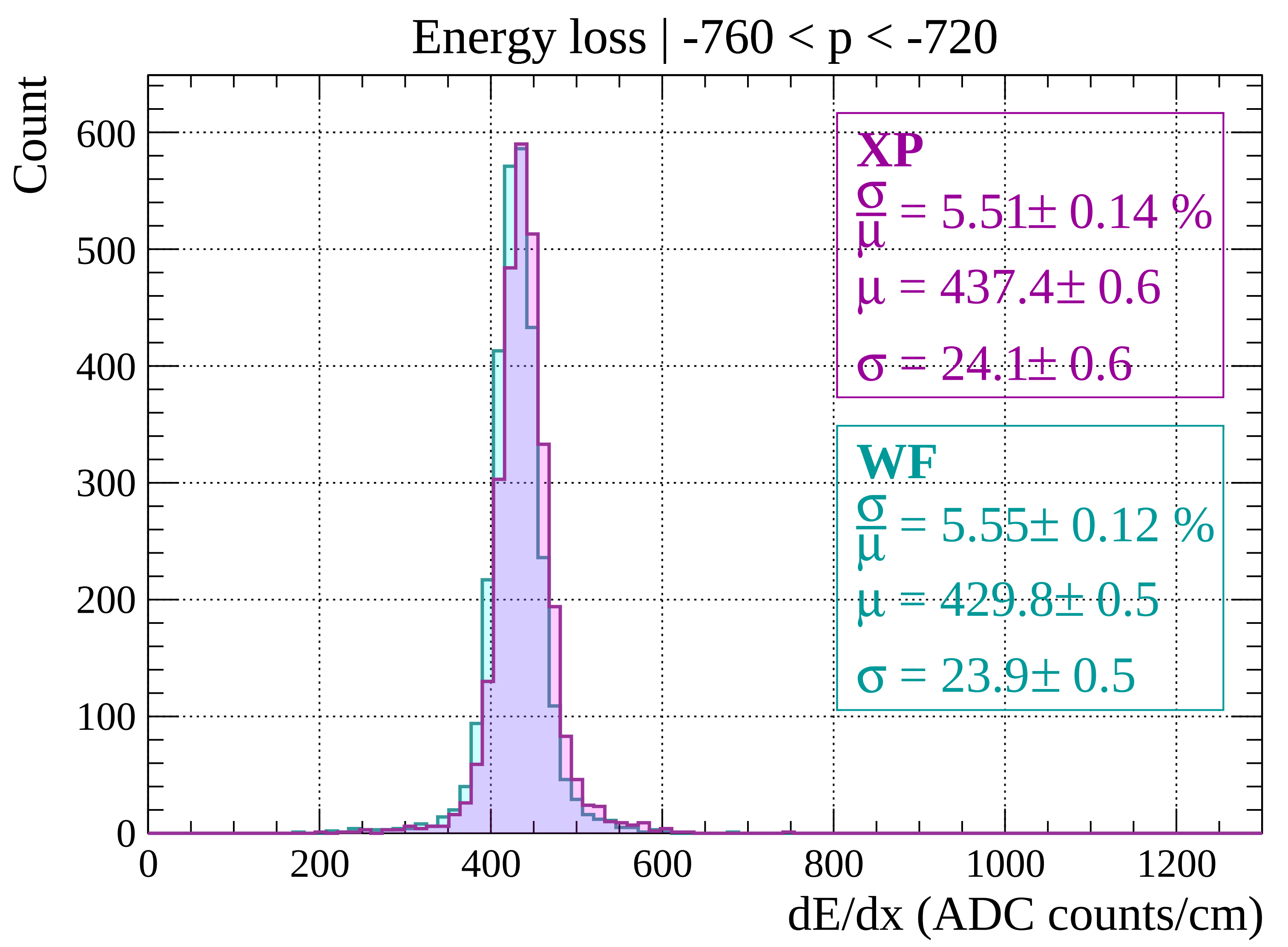


Energy loss | -880 < p < -840
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.53± 0.14 %
μ = 445.2± 0.6
σ = 24.6± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.43± 0.13 %
μ = 437.2± 0.5
σ = 23.7± 0.5

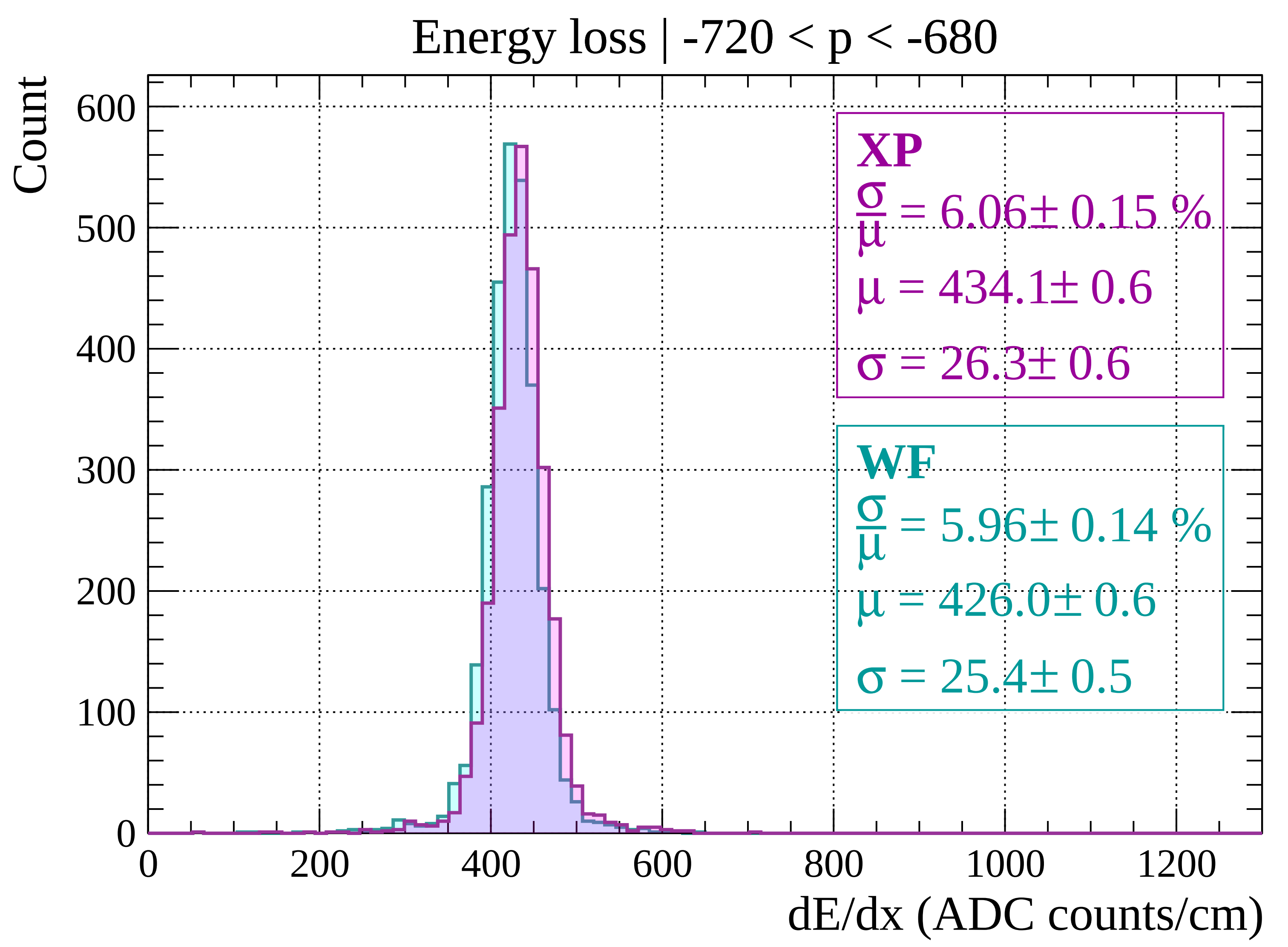


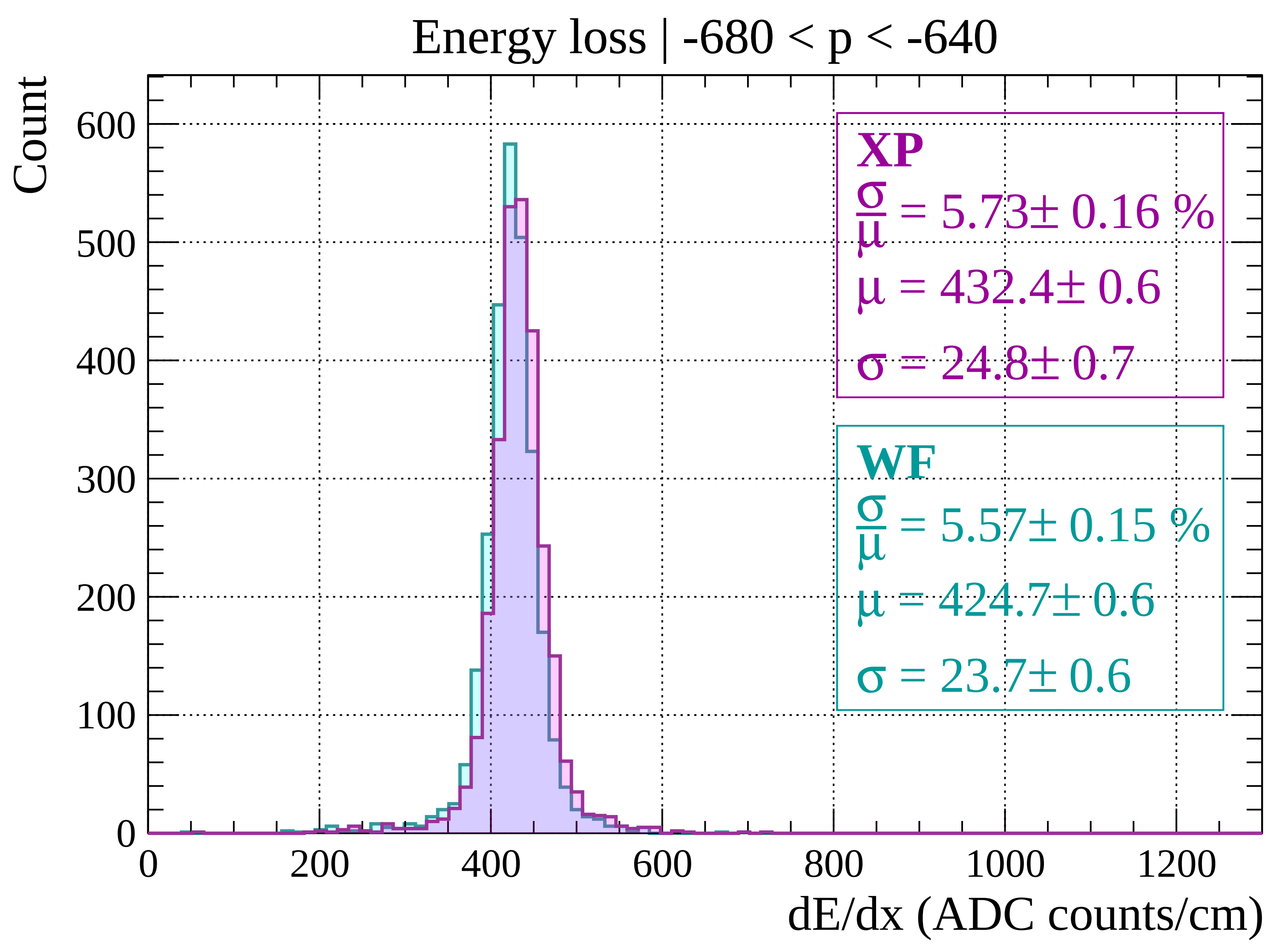
Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 5.58± 0.12 %
μ = 442.0± 0.5
σ = 24.7± 0.5
WF
σ/μ = 5.54± 0.14 %
μ = 434.5± 0.5
σ = 24.1± 0.6

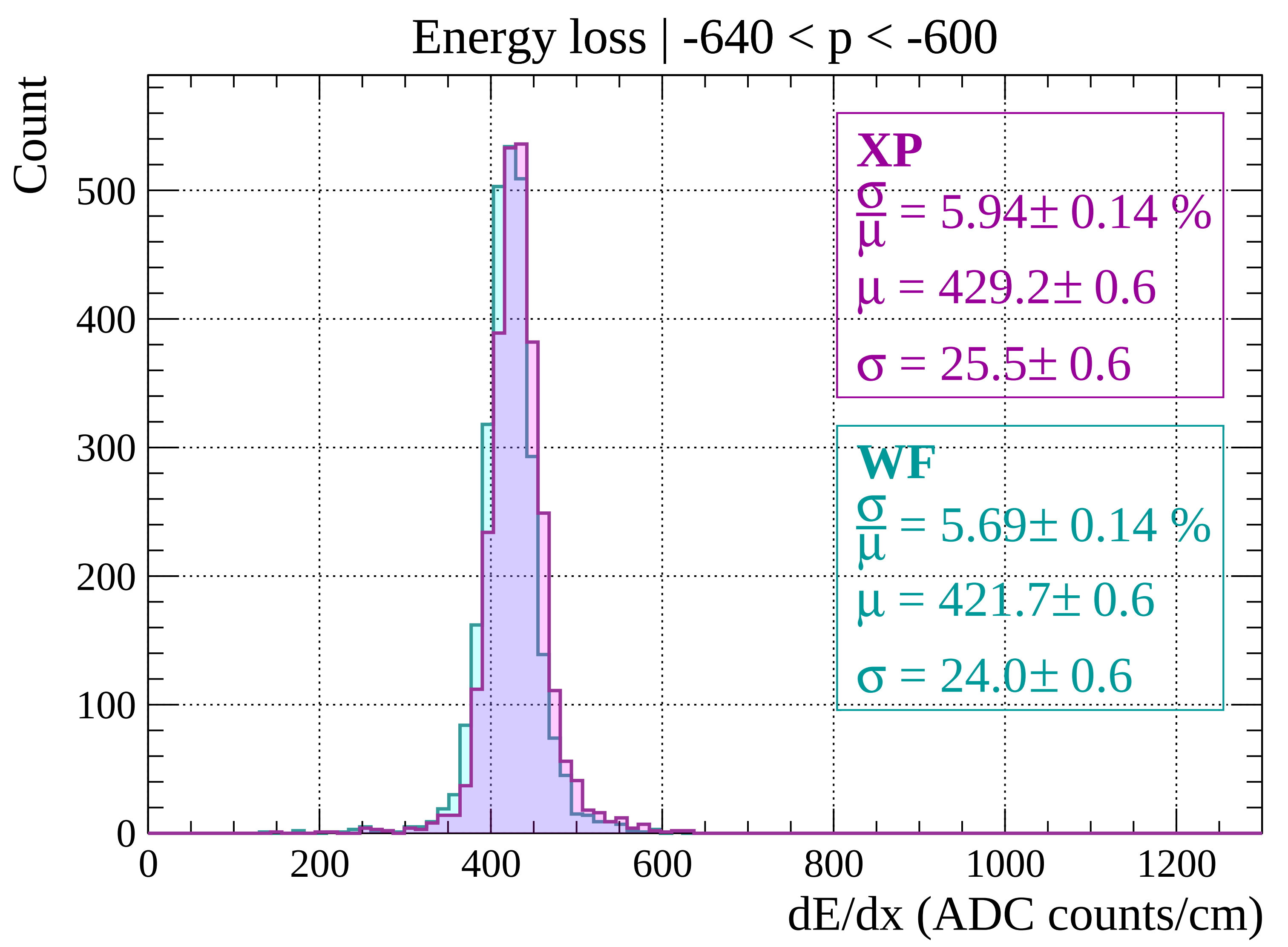

Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.94± 0.14 %
μ = 440.4± 0.6
σ = 26.1± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.89± 0.13 %
μ = 432.7± 0.5
σ = 25.5± 0.5

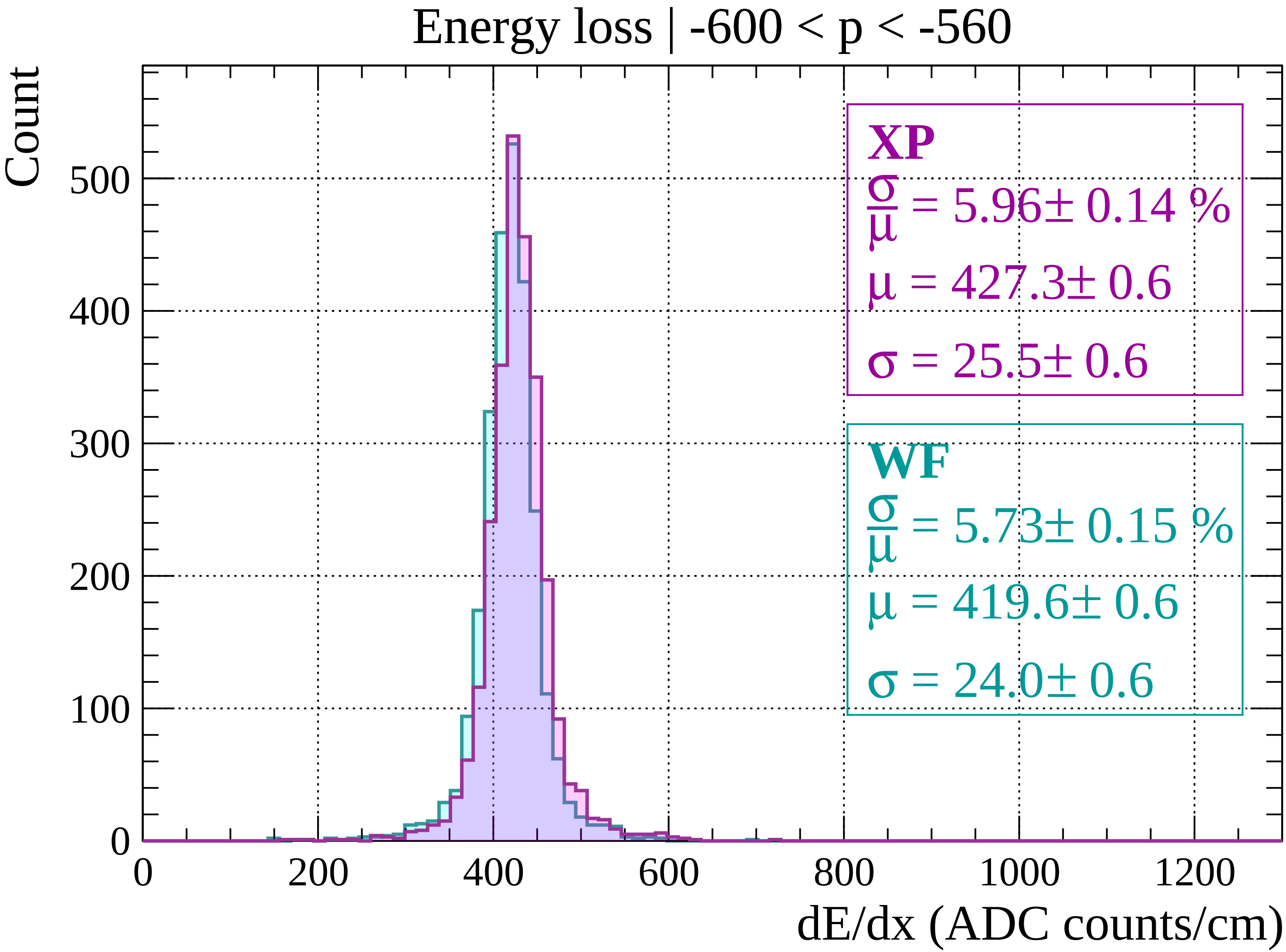


Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.51± 0.14 %
μ = 437.4± 0.6
σ = 24.1± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.55± 0.12 %
μ = 429.8± 0.5
σ = 23.9± 0.5

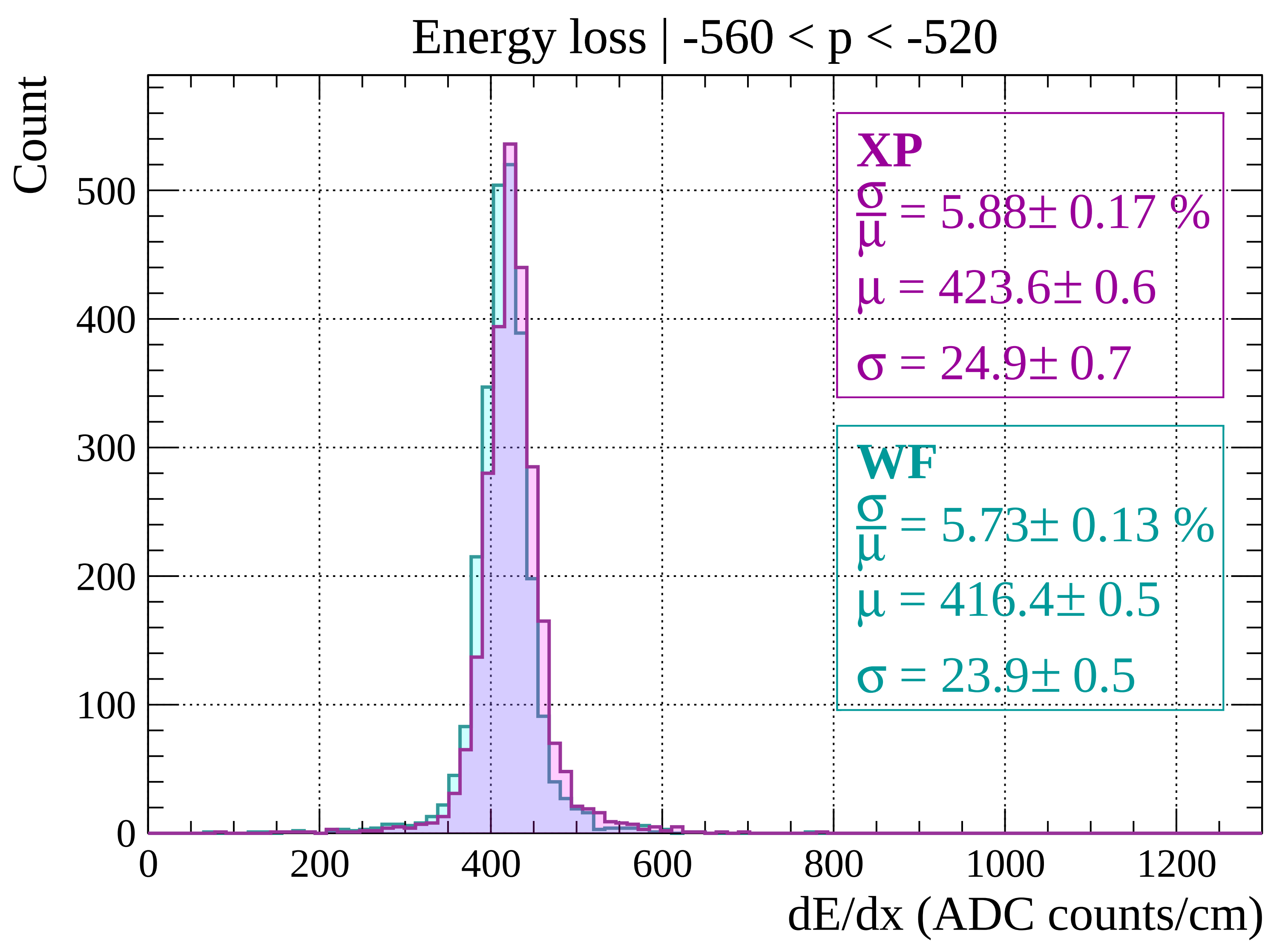

Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.06± 0.15 %
μ = 434.1± 0.6
σ = 26.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.96± 0.14 %
μ = 426.0± 0.6
σ = 25.4± 0.5


Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.73± 0.16 %
μ = 432.4± 0.6
σ = 24.8± 0.7
WF
σ/μ = 5.57± 0.15 %
μ = 424.7± 0.6
σ = 23.7± 0.6
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

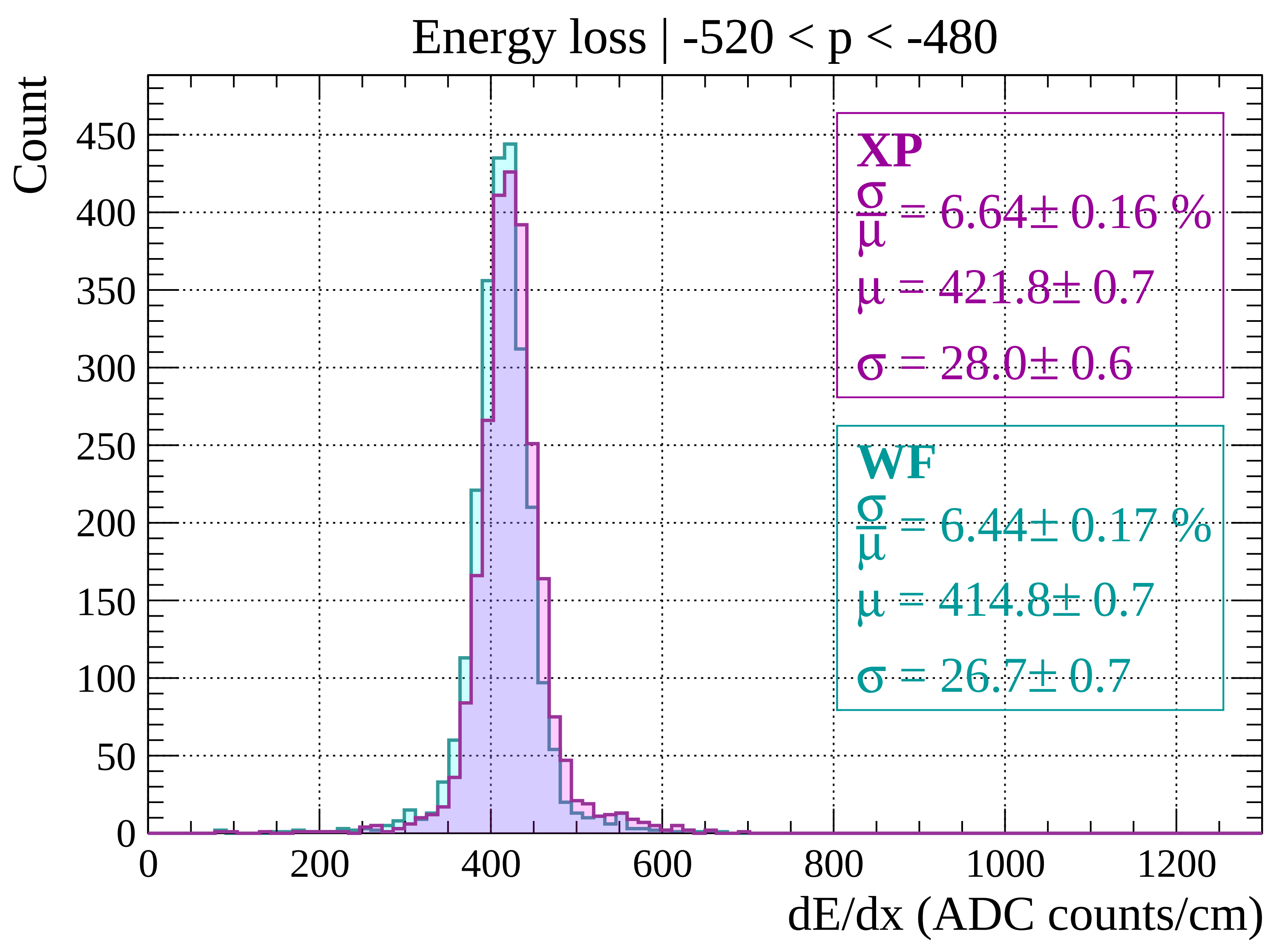

Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.94± 0.14 %
μ = 429.2± 0.6
σ = 25.5± 0.6
WF
$\frac{\sigma}{\mu}$ = 5.69± 0.14 %
μ = 421.7± 0.6
σ = 24.0± 0.6
0
100
200
300
400
500
0
200
400
600
800
1000
1200



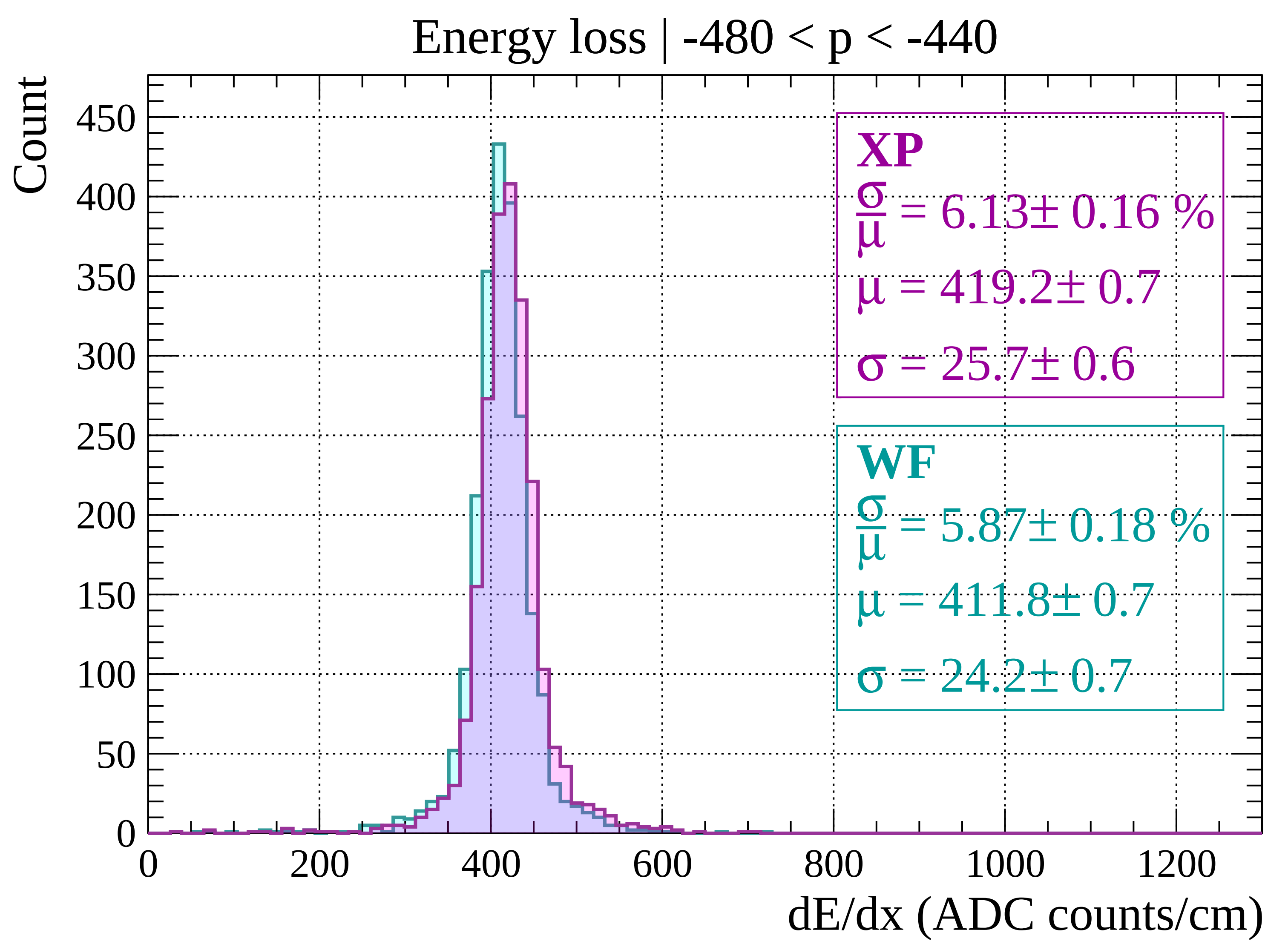
Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 5.96± 0.14 %
μ = 427.3± 0.6
σ = 25.5± 0.6
WF
σ/μ = 5.73± 0.15 %
μ = 419.6± 0.6
σ = 24.0± 0.6



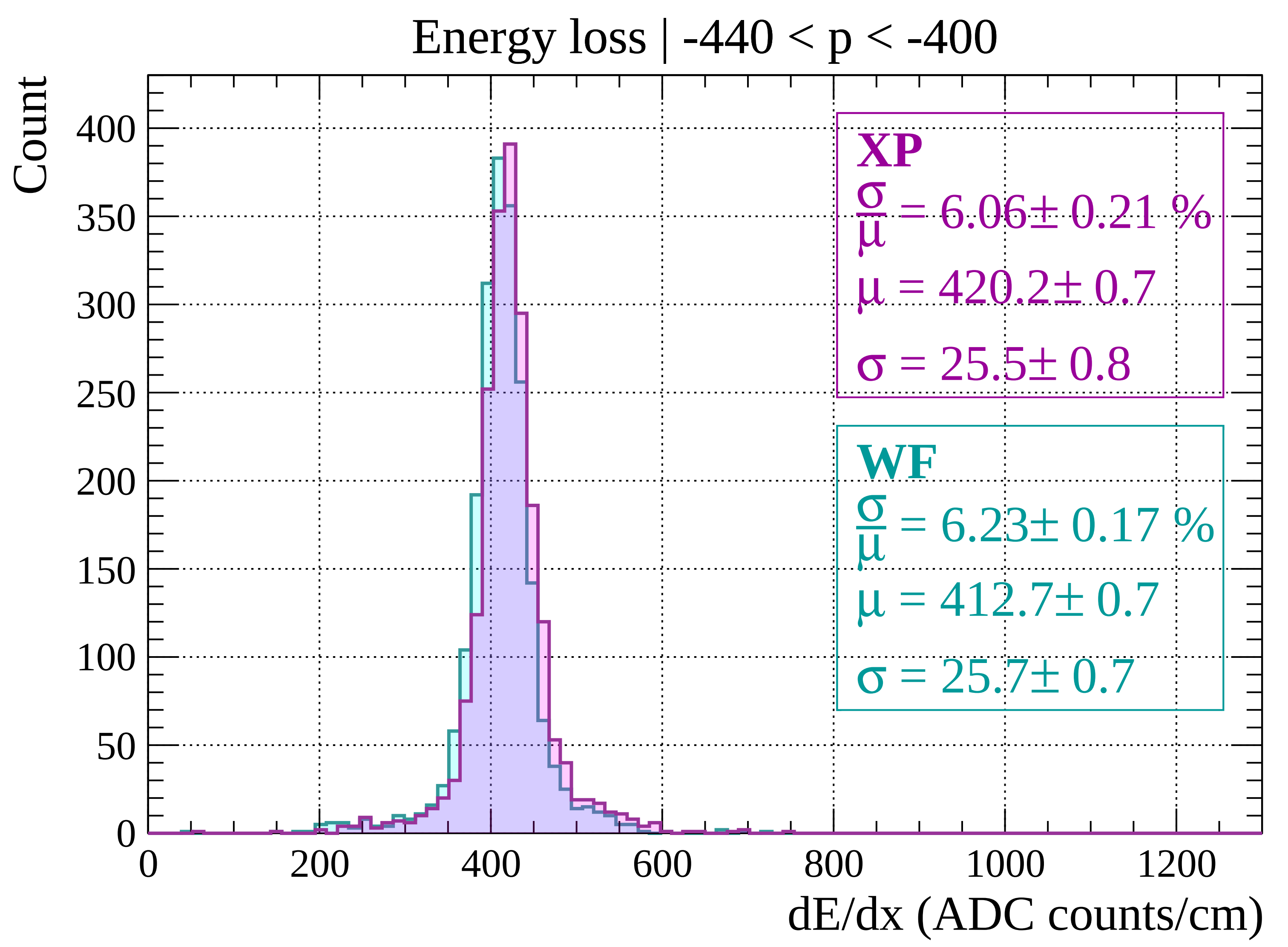
Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.88± 0.17 %
μ = 423.6± 0.6
σ = 24.9± 0.7
WF
$\frac{\sigma}{\mu}$ = 5.73± 0.13 %
μ = 416.4± 0.5
σ = 23.9± 0.5



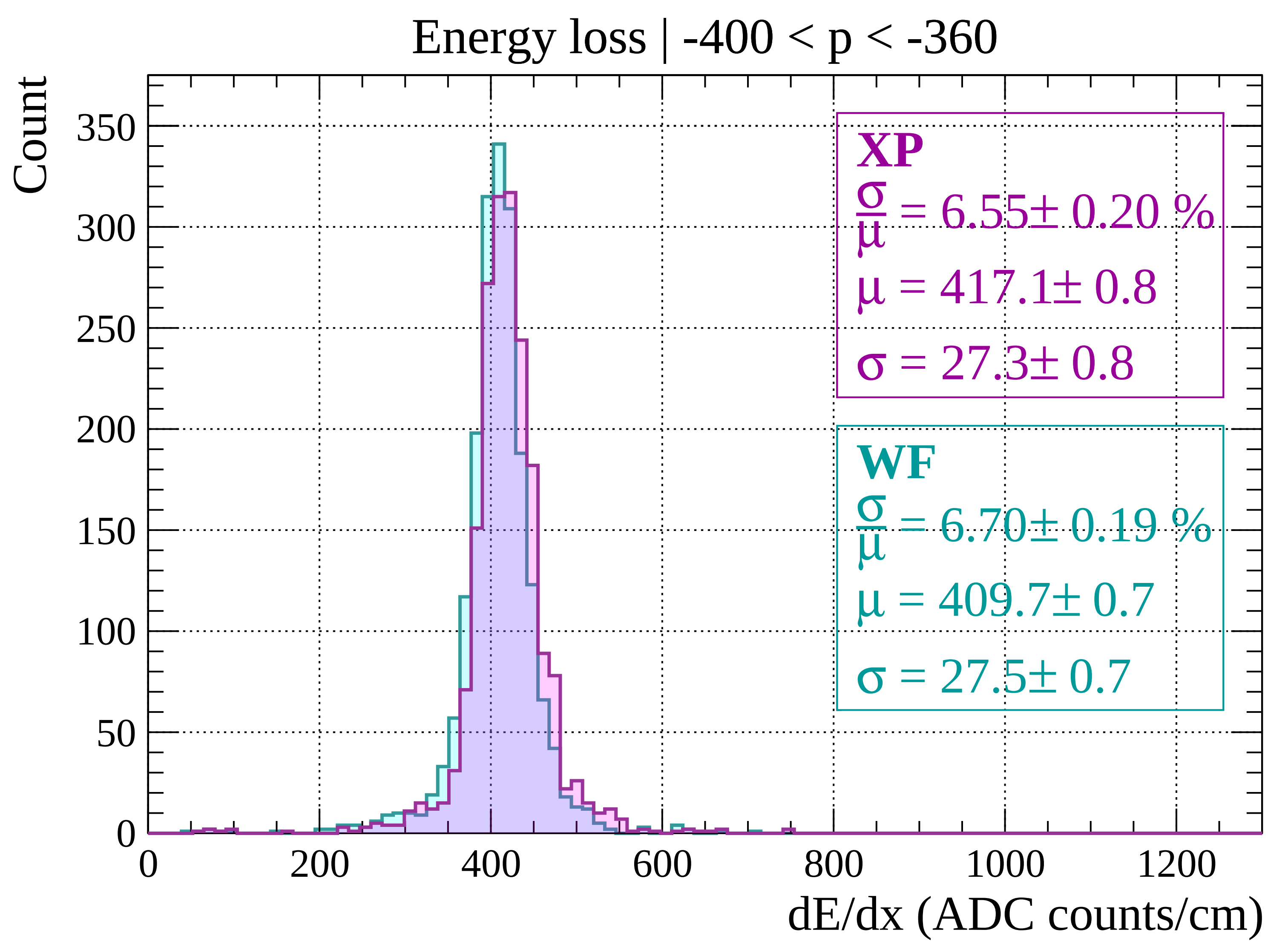
Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 6.64 ± 0.16 %
μ = 421.8 ± 0.7
σ = 28.0 ± 0.6
WF
σ/μ = 6.44 ± 0.17 %
μ = 414.8 ± 0.7
σ = 26.7 ± 0.7



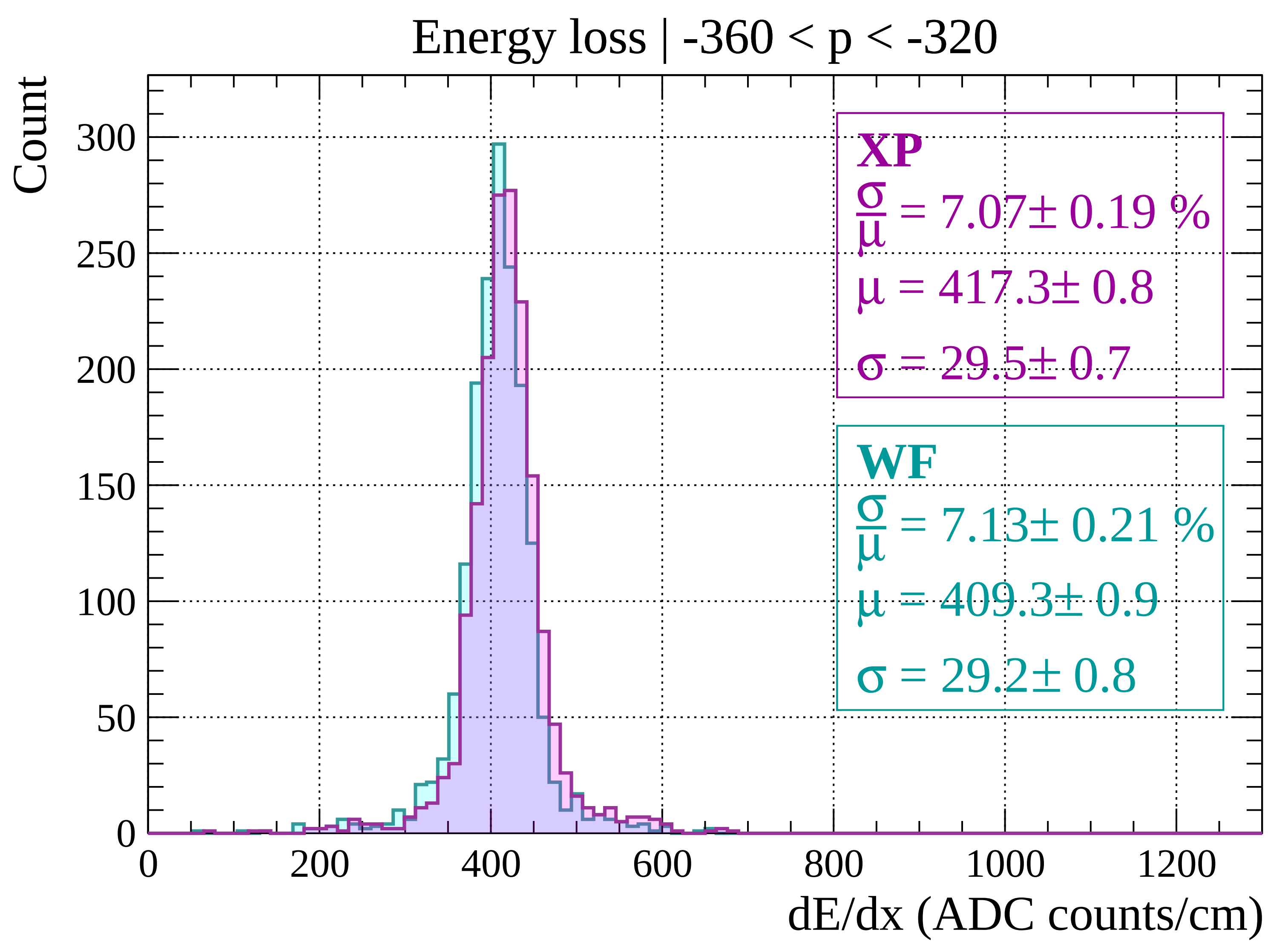
Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 6.13± 0.16 %
μ = 419.2± 0.7
σ = 25.7± 0.6
WF
σ/μ = 5.87± 0.18 %
μ = 411.8± 0.7
σ = 24.2± 0.7



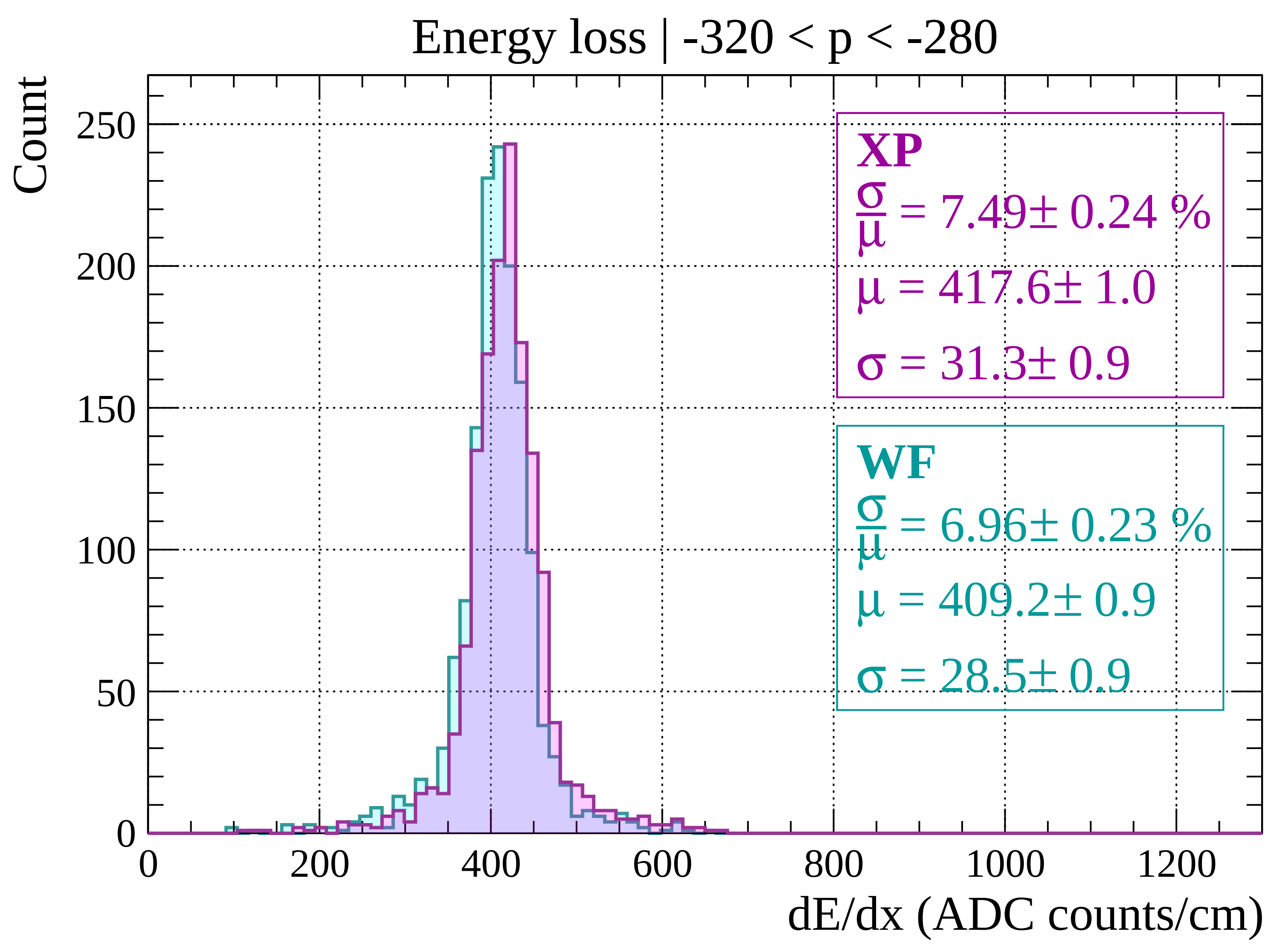
Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 6.06± 0.21 %
μ = 420.2± 0.7
σ = 25.5± 0.8
WF
σ/μ = 6.23± 0.17 %
μ = 412.7± 0.7
σ = 25.7± 0.7

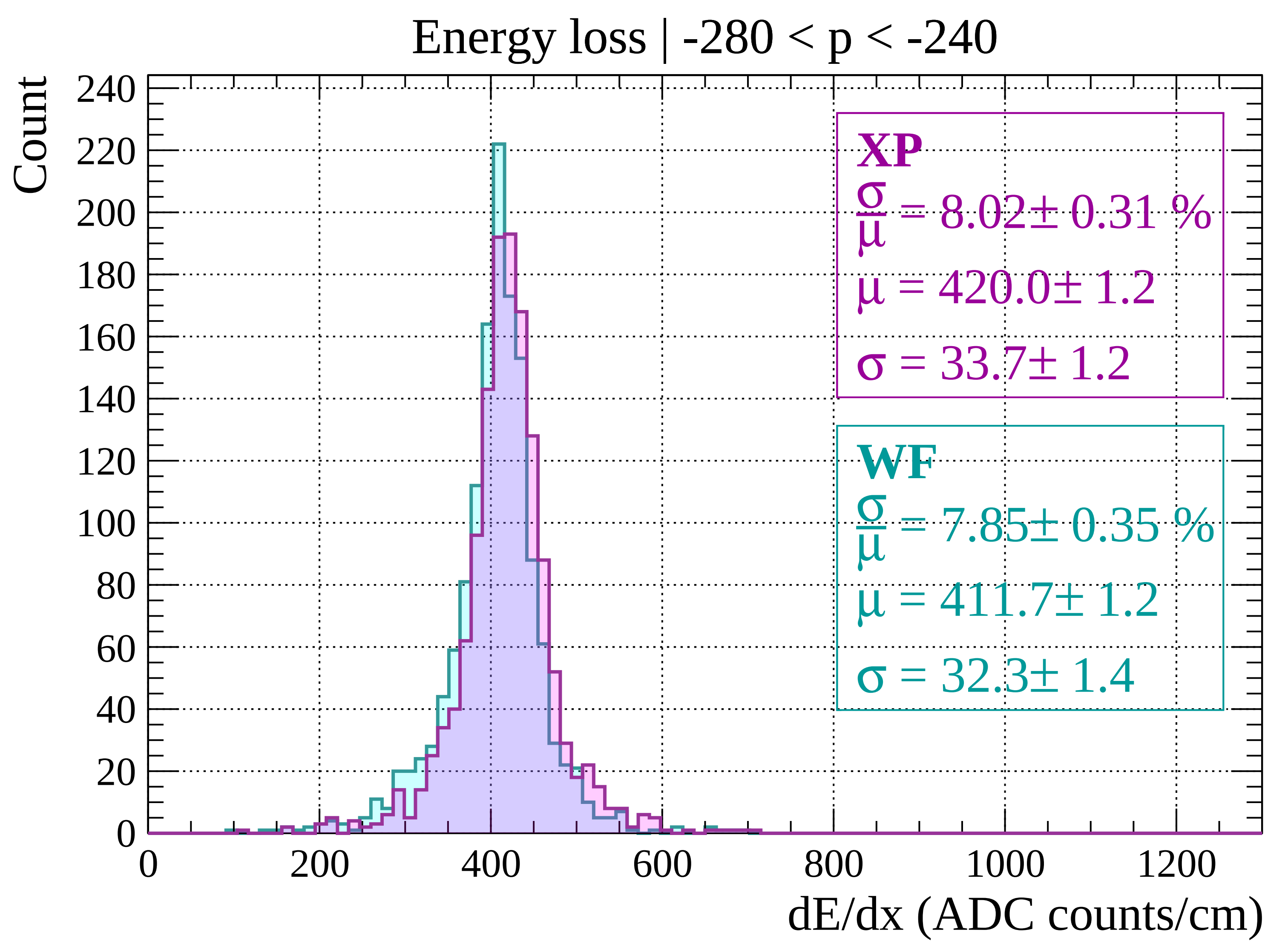


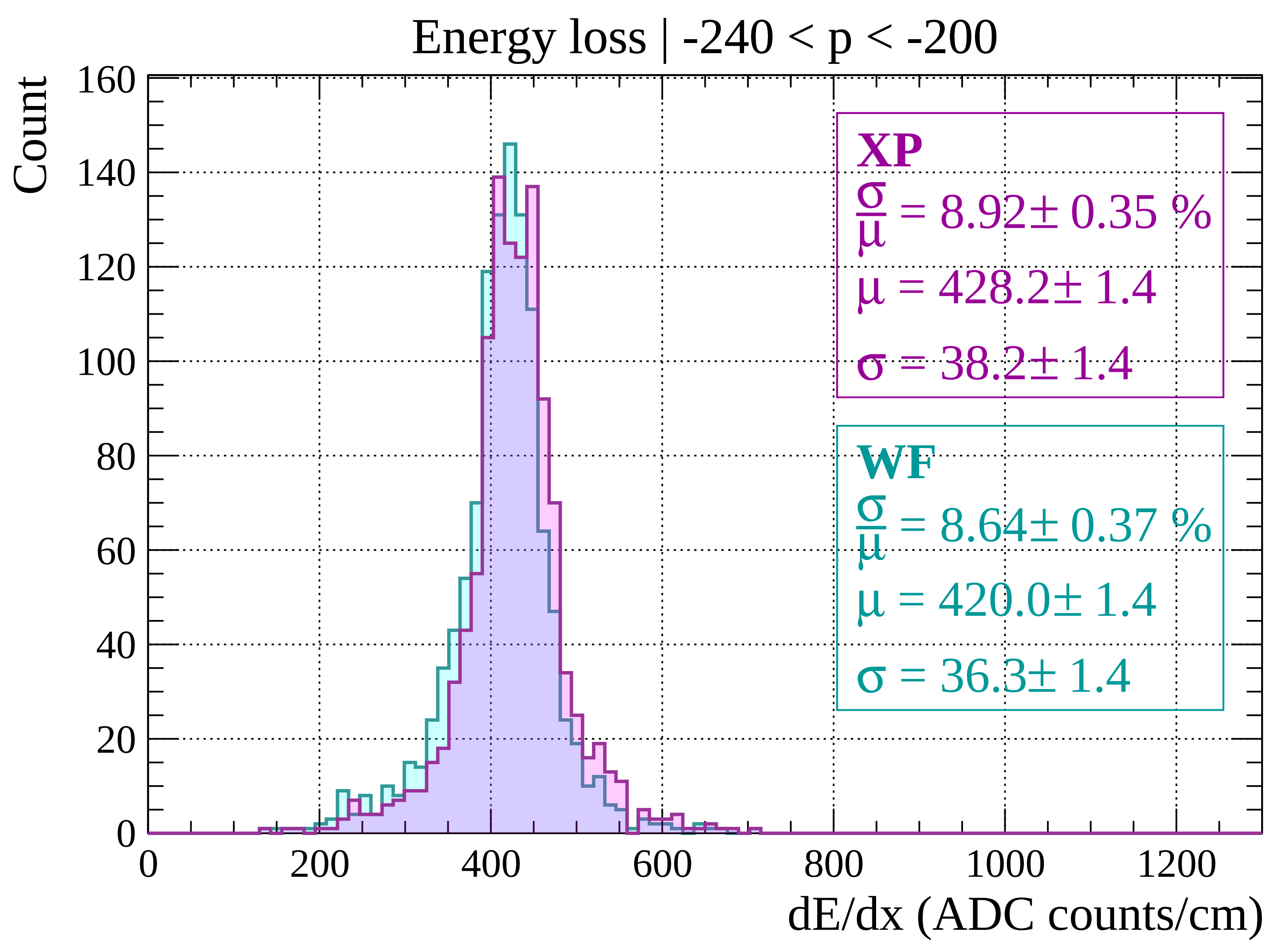
Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.55± 0.20 %
μ = 417.1± 0.8
σ = 27.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 6.70± 0.19 %
μ = 409.7± 0.7
σ = 27.5± 0.7



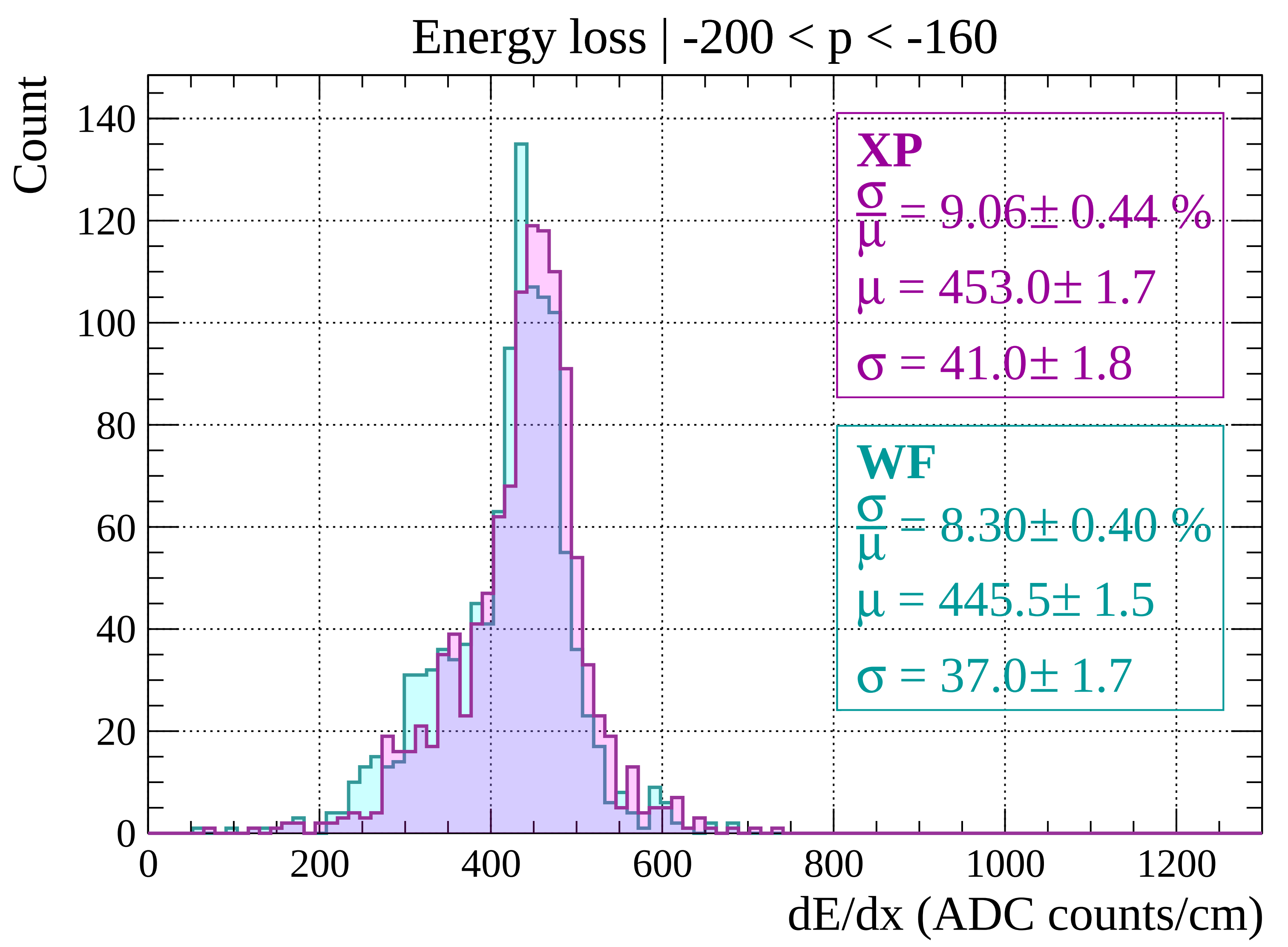
Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 7.07± 0.19 %
μ = 417.3± 0.8
σ = 29.5± 0.7
WF
σ/μ = 7.13± 0.21 %
μ = 409.3± 0.9
σ = 29.2± 0.8

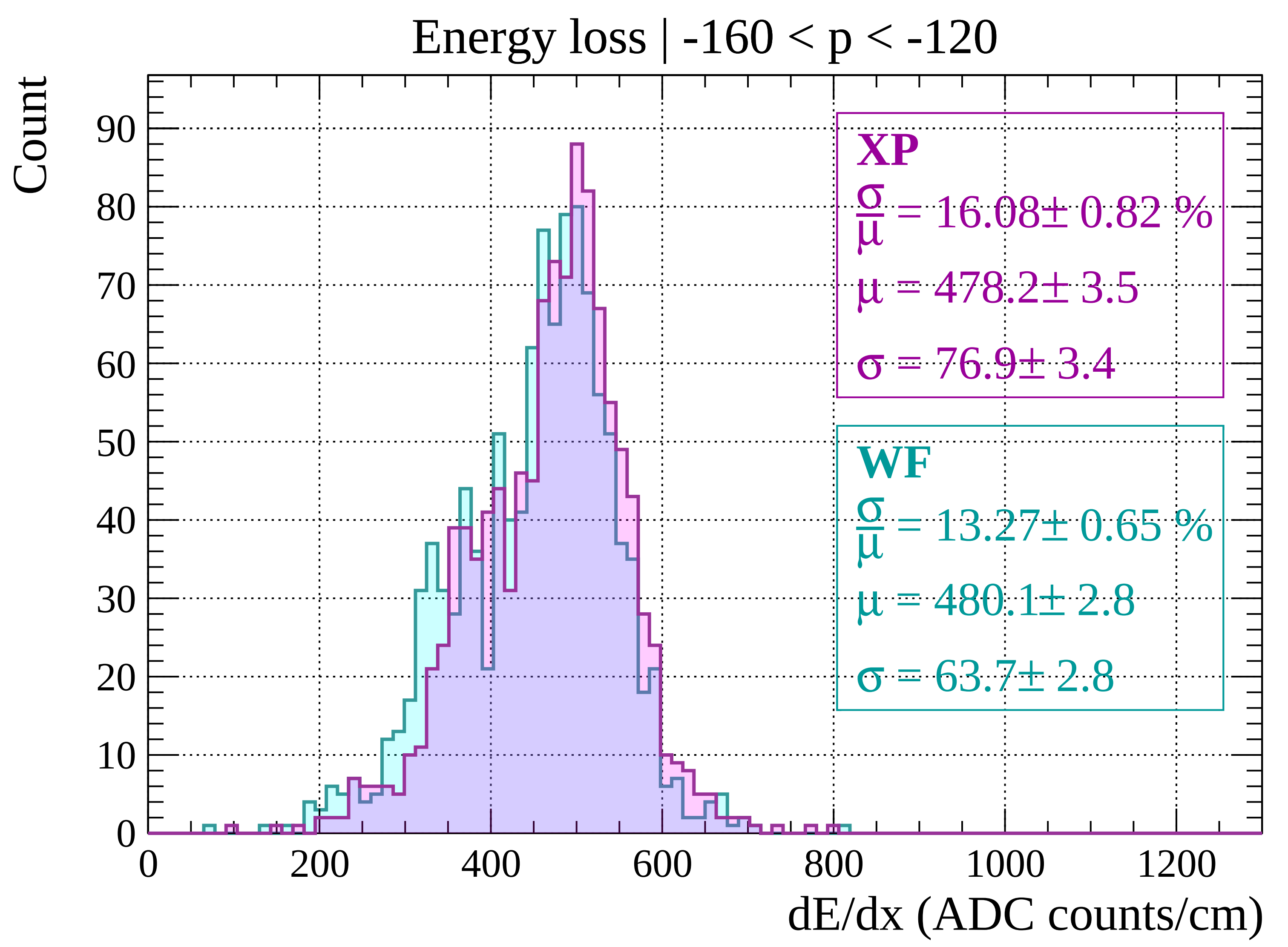

Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.49± 0.24 %
μ = 417.6± 1.0
σ = 31.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.96± 0.23 %
μ = 409.2± 0.9
σ = 28.5± 0.9


Energy loss | -280 < p < -240
Count
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.02± 0.31 %
μ = 420.0± 1.2
σ = 33.7± 1.2
WF
σ/μ = 7.85± 0.35 %
μ = 411.7± 1.2
σ = 32.3± 1.4

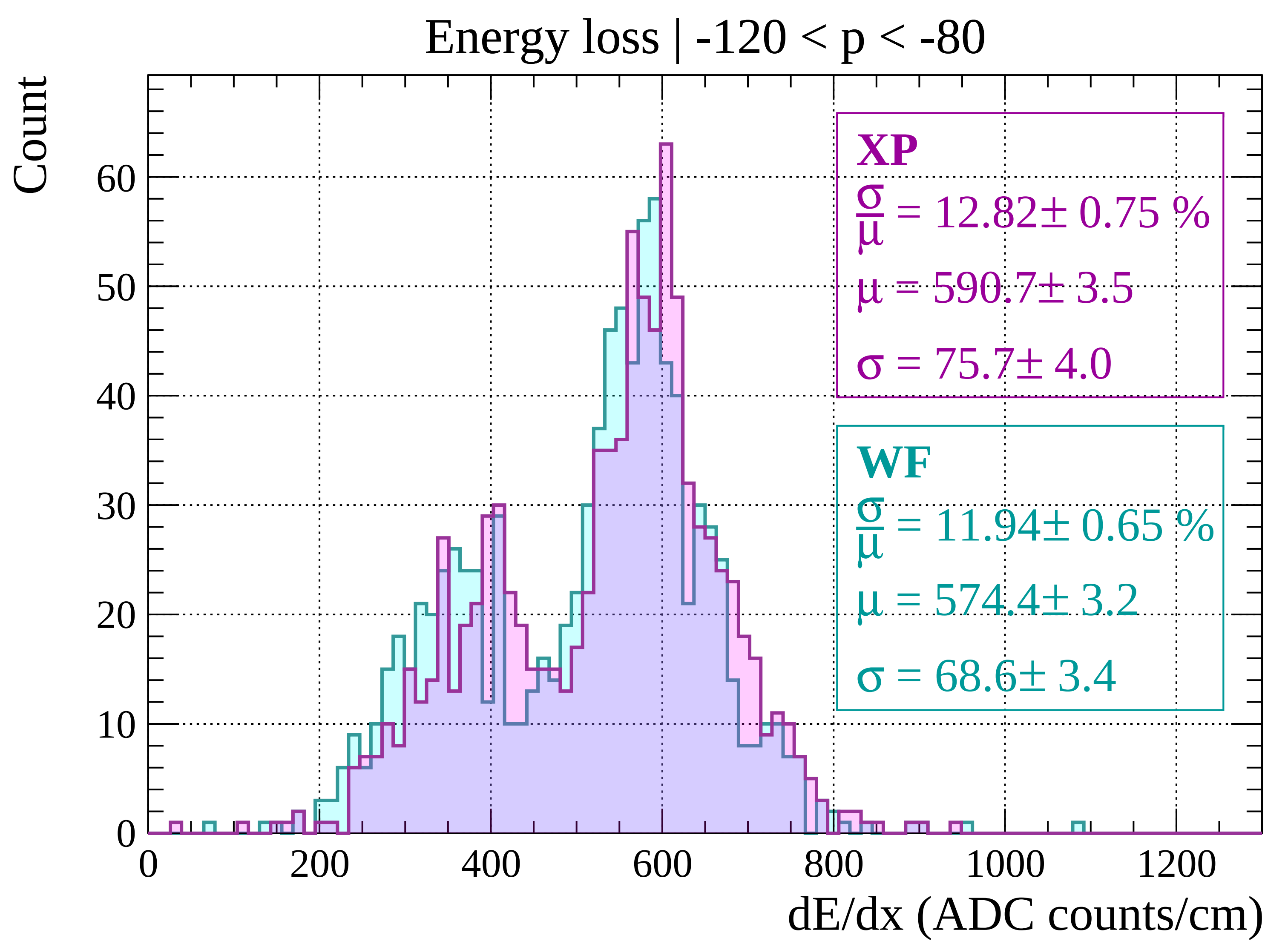


Energy loss | -240 < p < -200
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.92± 0.35 %
μ = 428.2± 1.4
σ = 38.2± 1.4
WF
σ/μ = 8.64± 0.37 %
μ = 420.0± 1.4
σ = 36.3± 1.4

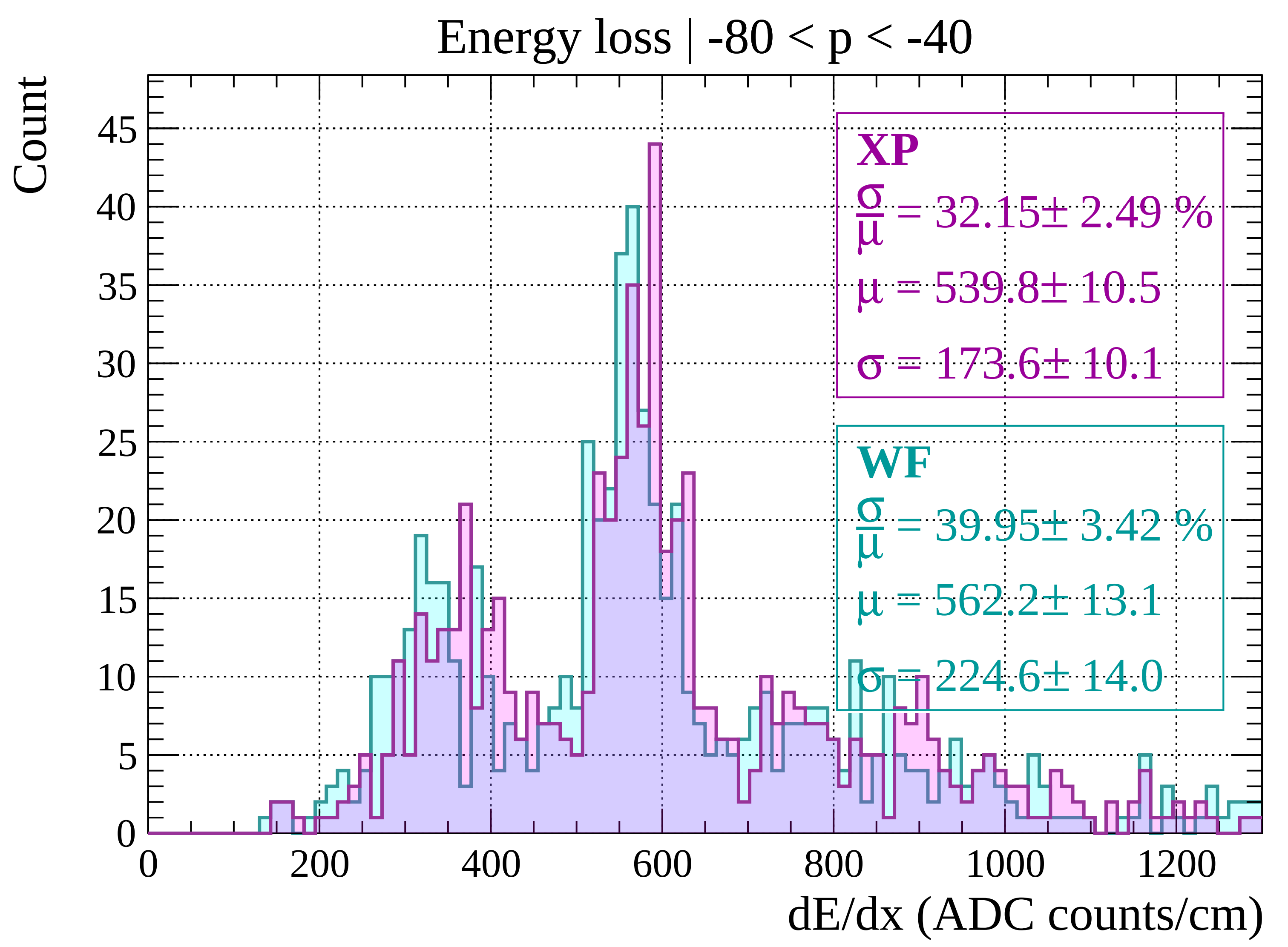

Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 9.06 ± 0.44 %
μ = 453.0 ± 1.7
σ = 41.0 ± 1.8
WF
σ/μ = 8.30 ± 0.40 %
μ = 445.5 ± 1.5
σ = 37.0 ± 1.7

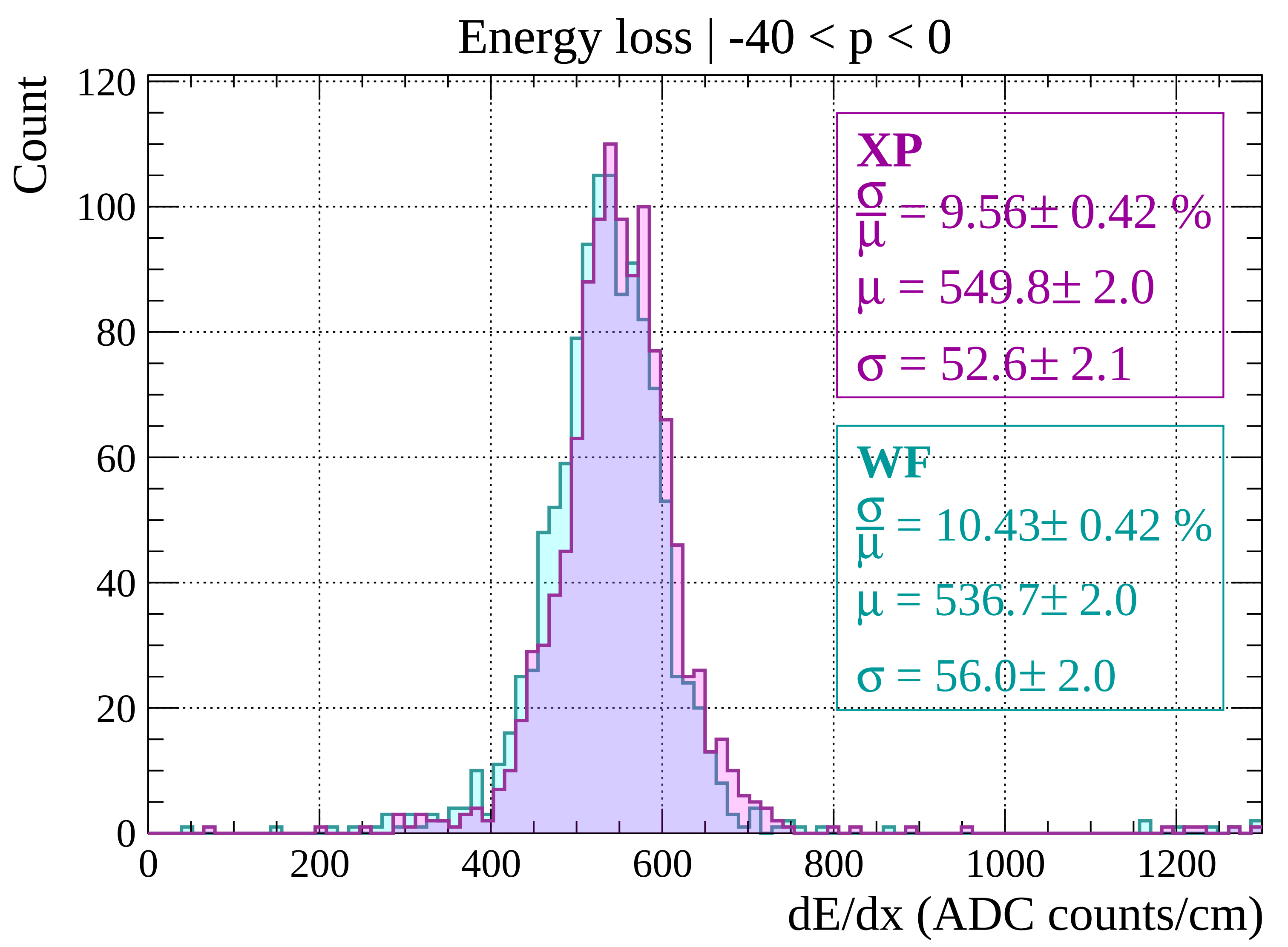


Energy loss | -160 < p < -120
Count
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.08± 0.82 %
μ = 478.2± 3.5
σ = 76.9± 3.4
WF
σ/μ = 13.27± 0.65 %
μ = 480.1± 2.8
σ = 63.7± 2.8

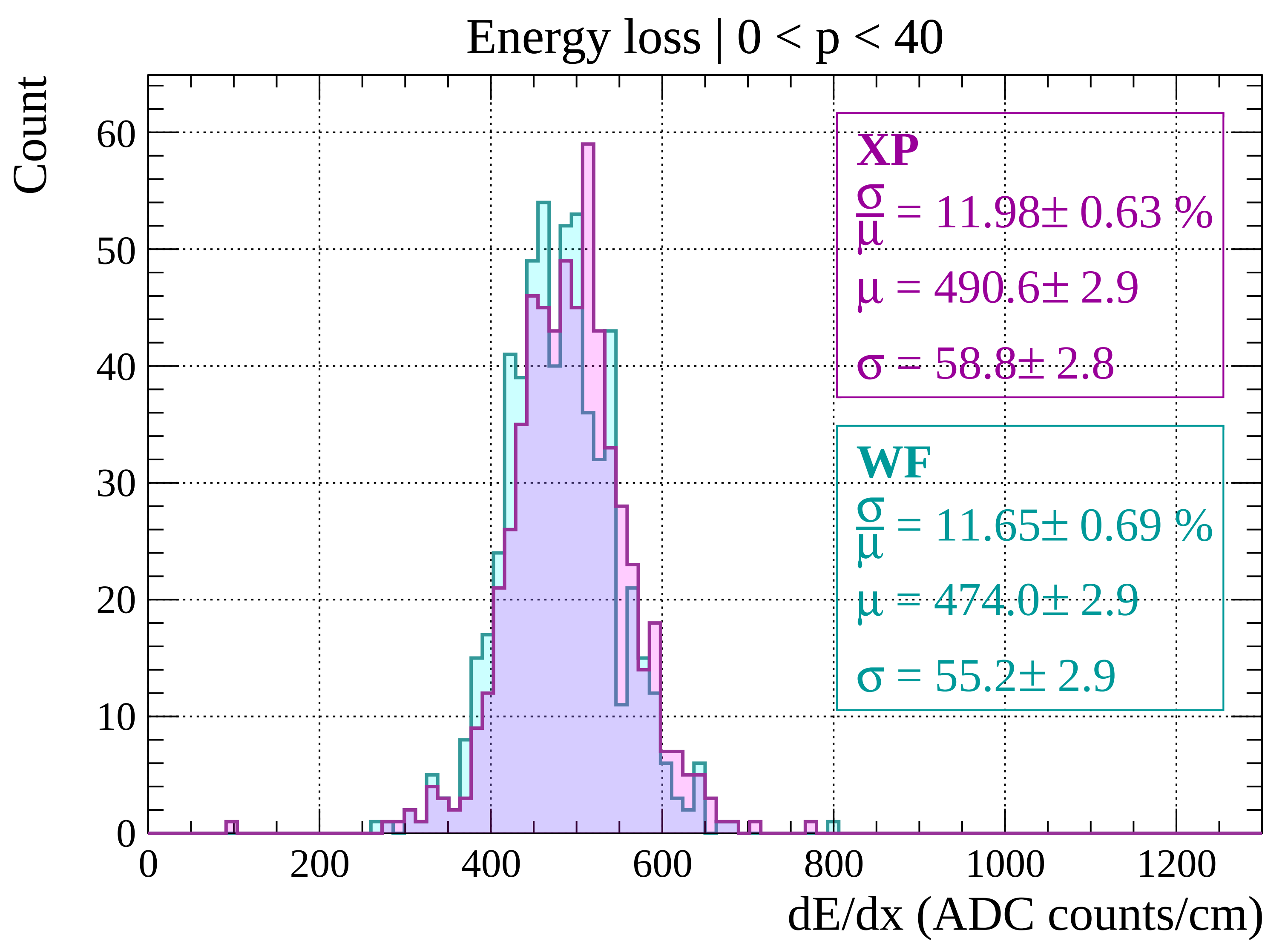


Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 12.82± 0.75 %
μ = 590.7± 3.5
σ = 75.7± 4.0
WF
σ/μ = 11.94± 0.65 %
μ = 574.4± 3.2
σ = 68.6± 3.4

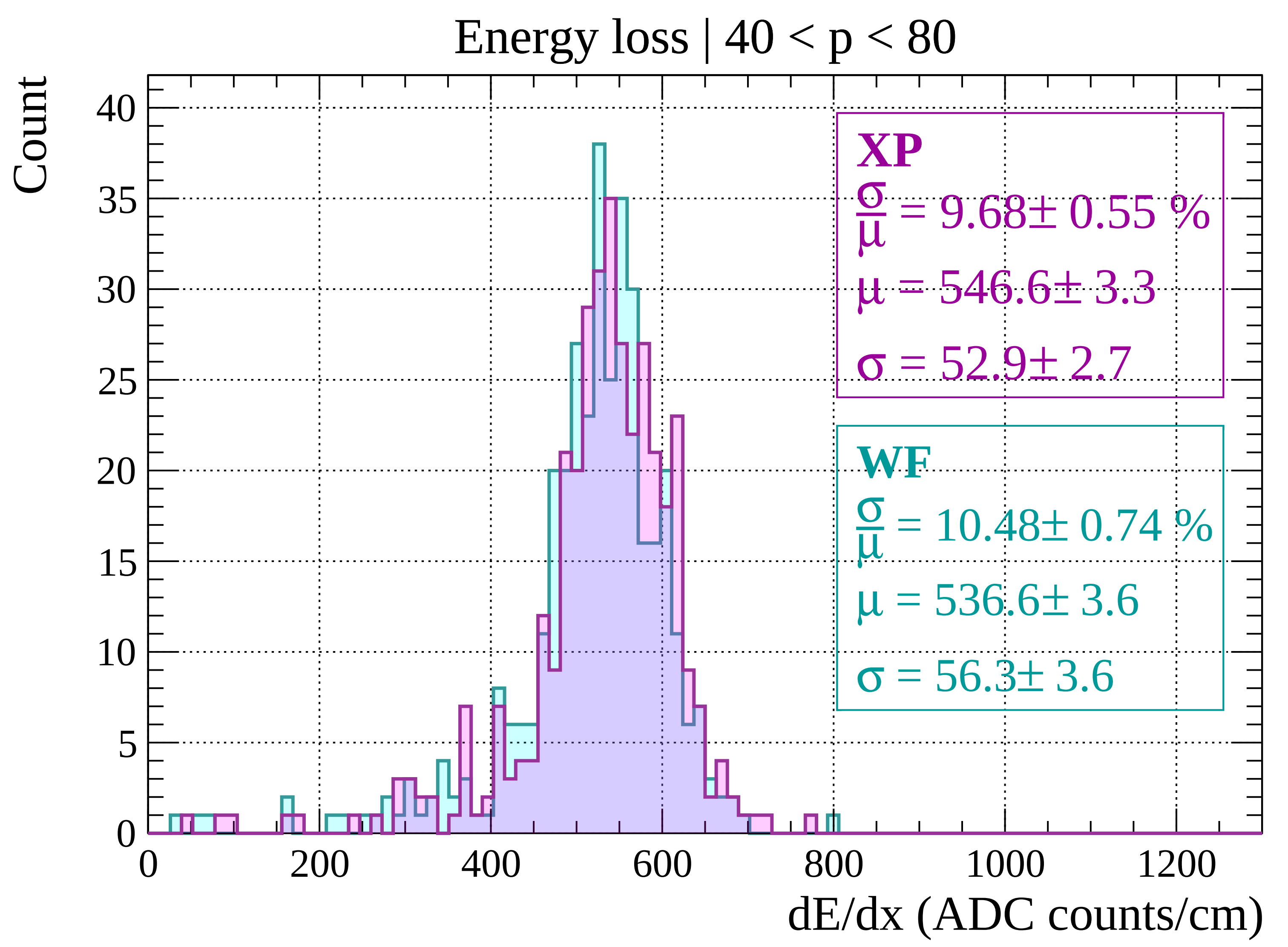

Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 32.15± 2.49 %
μ = 539.8± 10.5
σ = 173.6± 10.1
WF
σ/μ = 39.95± 3.42 %
μ = 562.2± 13.1
σ = 224.6± 14.0
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45



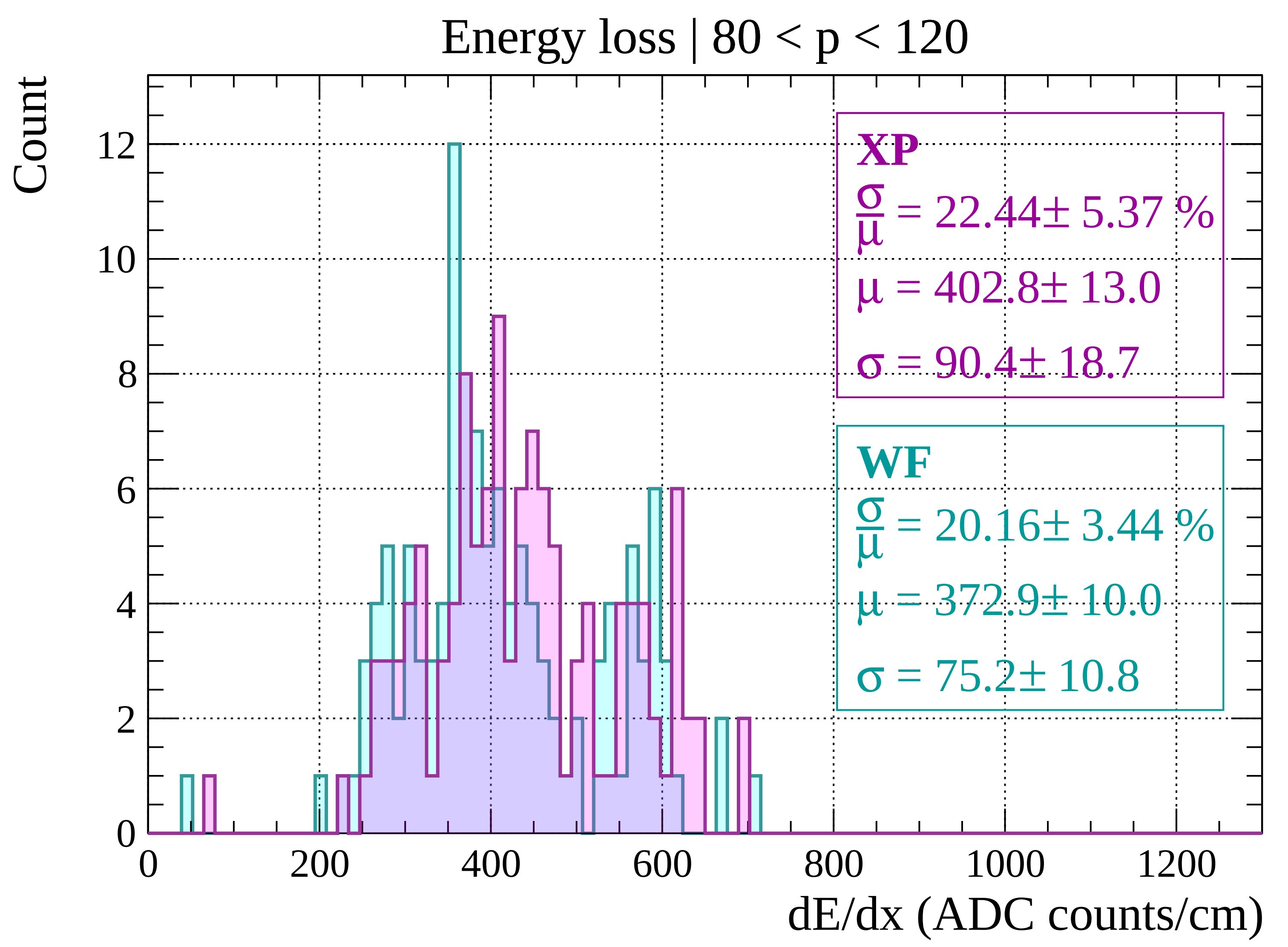
Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 9.56± 0.42 %
μ = 549.8± 2.0
σ = 52.6± 2.1
WF
σ/μ = 10.43± 0.42 %
μ = 536.7± 2.0
σ = 56.0± 2.0



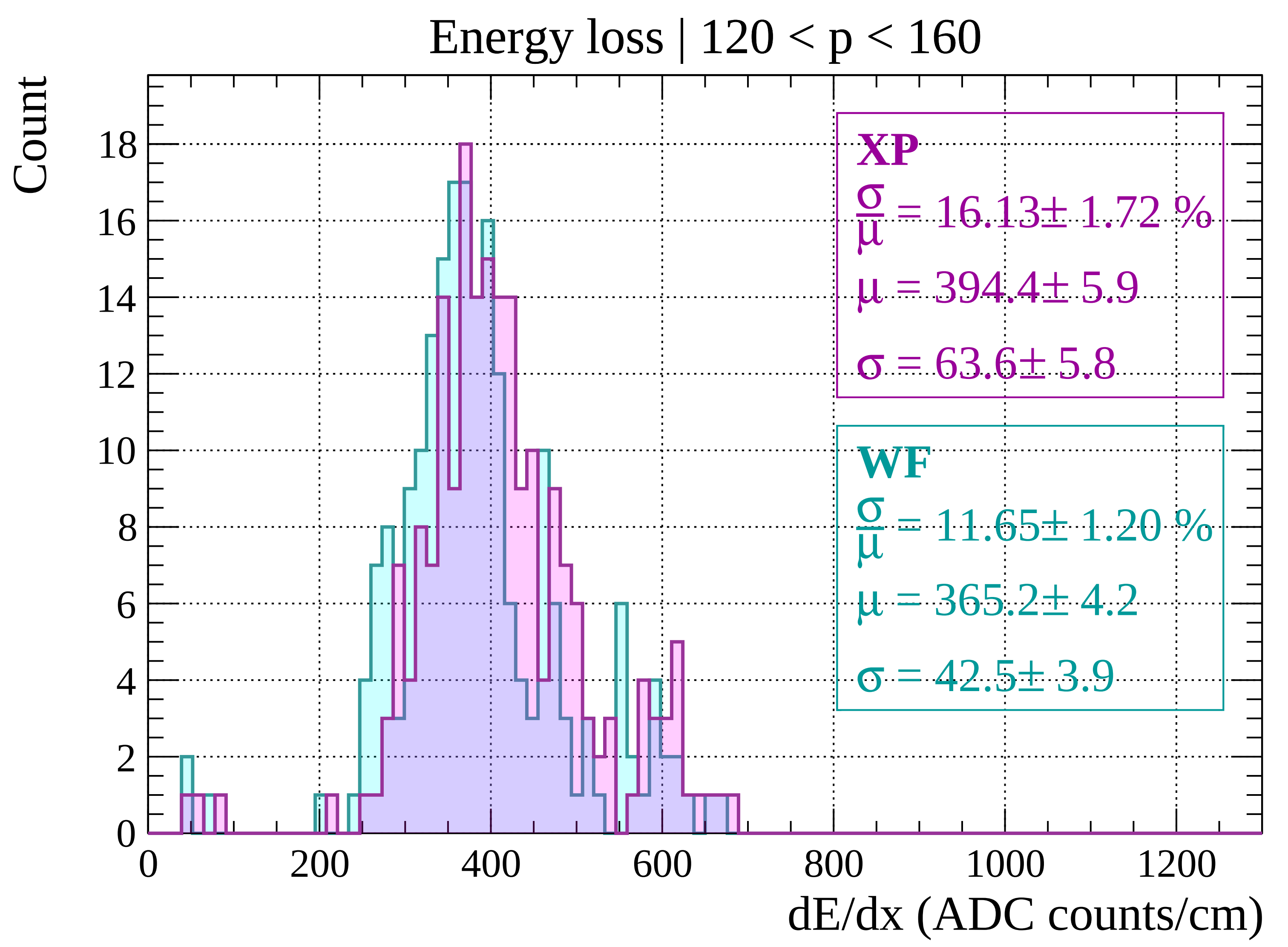
Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 11.98± 0.63 %
μ = 490.6± 2.9
σ = 58.8± 2.8
WF
σ/μ = 11.65± 0.69 %
μ = 474.0± 2.9
σ = 55.2± 2.9



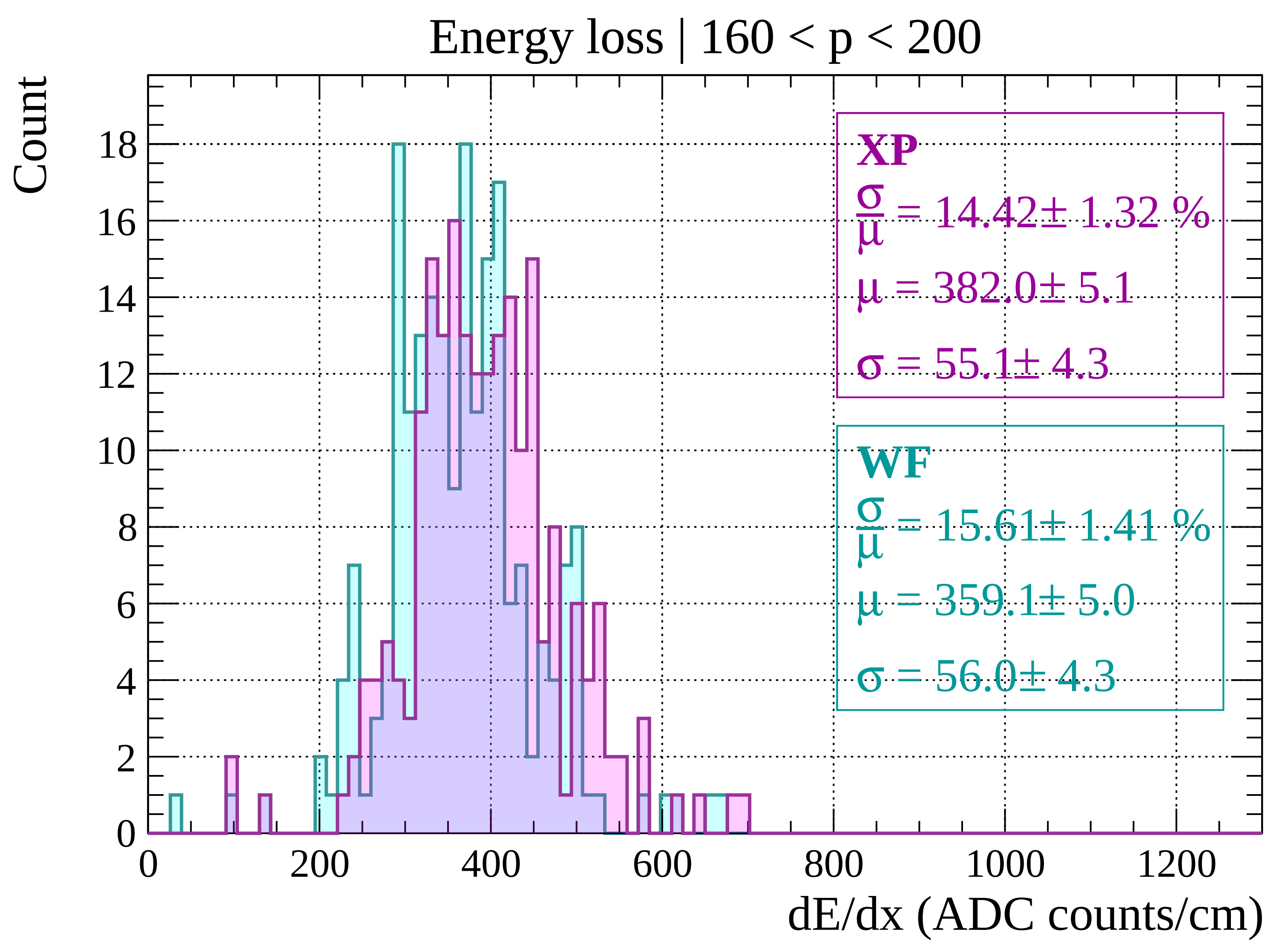
Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.68± 0.55 %
μ = 546.6± 3.3
σ = 52.9± 2.7
WF
$\frac{\sigma}{\mu}$ = 10.48± 0.74 %
μ = 536.6± 3.6
σ = 56.3± 3.6



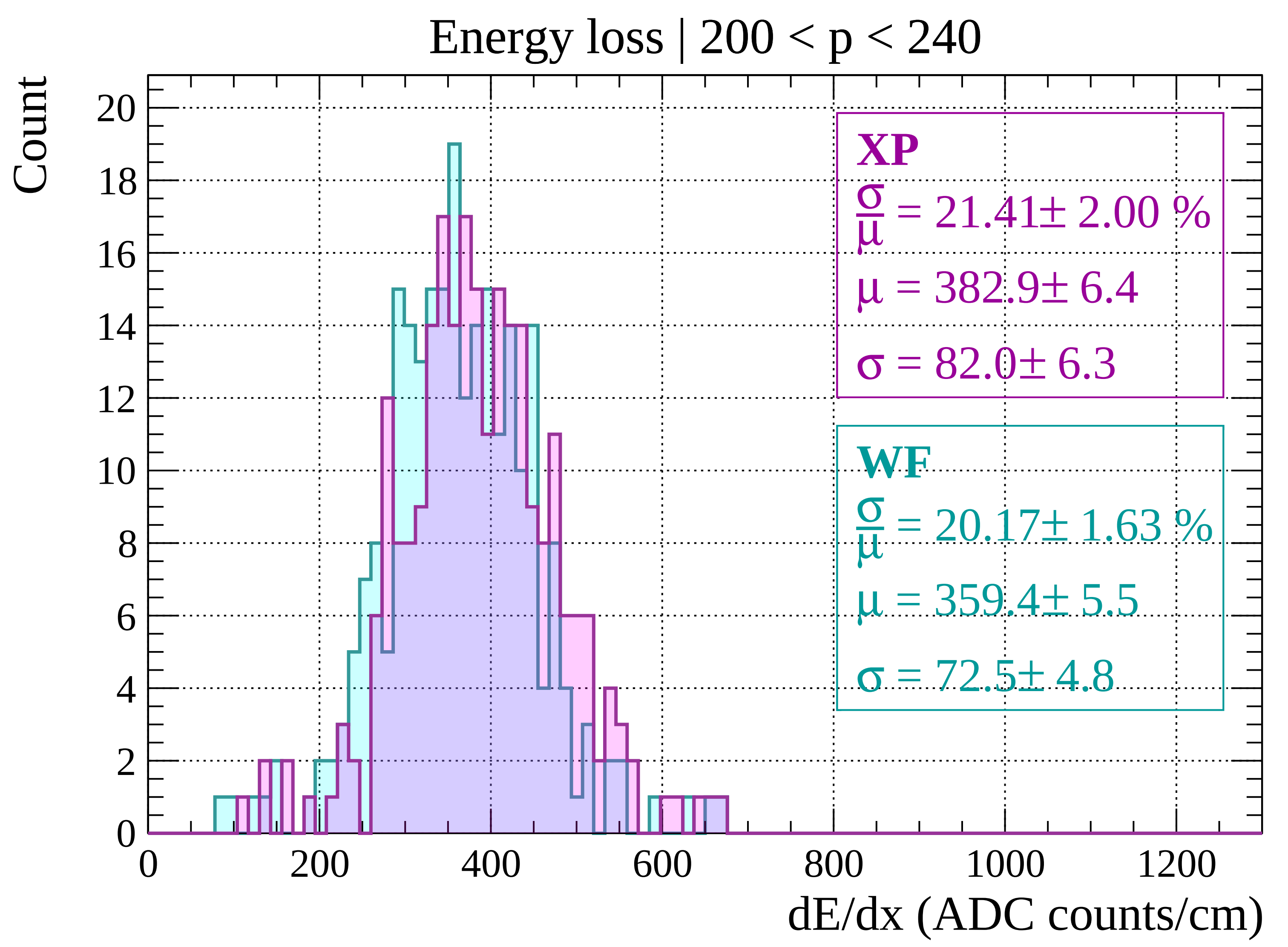
Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
XP
σ/μ = 22.44± 5.37 %
μ = 402.8± 13.0
σ = 90.4± 18.7
WF
σ/μ = 20.16± 3.44 %
μ = 372.9± 10.0
σ = 75.2± 10.8



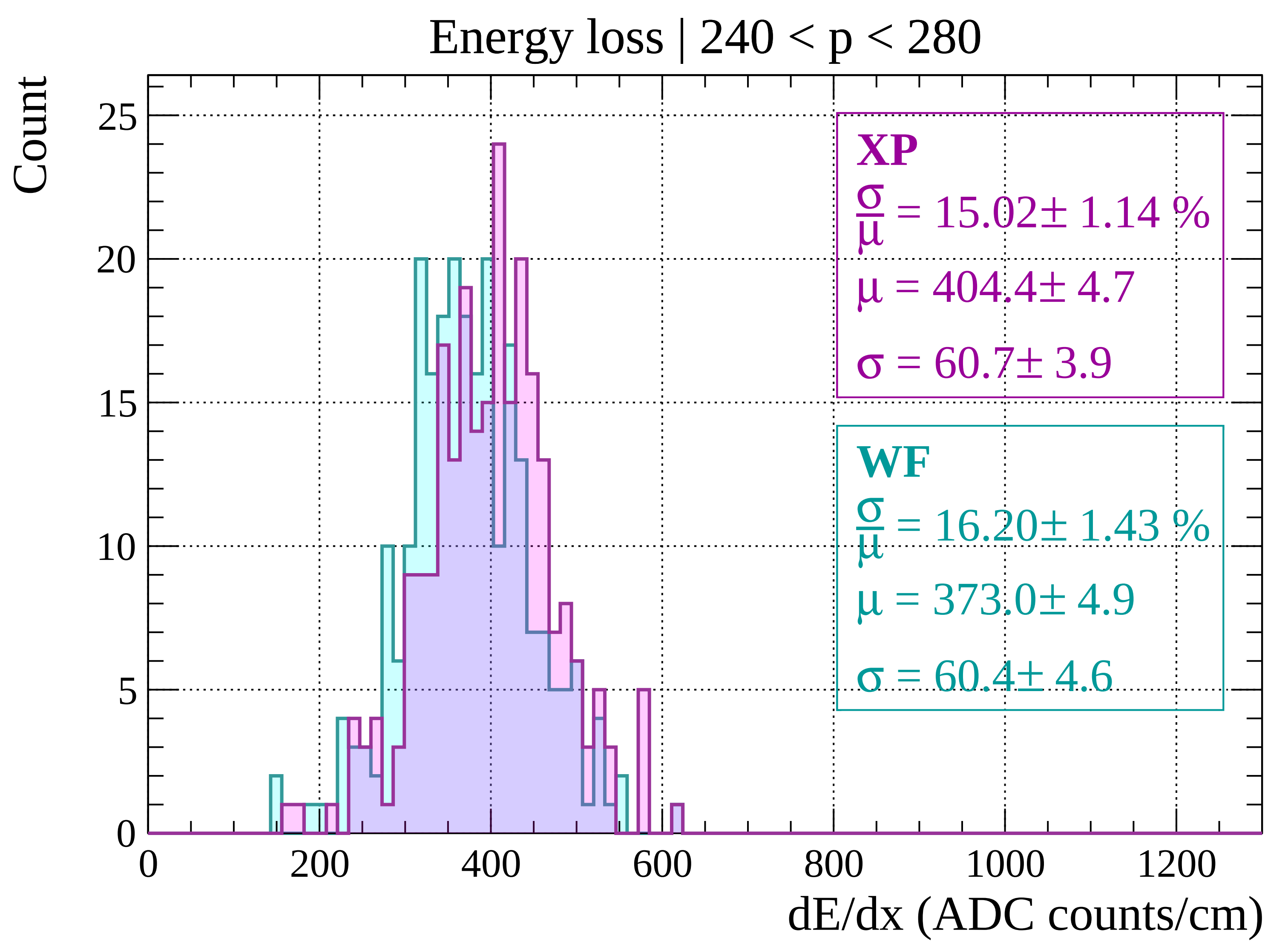
Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
XP
σ/μ = 16.13± 1.72 %
μ = 394.4± 5.9
σ = 63.6± 5.8
WF
σ/μ = 11.65± 1.20 %
μ = 365.2± 4.2
σ = 42.5± 3.9



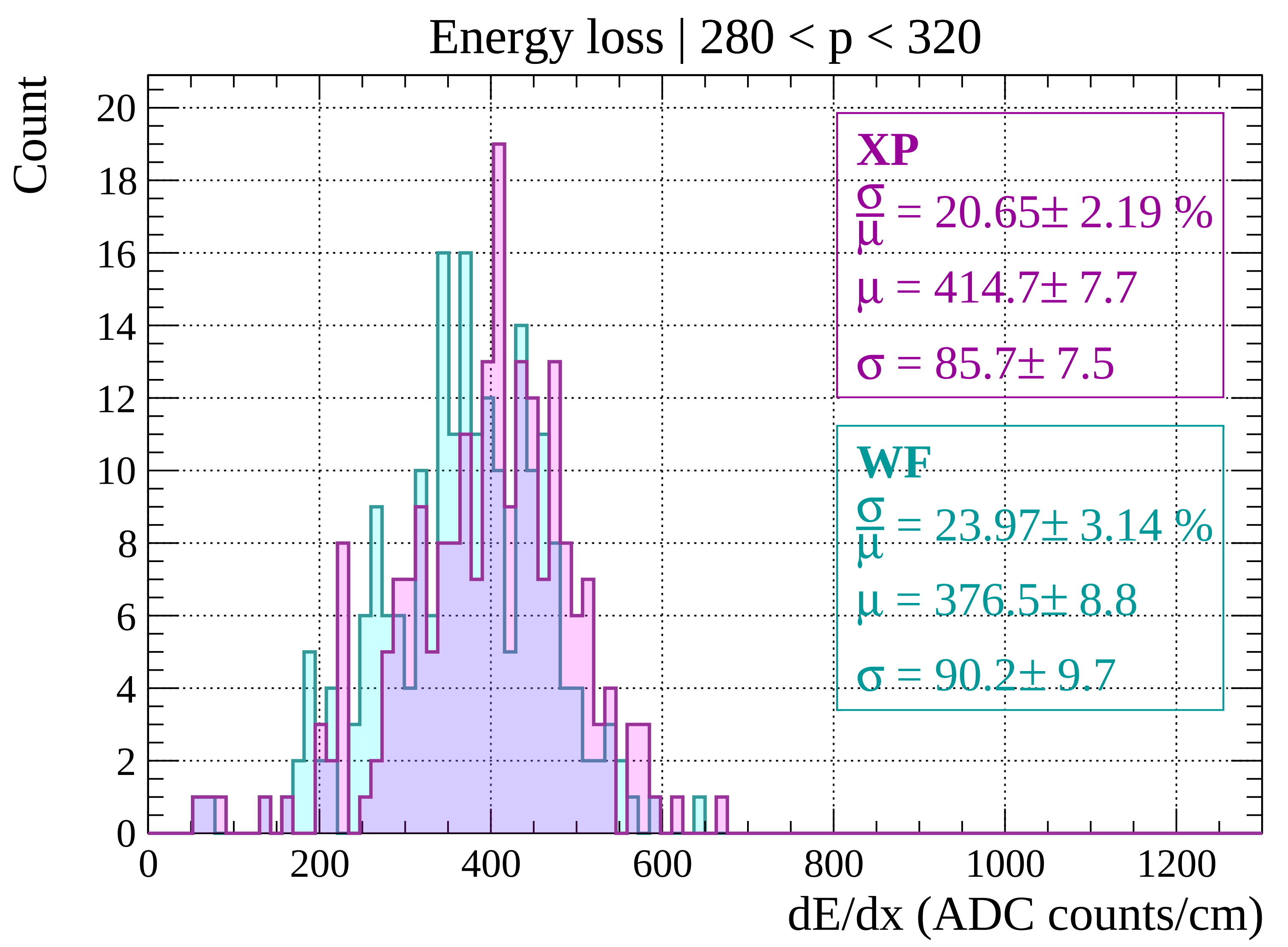
Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 14.42± 1.32 %
μ = 382.0± 5.1
σ = 55.1± 4.3
WF
σ/μ = 15.61± 1.41 %
μ = 359.1± 5.0
σ = 56.0± 4.3



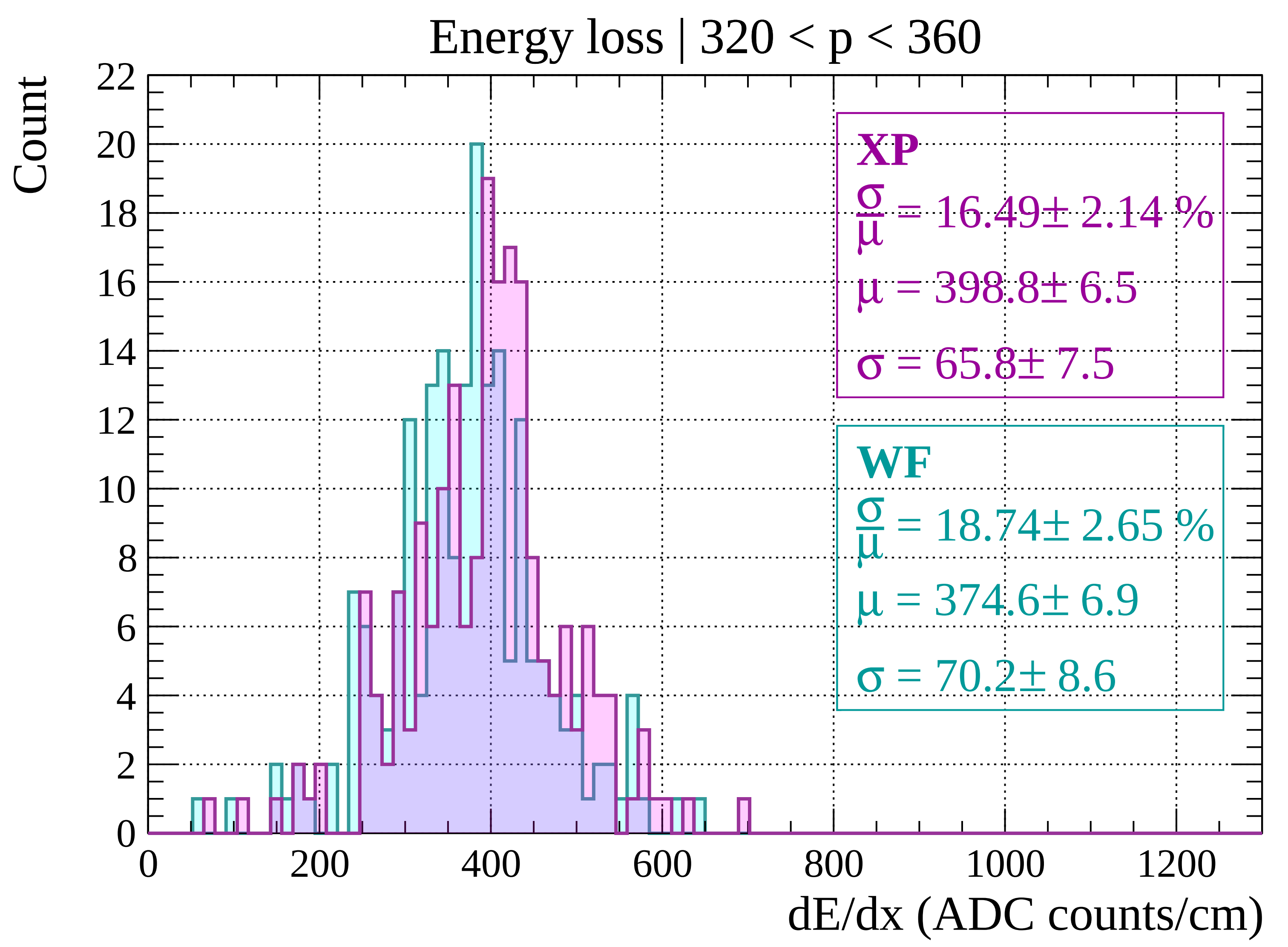
Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
0
200
400
600
800
1000
1200
XP
σ/μ = 21.41± 2.00 %
μ = 382.9± 6.4
σ = 82.0± 6.3
WF
σ/μ = 20.17± 1.63 %
μ = 359.4± 5.5
σ = 72.5± 4.8

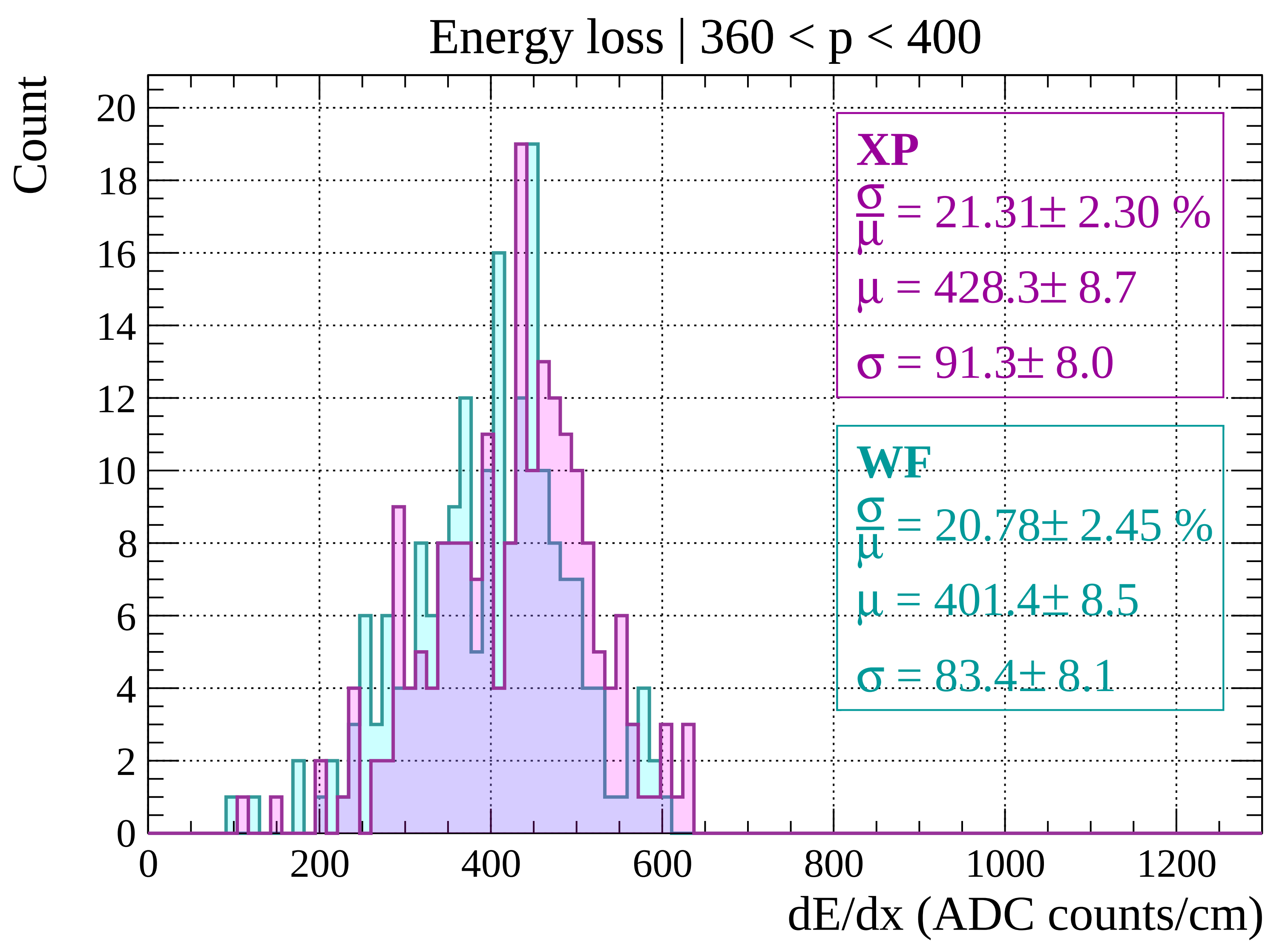


Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.02± 1.14 %
μ = 404.4± 4.7
σ = 60.7± 3.9
WF
σ/μ = 16.20± 1.43 %
μ = 373.0± 4.9
σ = 60.4± 4.6
0
200
400
600
800
1000
1200
0
5
10
15
20
25

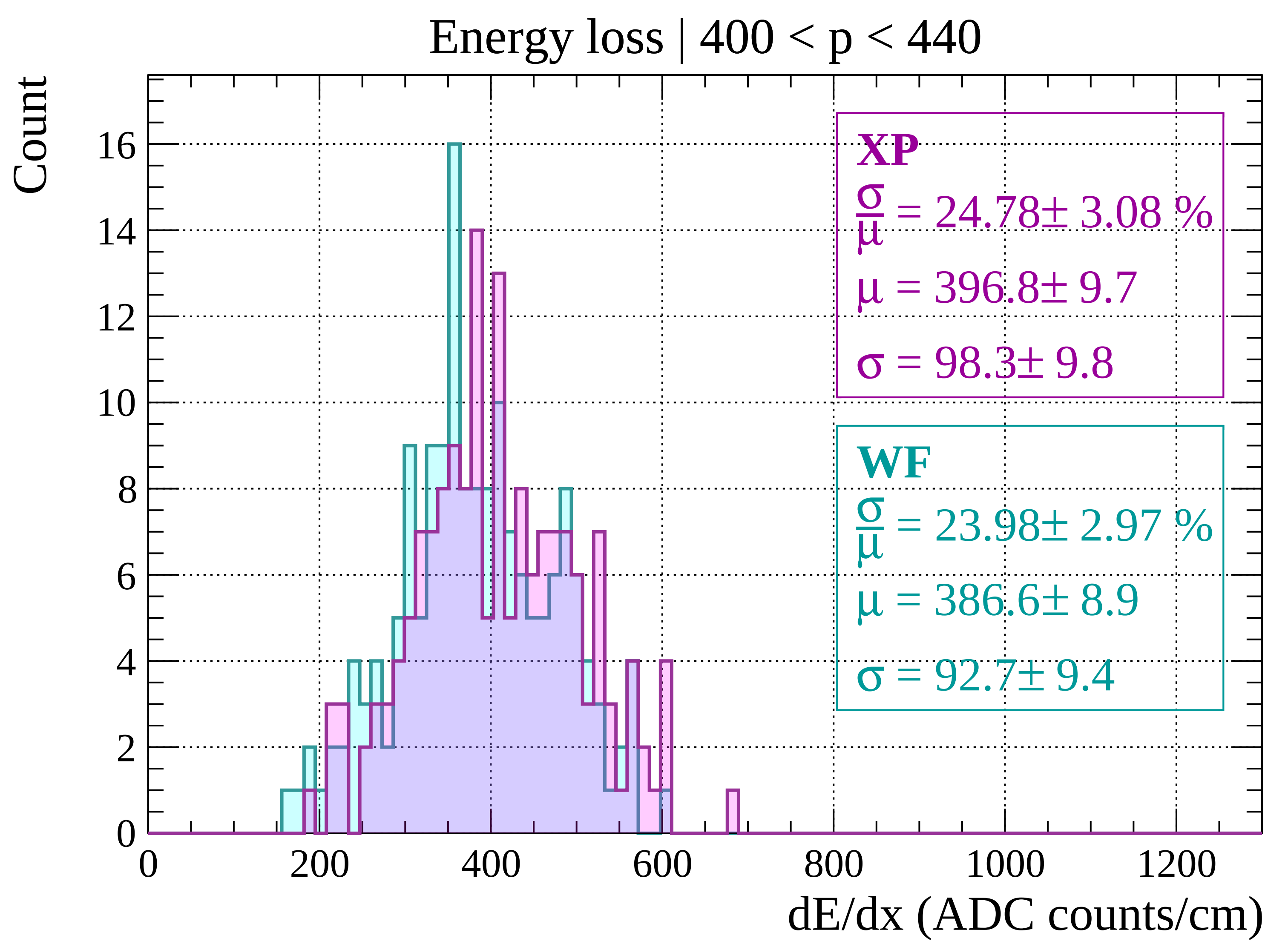


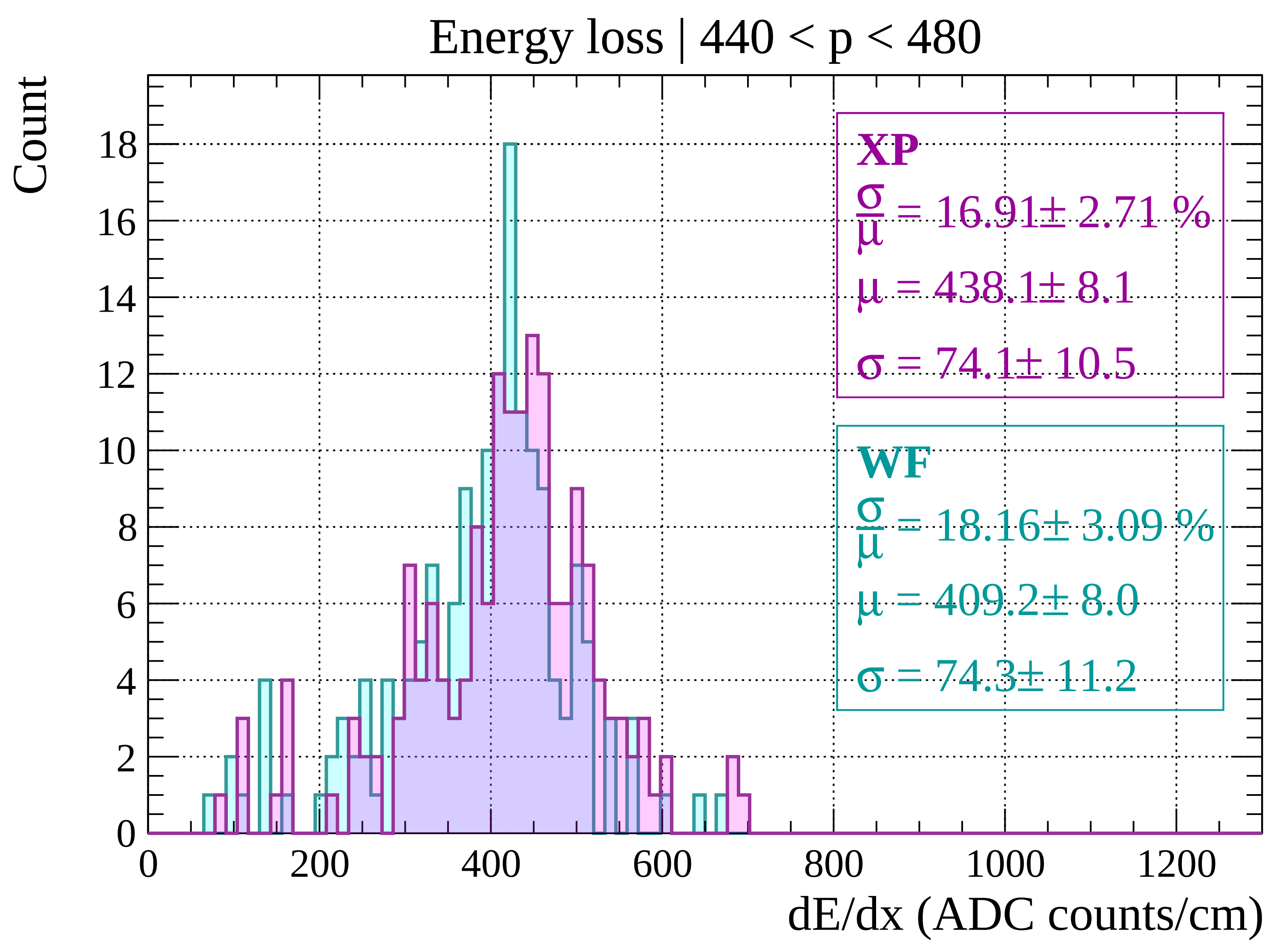
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
XP
σ/μ = 20.65± 2.19 %
μ = 414.7± 7.7
σ = 85.7± 7.5
WF
σ/μ = 23.97± 3.14 %
μ = 376.5± 8.8
σ = 90.2± 9.7


Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.49± 2.14 %
μ = 398.8± 6.5
σ = 65.8± 7.5
WF
σ/μ = 18.74± 2.65 %
μ = 374.6± 6.9
σ = 70.2± 8.6

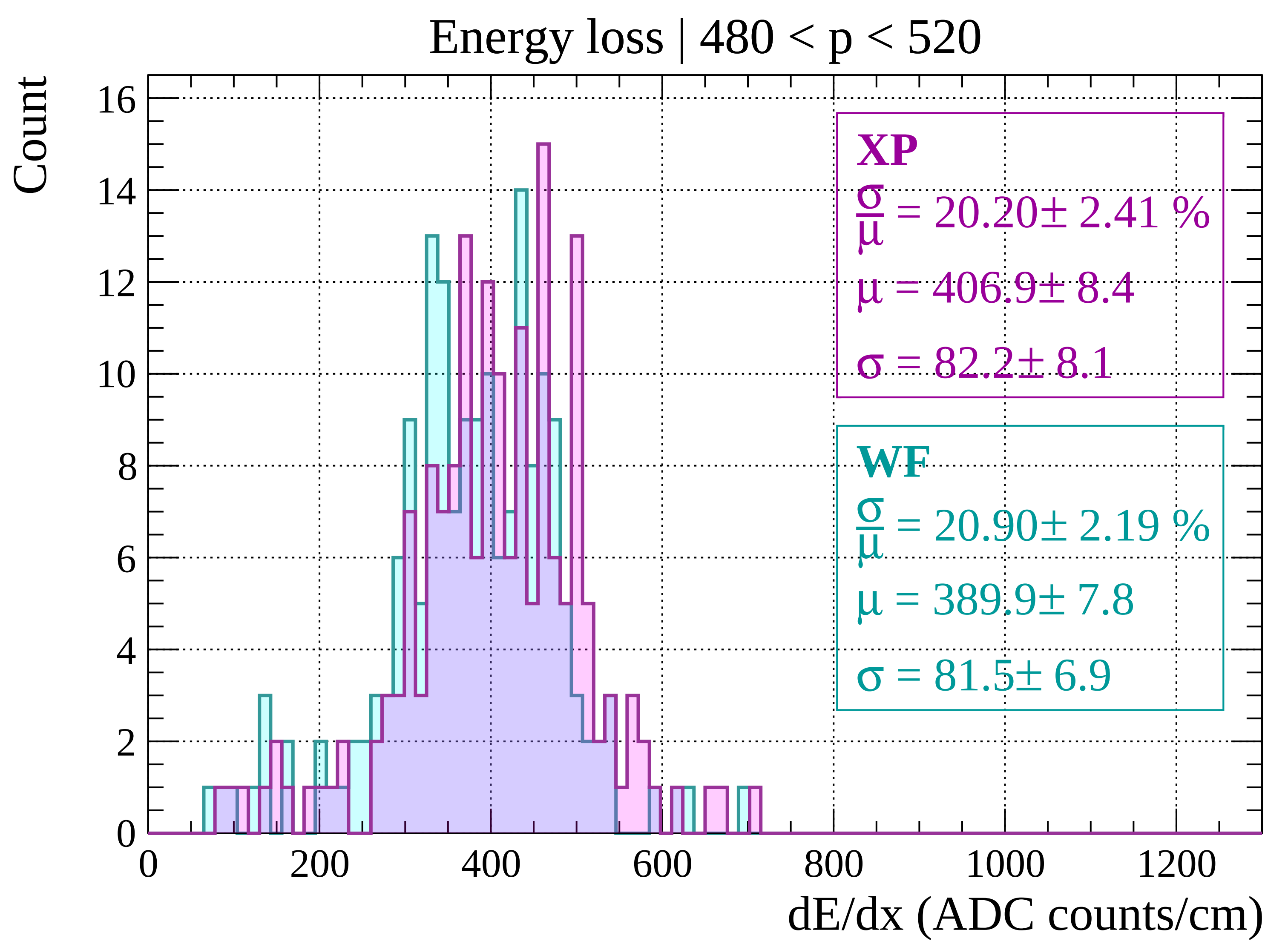


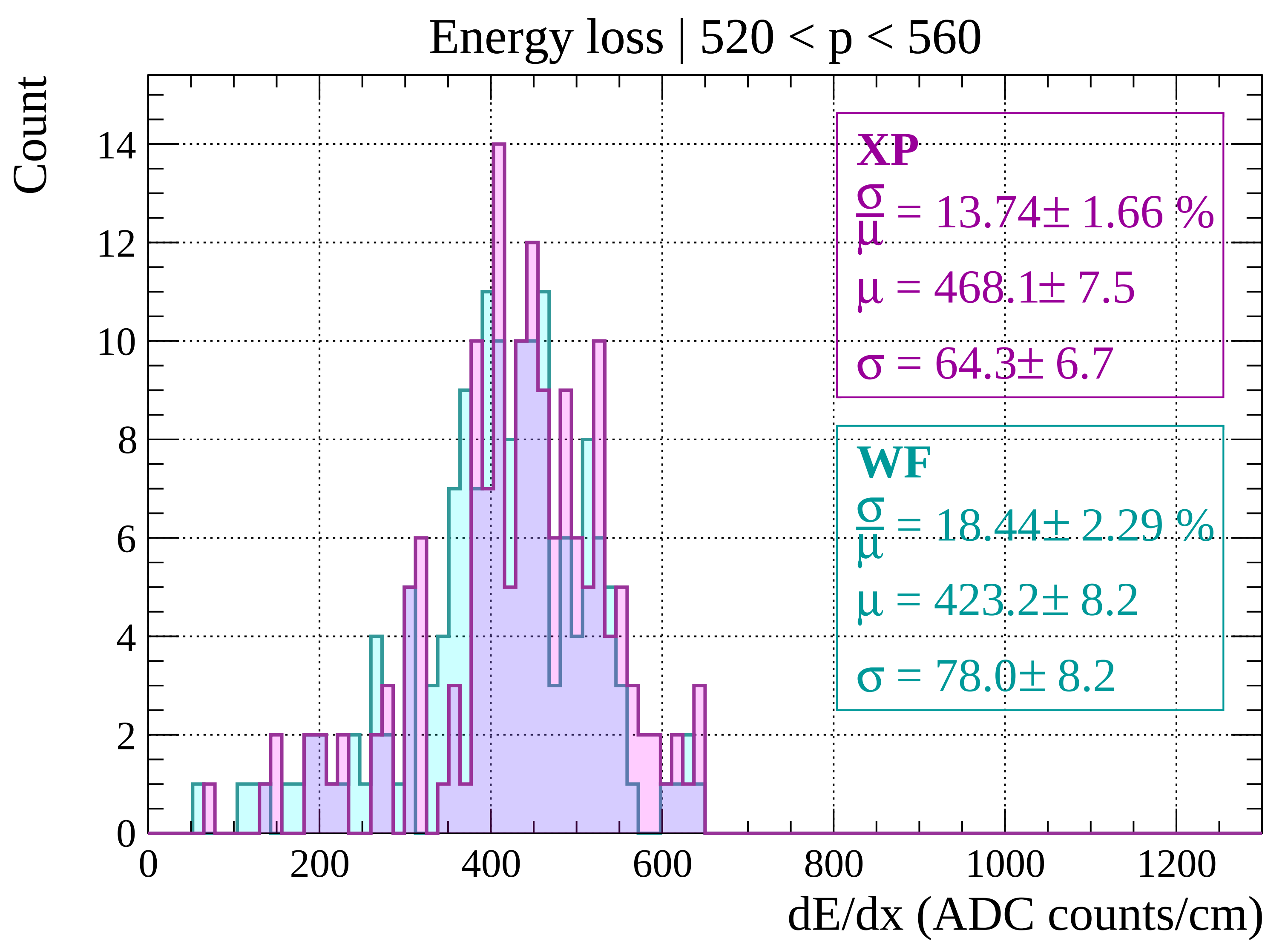
Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
XP
$\frac{\sigma}{\mu}$ = 21.31± 2.30 %
μ = 428.3± 8.7
σ = 91.3± 8.0
WF
$\frac{\sigma}{\mu}$ = 20.78± 2.45 %
μ = 401.4± 8.5
σ = 83.4± 8.1

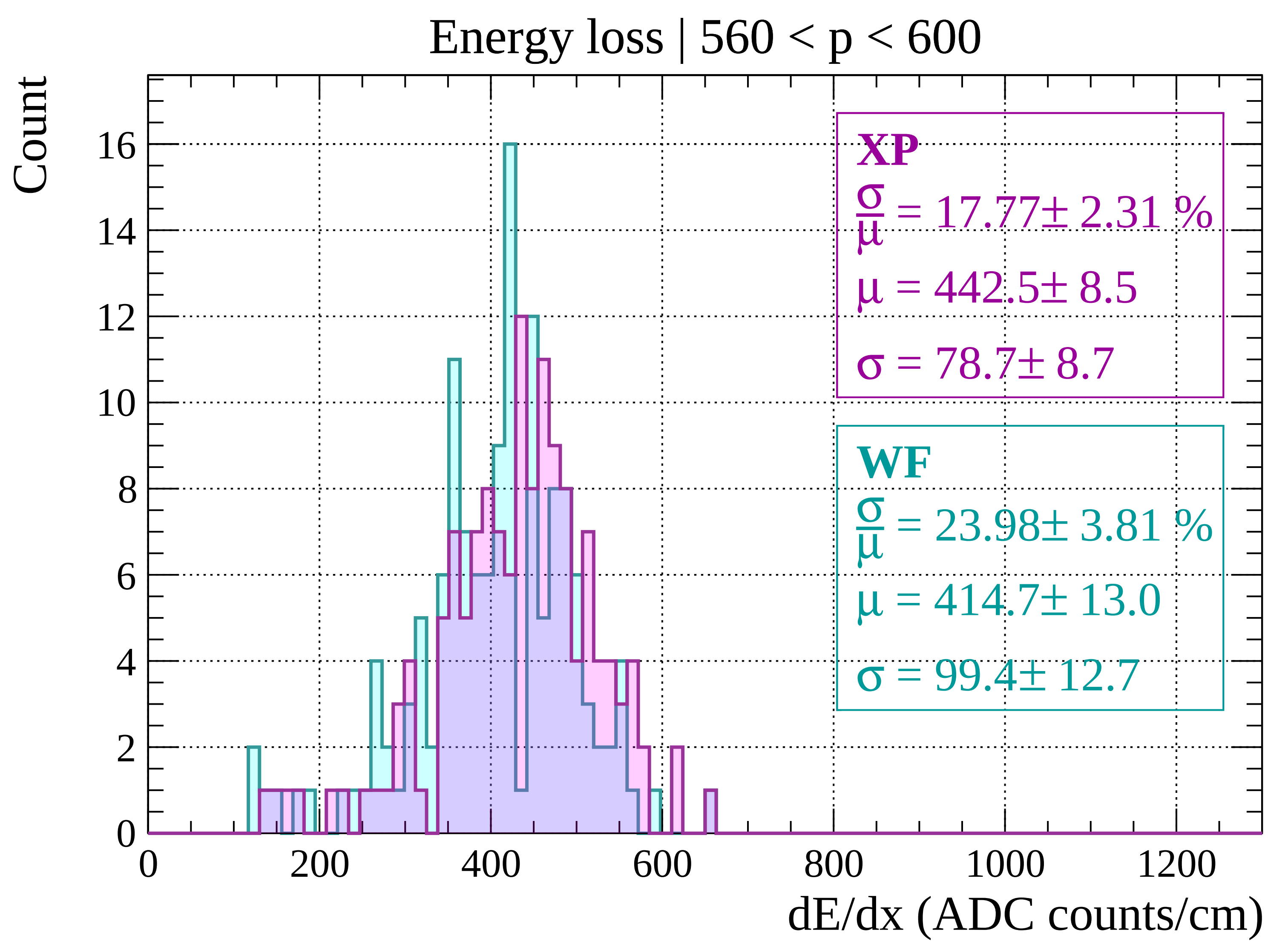

Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 24.78± 3.08 %
μ = 396.8± 9.7
σ = 98.3± 9.8
WF
σ/μ = 23.98± 2.97 %
μ = 386.6± 8.9
σ = 92.7± 9.4
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16

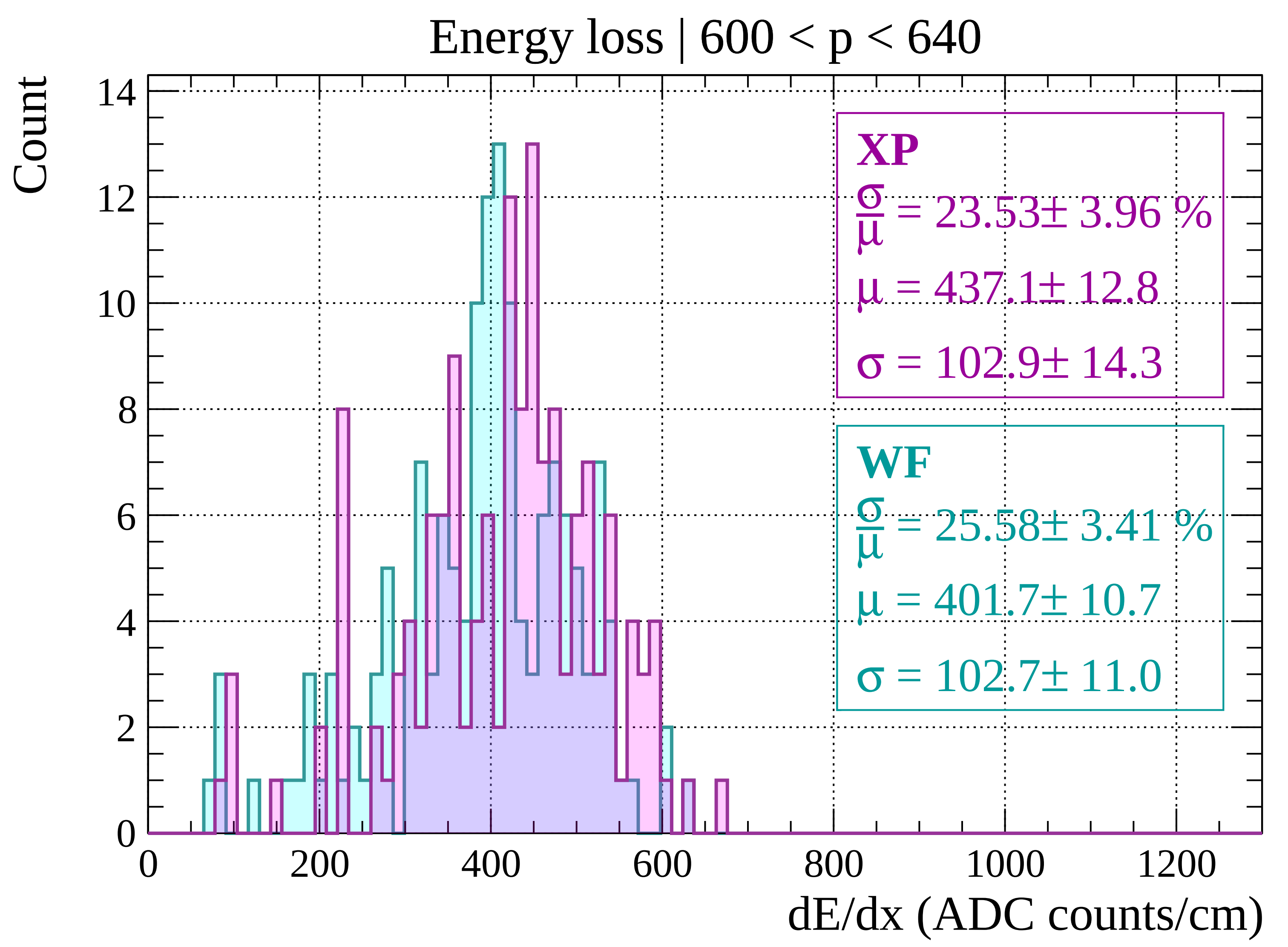


Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.91± 2.71 %
μ = 438.1± 8.1
σ = 74.1± 10.5
WF
σ/μ = 18.16± 3.09 %
μ = 409.2± 8.0
σ = 74.3± 11.2
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18

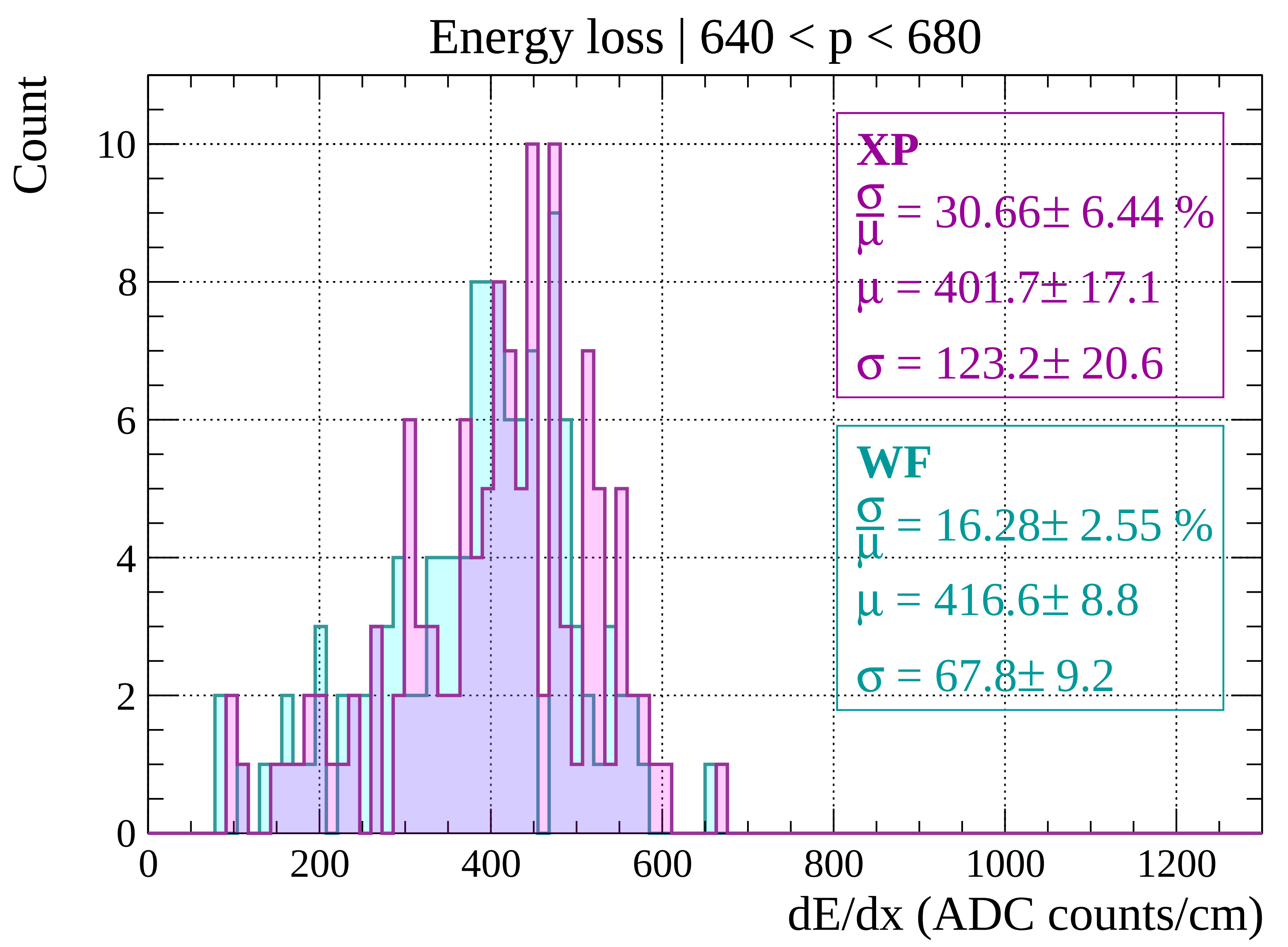

Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 20.20± 2.41 %
μ = 406.9± 8.4
σ = 82.2± 8.1
WF
$\frac{\sigma}{\mu}$ = 20.90± 2.19 %
μ = 389.9± 7.8
σ = 81.5± 6.9
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16

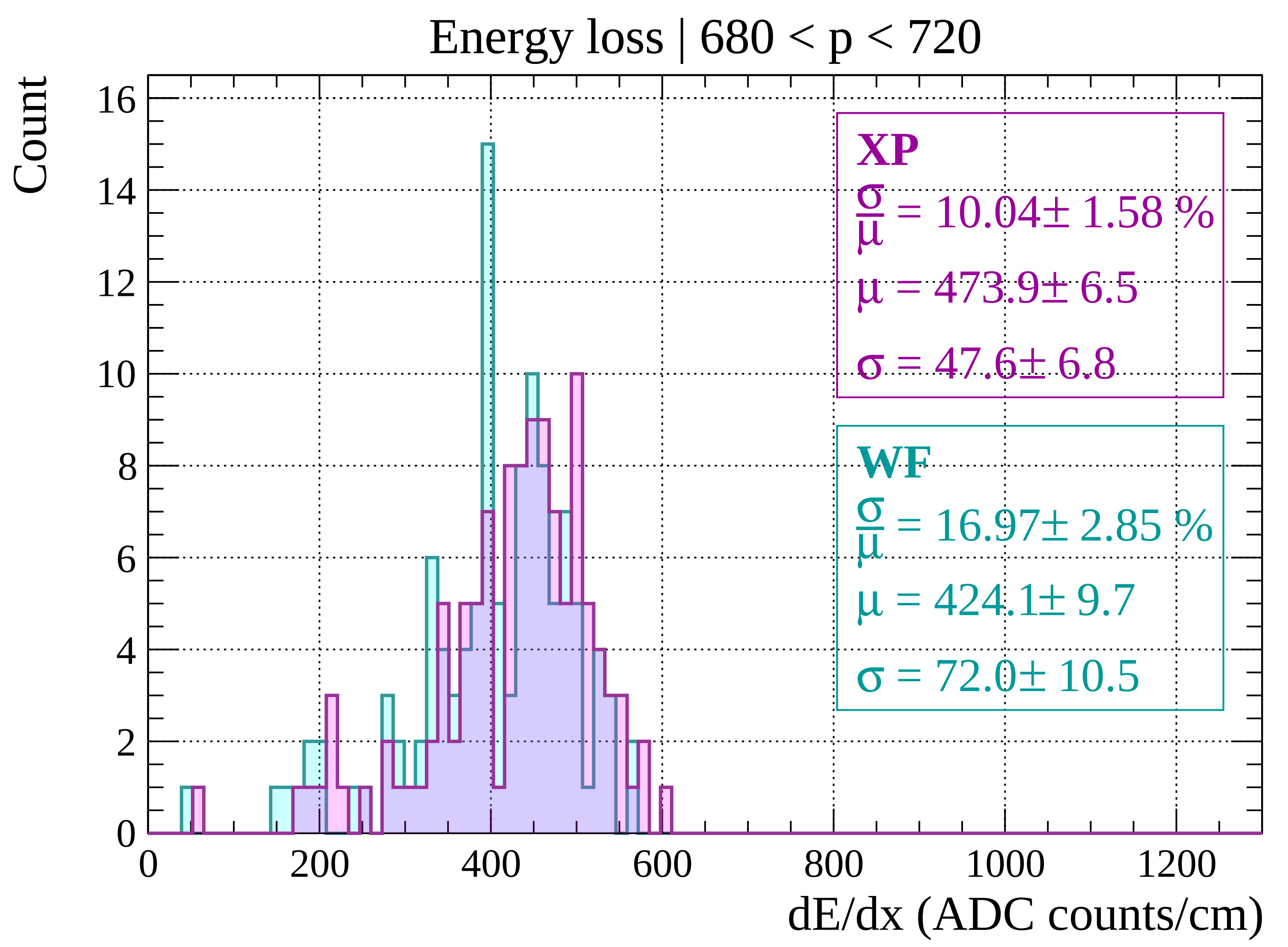

Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
XP
$\frac{\sigma}{\mu}$ = 13.74± 1.66 %
μ = 468.1± 7.5
σ = 64.3± 6.7
WF
$\frac{\sigma}{\mu}$ = 18.44± 2.29 %
μ = 423.2± 8.2
σ = 78.0± 8.2


Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 17.77± 2.31 %
μ = 442.5± 8.5
σ = 78.7± 8.7
WF
σ/μ = 23.98± 3.81 %
μ = 414.7± 13.0
σ = 99.4± 12.7
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16

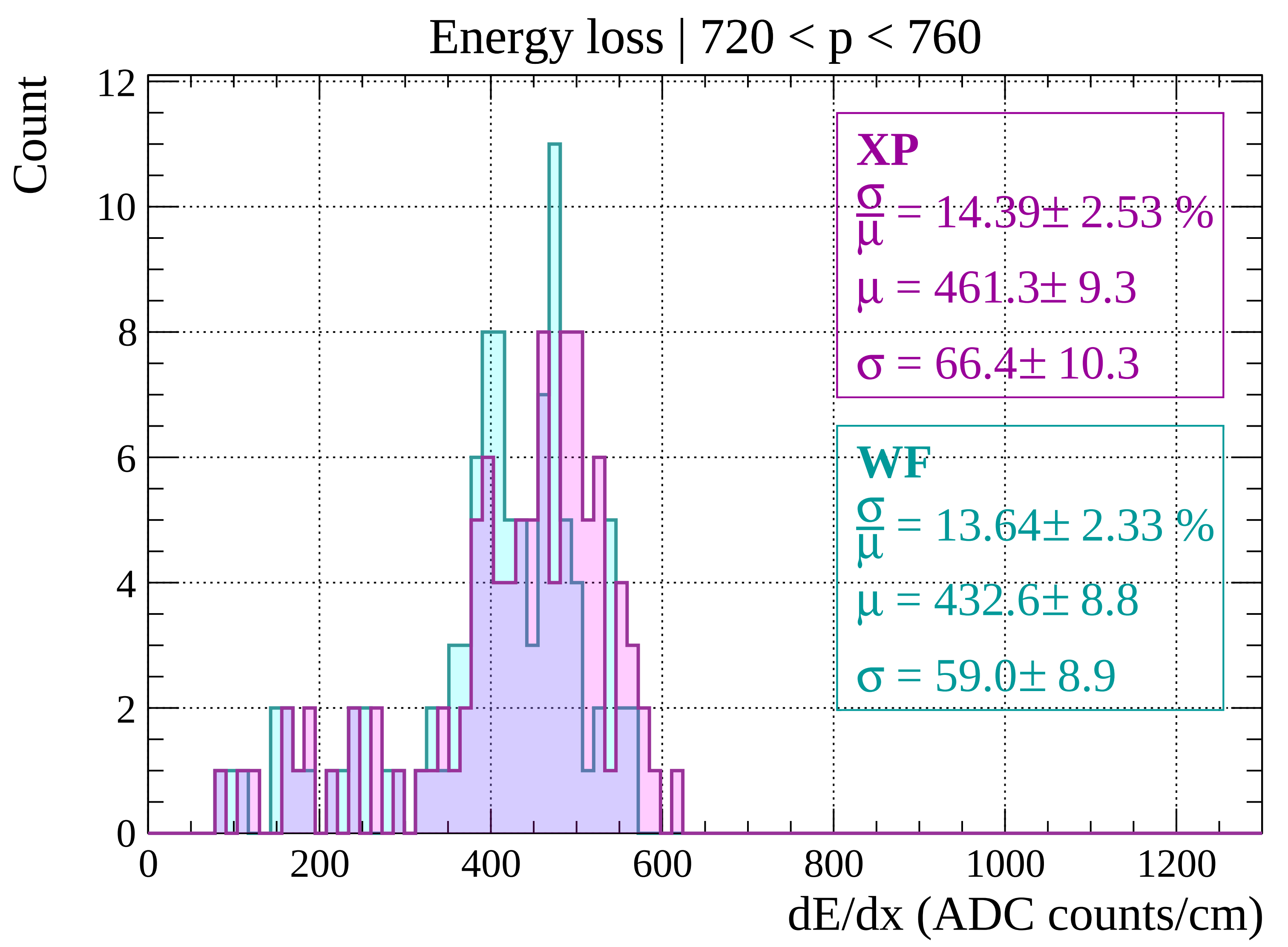


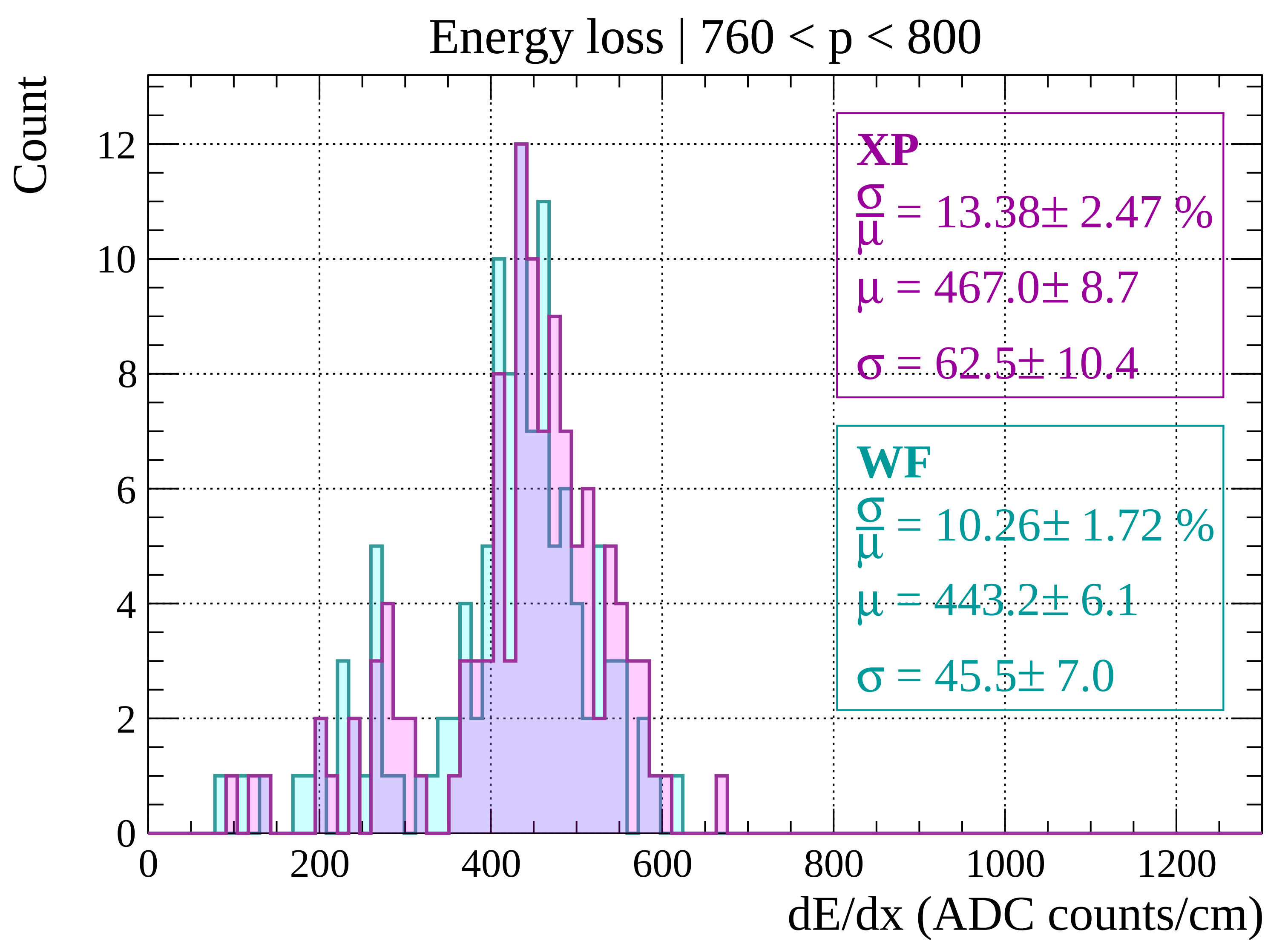
Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 23.53± 3.96 %
μ = 437.1± 12.8
σ = 102.9± 14.3
WF
σ/μ = 25.58± 3.41 %
μ = 401.7± 10.7
σ = 102.7± 11.0
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14

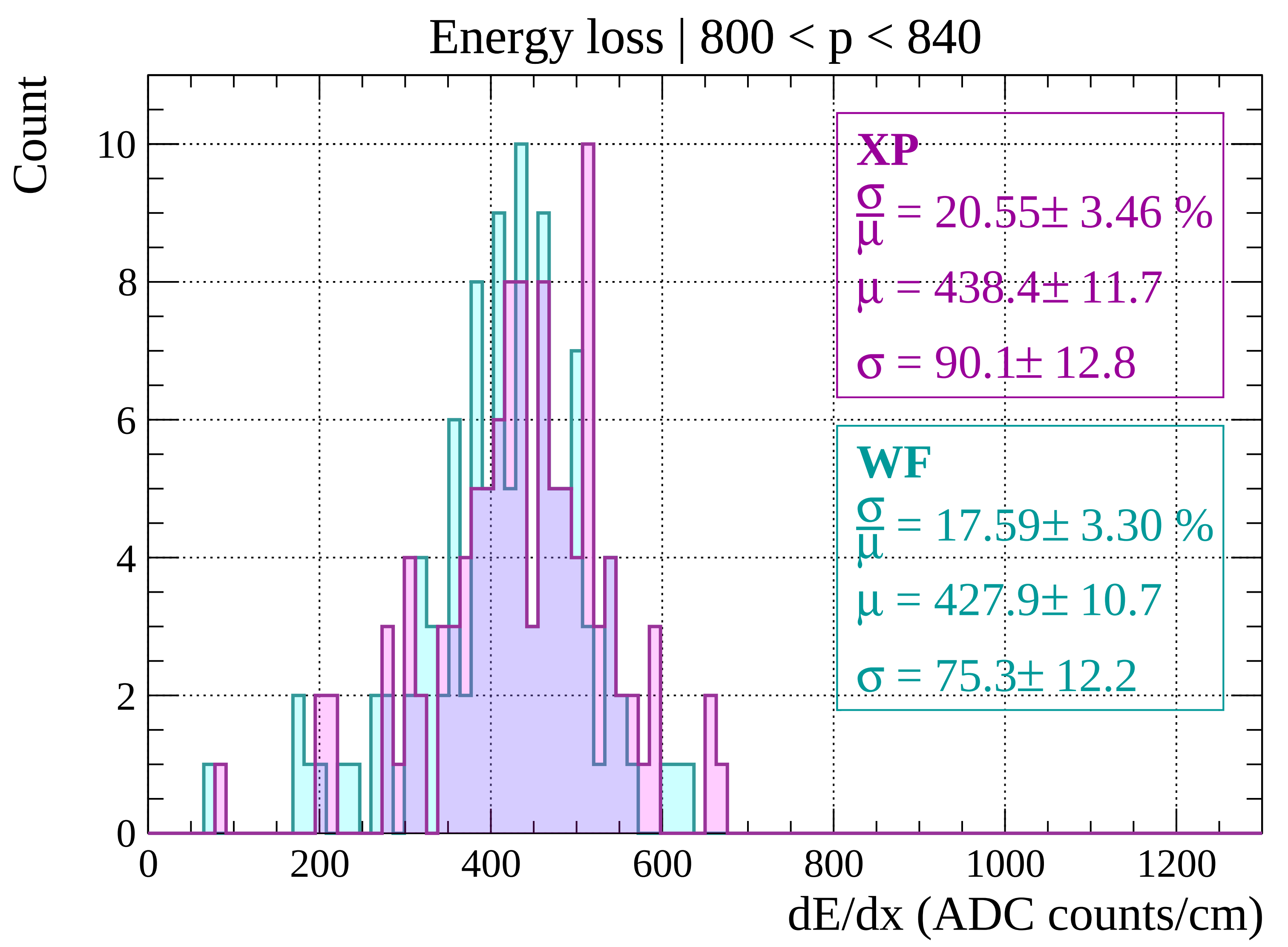

Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 30.66± 6.44 %
μ = 401.7± 17.1
σ = 123.2± 20.6
WF
$\frac{\sigma}{\mu}$ = 16.28± 2.55 %
μ = 416.6± 8.8
σ = 67.8± 9.2



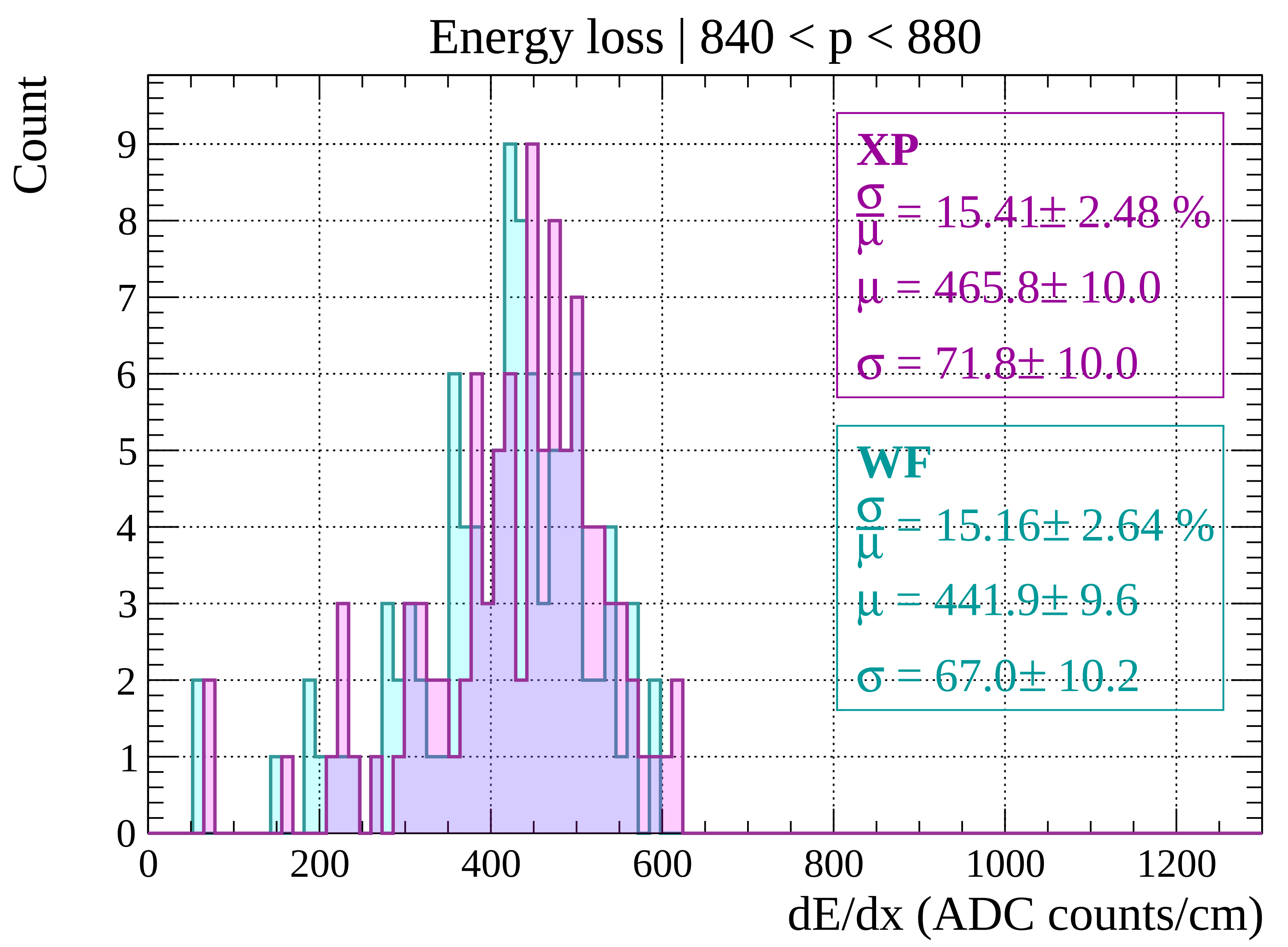
Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.04± 1.58 %
μ = 473.9± 6.5
σ = 47.6± 6.8
WF
σ/μ = 16.97± 2.85 %
μ = 424.1± 9.7
σ = 72.0± 10.5
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200


Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 14.39± 2.53 %
μ = 461.3± 9.3
σ = 66.4± 10.3
WF
σ/μ = 13.64± 2.33 %
μ = 432.6± 8.8
σ = 59.0± 8.9

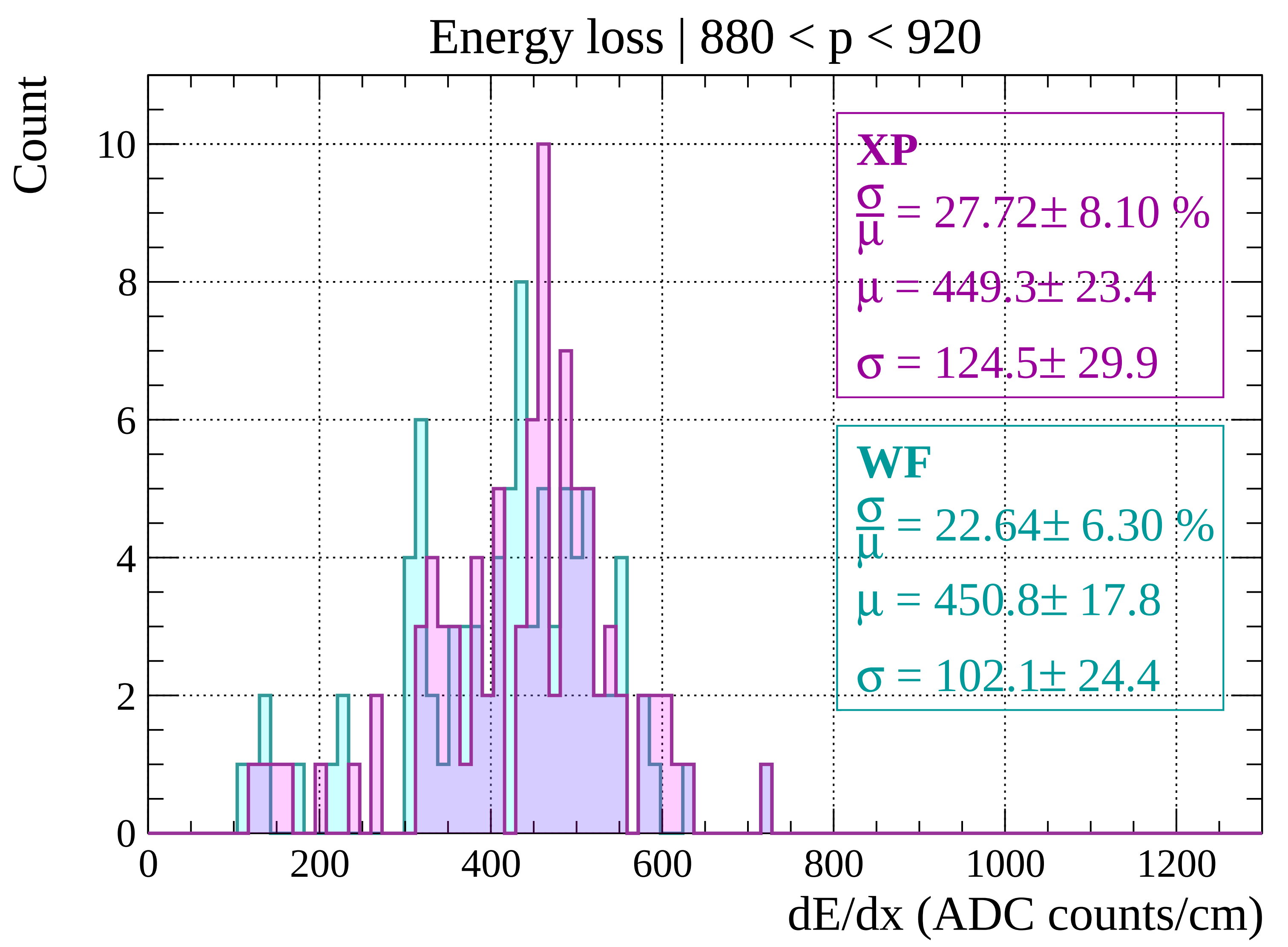

Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
XP
$\frac{\sigma}{\mu}$ = 13.38± 2.47 %
μ = 467.0± 8.7
σ = 62.5± 10.4
WF
$\frac{\sigma}{\mu}$ = 10.26± 1.72 %
μ = 443.2± 6.1
σ = 45.5± 7.0



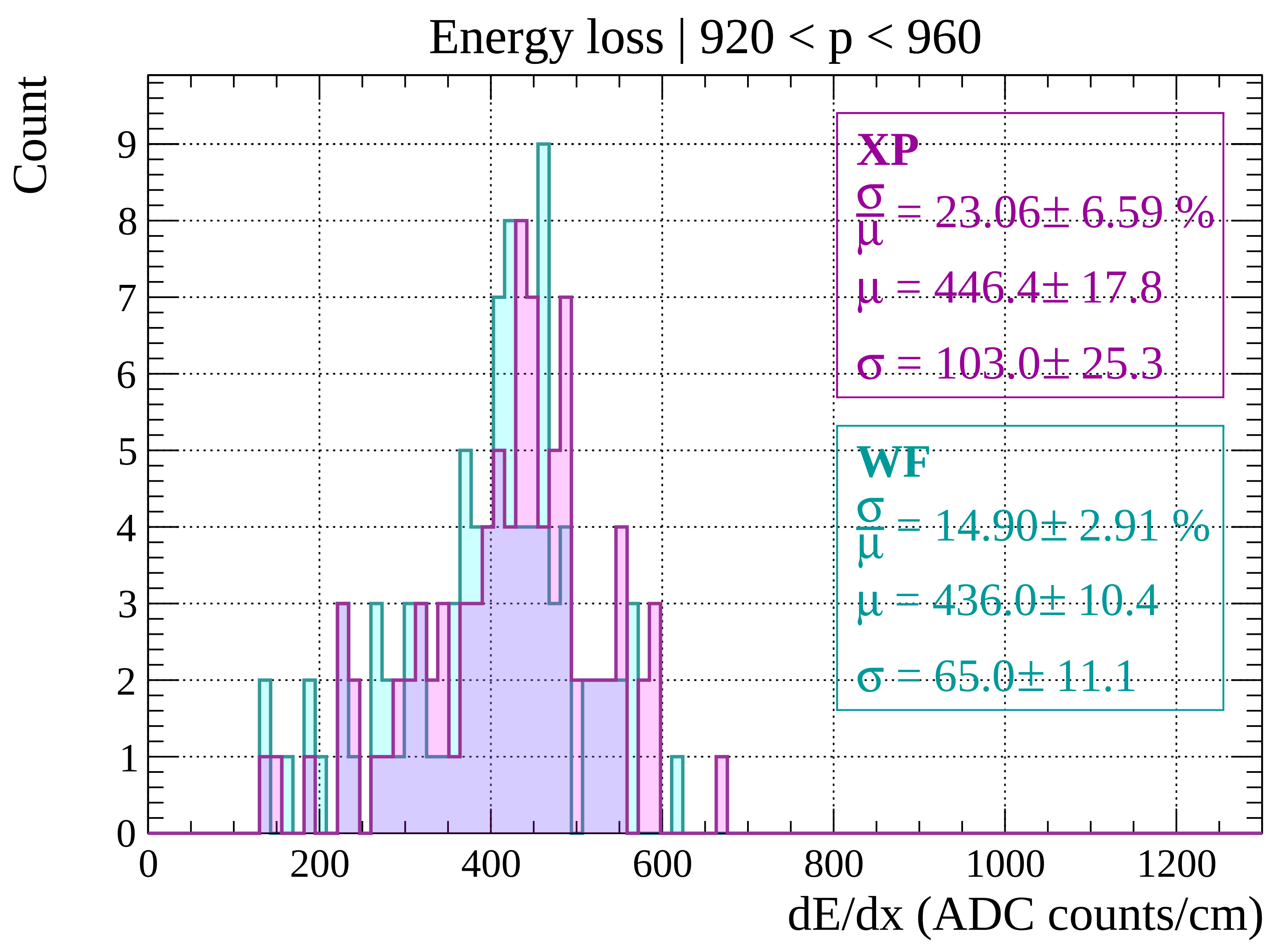
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
σ/μ = 20.55± 3.46 %
μ = 438.4± 11.7
σ = 90.1± 12.8
WF
σ/μ = 17.59± 3.30 %
μ = 427.9± 10.7
σ = 75.3± 12.2



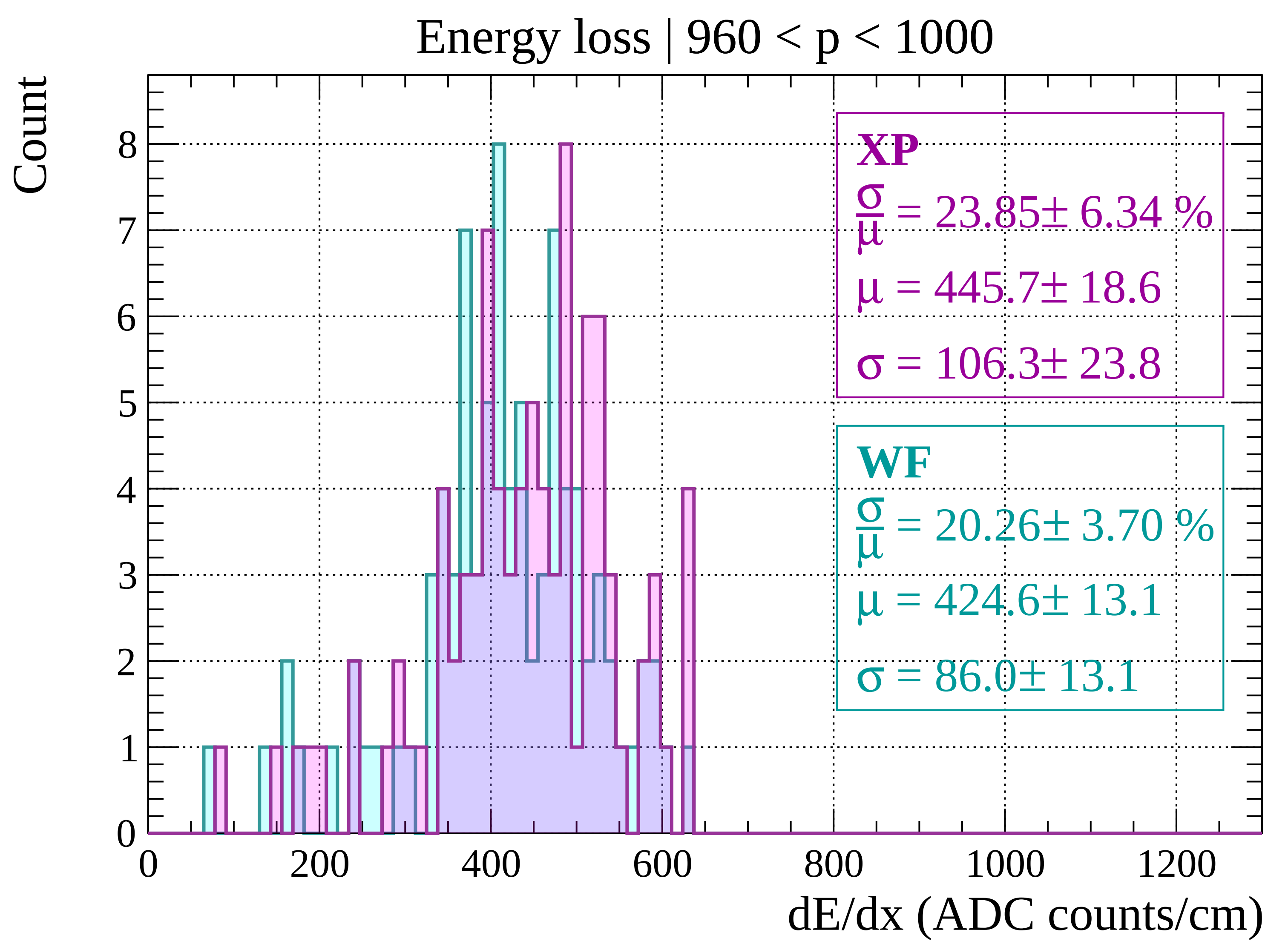
Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 15.41± 2.48 %
μ = 465.8± 10.0
σ = 71.8± 10.0
WF
σ/μ = 15.16± 2.64 %
μ = 441.9± 9.6
σ = 67.0± 10.2

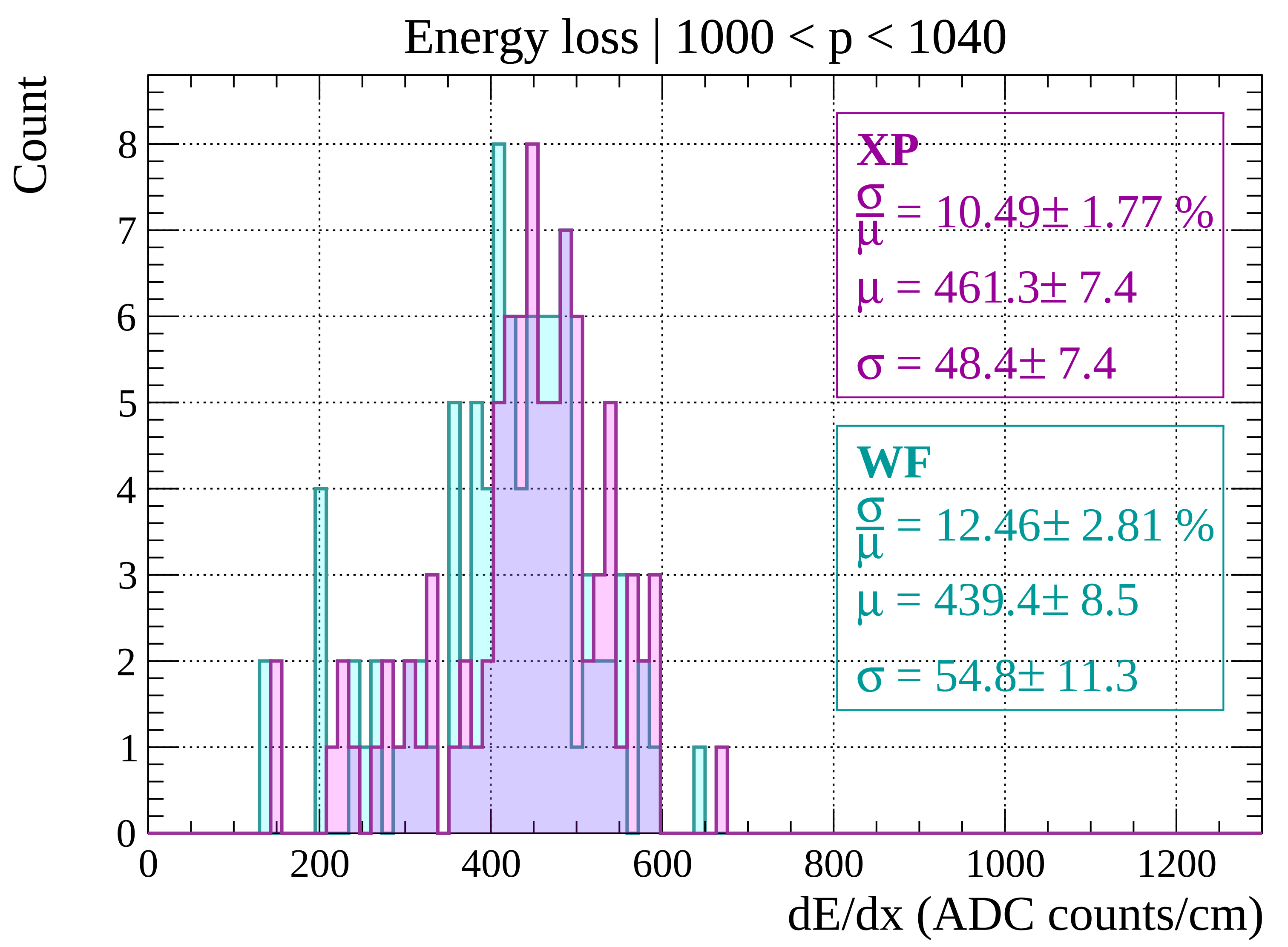


Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 27.72± 8.10 %
μ = 449.3± 23.4
σ = 124.5± 29.9
WF
$\frac{\sigma}{\mu}$ = 22.64± 6.30 %
μ = 450.8± 17.8
σ = 102.1± 24.4

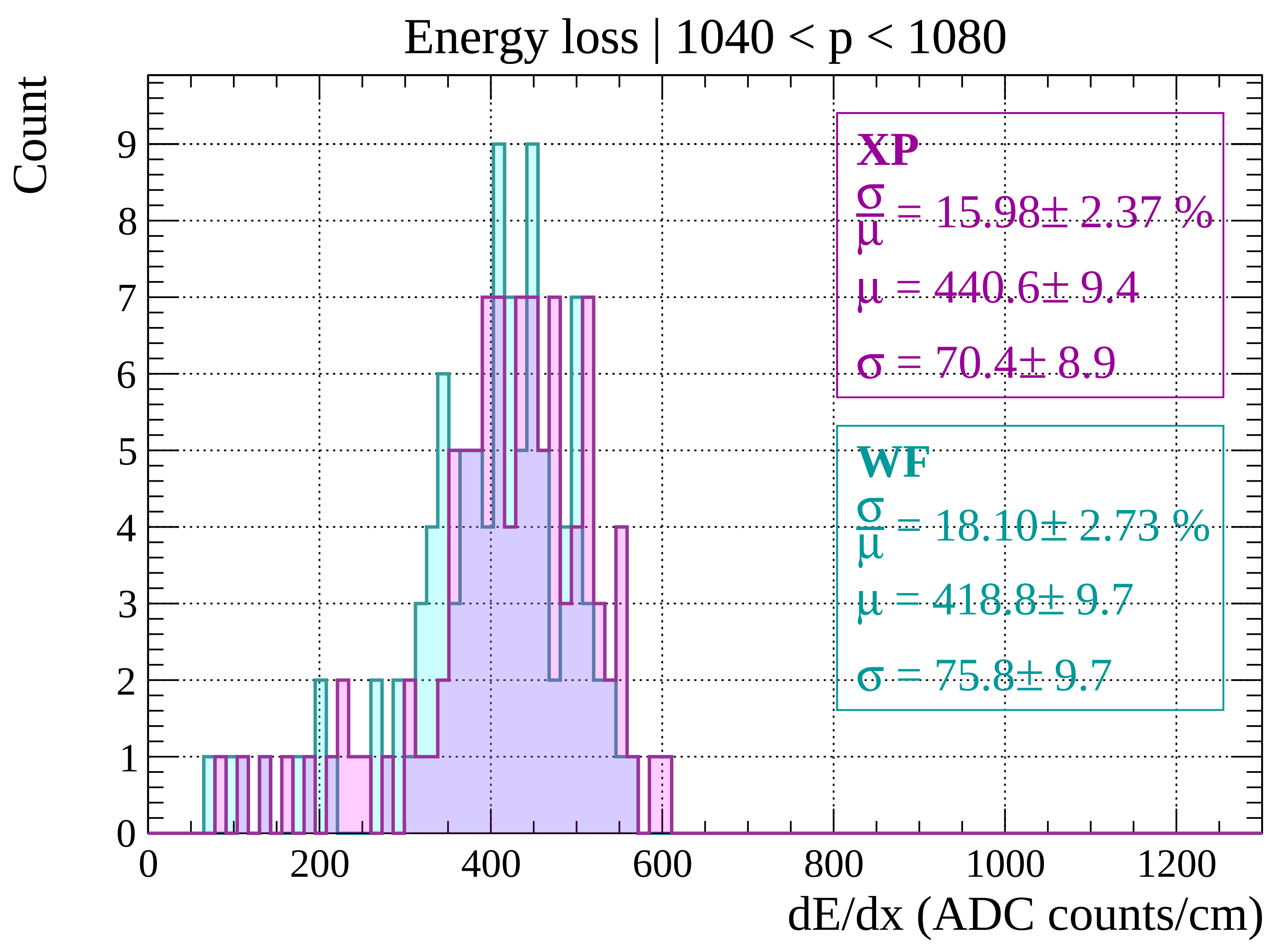


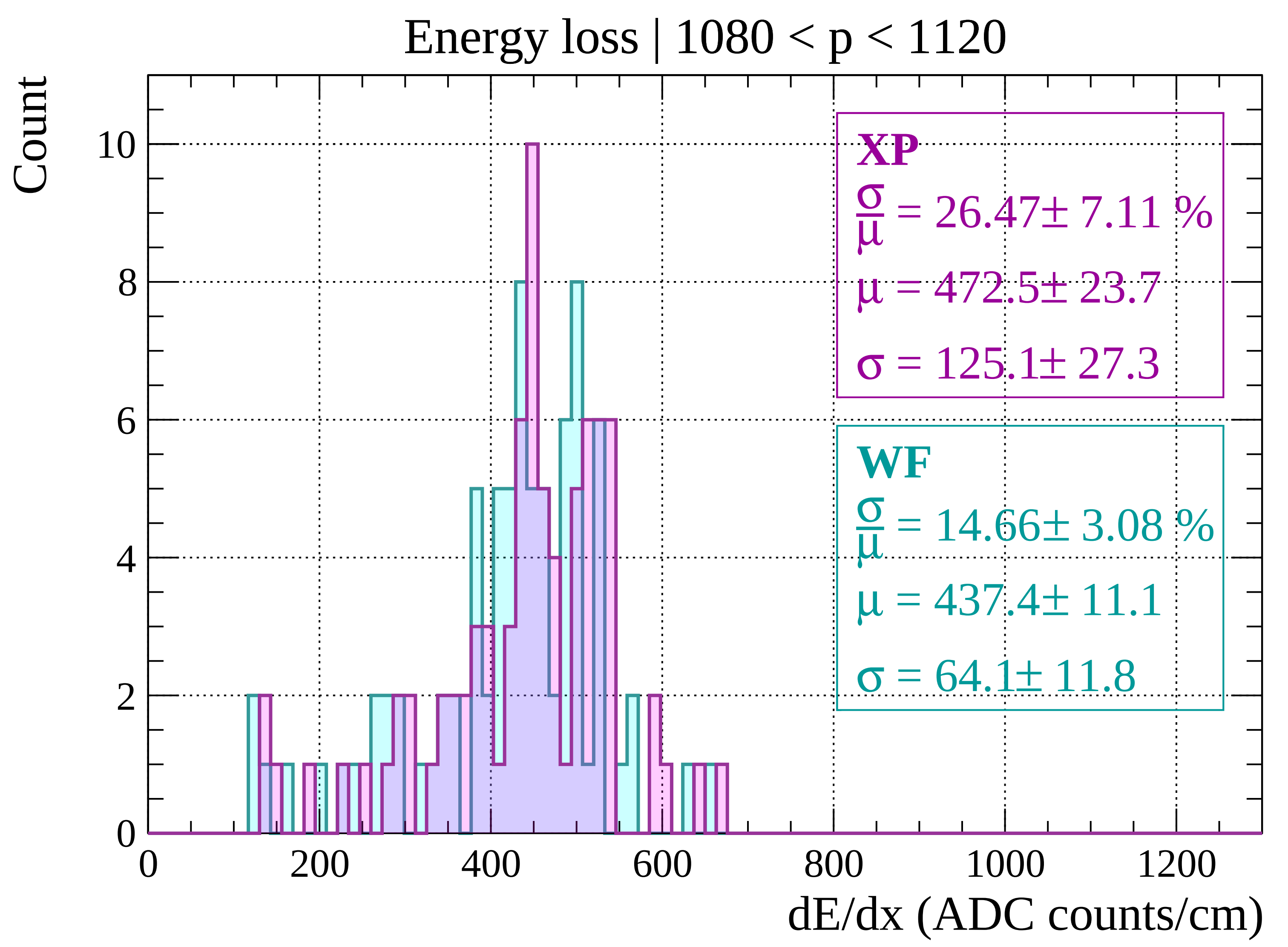
Energy loss | 920 < p < 960
Count
0
1
2
3
4
5
6
7
8
9
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 23.06± 6.59 %
μ = 446.4± 17.8
σ = 103.0± 25.3
WF
$\frac{\sigma}{\mu}$ = 14.90± 2.91 %
μ = 436.0± 10.4
σ = 65.0± 11.1


Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200
XP
σ/μ = 23.85± 6.34 %
μ = 445.7± 18.6
σ = 106.3± 23.8
WF
σ/μ = 20.26± 3.70 %
μ = 424.6± 13.1
σ = 86.0± 13.1

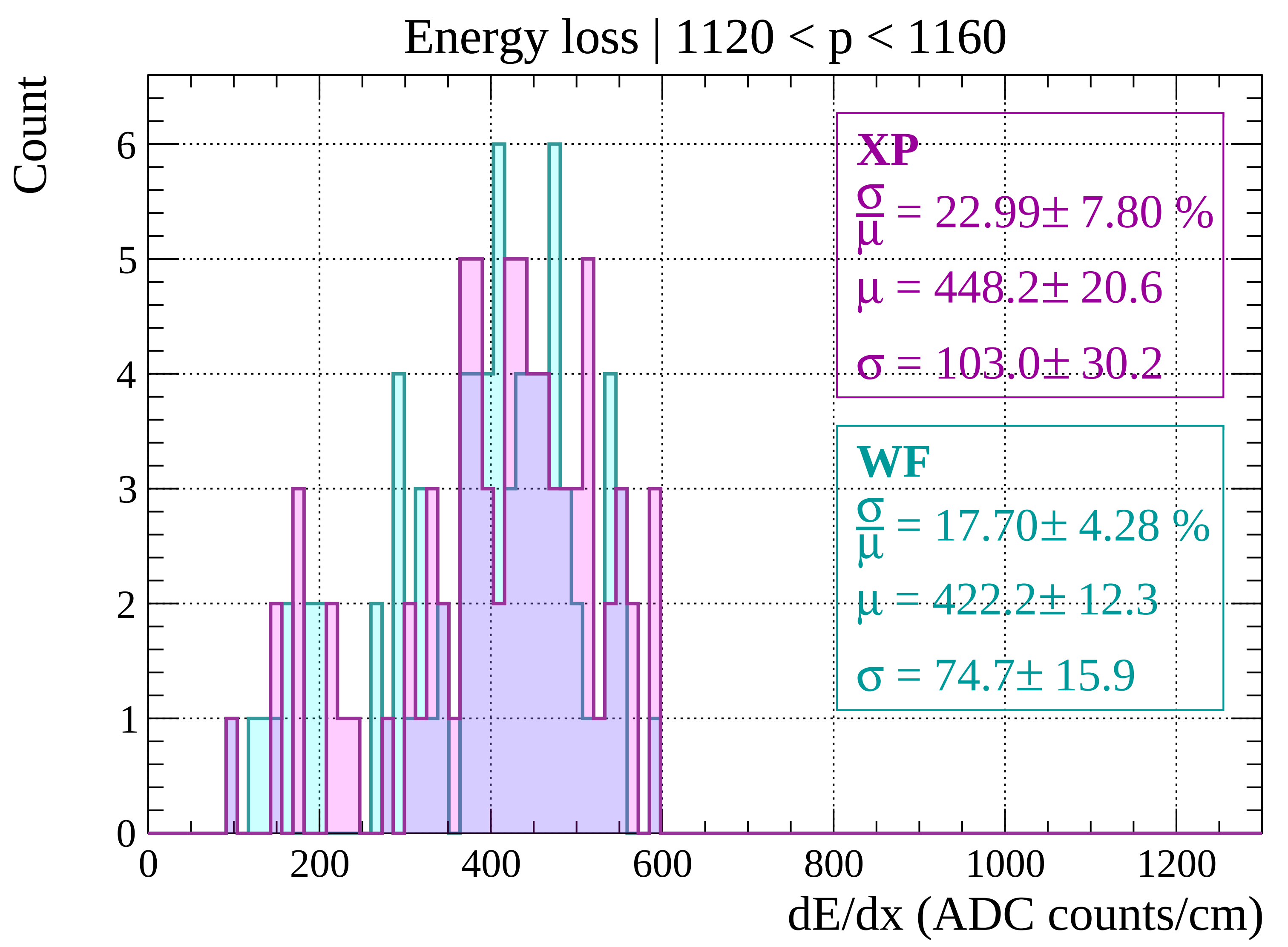


Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP
σ/μ = 10.49± 1.77 %
μ = 461.3± 7.4
σ = 48.4± 7.4
WF
σ/μ = 12.46± 2.81 %
μ = 439.4± 8.5
σ = 54.8± 11.3

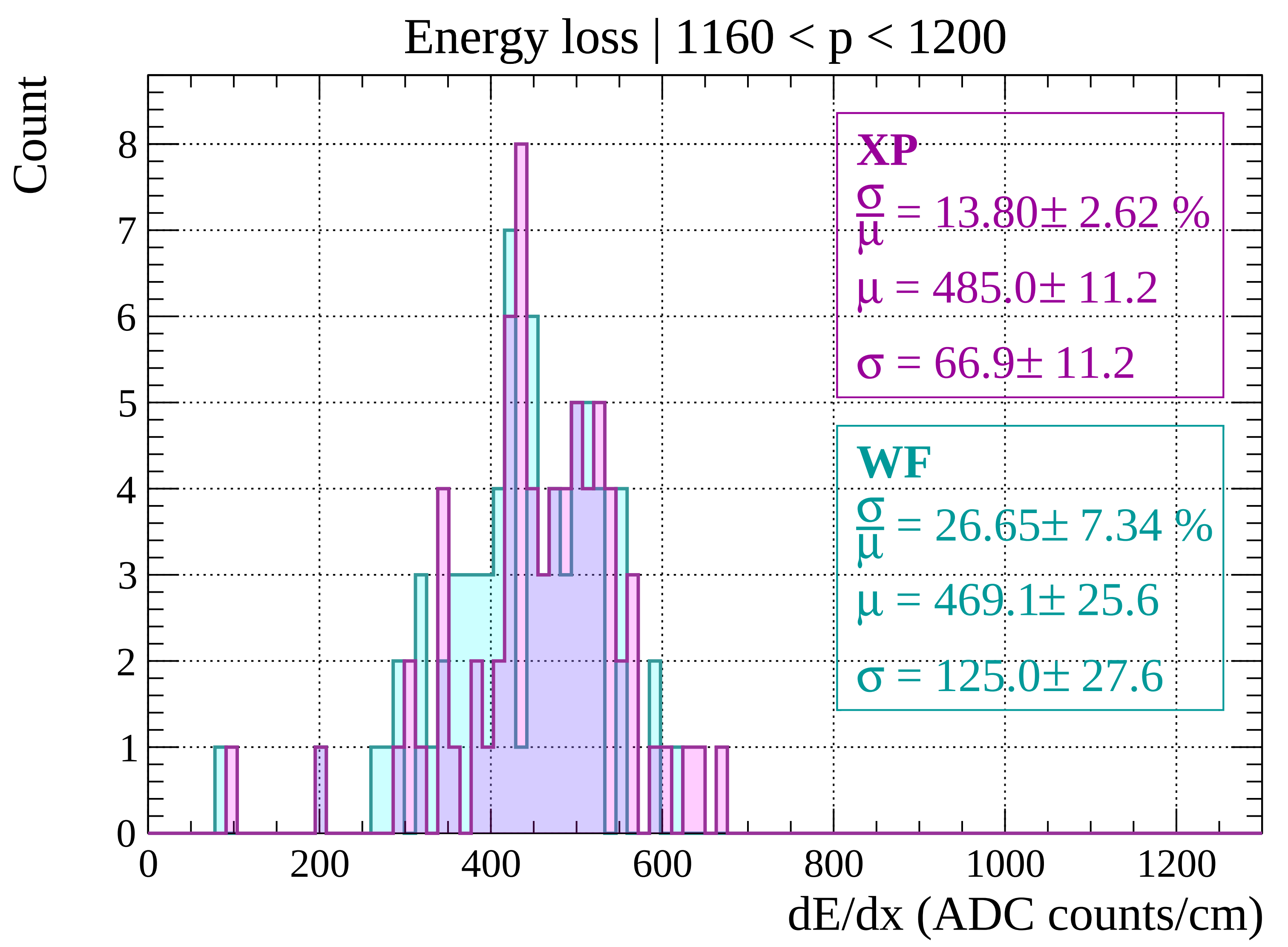

Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
$\frac{\sigma}{\mu}$ = 15.98± 2.37 %
μ = 440.6± 9.4
σ = 70.4± 8.9
WF
$\frac{\sigma}{\mu}$ = 18.10± 2.73 %
μ = 418.8± 9.7
σ = 75.8± 9.7

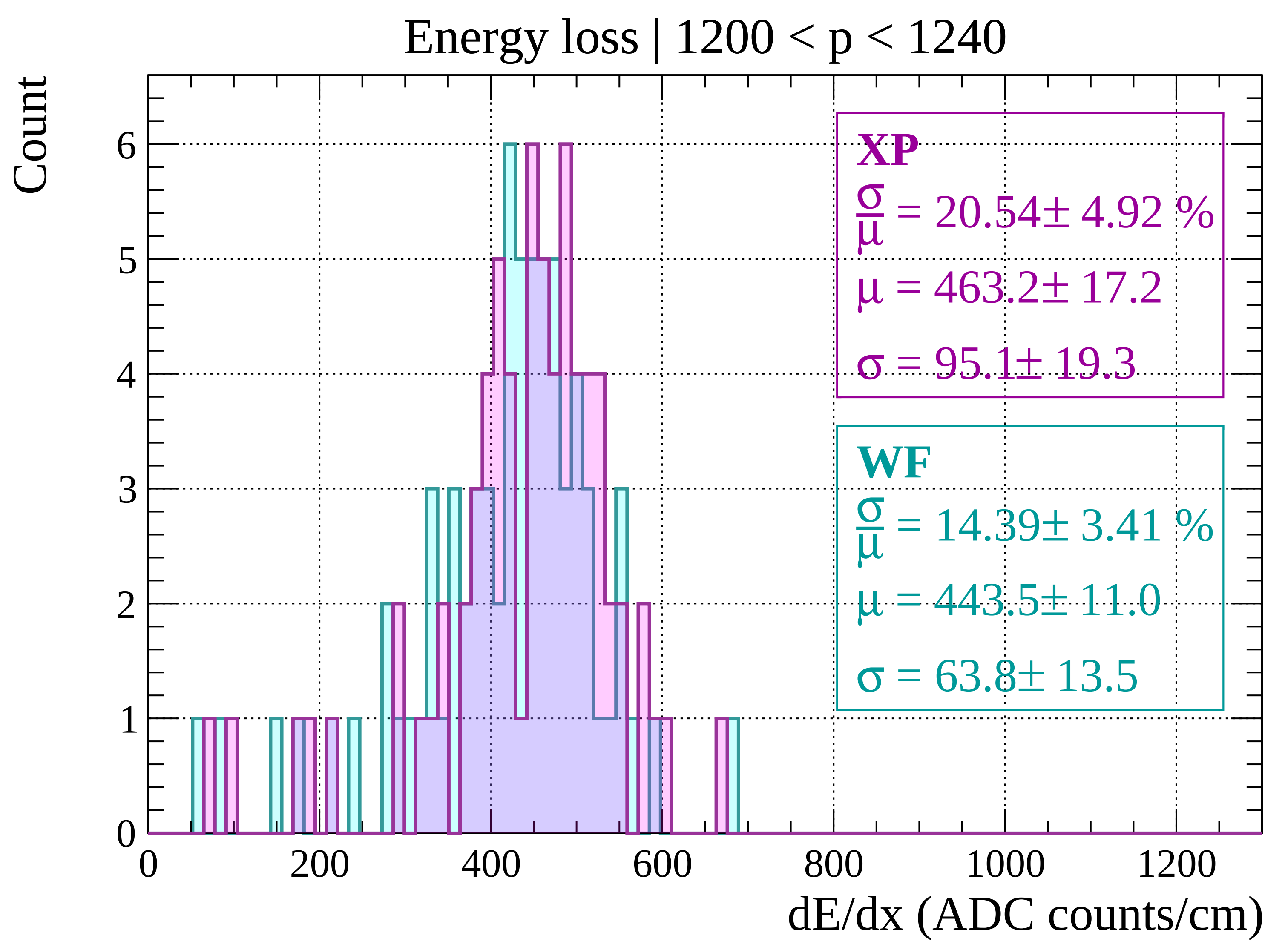


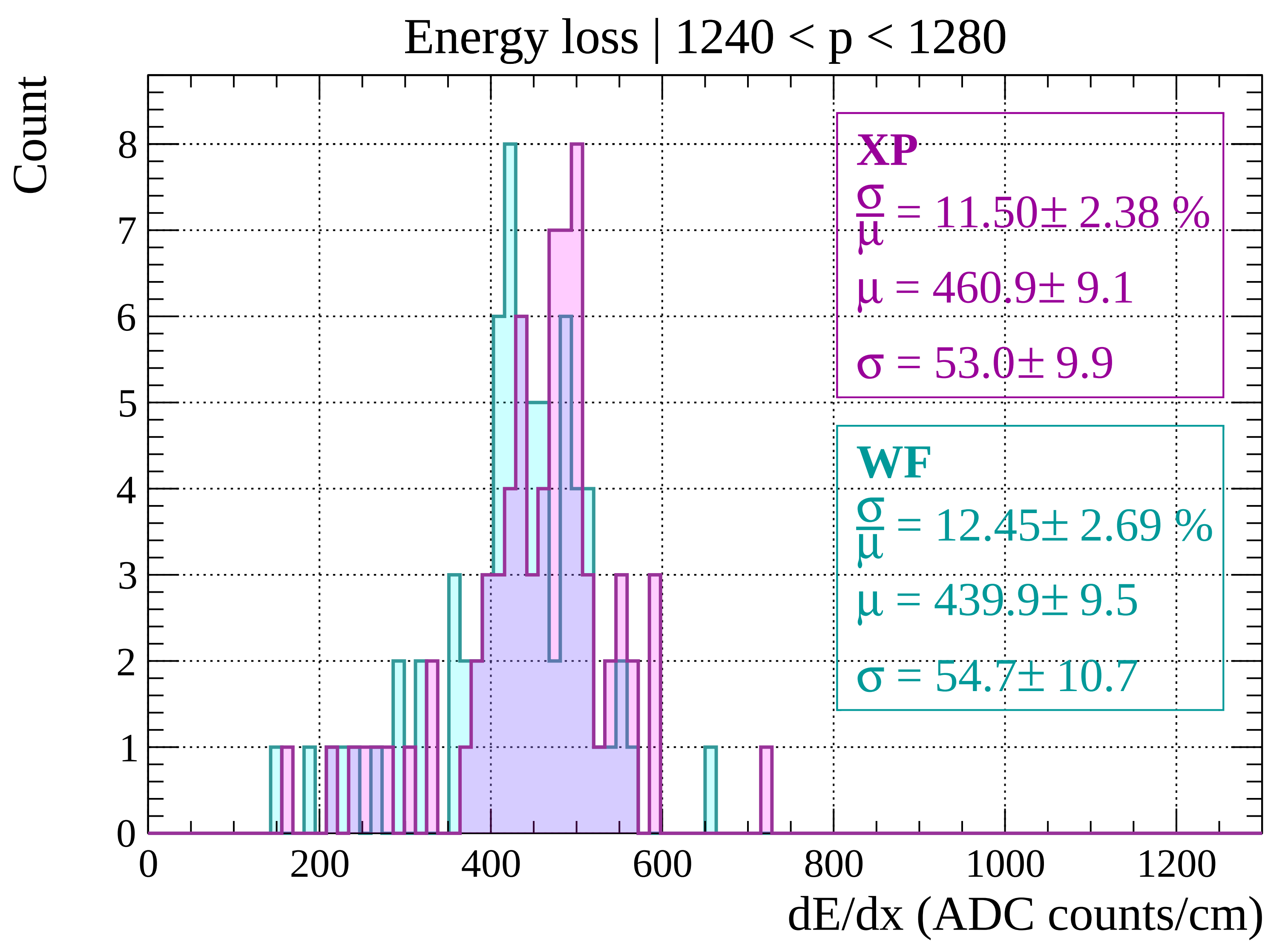
Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 26.47± 7.11 %
μ = 472.5± 23.7
σ = 125.1± 27.3
WF
$\frac{\sigma}{\mu}$ = 14.66± 3.08 %
μ = 437.4± 11.1
σ = 64.1± 11.8
0
2
4
6
8
10
0
200
400
600
800
1000
1200



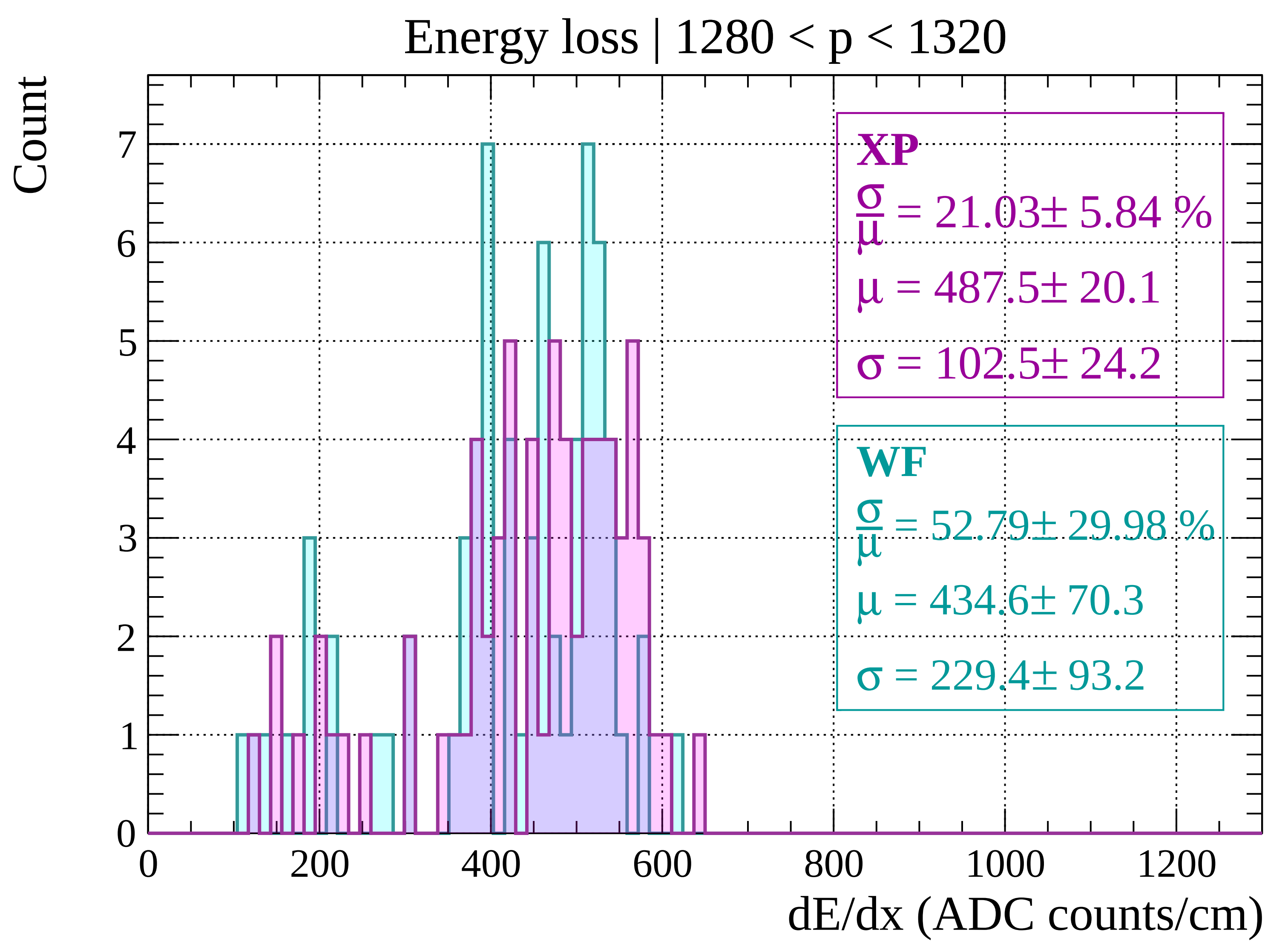
Energy loss | 1120 < p < 1160
Count
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 22.99± 7.80 %
μ = 448.2± 20.6
σ = 103.0± 30.2
WF
σ/μ = 17.70± 4.28 %
μ = 422.2± 12.3
σ = 74.7± 15.9

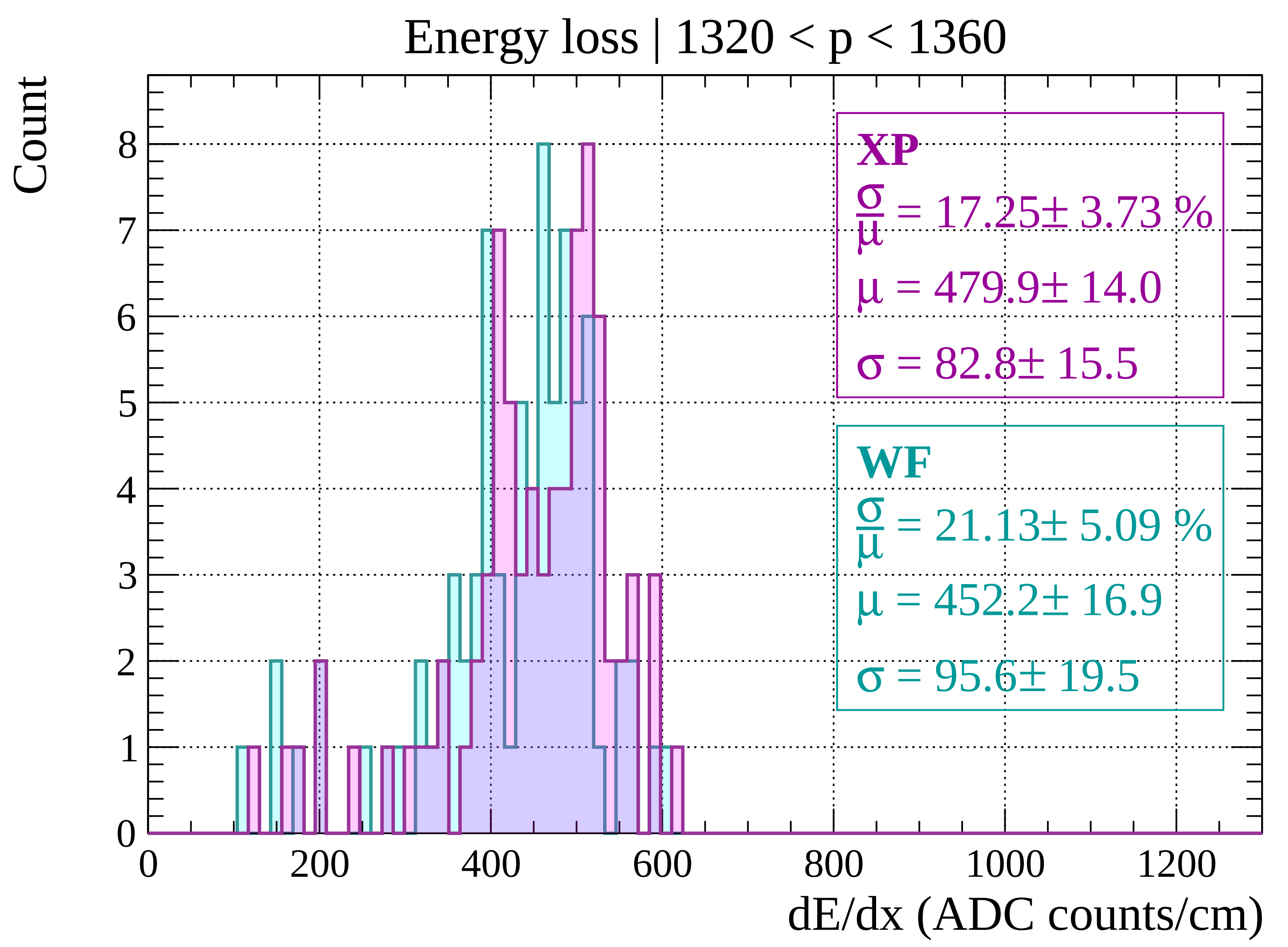

Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP
$\frac{\sigma}{\mu}$ = 13.80± 2.62 %
μ = 485.0± 11.2
σ = 66.9± 11.2
WF
$\frac{\sigma}{\mu}$ = 26.65± 7.34 %
μ = 469.1± 25.6
σ = 125.0± 27.6


Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
σ/μ = 20.54± 4.92 %
μ = 463.2± 17.2
σ = 95.1± 19.3
WF
σ/μ = 14.39± 3.41 %
μ = 443.5± 11.0
σ = 63.8± 13.5

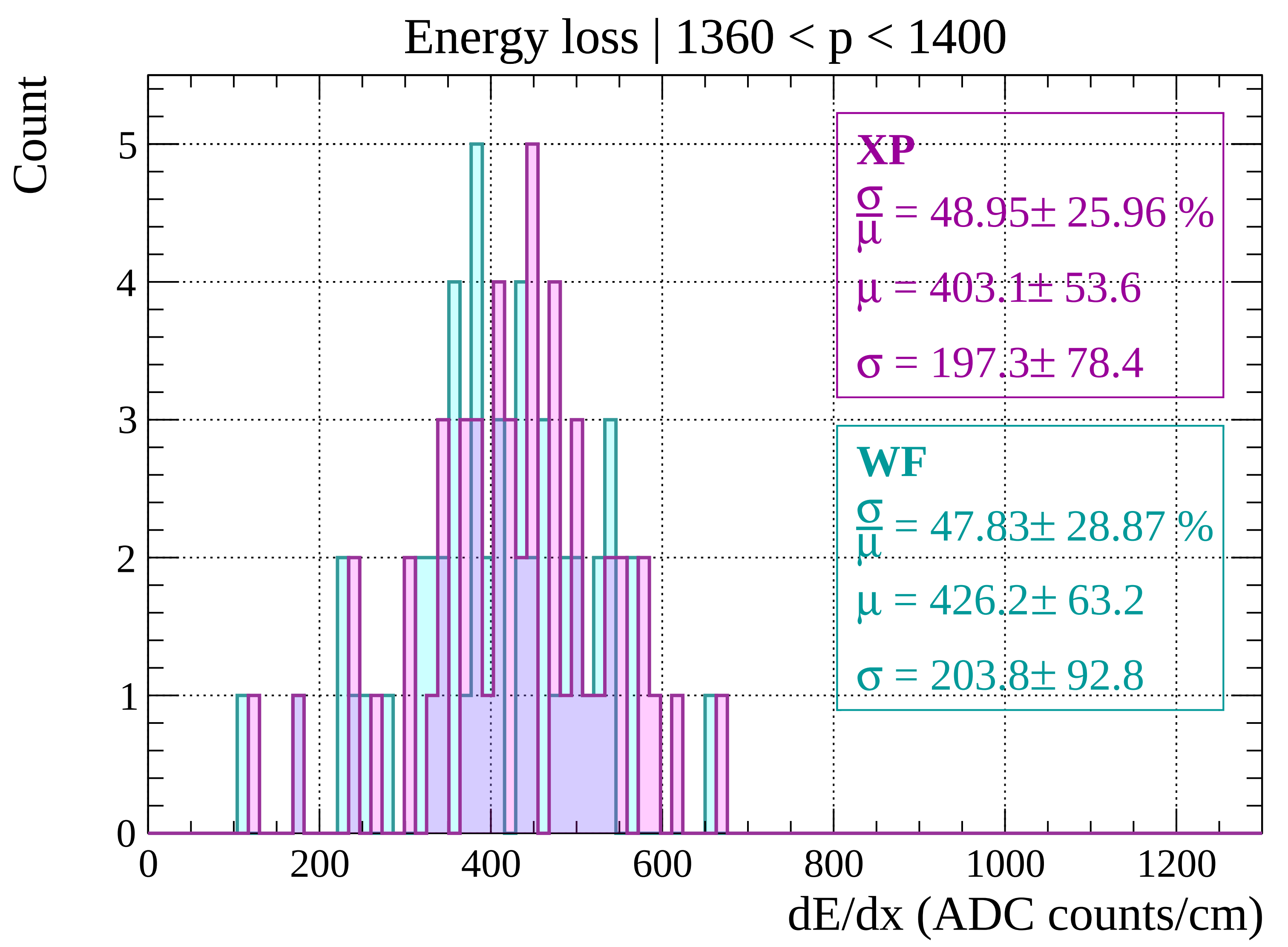


Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
$\frac{\sigma}{\mu}$ = 11.50± 2.38 %
μ = 460.9± 9.1
σ = 53.0± 9.9
WF
$\frac{\sigma}{\mu}$ = 12.45± 2.69 %
μ = 439.9± 9.5
σ = 54.7± 10.7

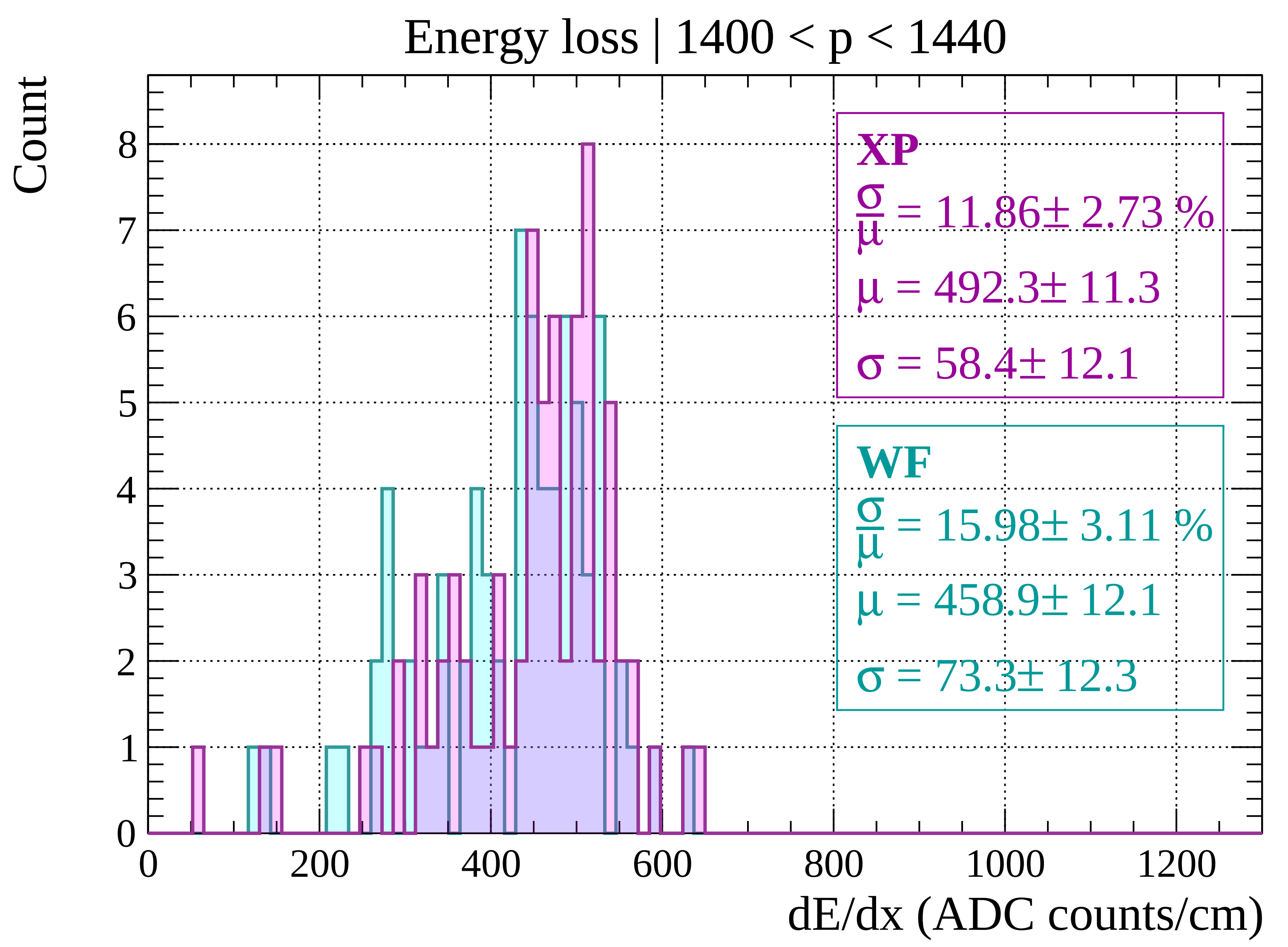

Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
XP
$\frac{\sigma}{\mu}$ = 21.03± 5.84 %
μ = 487.5± 20.1
σ = 102.5± 24.2
WF
$\frac{\sigma}{\mu}$ = 52.79± 29.98 %
μ = 434.6± 70.3
σ = 229.4± 93.2



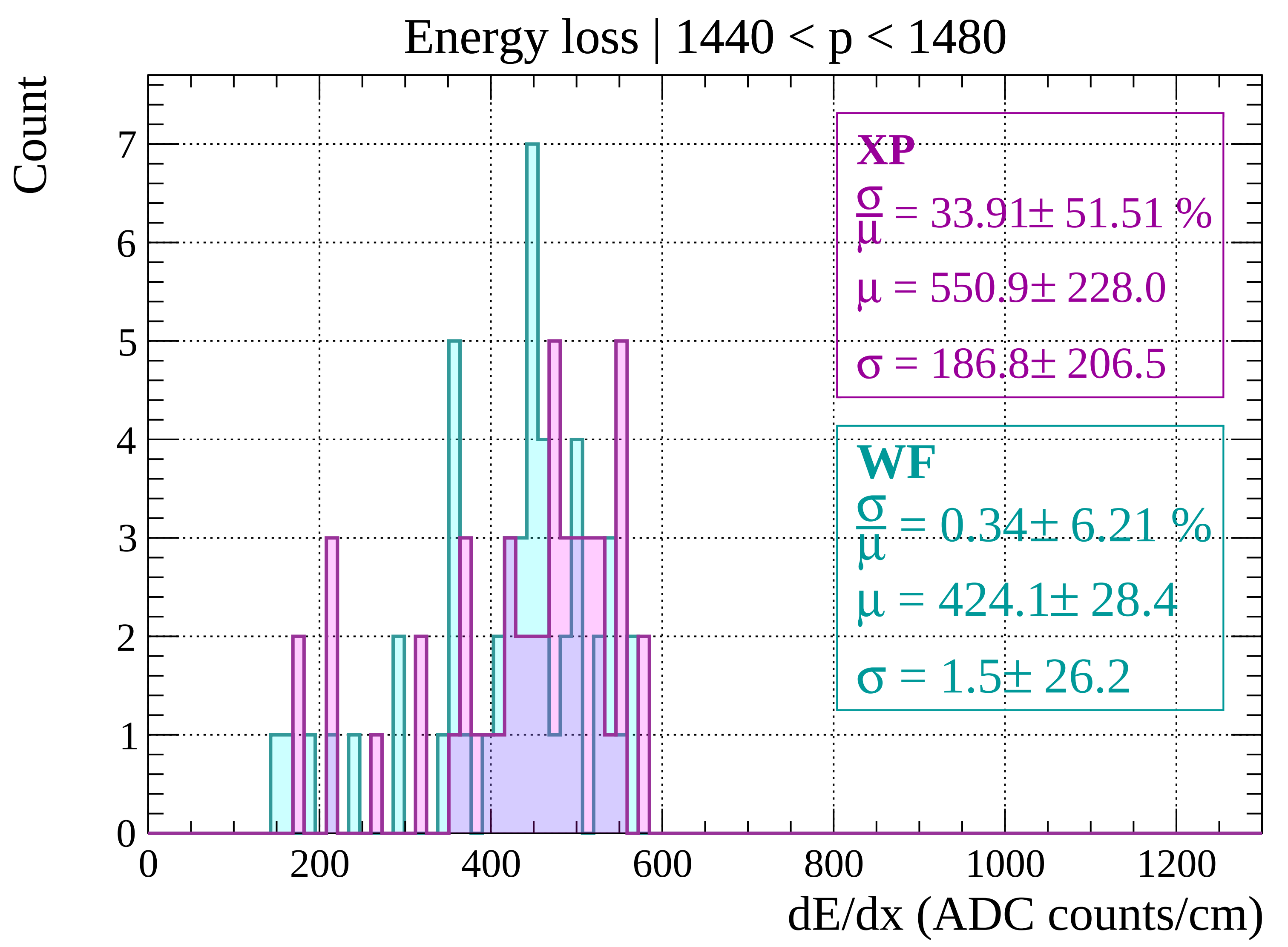
Energy loss | 1320 < p < 1360
Count
8
7
6
5
4
3
2
1
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 17.25± 3.73 %
μ = 479.9± 14.0
σ = 82.8± 15.5
WF
σ/μ = 21.13± 5.09 %
μ = 452.2± 16.9
σ = 95.6± 19.5

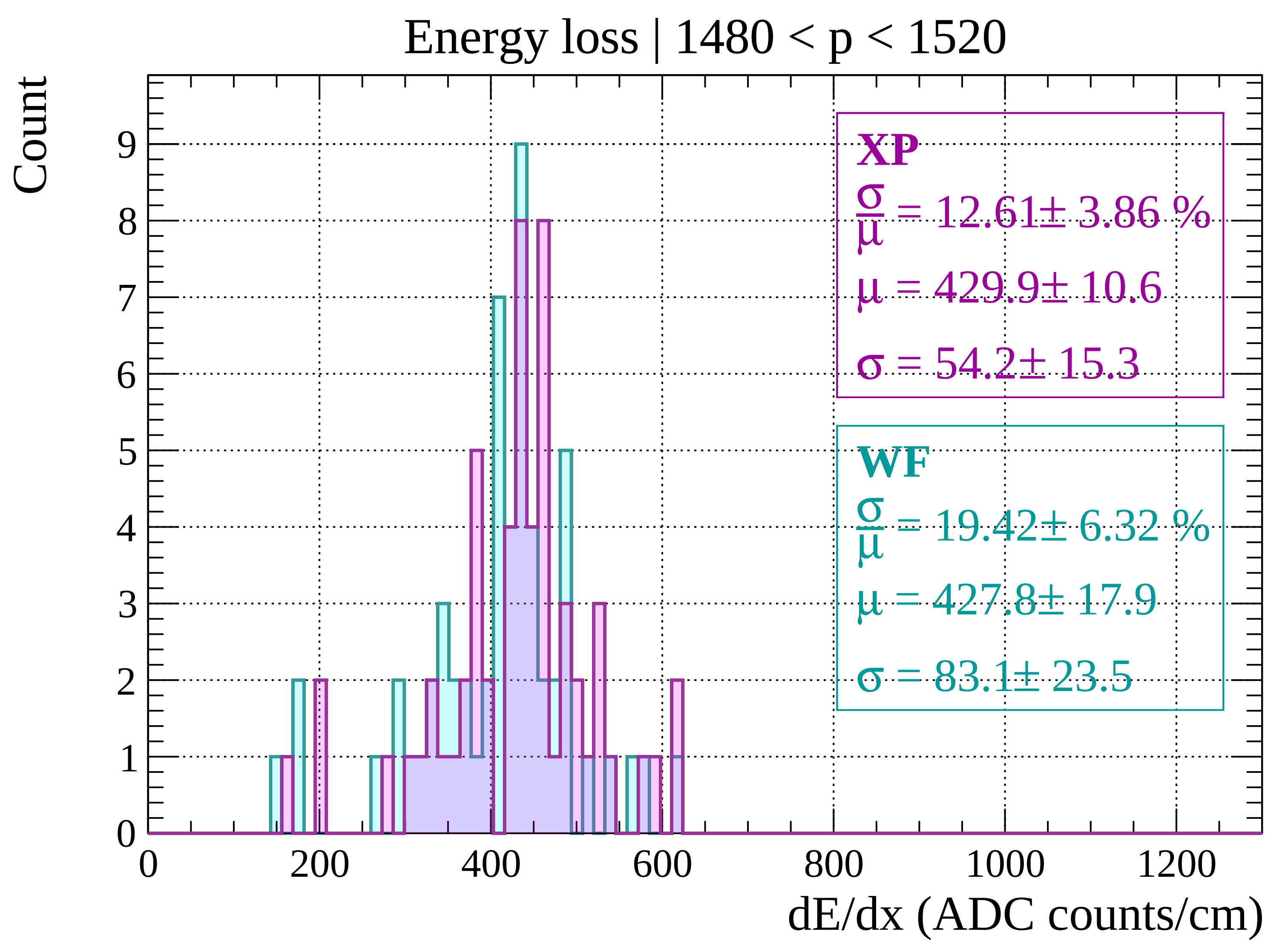


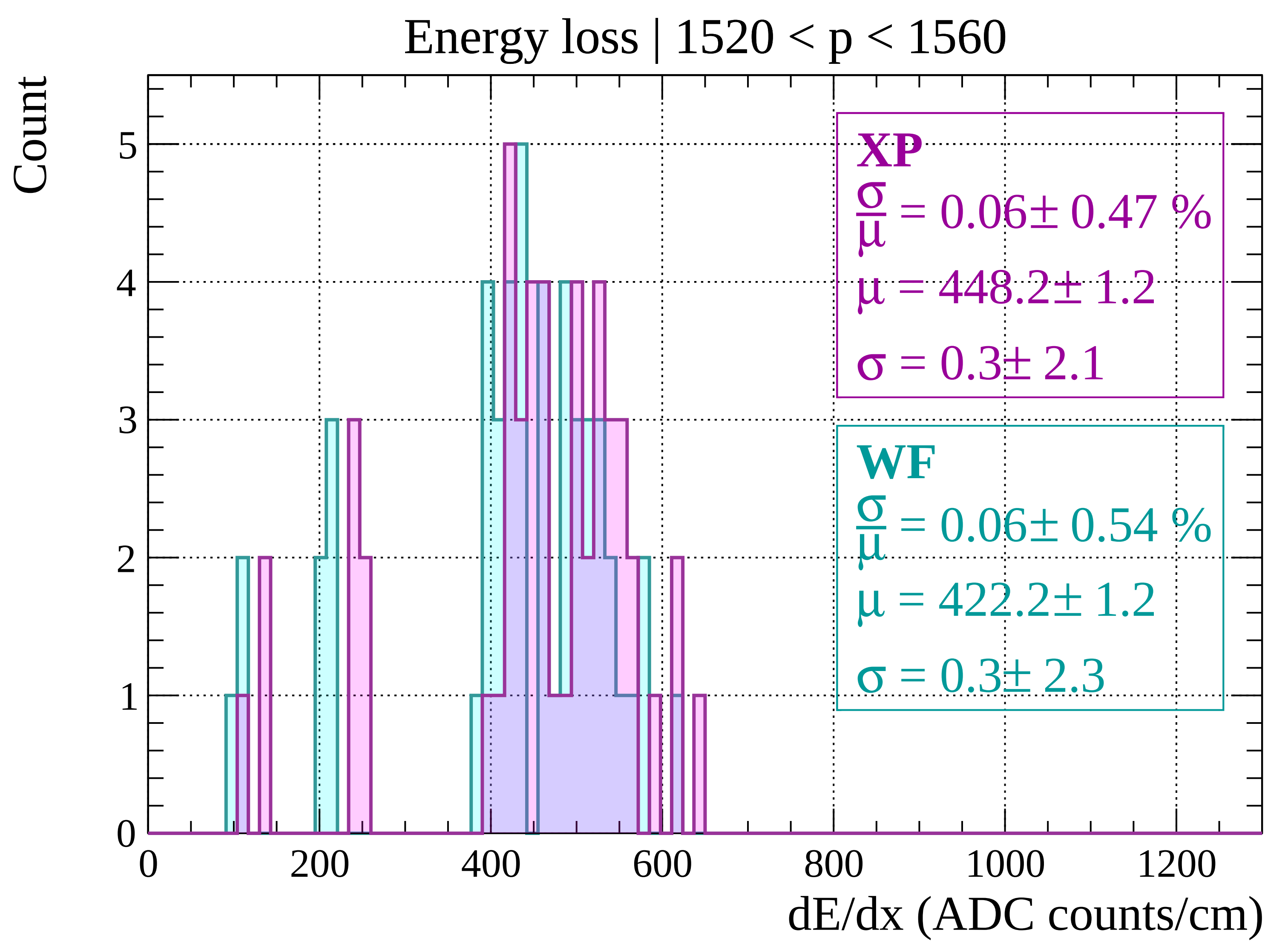
Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
$\frac{\sigma}{\mu}$ = 48.95± 25.96 %
μ = 403.1± 53.6
σ = 197.3± 78.4
WF
$\frac{\sigma}{\mu}$ = 47.83± 28.87 %
μ = 426.2± 63.2
σ = 203.8± 92.8

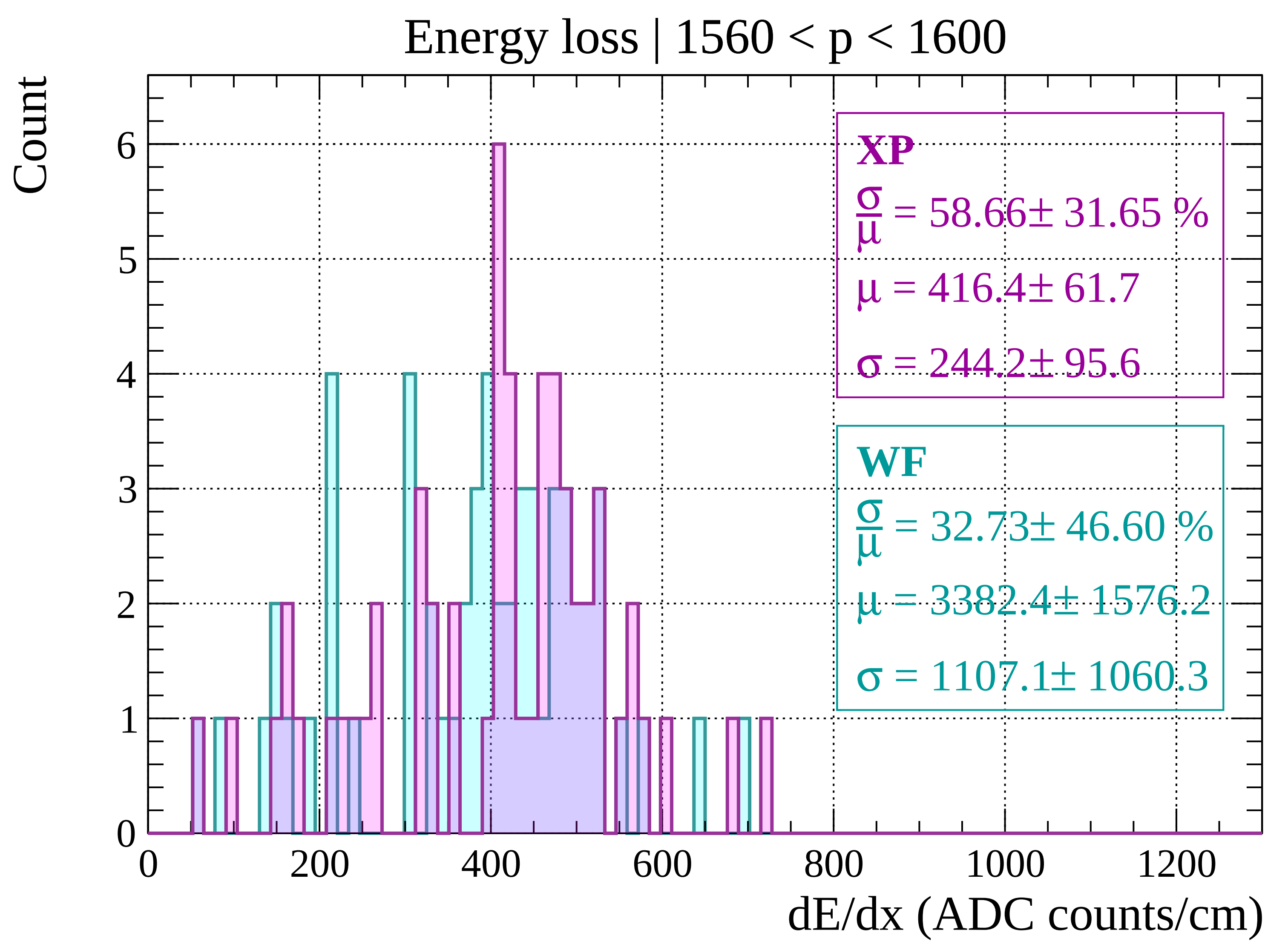


Energy loss | 1400 < p < 1440
Count
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.86± 2.73 %
μ = 492.3± 11.3
σ = 58.4± 12.1
WF
σ/μ = 15.98± 3.11 %
μ = 458.9± 12.1
σ = 73.3± 12.3

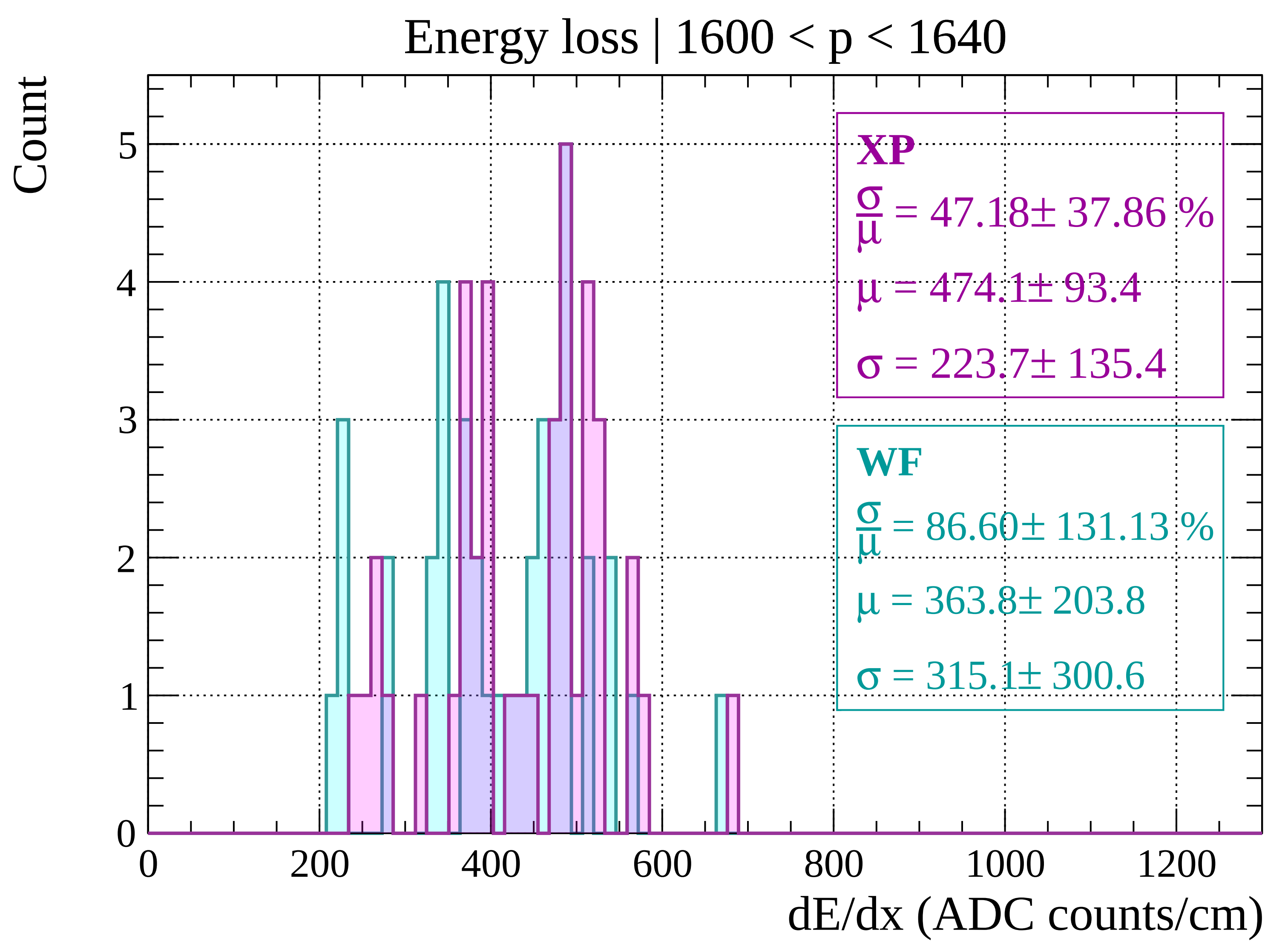

Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
XP
σ/μ = 33.91± 51.51 %
μ = 550.9± 228.0
σ = 186.8± 206.5
WF
σ/μ = 0.34± 6.21 %
μ = 424.1± 28.4
σ = 1.5± 26.2

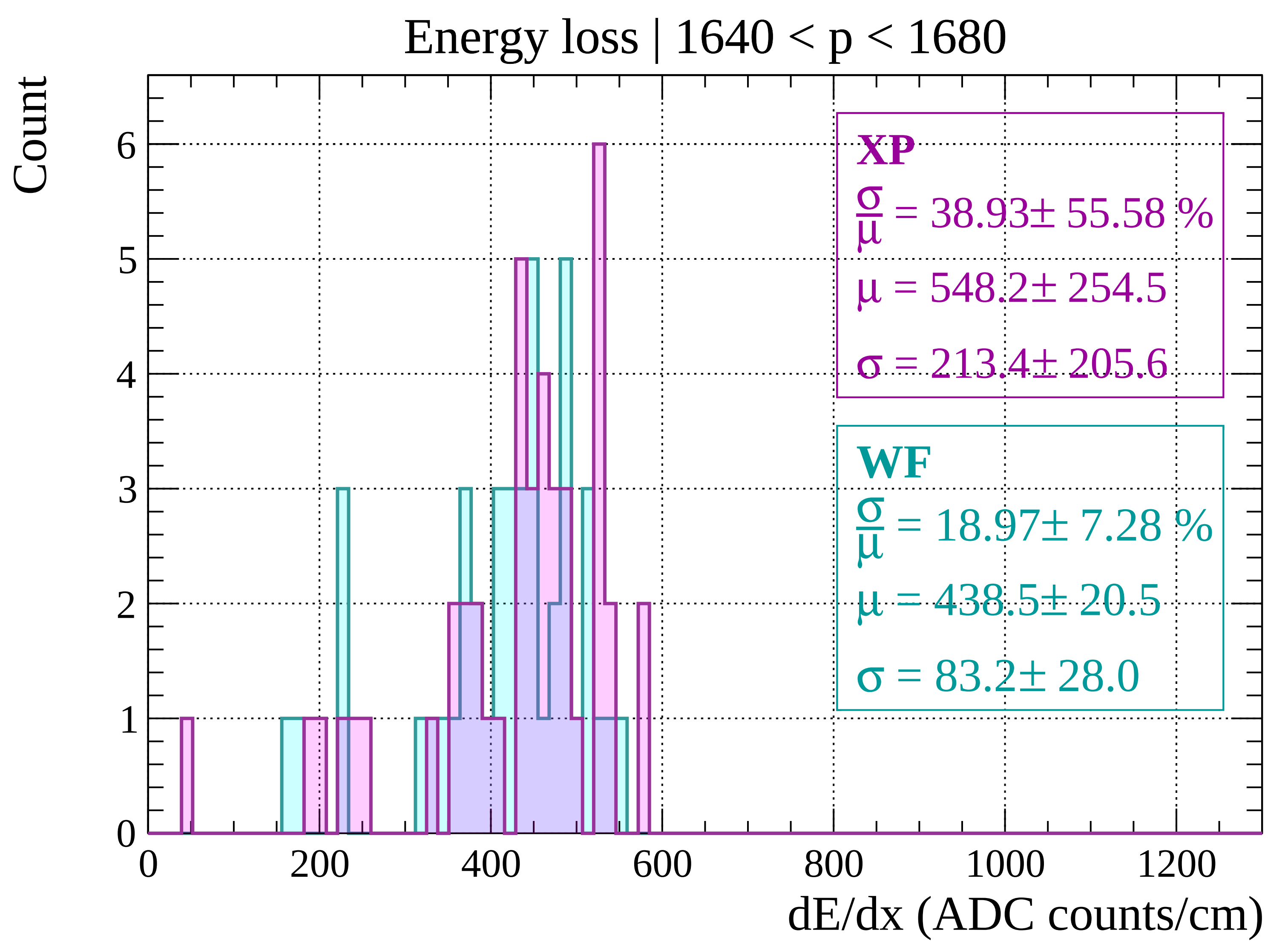

Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
$\frac{\sigma}{\mu}$ = 12.61± 3.86 %
μ = 429.9± 10.6
σ = 54.2± 15.3
WF
$\frac{\sigma}{\mu}$ = 19.42± 6.32 %
μ = 427.8± 17.9
σ = 83.1± 23.5


Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.06± 0.47 %
μ = 448.2± 1.2
σ = 0.3± 2.1
WF
σ/μ = 0.06± 0.54 %
μ = 422.2± 1.2
σ = 0.3± 2.3
0
1
2
3
4
5
0
200
400
600
800
1000
1200

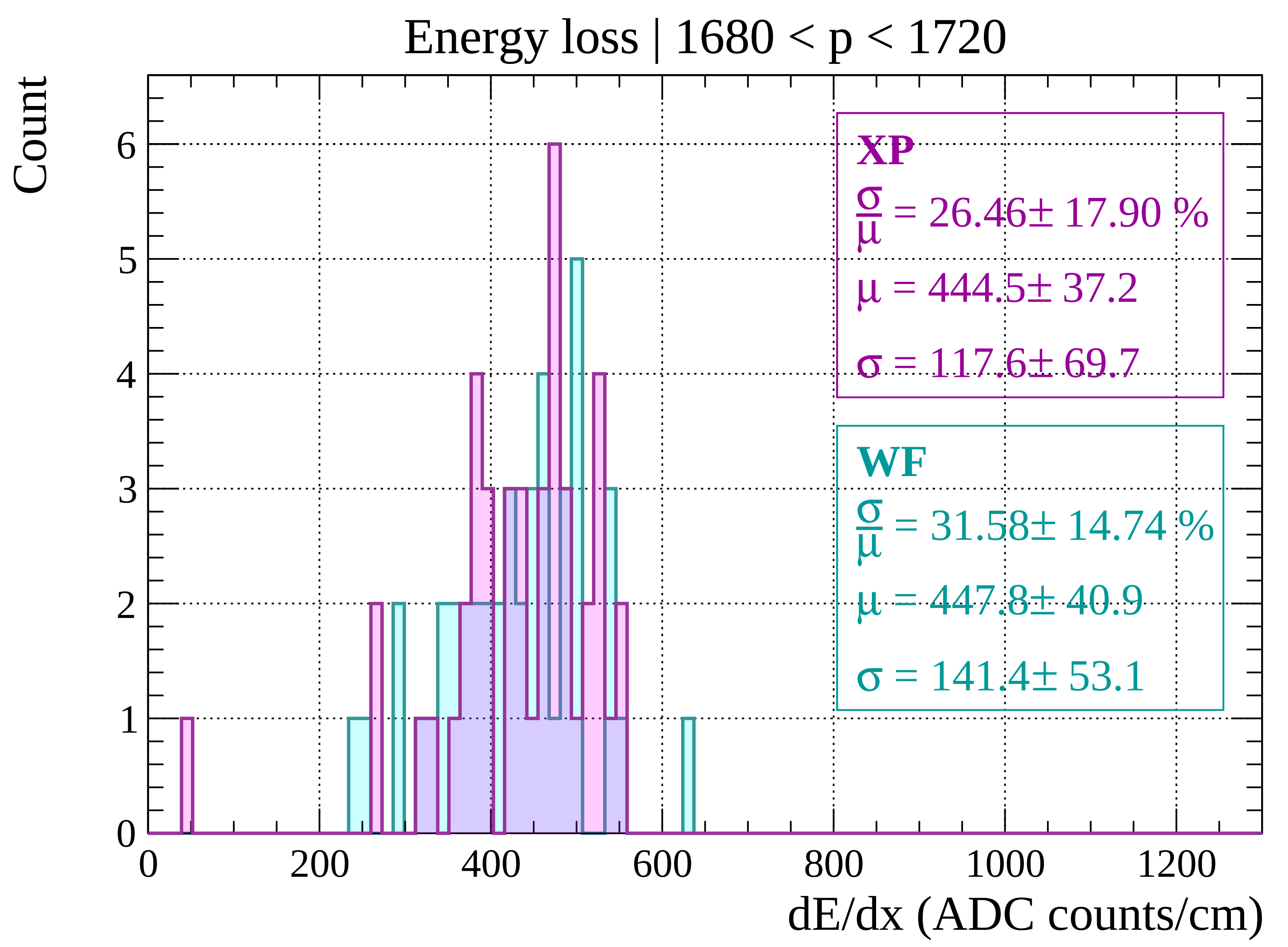


Energy loss | 1560 < p < 1600
Count
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 58.66± 31.65 %
μ = 416.4± 61.7
σ = 244.2± 95.6
WF
σ/μ = 32.73± 46.60 %
μ = 3382.4± 1576.2
σ = 1107.1± 1060.3

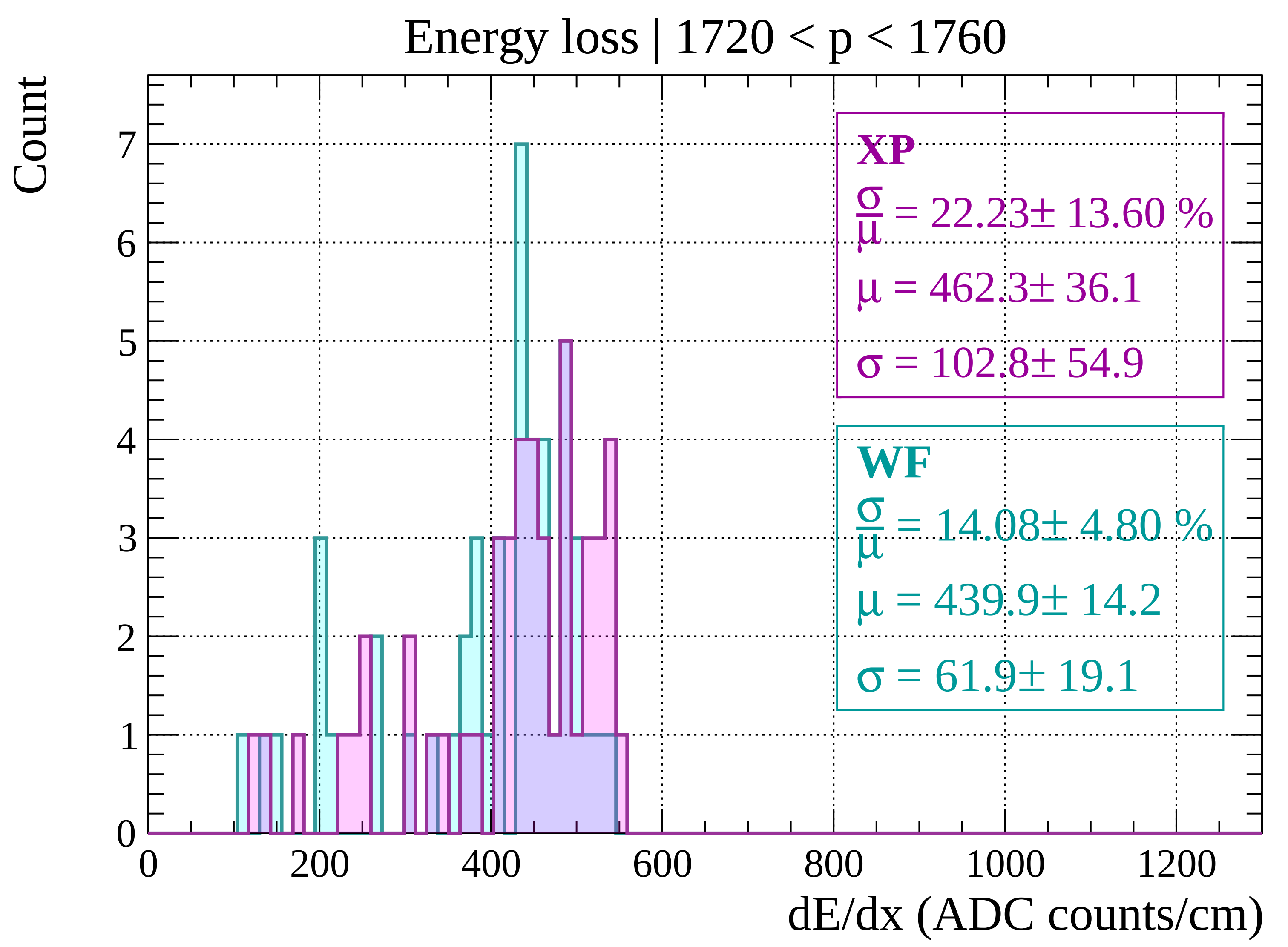

Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 47.18± 37.86 %
μ = 474.1± 93.4
σ = 223.7± 135.4
WF
σ/μ = 86.60± 131.13 %
μ = 363.8± 203.8
σ = 315.1± 300.6



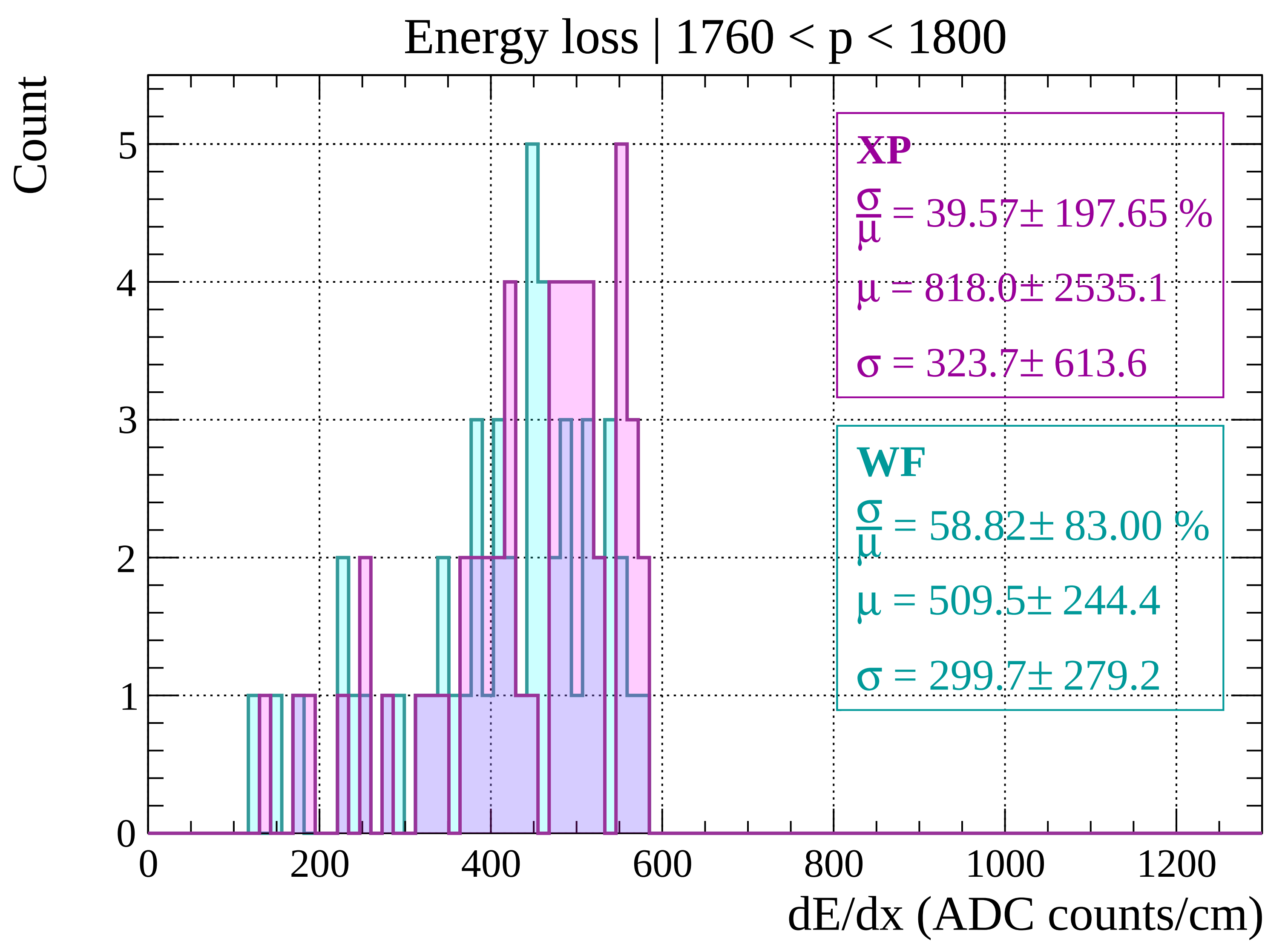
Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 38.93± 55.58 %
μ = 548.2± 254.5
σ = 213.4± 205.6
WF
$\frac{\sigma}{\mu}$ = 18.97± 7.28 %
μ = 438.5± 20.5
σ = 83.2± 28.0



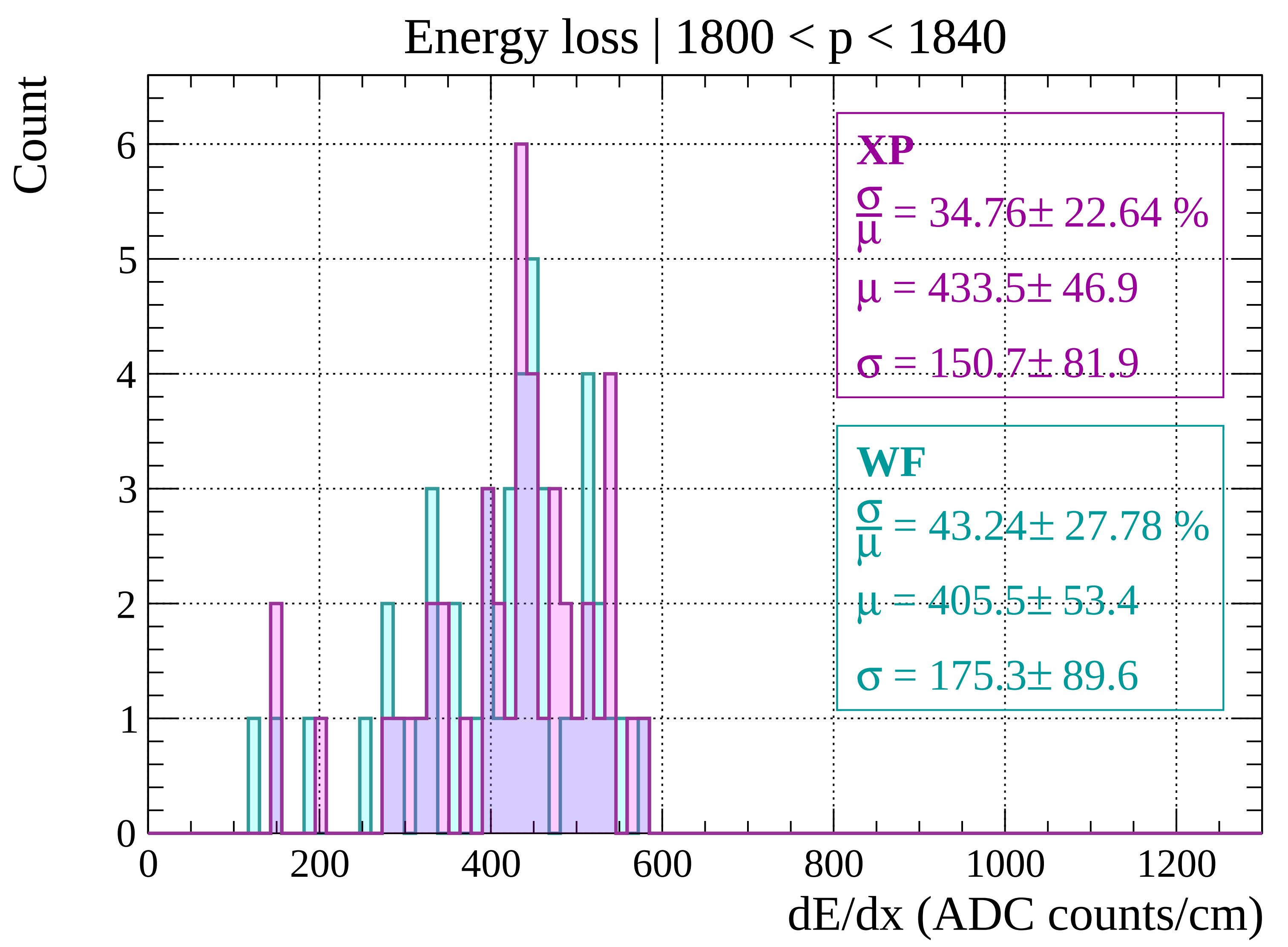
Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 26.46± 17.90 %
μ = 444.5± 37.2
σ = 117.6± 69.7
WF
σ/μ = 31.58± 14.74 %
μ = 447.8± 40.9
σ = 141.4± 53.1



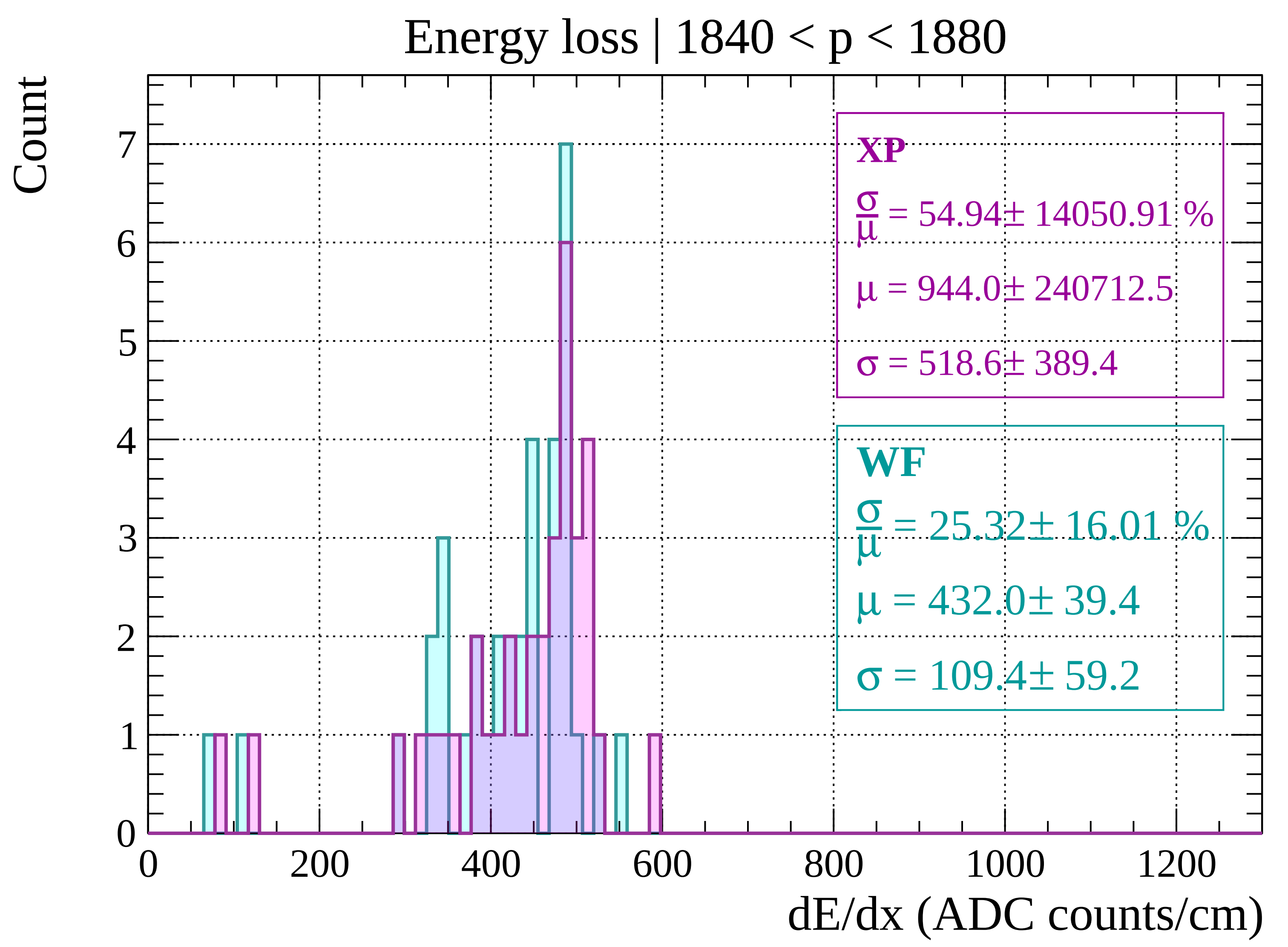
Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 22.23± 13.60 %
μ = 462.3± 36.1
σ = 102.8± 54.9
WF
$\frac{\sigma}{\mu}$ = 14.08± 4.80 %
μ = 439.9± 14.2
σ = 61.9± 19.1



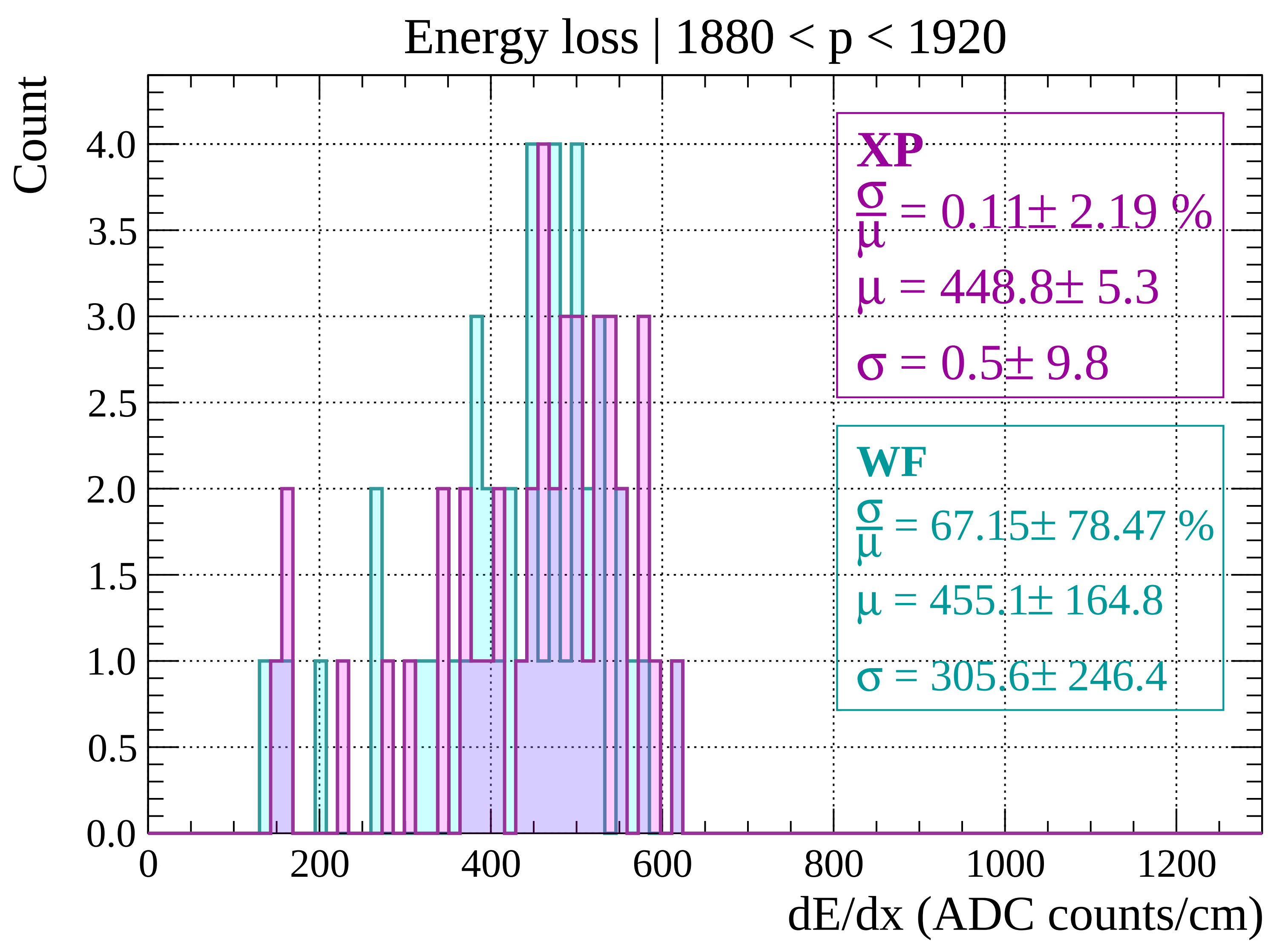
Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 39.57± 197.65 %
μ = 818.0± 2535.1
σ = 323.7± 613.6
WF
σ/μ = 58.82± 83.00 %
μ = 509.5± 244.4
σ = 299.7± 279.2

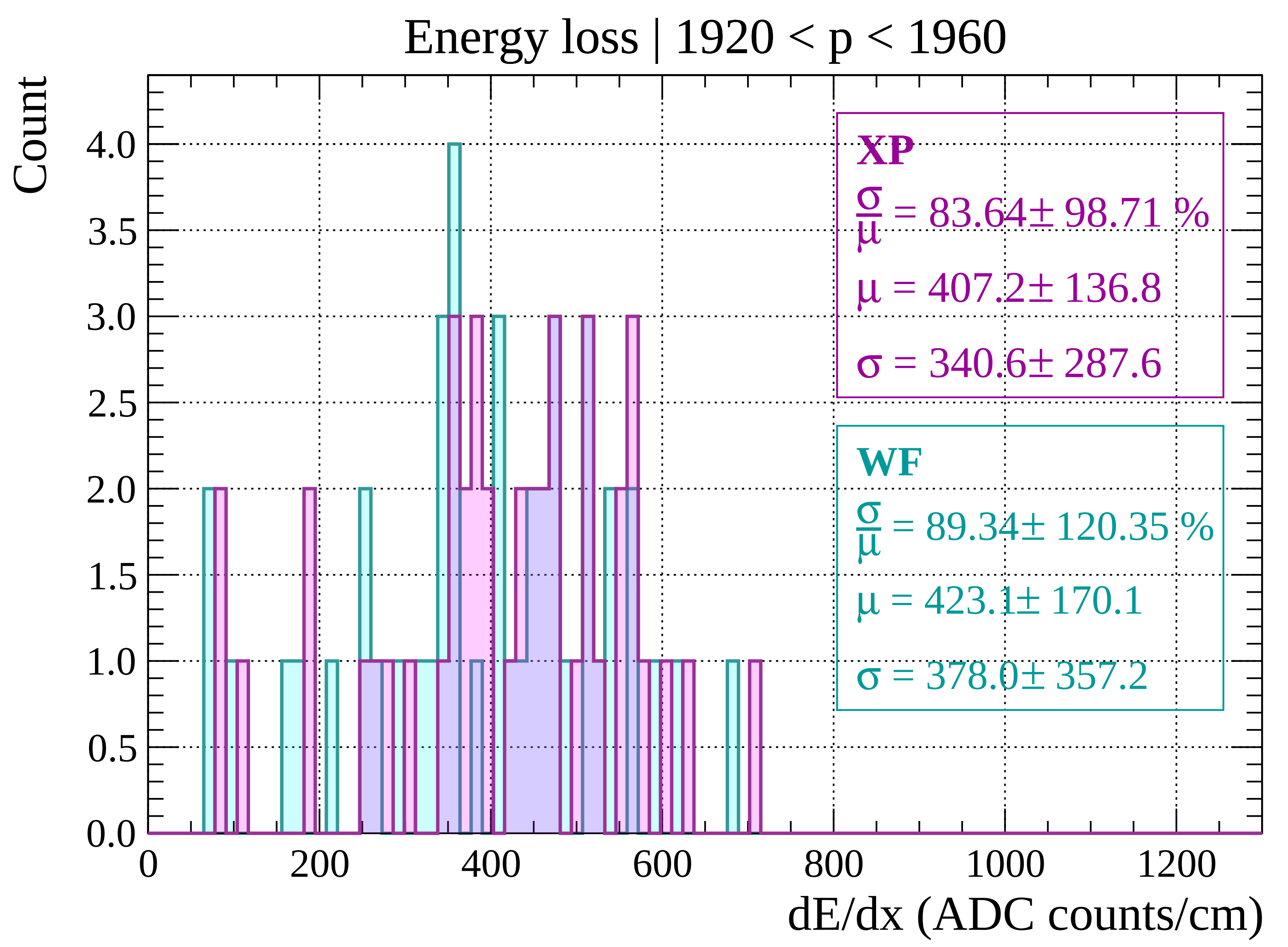


Energy loss | 1800 < p < 1840
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 34.76± 22.64 %
μ = 433.5± 46.9
σ = 150.7± 81.9
WF
$\frac{\sigma}{\mu}$ = 43.24± 27.78 %
μ = 405.5± 53.4
σ = 175.3± 89.6

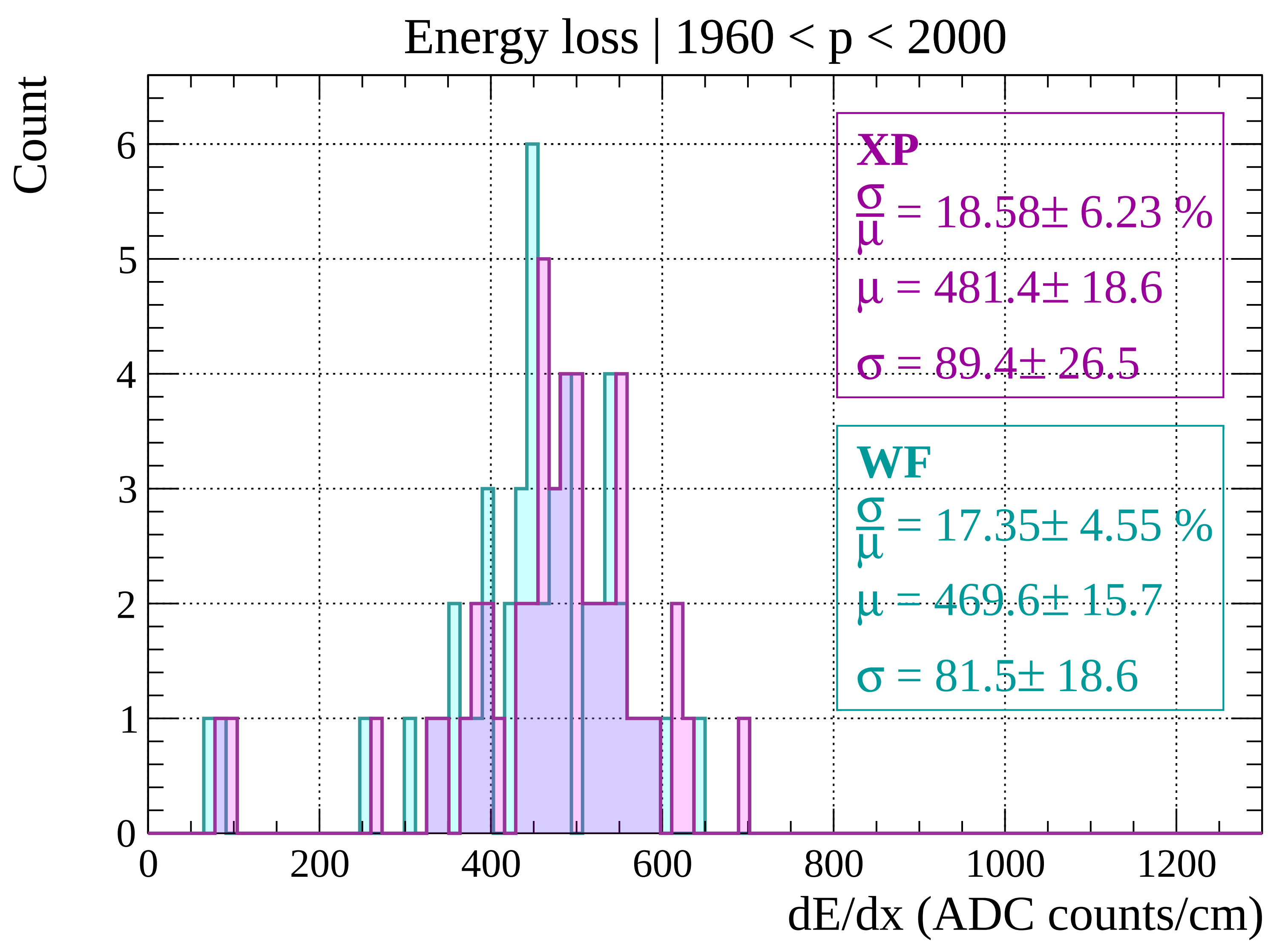


Energy loss | 1840 < p < 1880
Count
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 54.94± 14050.91 %
μ = 944.0± 240712.5
σ = 518.6± 389.4
WF
σ/μ = 25.32± 16.01 %
μ = 432.0± 39.4
σ = 109.4± 59.2

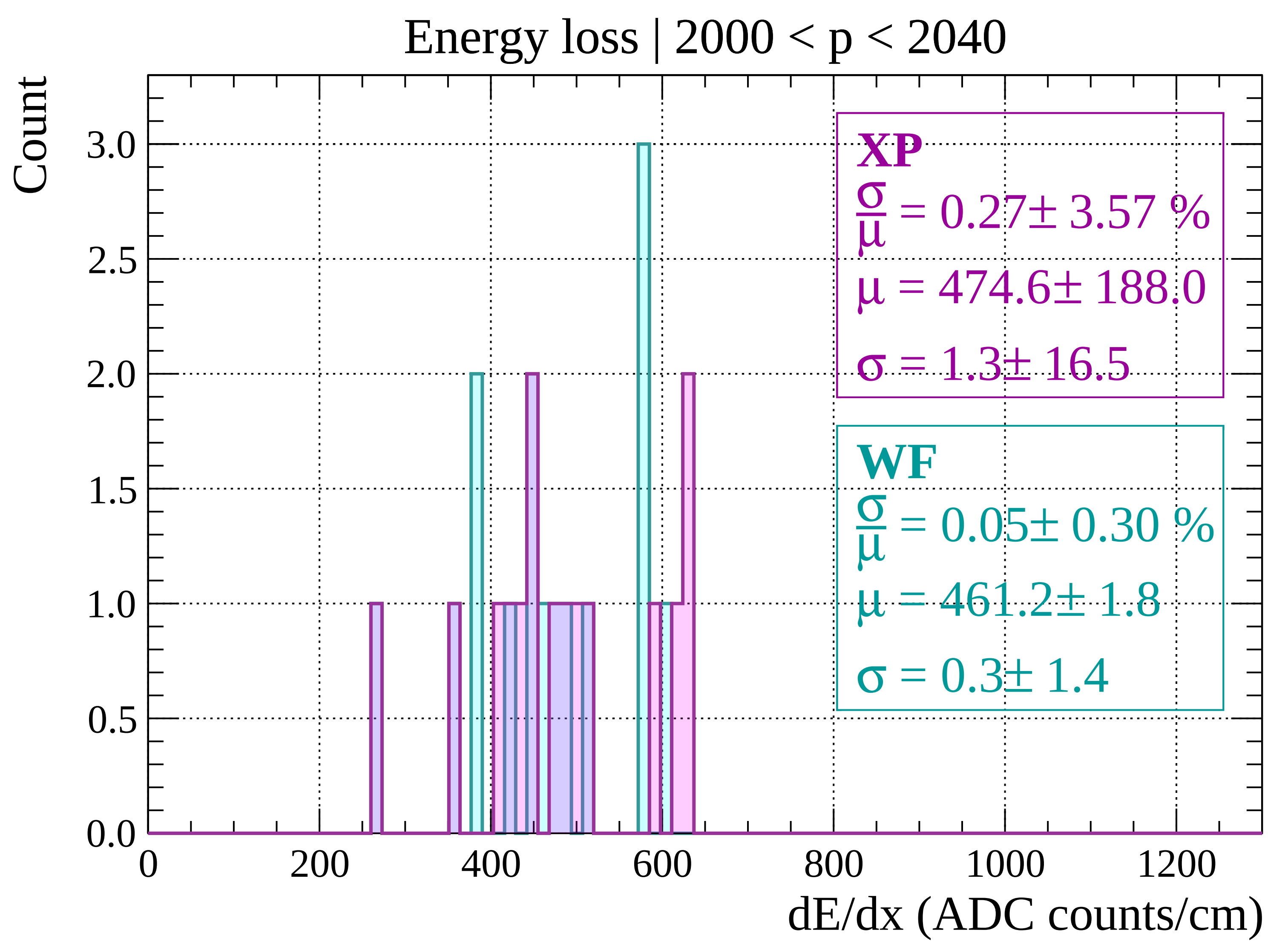

Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
σ/μ = 0.11± 2.19 %
μ = 448.8± 5.3
σ = 0.5± 9.8
WF
σ/μ = 67.15± 78.47 %
μ = 455.1± 164.8
σ = 305.6± 246.4



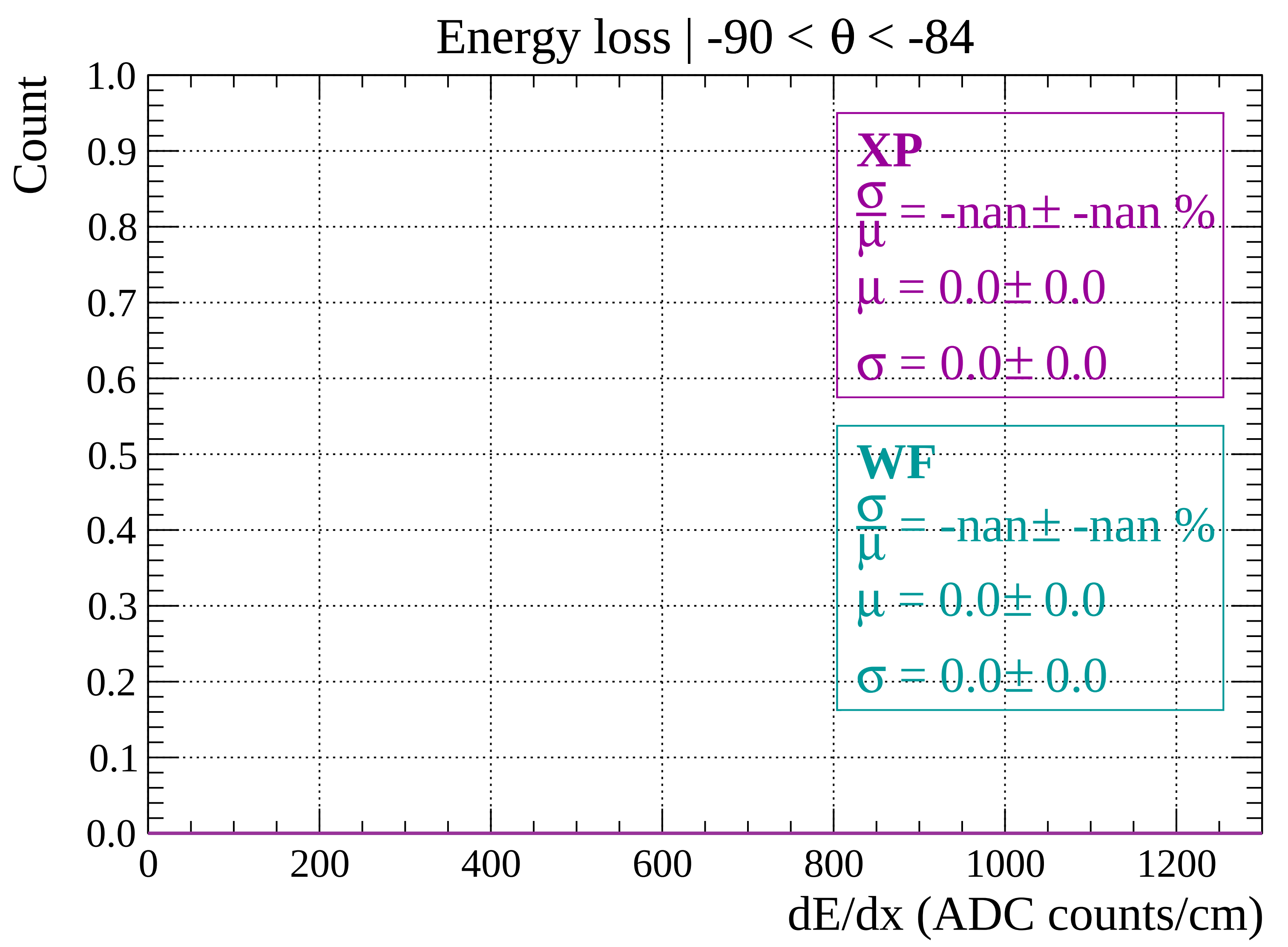
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 83.64± 98.71 %
μ = 407.2± 136.8
σ = 340.6± 287.6
WF
$\frac{\sigma}{\mu}$ = 89.34± 120.35 %
μ = 423.1± 170.1
σ = 378.0± 357.2

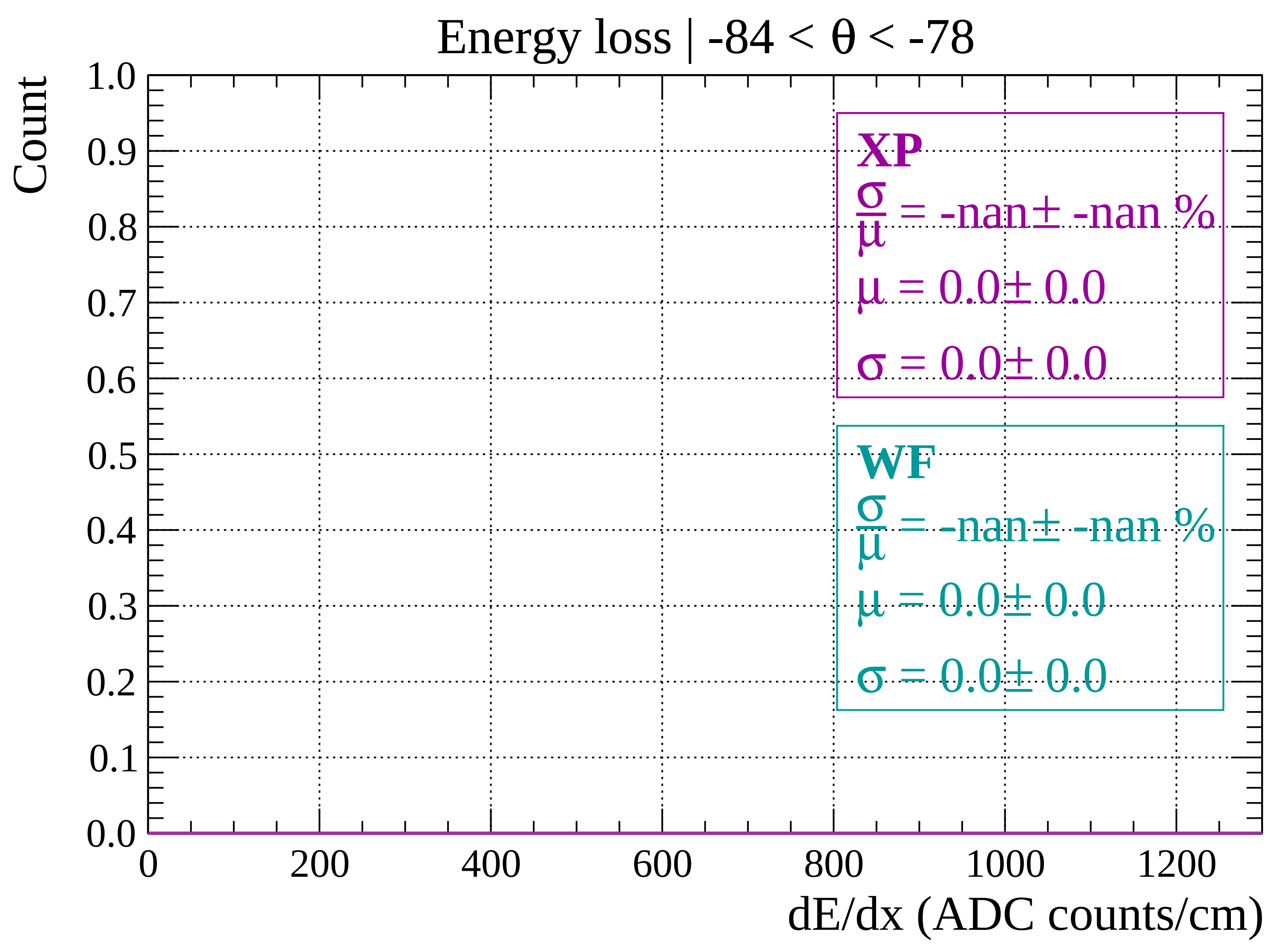


Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 18.58± 6.23 %
μ = 481.4± 18.6
σ = 89.4± 26.5
WF
σ/μ = 17.35± 4.55 %
μ = 469.6± 15.7
σ = 81.5± 18.6

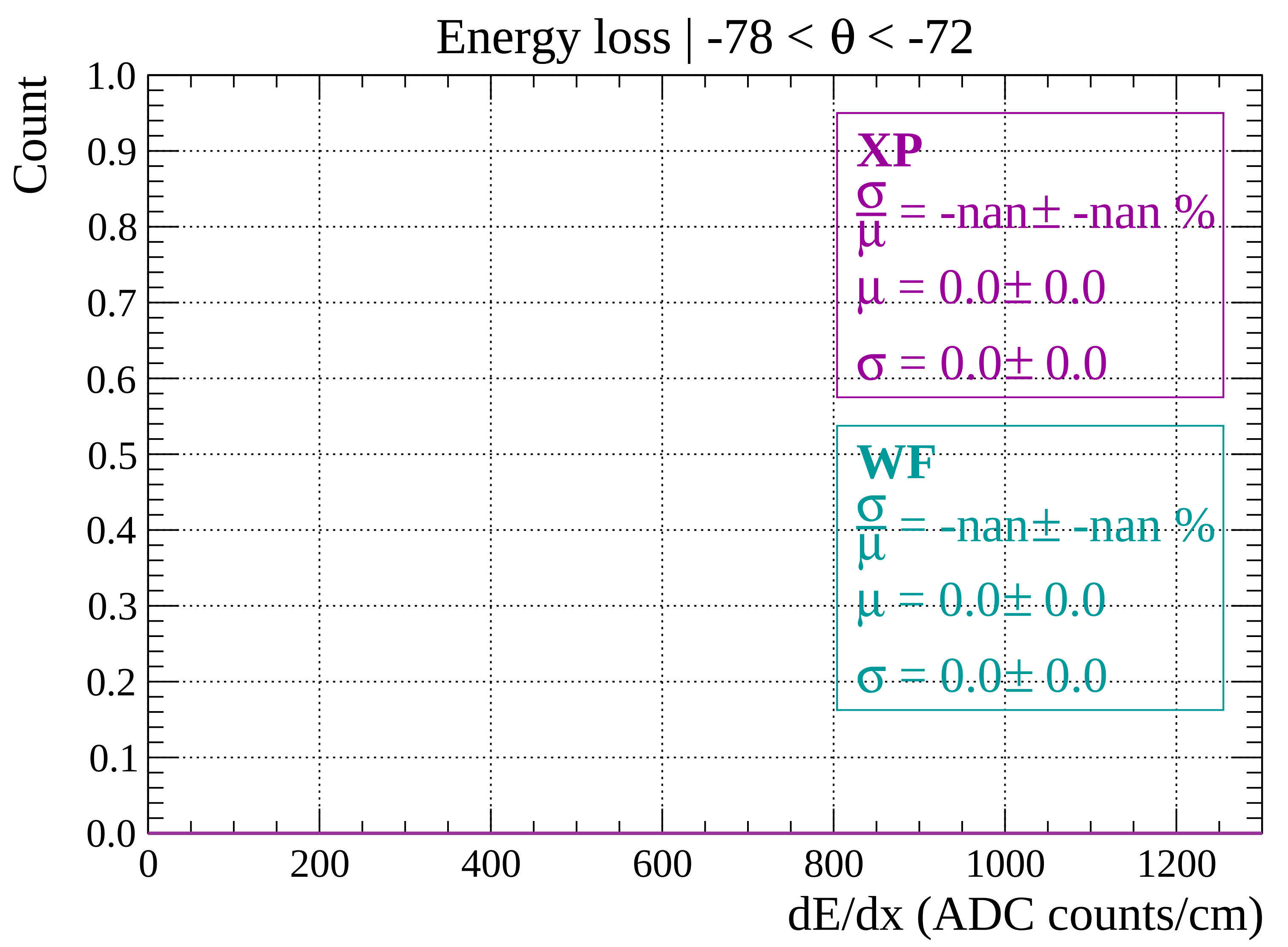


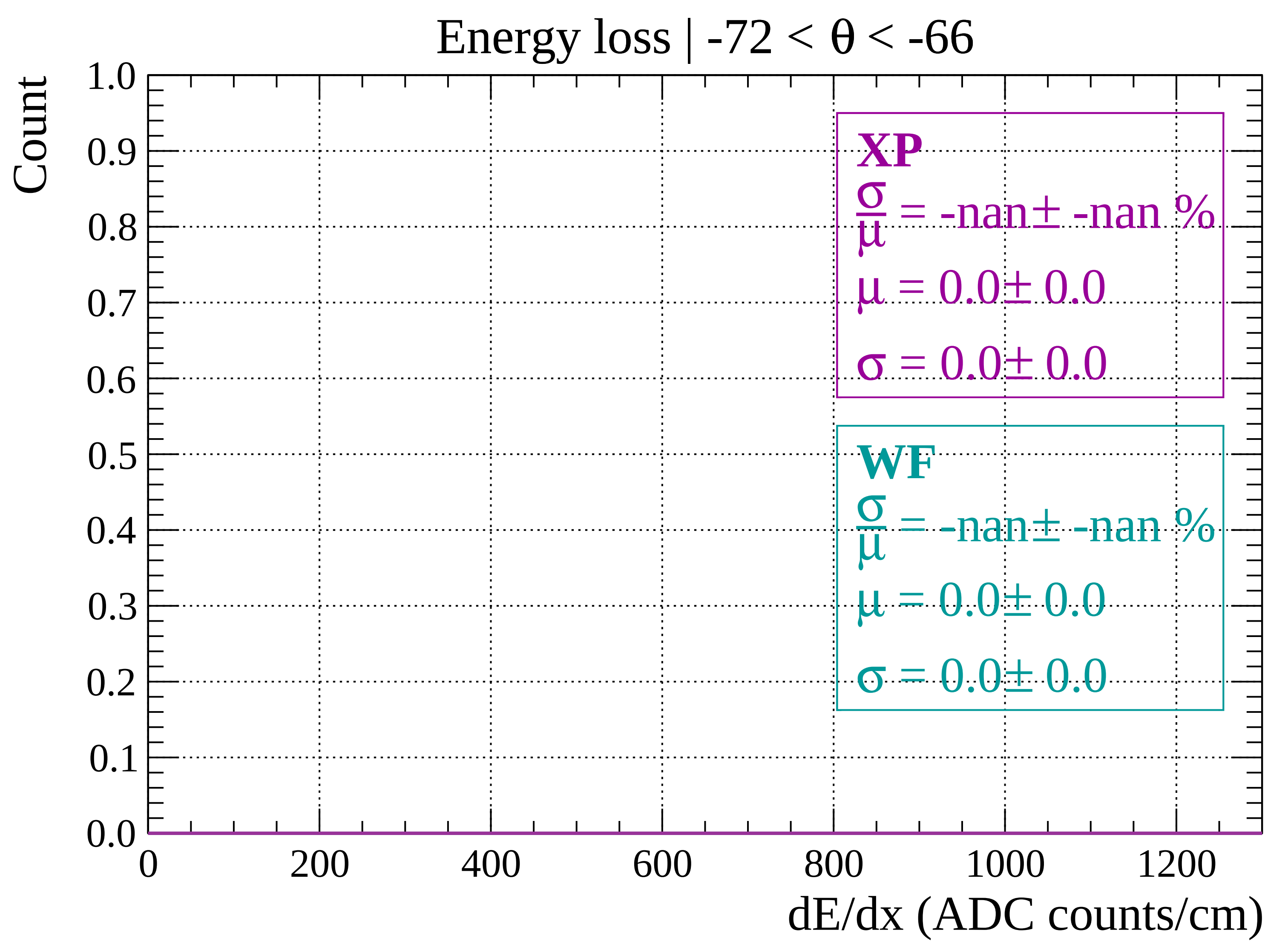
Energy loss | 2000 < p < 2040
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.27± 3.57 %
μ = 474.6± 188.0
σ = 1.3± 16.5
WF
$\frac{\sigma}{\mu}$ = 0.05± 0.30 %
μ = 461.2± 1.8
σ = 0.3± 1.4


Energy loss | -90 < θ < -84
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

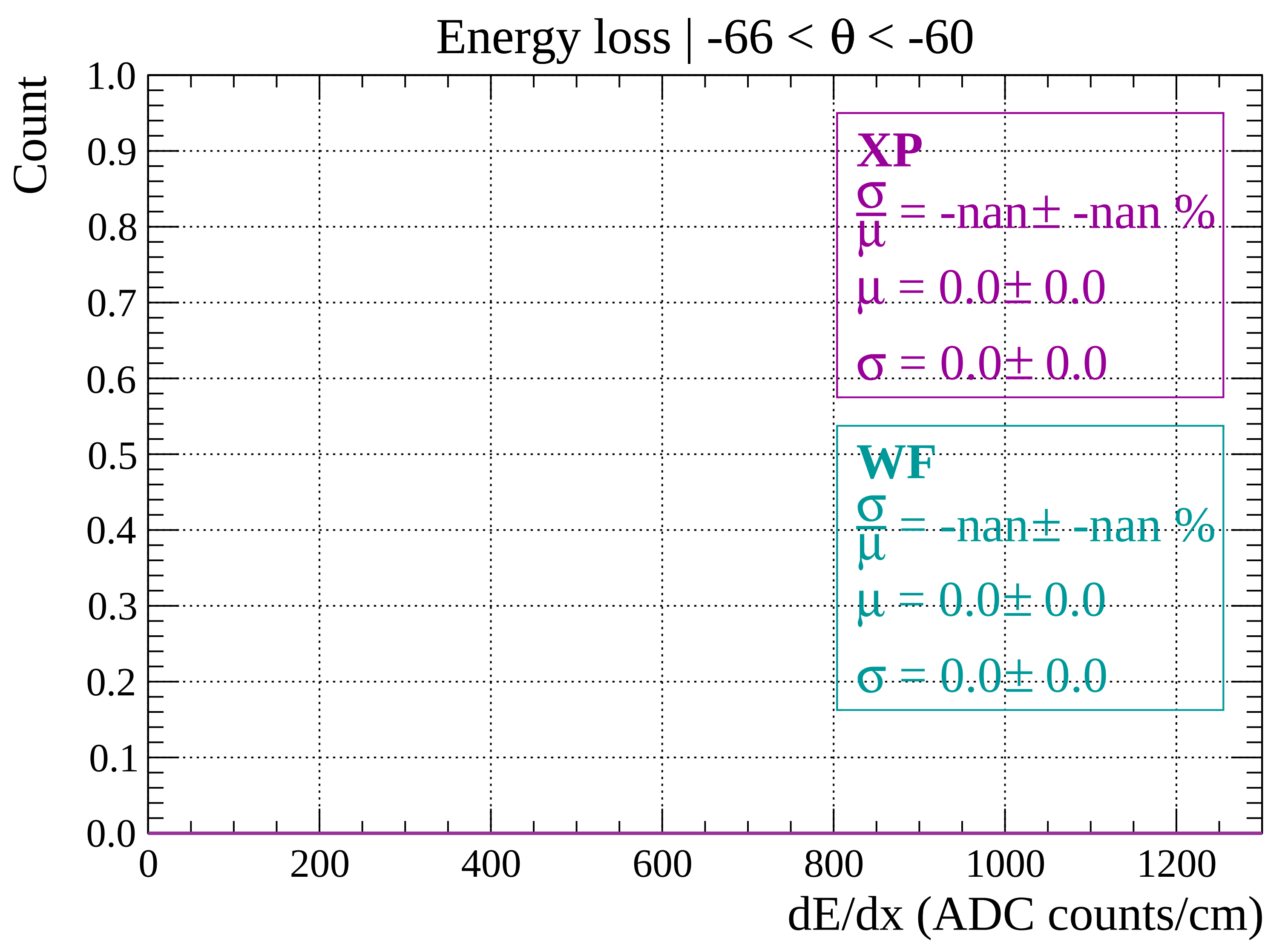


Energy loss | -84 < θ < -78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

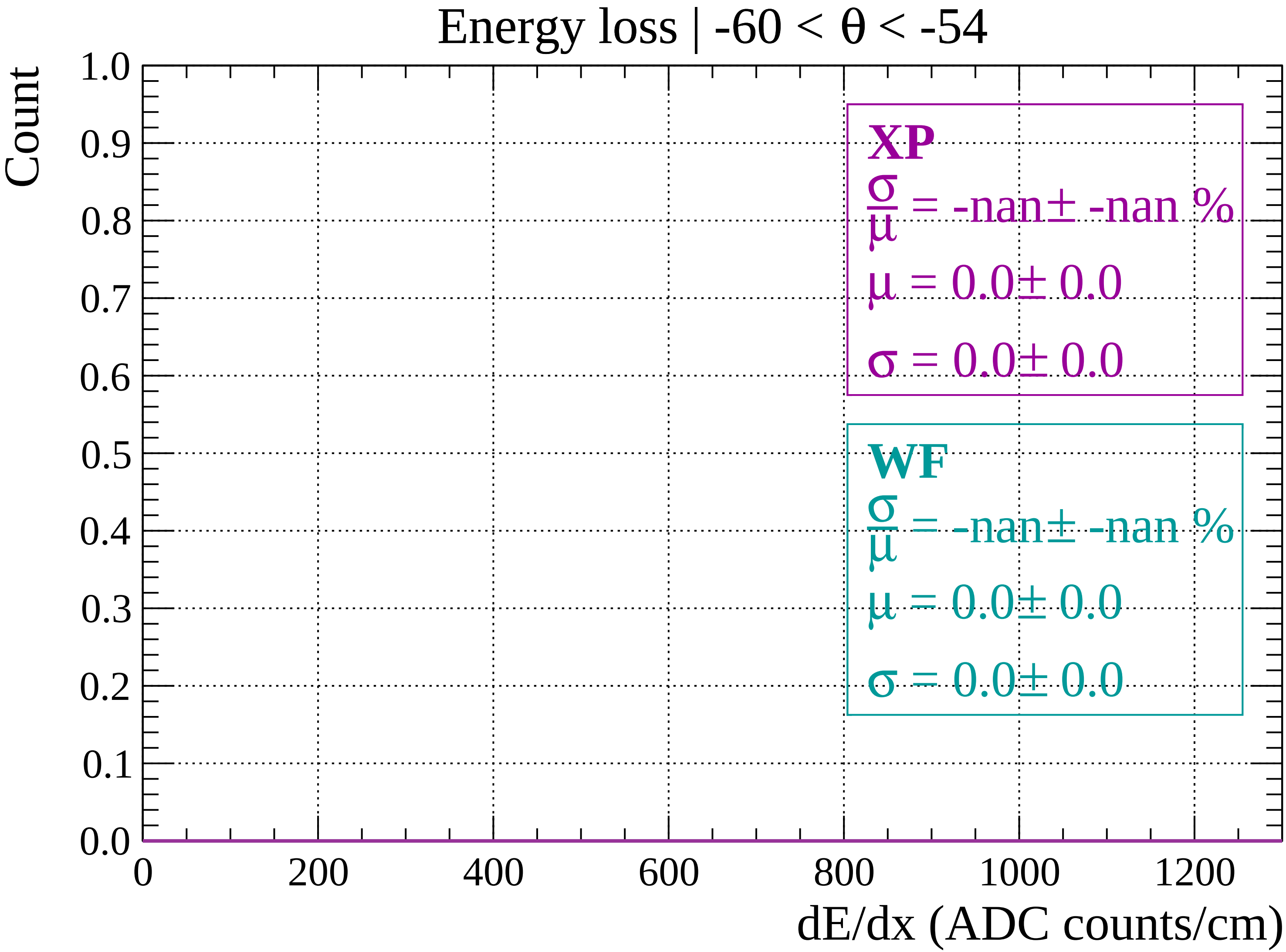

Energy loss | -78 < θ < -72
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



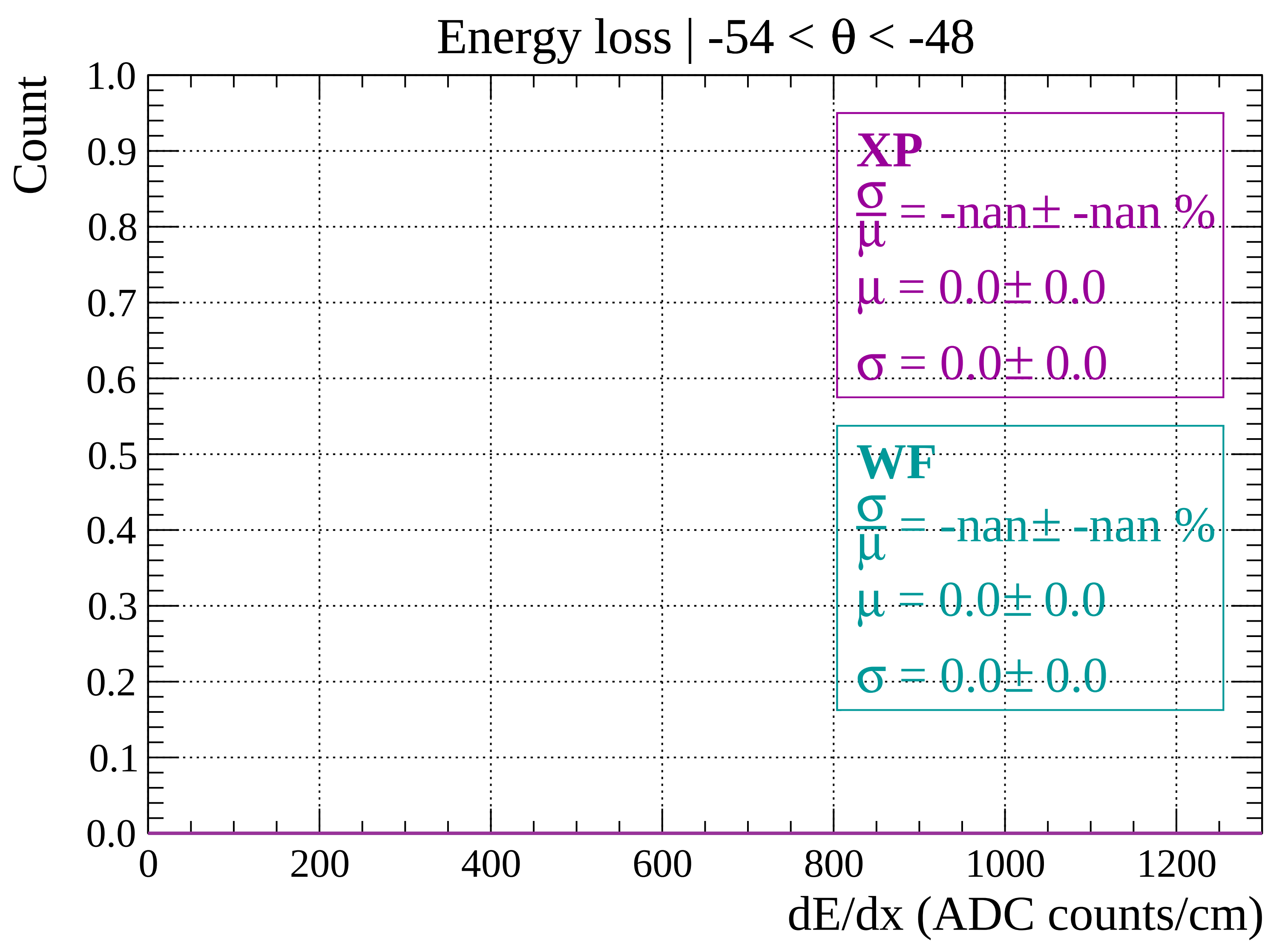
Energy loss | -72 < θ < -66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



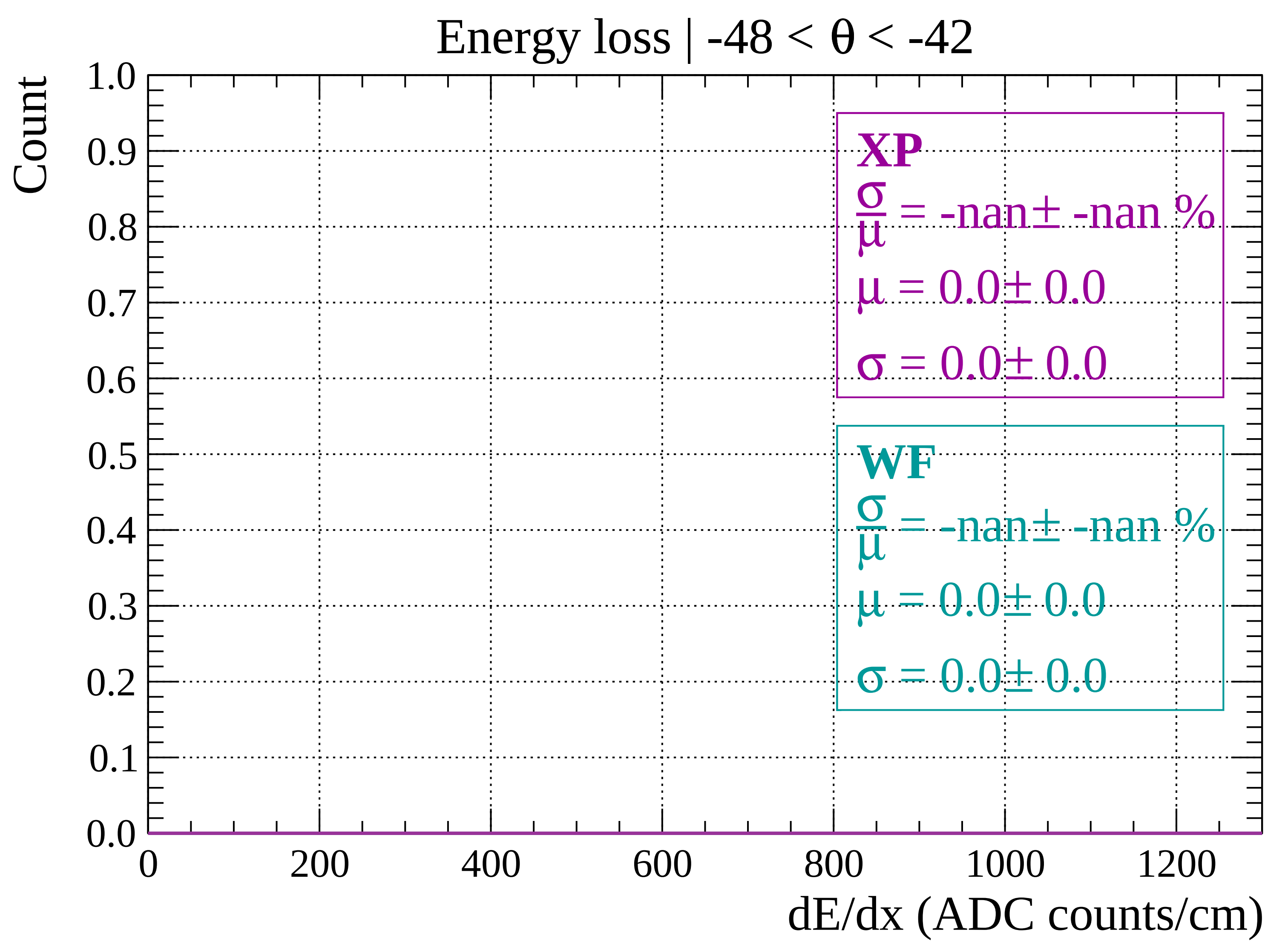
Energy loss | -66 < θ < -60
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

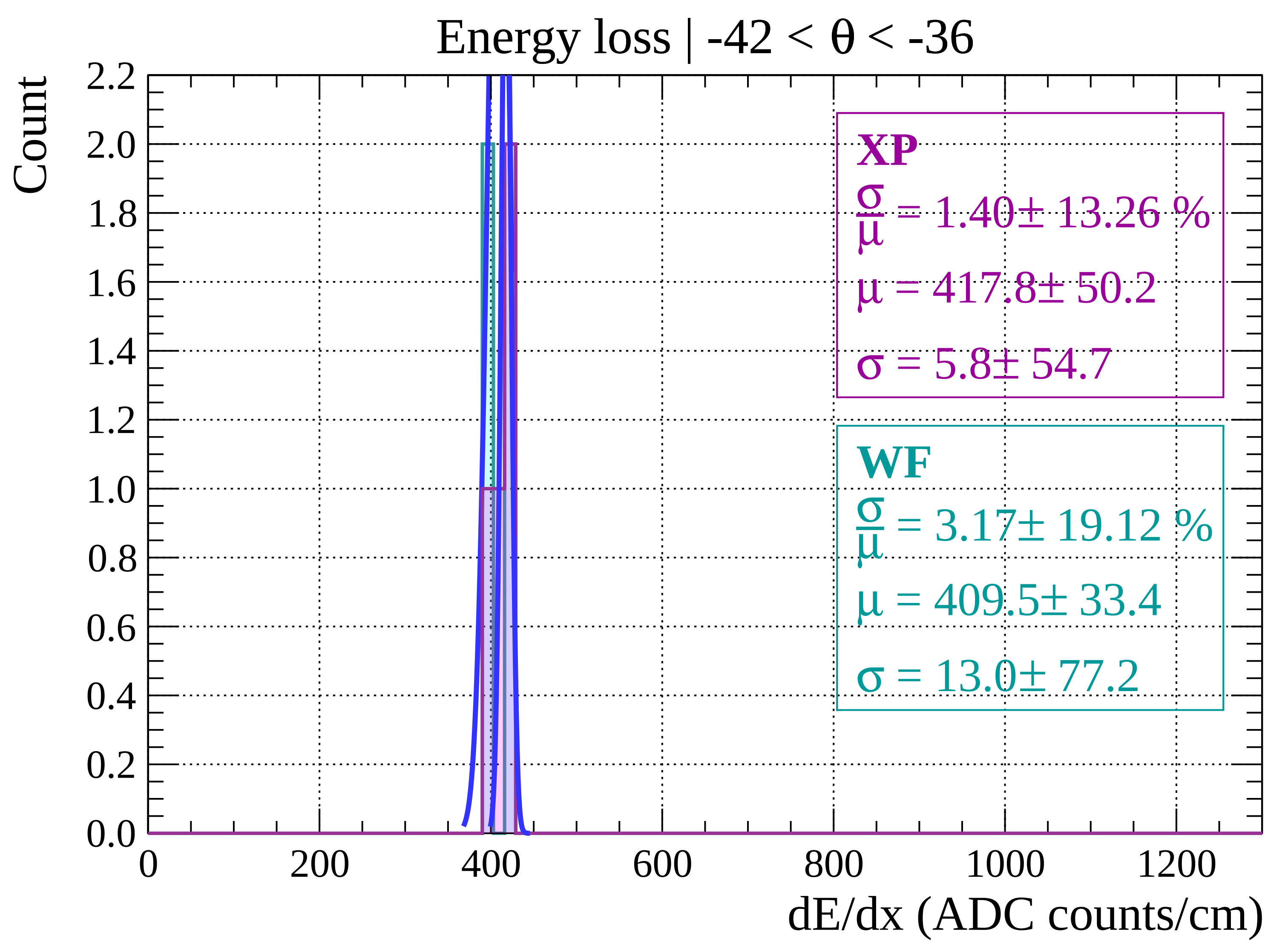


Energy loss | -60 < θ < -54
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

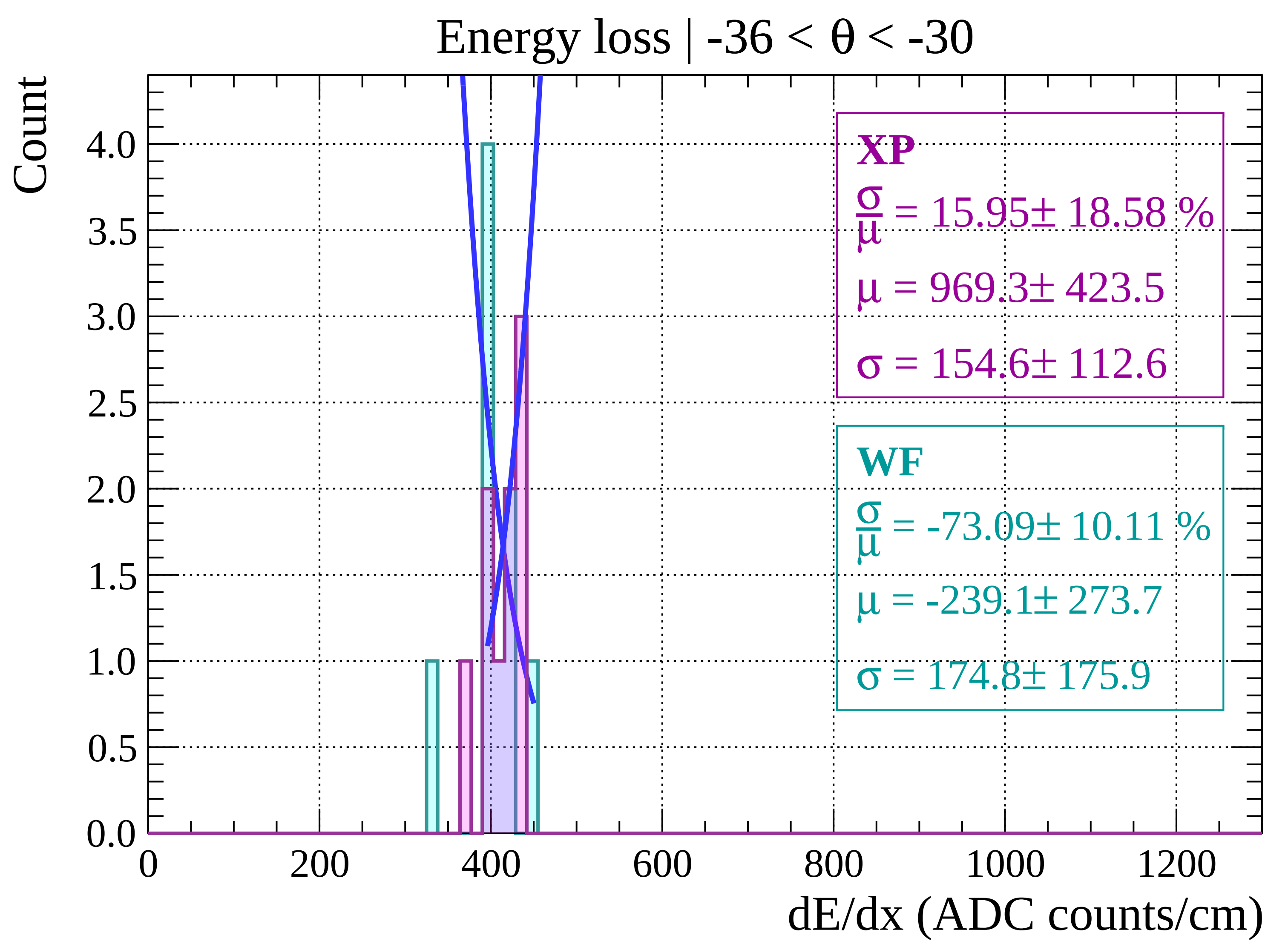


Energy loss | -54 < θ < -48
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

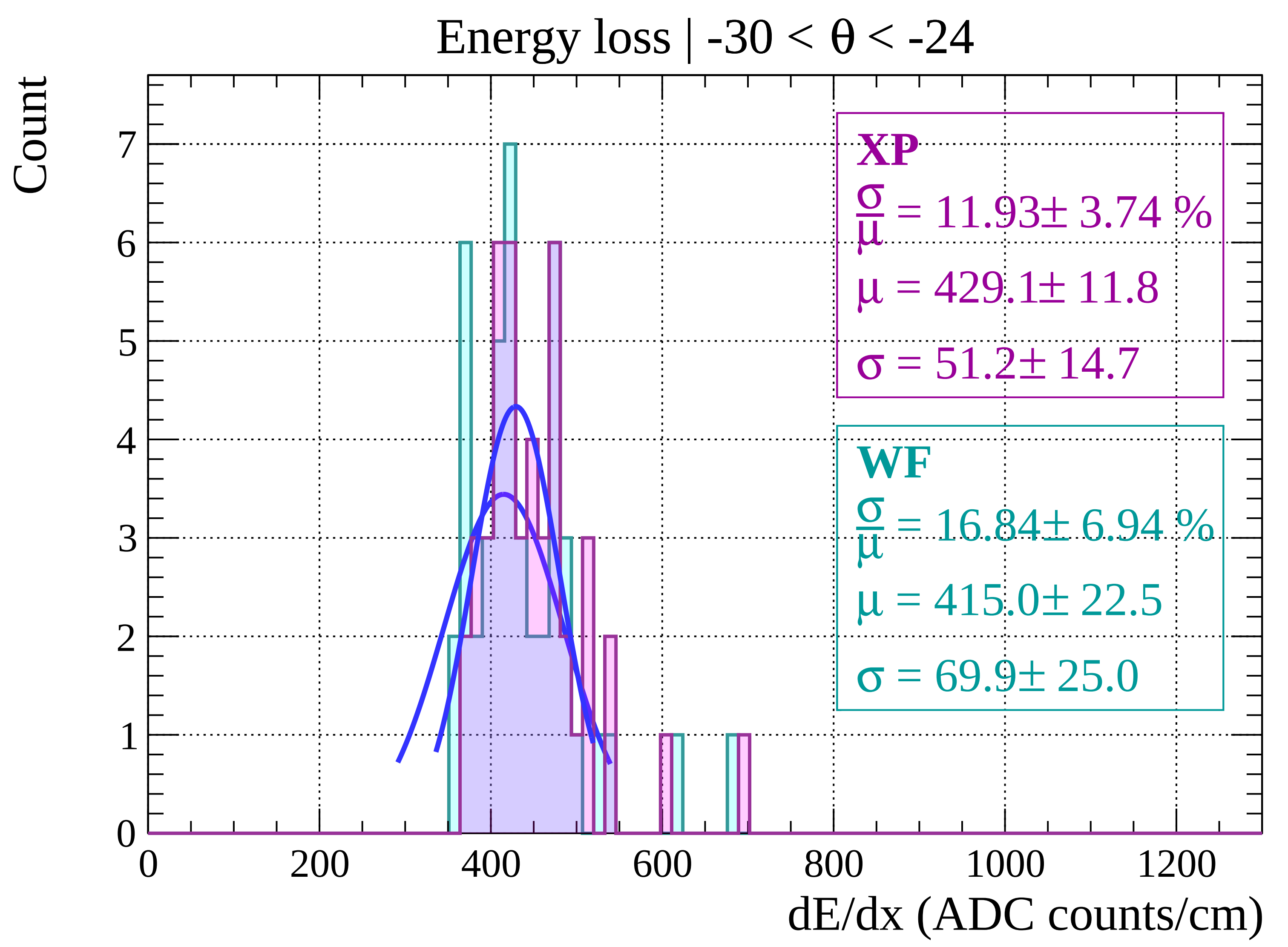

Energy loss | -48 < θ < -42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0

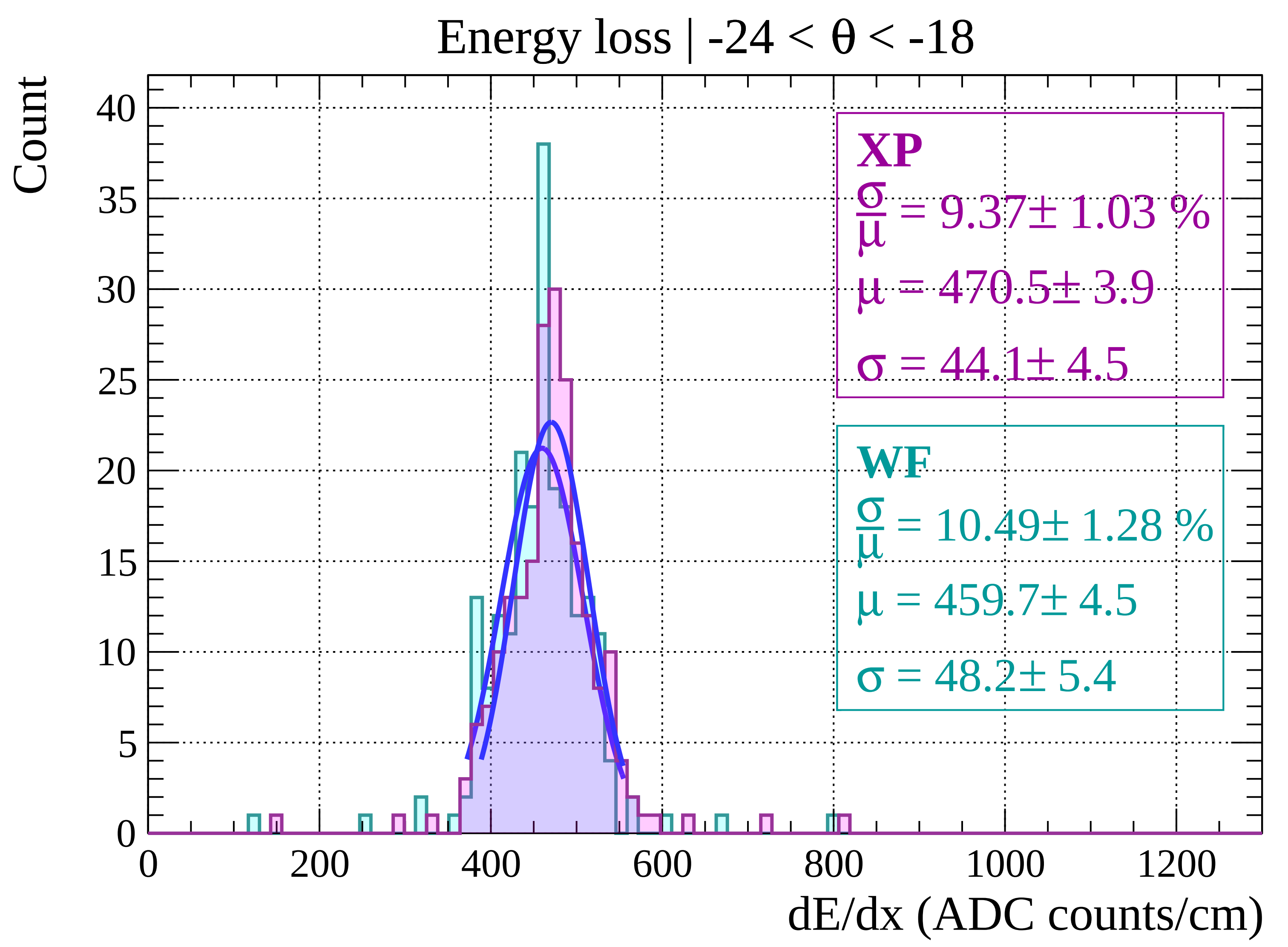


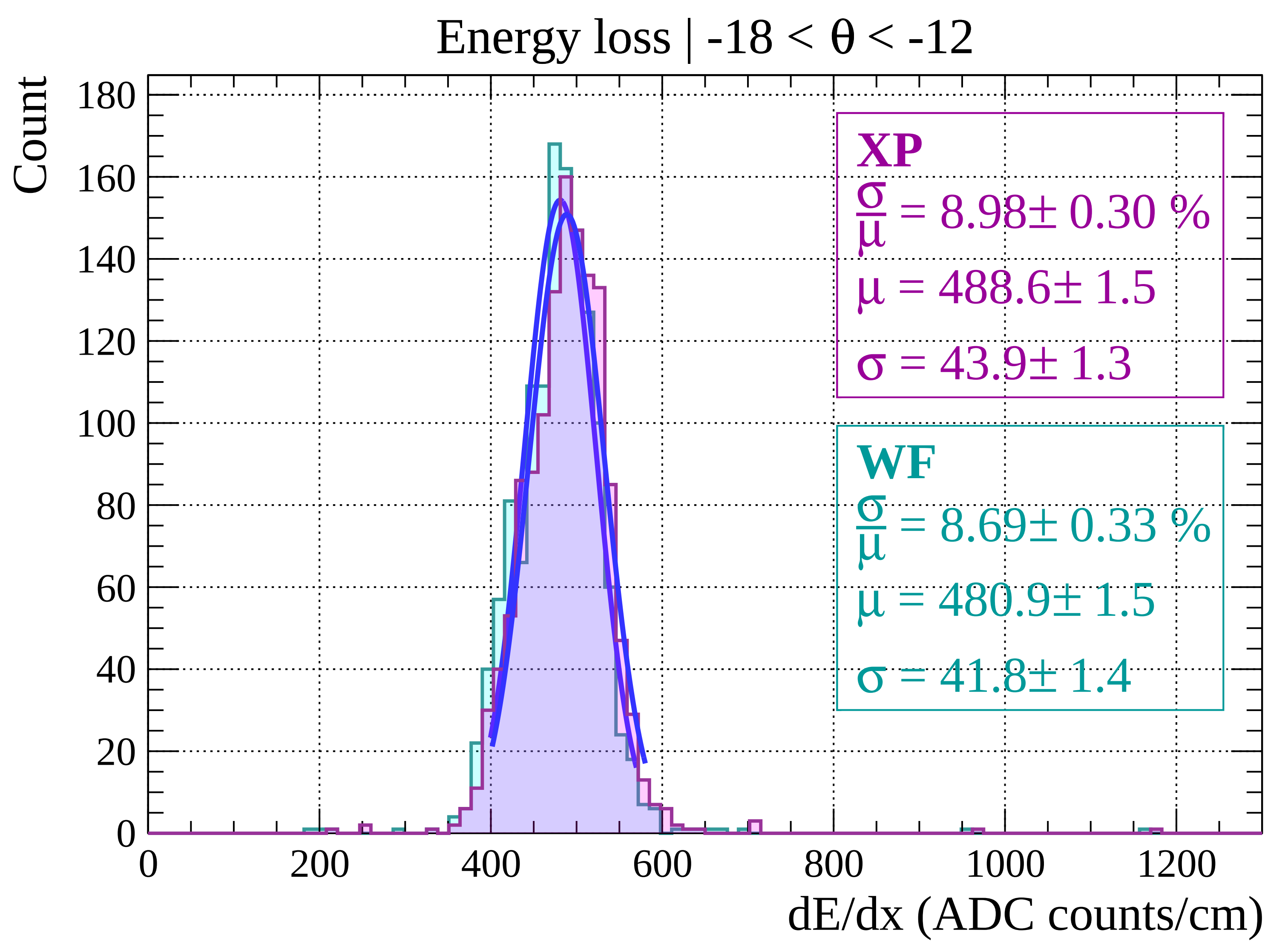
Energy loss | -42 < θ < -36
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 1.40± 13.26 %
μ = 417.8± 50.2
σ = 5.8± 54.7
WF
σ/μ = 3.17± 19.12 %
μ = 409.5± 33.4
σ = 13.0± 77.2



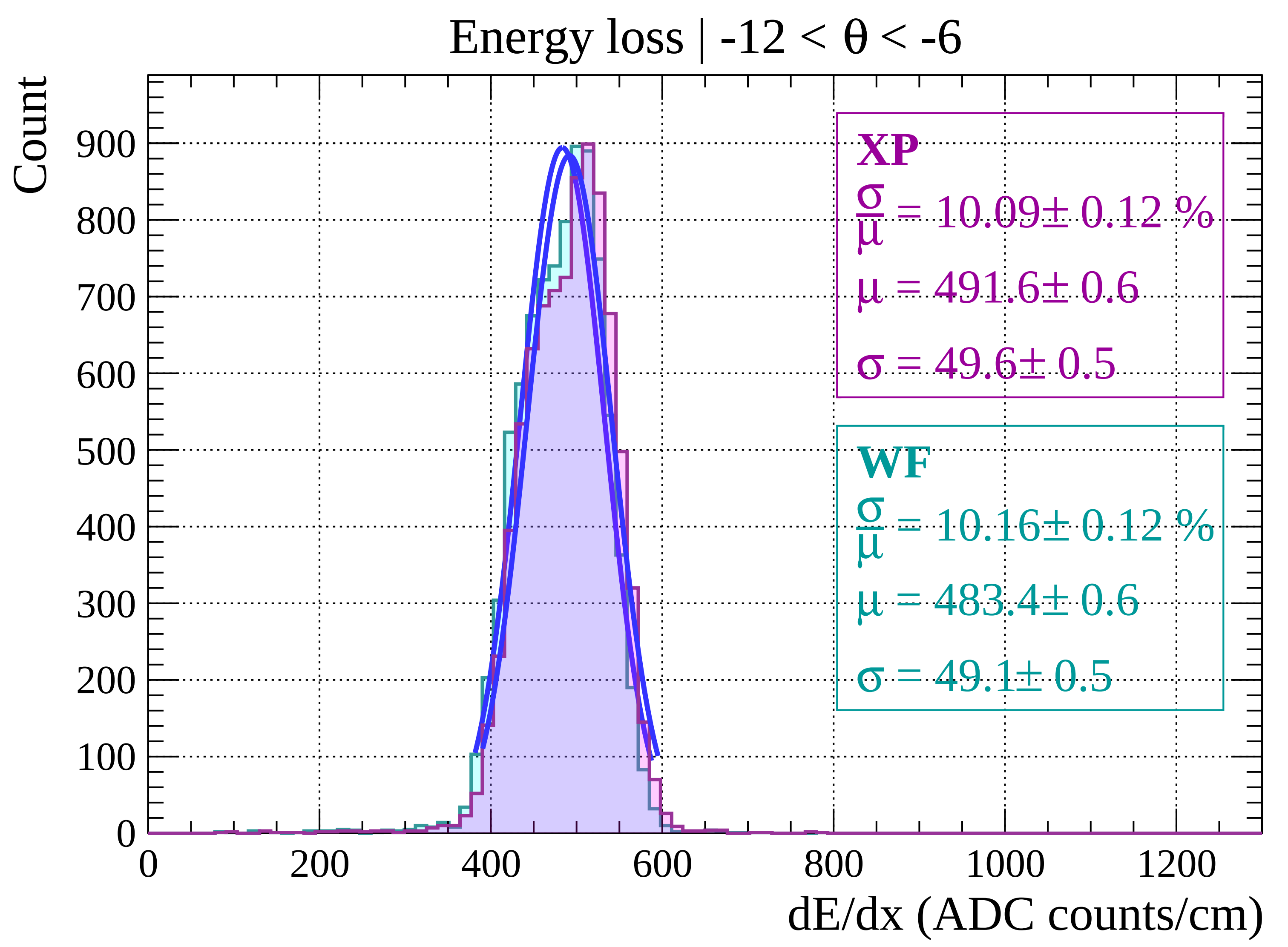
Energy loss | -36 < θ < -30
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.95± 18.58 %
μ = 969.3± 423.5
σ = 154.6± 112.6
WF
σ/μ = -73.09± 10.11 %
μ = -239.1± 273.7
σ = 174.8± 175.9

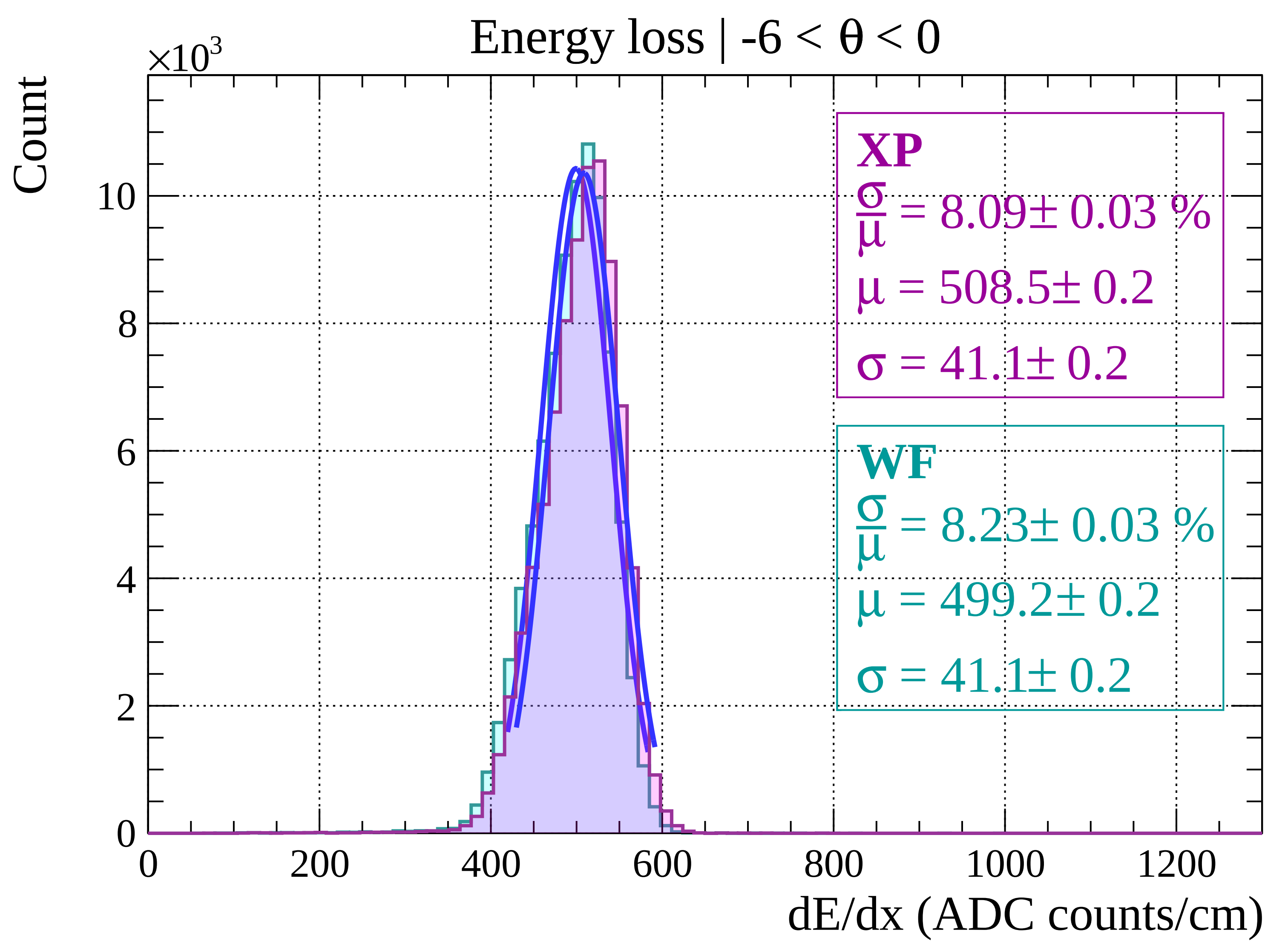

Energy loss | -30 < θ < -24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
XP
σ/μ = 11.93± 3.74 %
μ = 429.1± 11.8
σ = 51.2± 14.7
WF
σ/μ = 16.84± 6.94 %
μ = 415.0± 22.5
σ = 69.9± 25.0


Energy loss | -24 < θ < -18
Count
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.37± 1.03 %
μ = 470.5± 3.9
σ = 44.1± 4.5
WF
σ/μ = 10.49± 1.28 %
μ = 459.7± 4.5
σ = 48.2± 5.4

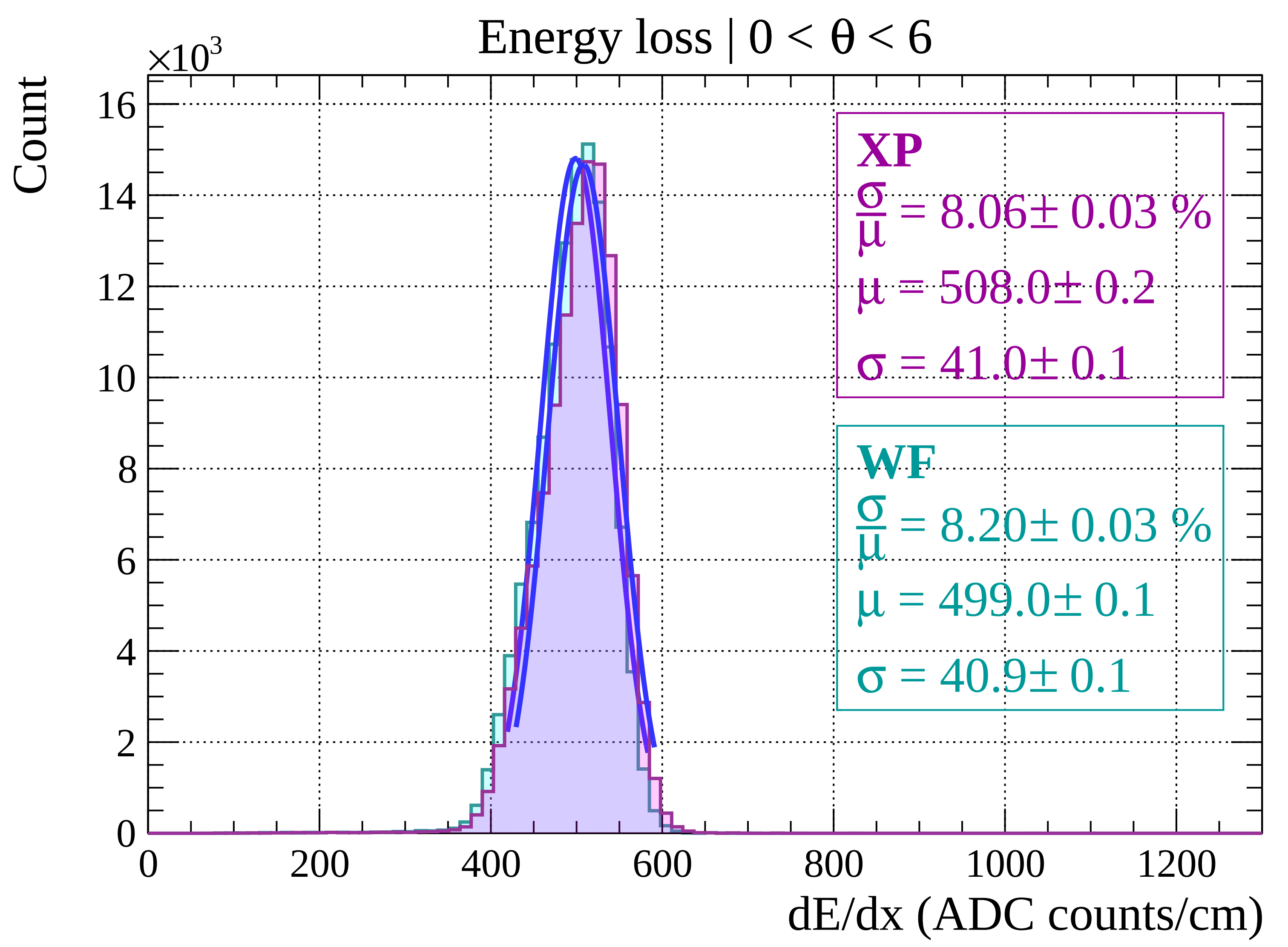


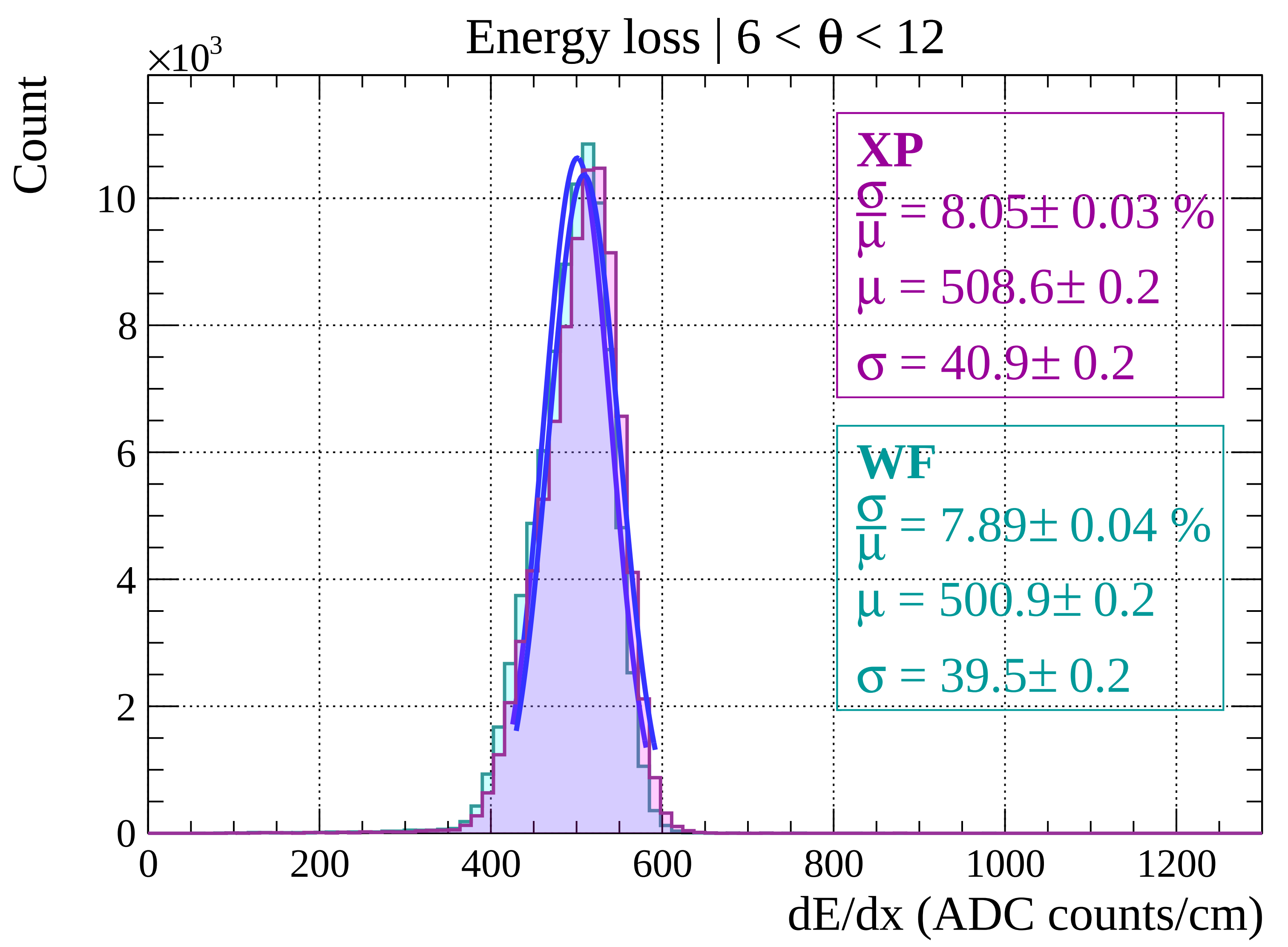
Energy loss | -18 < θ < -12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.98± 0.30 %
μ = 488.6± 1.5
σ = 43.9± 1.3
WF
σ/μ = 8.69± 0.33 %
μ = 480.9± 1.5
σ = 41.8± 1.4


Energy loss | -12 < θ < -6
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 10.09 ± 0.12 %
μ = 491.6 ± 0.6
σ = 49.6 ± 0.5
WF
σ/μ = 10.16 ± 0.12 %
μ = 483.4 ± 0.6
σ = 49.1 ± 0.5

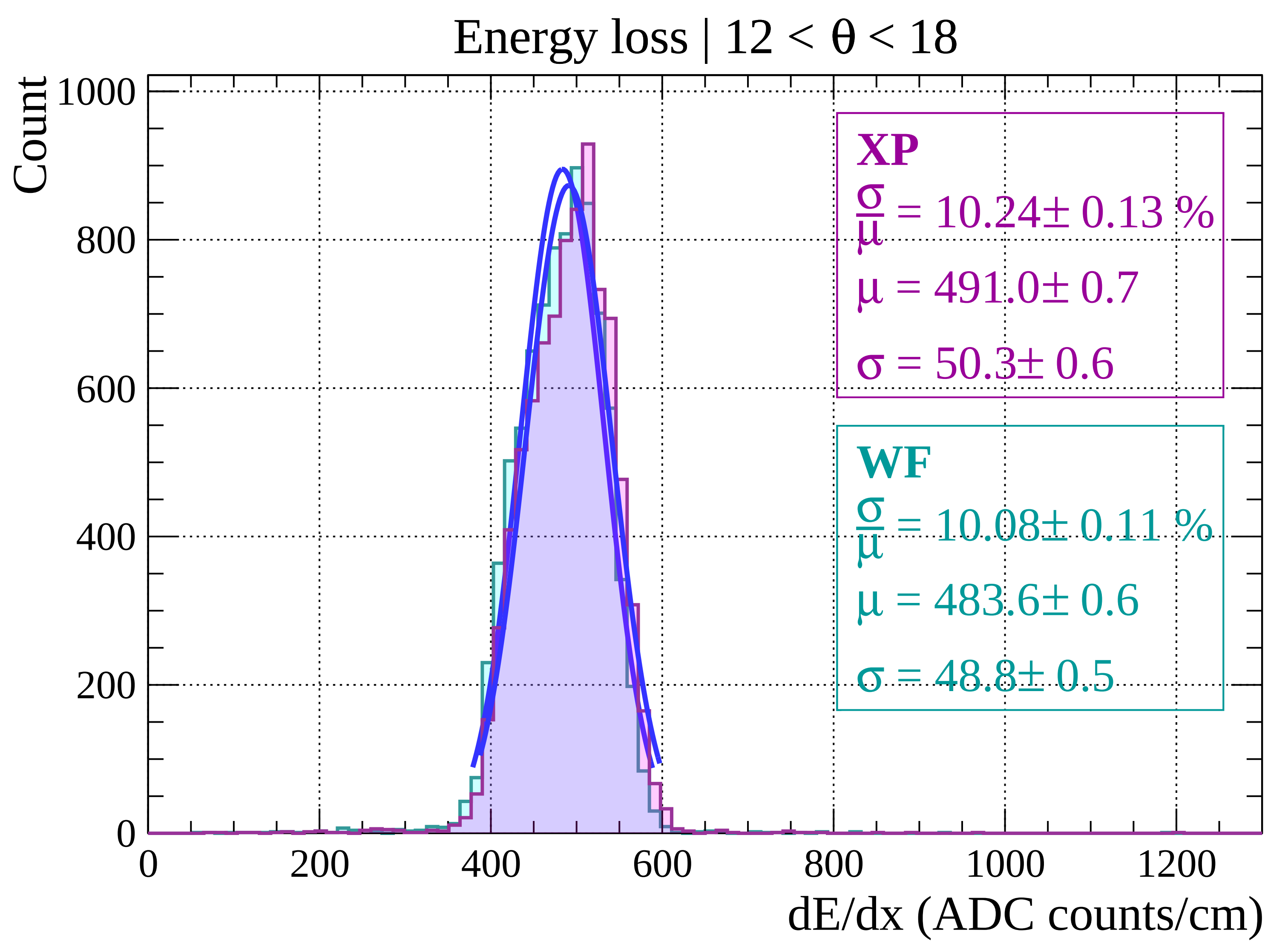


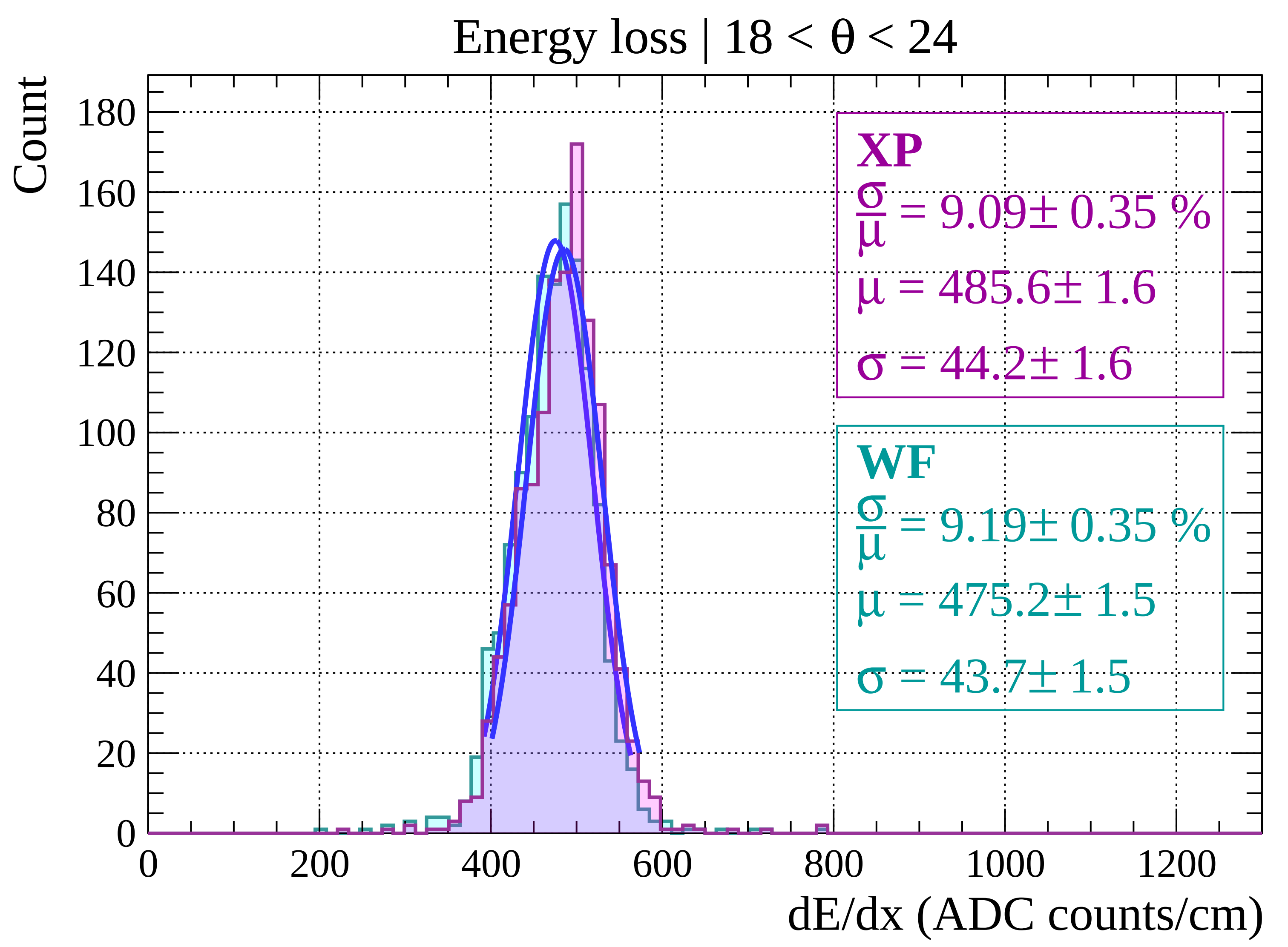
Energy loss | -6 < θ < 0
×10^3
Count
XP
σ/μ = 8.09± 0.03 %
μ = 508.5± 0.2
σ = 41.1± 0.2
WF
σ/μ = 8.23± 0.03 %
μ = 499.2± 0.2
σ = 41.1± 0.2
dE/dx (ADC counts/cm)
0
2
4
6
8
10
200
400
600
800
1000
1200


Energy loss | 0 < θ < 6
×10³
Count
16
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.06± 0.03 %
μ = 508.0± 0.2
σ = 41.0± 0.1
WF
σ/μ = 8.20± 0.03 %
μ = 499.0± 0.1
σ = 40.9± 0.1

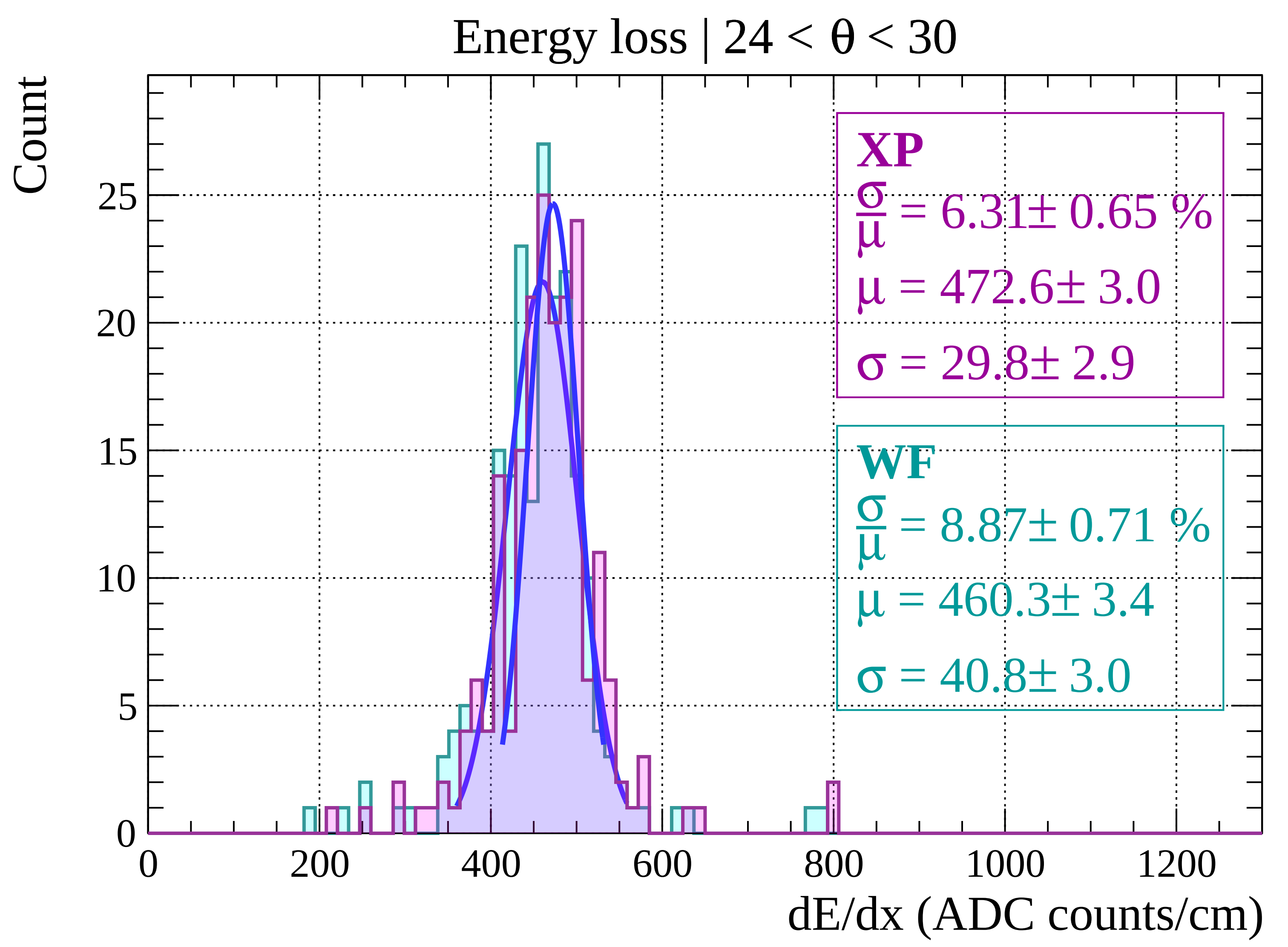


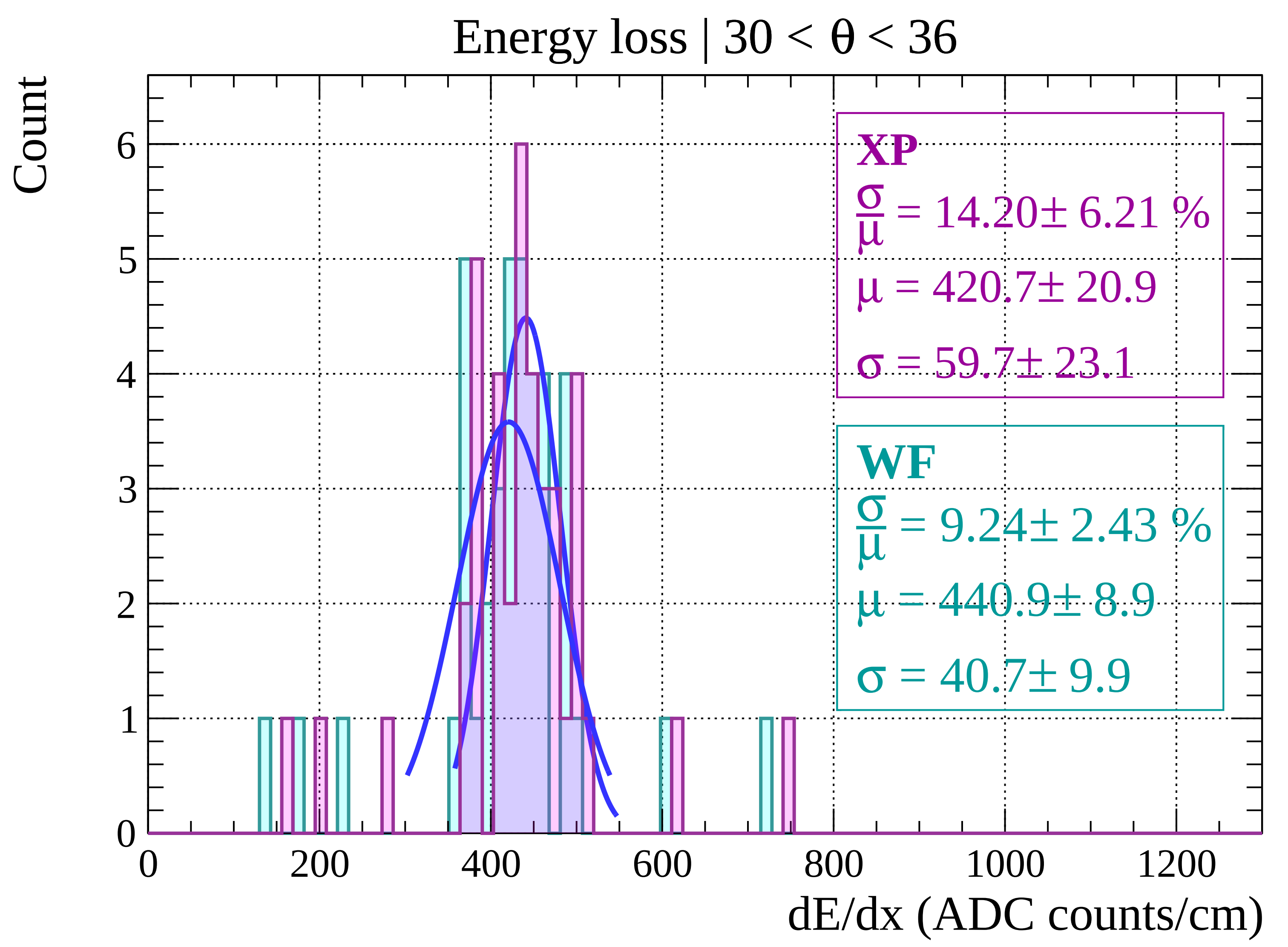
Energy loss | 6 < θ < 12
×10³
Count
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.05± 0.03 %
μ = 508.6± 0.2
σ = 40.9± 0.2
WF
σ/μ = 7.89± 0.04 %
μ = 500.9± 0.2
σ = 39.5± 0.2


Energy loss | 12 < θ < 18
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.24± 0.13 %
μ = 491.0± 0.7
σ = 50.3± 0.6
WF
σ/μ = 10.08± 0.11 %
μ = 483.6± 0.6
σ = 48.8± 0.5

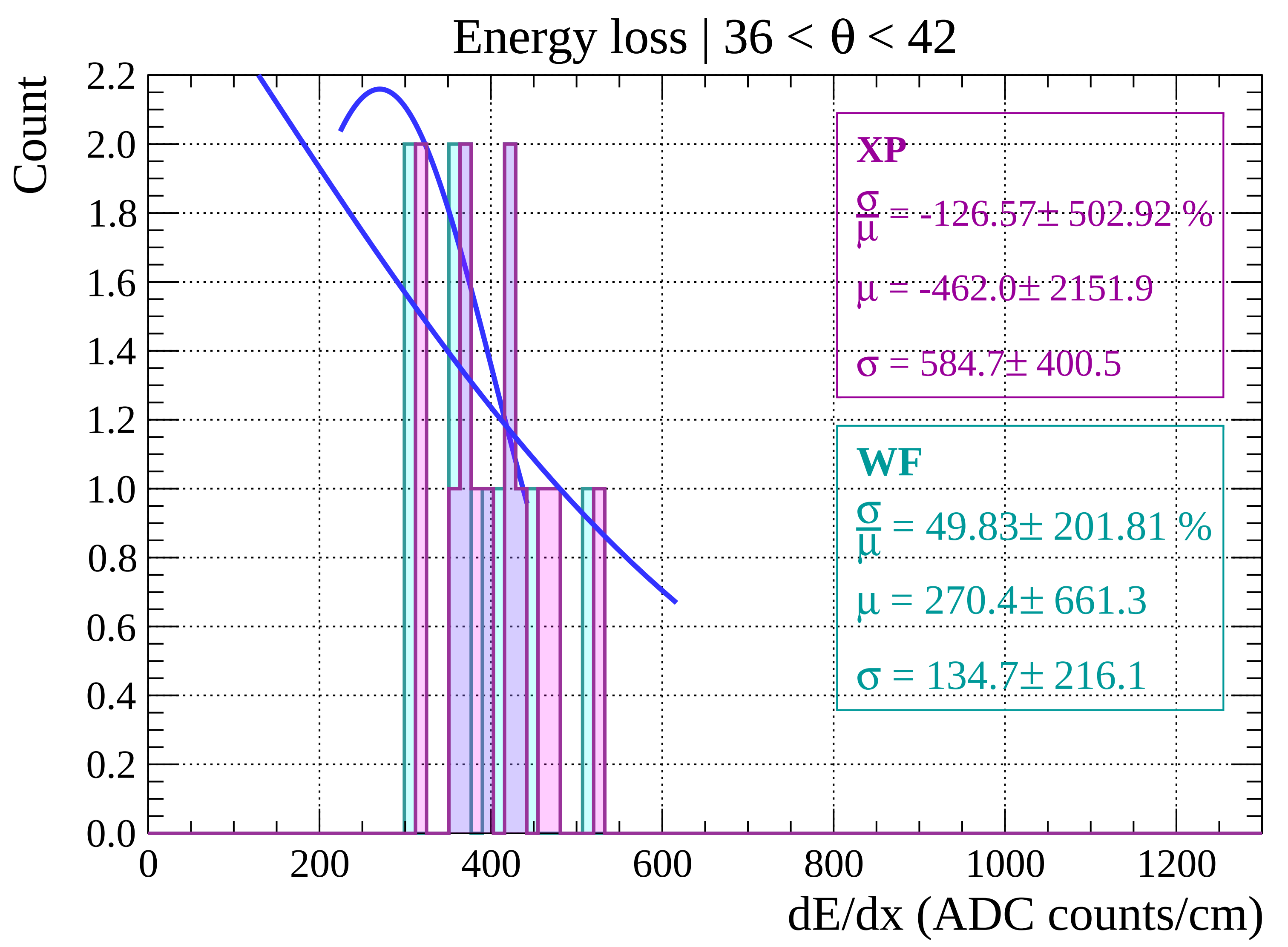


Energy loss | 18 < θ < 24
Count
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.09± 0.35 %
μ = 485.6± 1.6
σ = 44.2± 1.6
WF
$\frac{\sigma}{\mu}$ = 9.19± 0.35 %
μ = 475.2± 1.5
σ = 43.7± 1.5

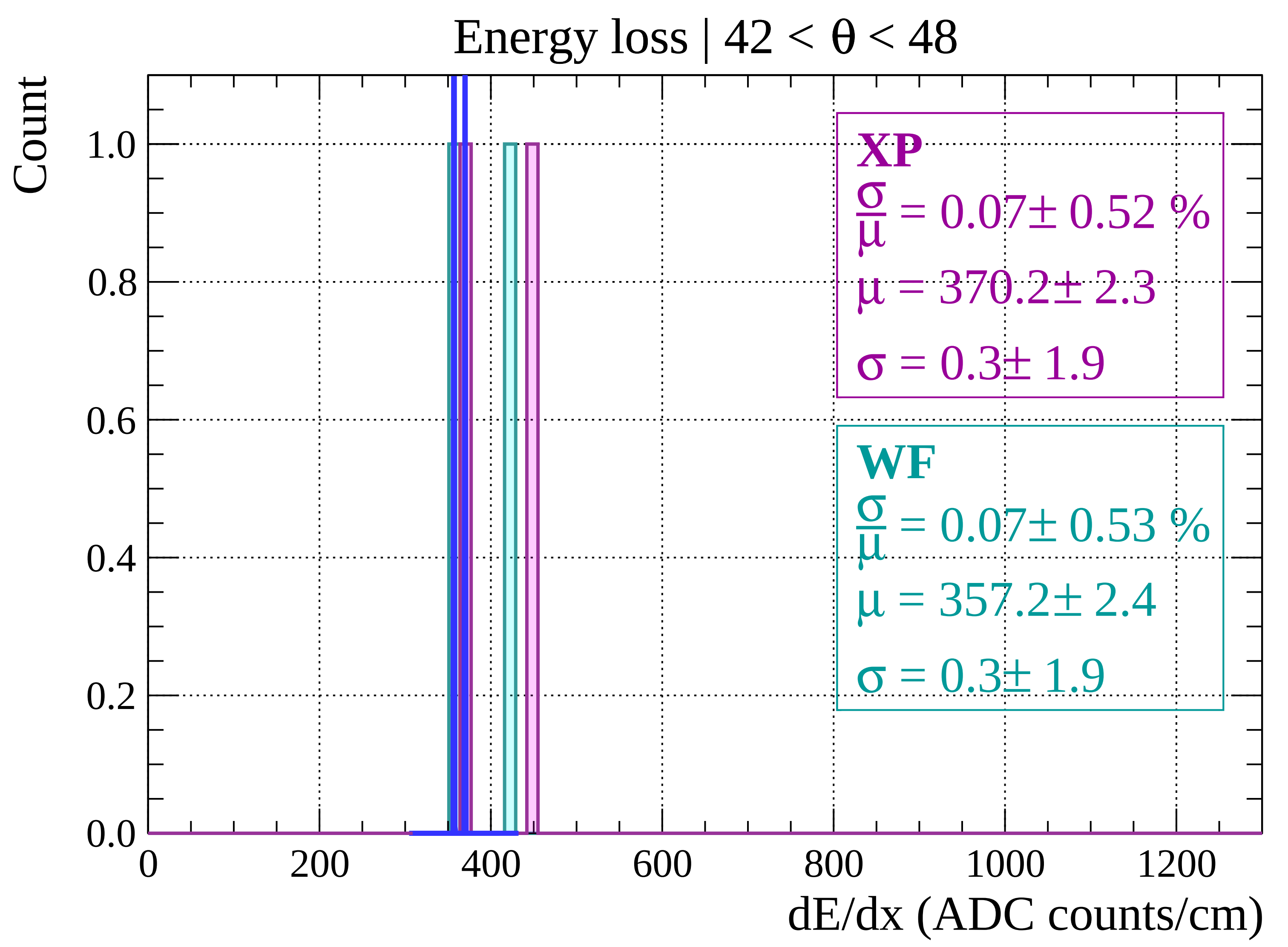

Energy loss | 24 < θ < 30
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 6.31± 0.65 %
μ = 472.6± 3.0
σ = 29.8± 2.9
WF
σ/μ = 8.87± 0.71 %
μ = 460.3± 3.4
σ = 40.8± 3.0

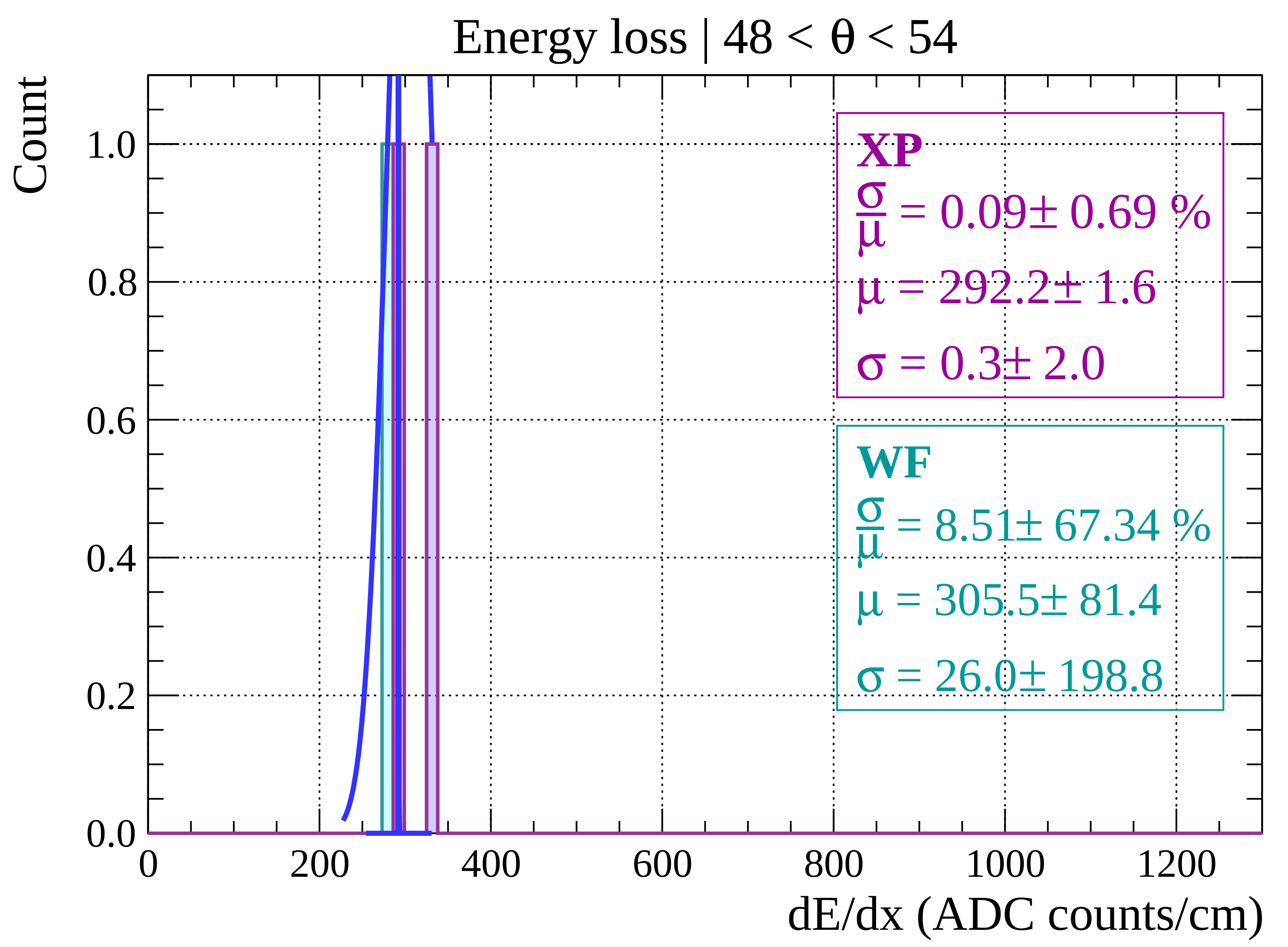


Energy loss | 30 < θ < 36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 14.20± 6.21 %
μ = 420.7± 20.9
σ = 59.7± 23.1
WF
σ/μ = 9.24± 2.43 %
μ = 440.9± 8.9
σ = 40.7± 9.9

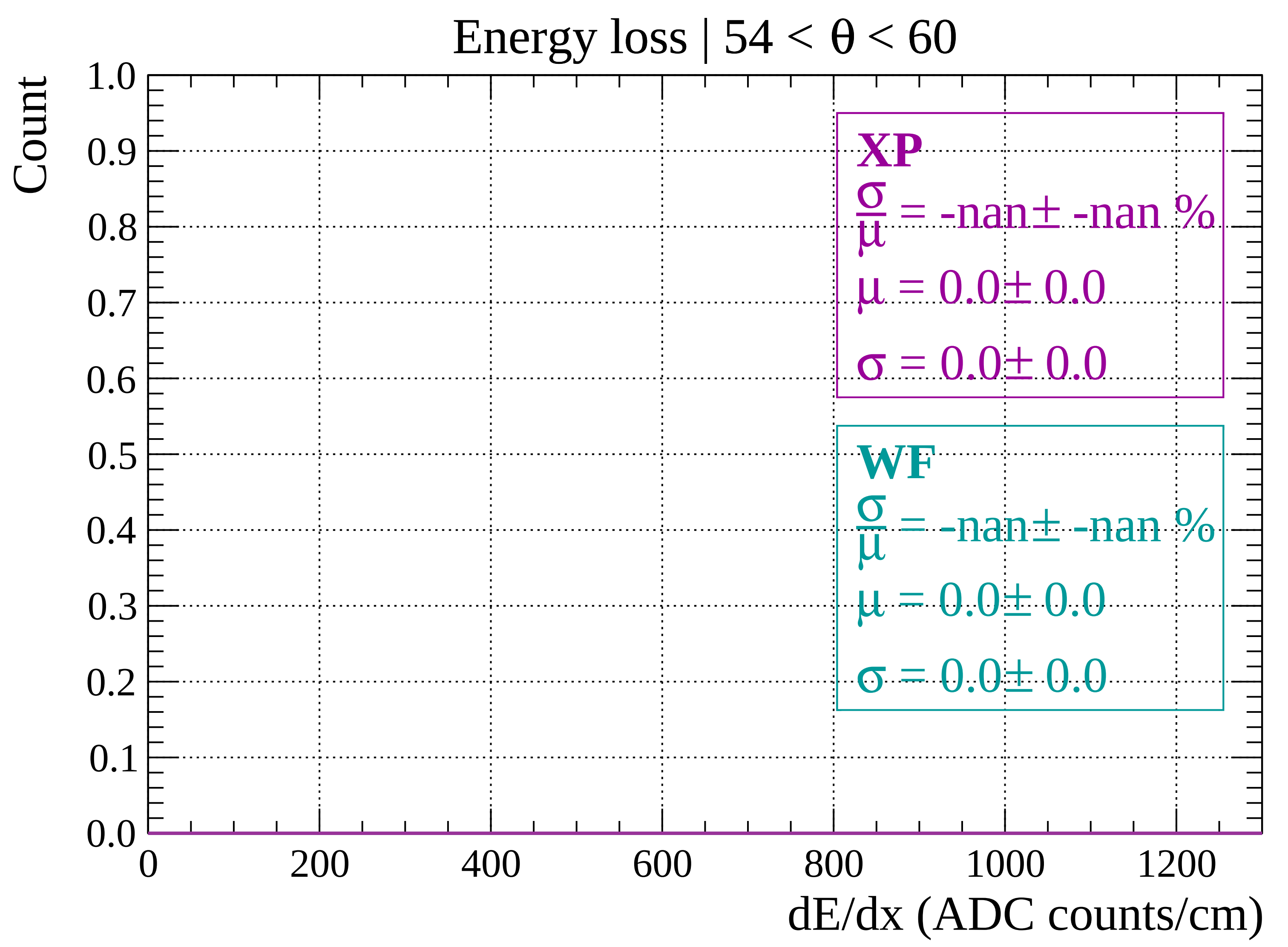


Energy loss | 36 < θ < 42
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = -126.57± 502.92 %
μ = -462.0± 2151.9
σ = 584.7± 400.5
WF
σ/μ = 49.83± 201.81 %
μ = 270.4± 661.3
σ = 134.7± 216.1

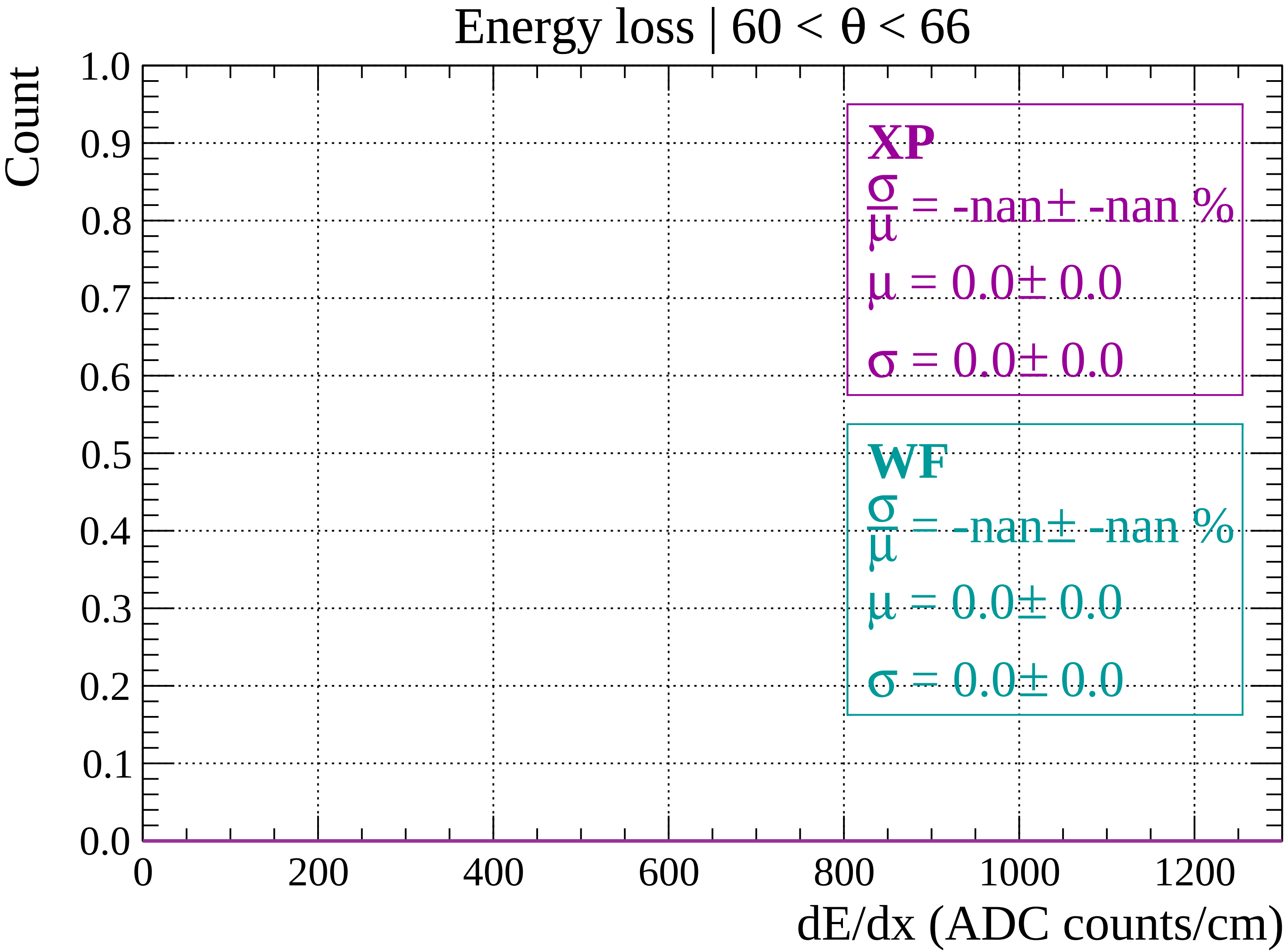

Energy loss | 42 < θ < 48
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = 0.07± 0.52 %
μ = 370.2± 2.3
σ = 0.3± 1.9
WF
σ/μ = 0.07± 0.53 %
μ = 357.2± 2.4
σ = 0.3± 1.9

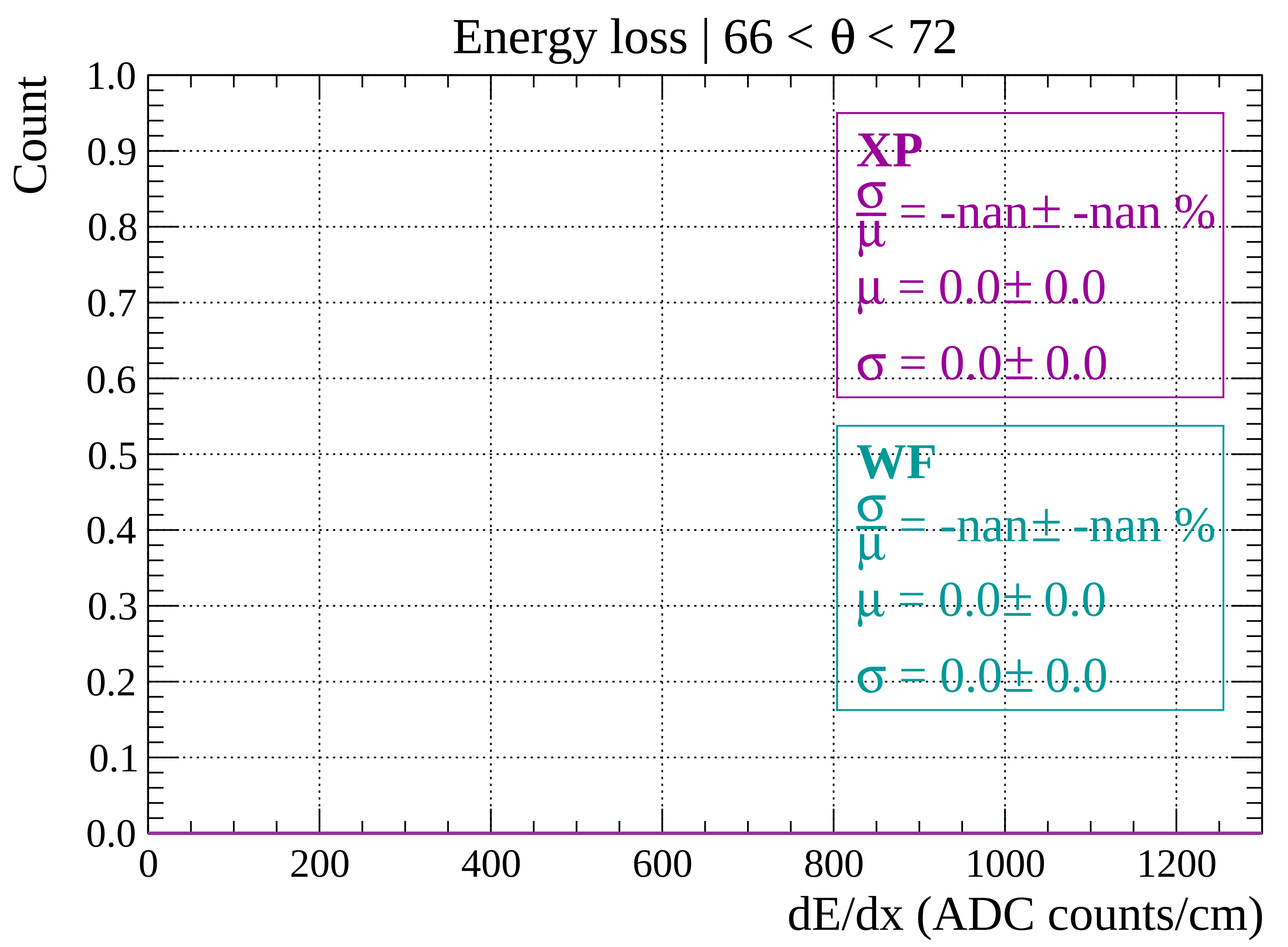


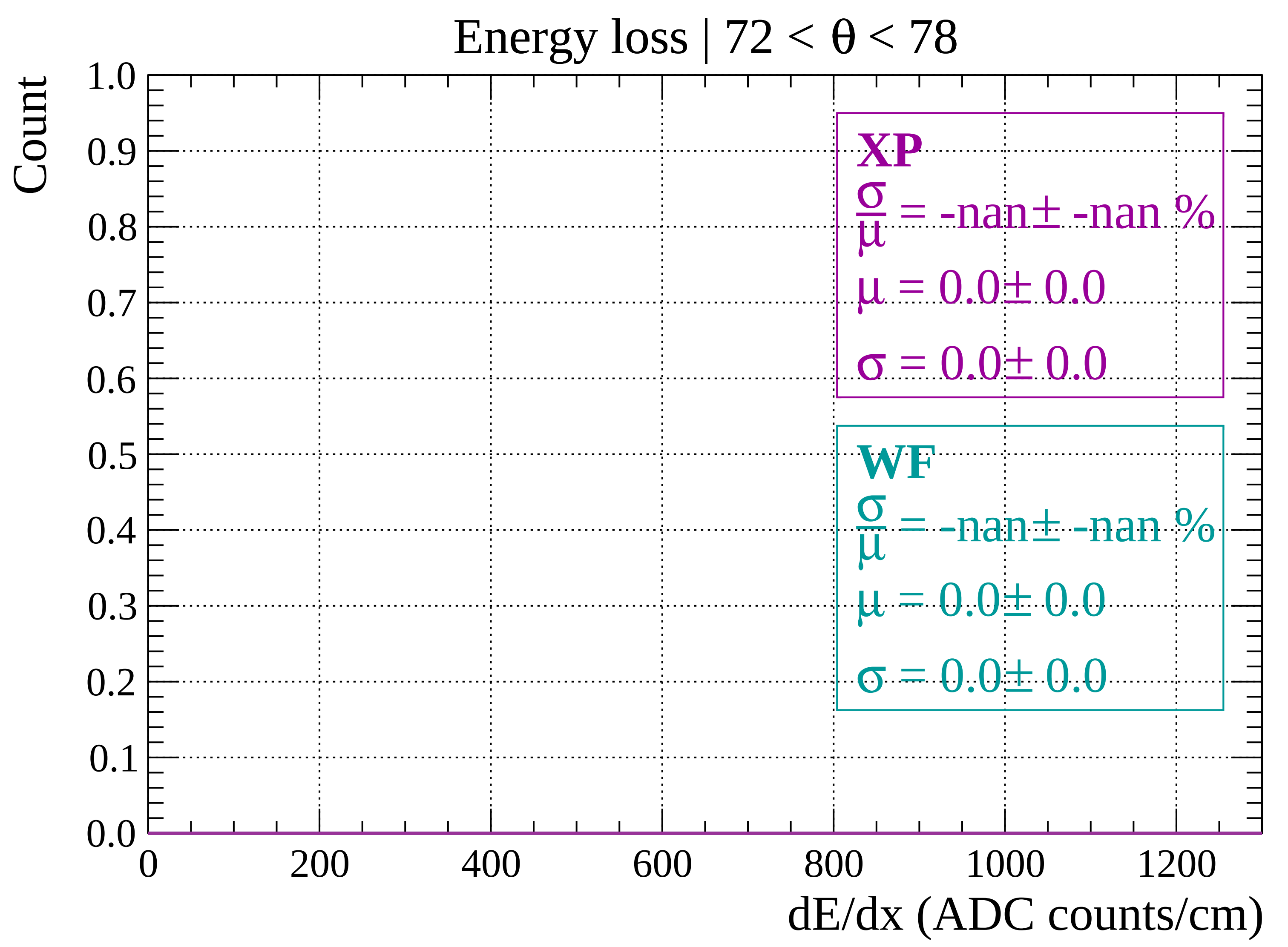
Energy loss | 48 < θ < 54
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.09± 0.69 %
μ = 292.2± 1.6
σ = 0.3± 2.0
WF
σ/μ = 8.51± 67.34 %
μ = 305.5± 81.4
σ = 26.0± 198.8

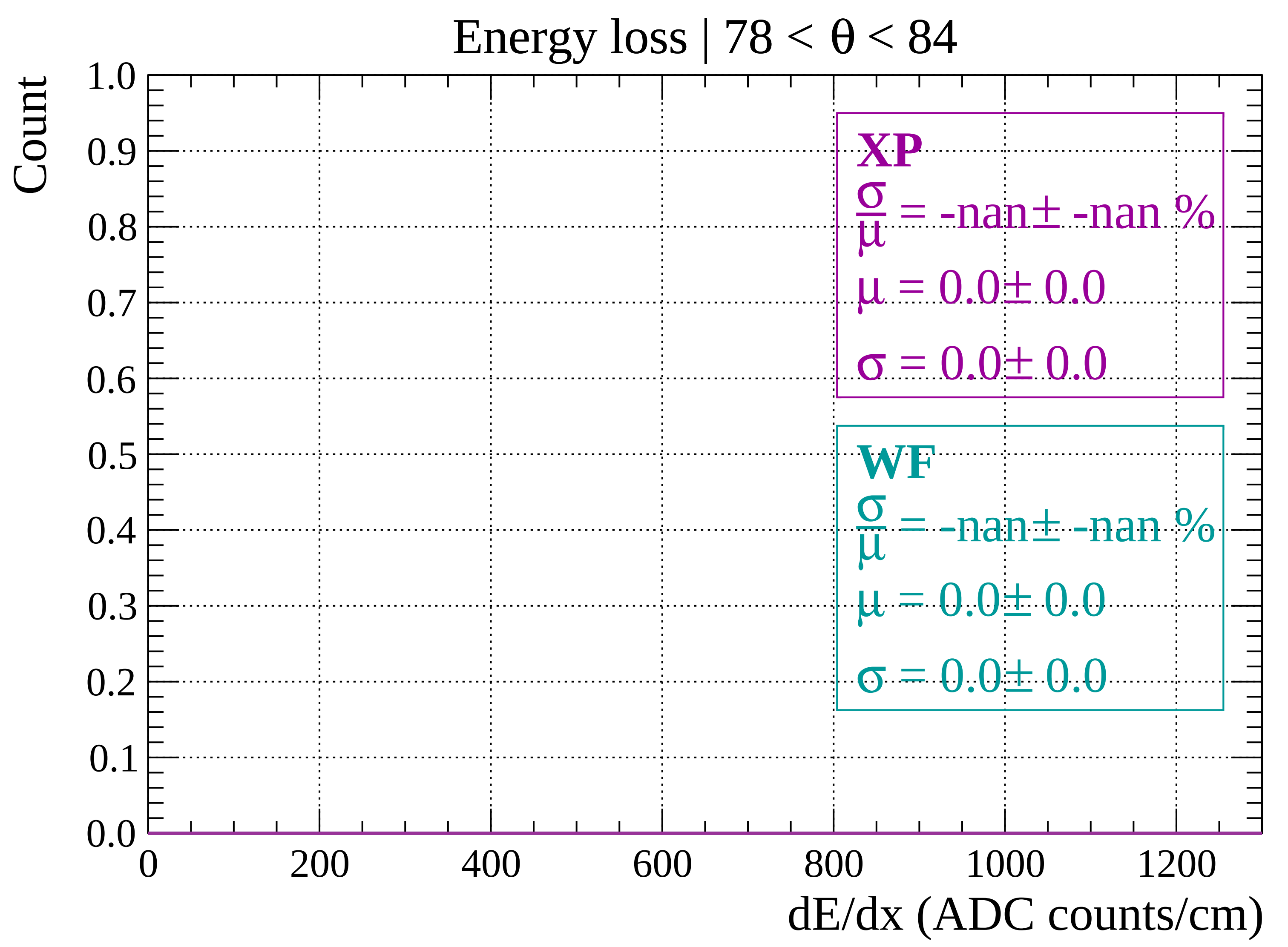


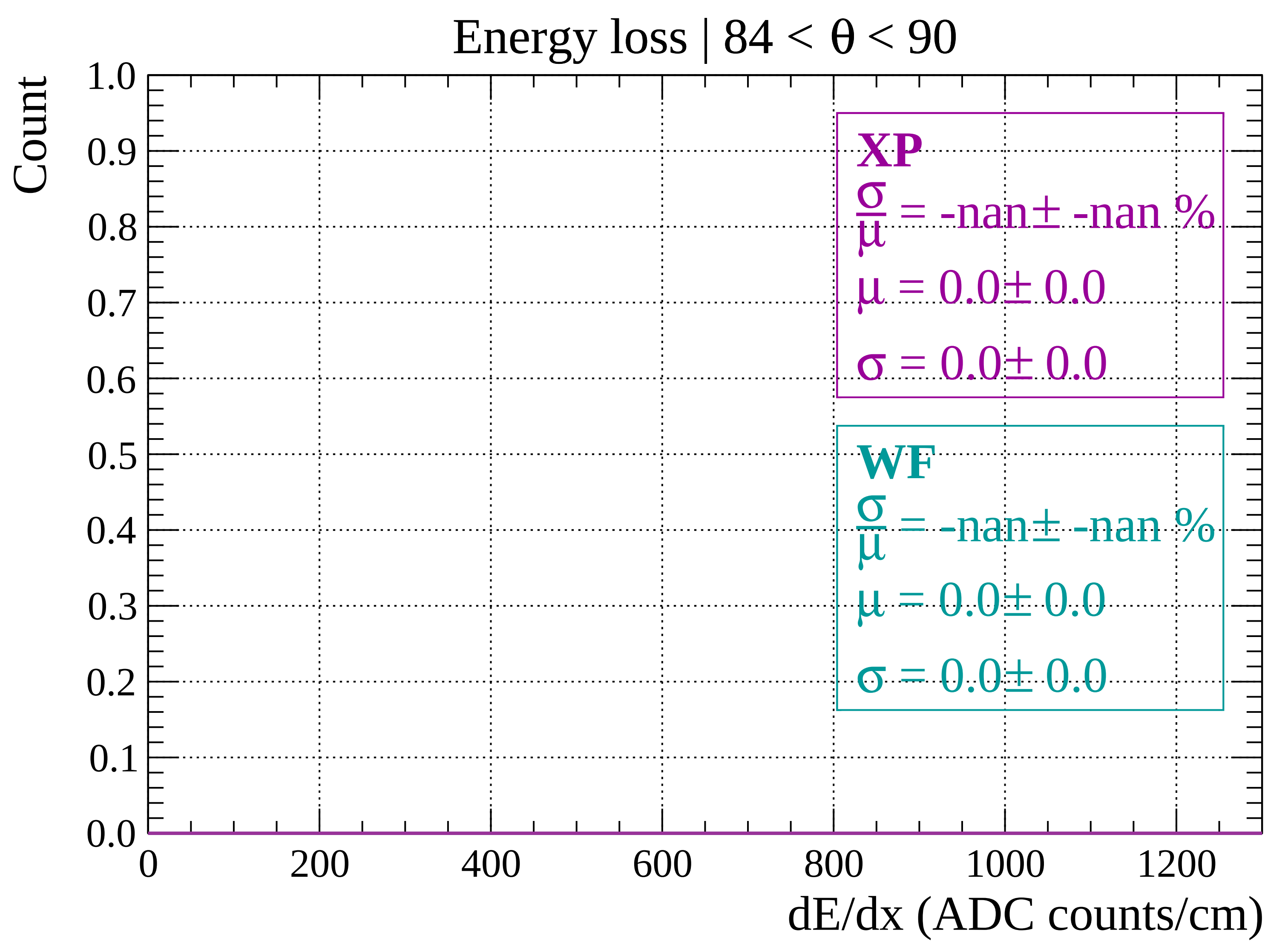
Energy loss | 54 < θ < 60
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 60 < θ < 66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

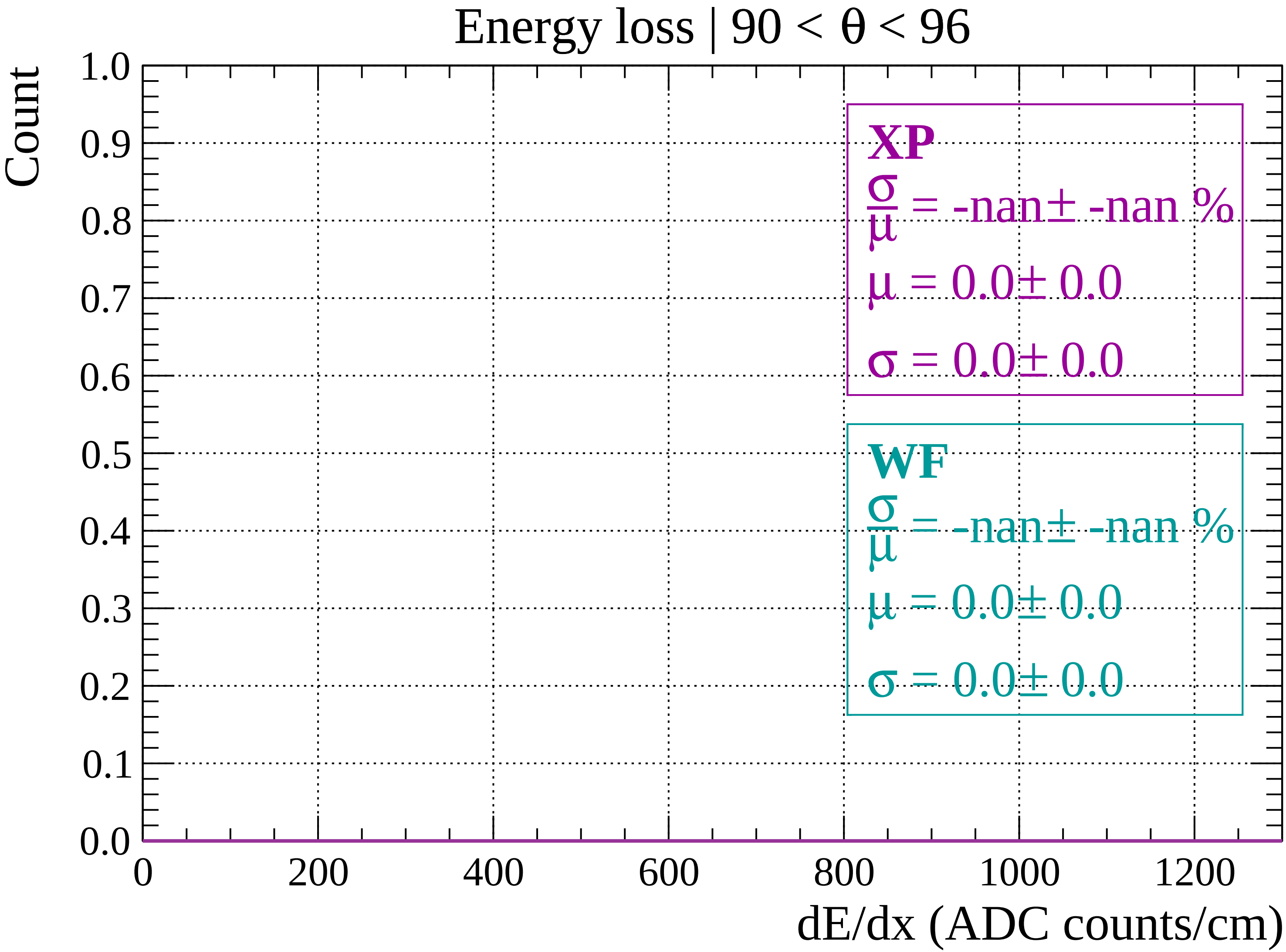

Energy loss | 66 < θ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

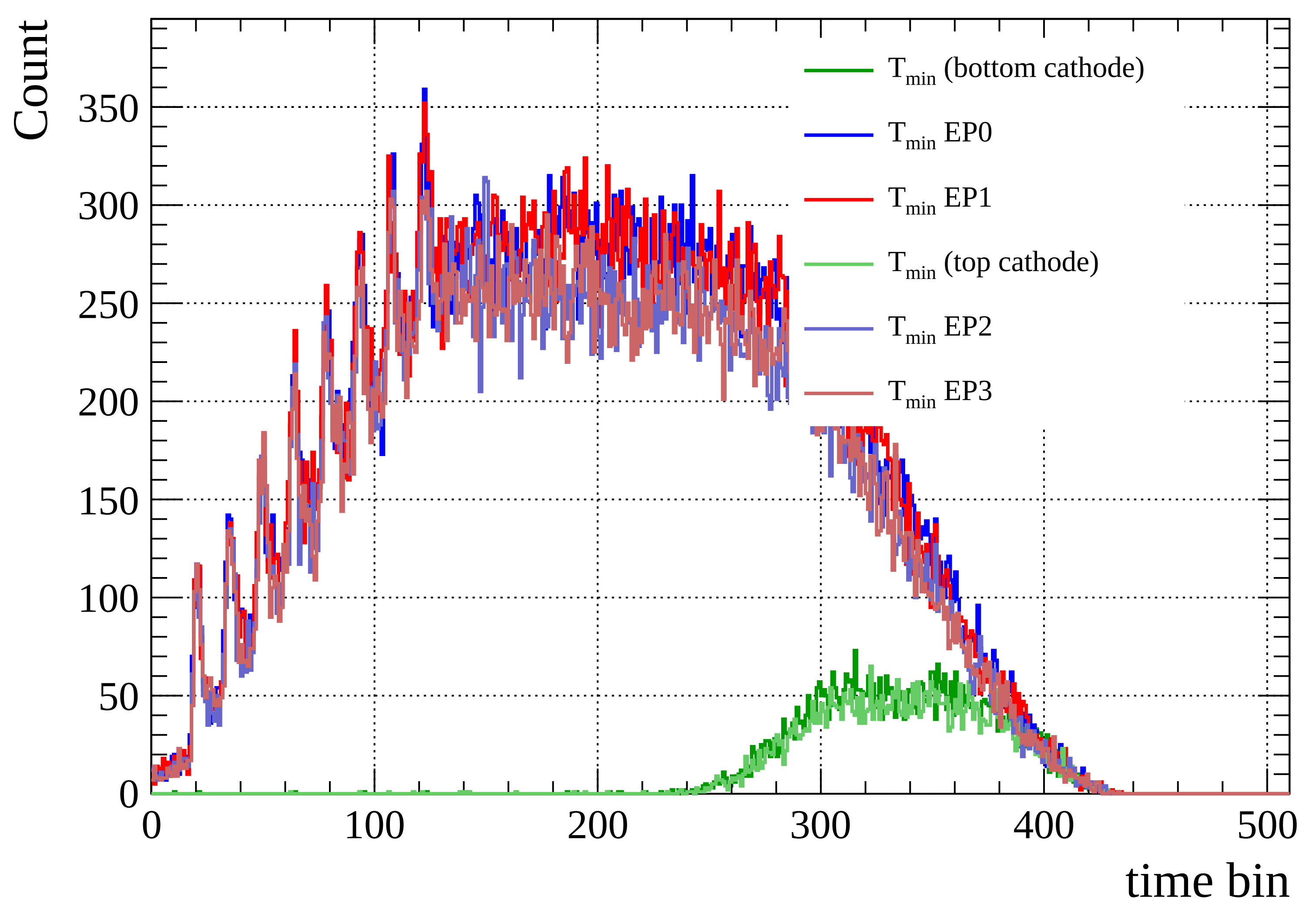

Energy loss | 72 < θ < 78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 78 < θ < 84
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

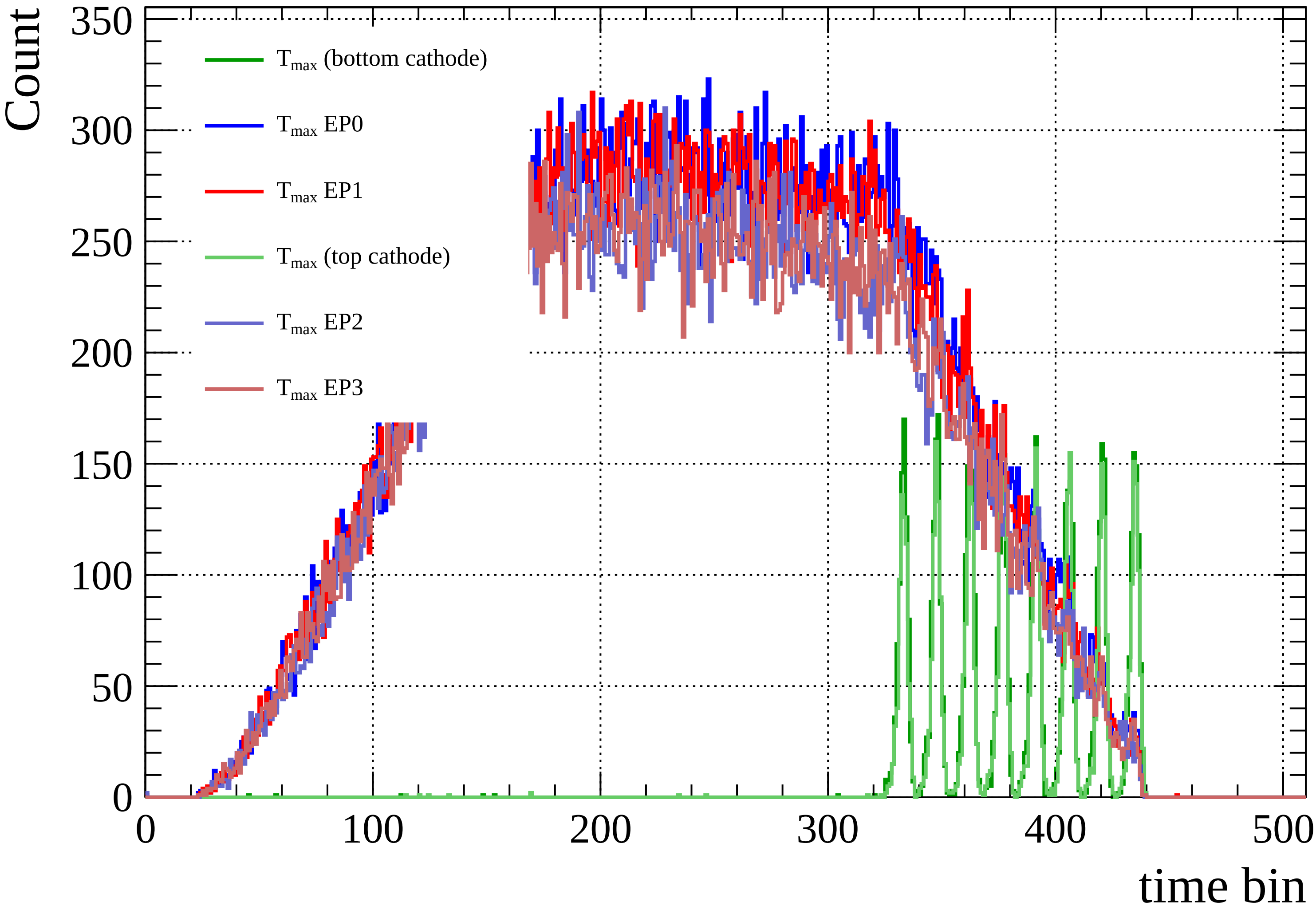


Energy loss | 84 < θ < 90
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 90 < θ < 96
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
50
100
150
200
250
300
350
0
100
200
300
400
500
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
350
300
250
200
150
100
50
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3